

CHAPTER 14

HOIST AND WINCHES

CHAPTER OVERVIEW

SECTION	TITLE
14-1	Equipment Description and Data
14-2	Troubleshooting
14-3	On Aircraft Inspections
14-4	Maintenance Procedures
14-5	Maintenance Procedures (BENCH)

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SECTION 14-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

PARAGRAPH	TITLE
14-1-1	Rescue Hoist System Description and Data HOIST, BL-27100-71 HOIST, 42315-10
14-1-2	Rescue Hoist System Description and Data HOIST BL-29900-30

14-1-1/(14-1-2 Blank)

14-1-1 RESCUE HOIST SYSTEM DESCRIPTION AND DATA

HOIST, BL-27100-71 **HOIST, 42315-10** .

14-1-1.1 System Description.

The rescue hoist is mounted outside and above the cabin door on the right side of the helicopter (Figure 14-1-1-1). The rescue hoist system consists of modular components, hydraulically driven and electrically controlled, to provide lift capacity of 600 pounds at an average speed of 250 feet per minute with a usable cable length of 200 feet. The components comprising the rescue hoist system are the rescue hoist assembly, the crewmember's hover grip, the rescue hoist/cargo hook pendant, the pilot's and copilot's cyclic sticks, the rescue hoist control box, hover trim control panel, and the auxiliary rescue hoist control panel. The control system allows for two modes of control with the pilot having the highest priority and the rescue hoist pendant having the lowest priority. The speed command from the pilot's and copilot's cyclic sticks and rescue hoist control box are single speed controls. The rescue hoist/cargo hook pendant and crewmember's hover grip control hoist speed via the deflection of the thumbwheel. All controls have automatic slow speed prior to the automatic stop at the cable extremes.

The rescue hoist cable can be controlled or sheared by using the pilot's and copilot's cyclic sticks, the crewmember's hover grip, the auxiliary rescue hoist control panel, the rescue hoist/cargo hook pendant, or the rescue hoist control box. The pilot's and copilot's cyclic sticks have two switches that can control the cable's direction or shear it when necessary. The auxiliary rescue hoist control panel, in cabin, provides hoist cable shear control when the HOIST SHEAR switch is placed to shear and the control of the rescue hoist light's brightness. The rescue hoist/cargo hook pendant, located in the cabin, operates with the rescue hoist control box and has a thumbwheel potentiometer for controlling the variable speeds of the rescue hoist cable. The rescue hoist control box, located in the cabin, is an additional control and indicator for rescue hoist operation.

14-1-1.2 Principles Of Operation.

14-1-1.2.1 Power Distribution.

The rescue hoist system requires 115 vac, 3-phase 400 Hz, 5 amp main power; 28 vdc 5 amp control power; and a separate 28 vdc 5 amp input for cable cutter power (Figure 14-1-1-2).

14-1-1.3 Component Description.

14-1-1.3.1 Rescue Hoist Assembly.

The rescue hoist assembly operates with the pilot's hoist control panel or the rescue hoist pendant in this order of precedence. The assembly includes an electric motor, a hoist, a controller section, and aerodynamic fairings. The hoist assembly contains a rotating drum, which is driven by the electric motor through a gear train and a clutch assembly to raise and lower a 290 foot cable terminated in a swivel hook. A brake assembly inside the drum locks the drum when input power is cut off or the load tries to overhaul the motor, and the clutch assembly allows slippage if the load on the hook exceeds 1800 pounds. A levelwind mechanism on the hoist assembly ensures even winding of the cable on the drum. Limit switches sense when the cable approaches the full-in or full-out position, and when the cable reaches these positions. A potentiometer coupled to the drum continuously senses cable position for cable pay out indication.

The hoist uses a tension roller located between the cable guide and the drum. The tension roller is gear driven by the drum through an over-running clutch such that when reeling out, the tension roller is rotating faster than the tangential speed of the drum, thereby maintaining a "pull" on the cable at no load. In like manner, the tension roller rotates slower than the tangential speed of the drum when reeling in, thus providing drag on the cable which ensures tight cable storage on the drum under all conditions.

The cable assembly for the rescue hoist consists of a wire rope with a swaged stainless steel ball/shank fitting. The wire rope is 3/16 inch diameter, corrosion resistant, preformed spin resistant type. The drum end of the cable assembly is cut off by an inert gas shielded electric arc process such that the cable end is sealed and made suitable for attachment at the drum anchor point. The swivel hook assembly prevents inadvertent load release and is operable with one hand.

The swivel hook assembly incorporates a lightweight, high-strength honeycomb aluminum, crushable bumper assembly. During normal hoist operations after the slow in limit switch has actuated, the crushable bumper triggers the "full-in" limit switch plunger to stop the hoist. In the event of a failure of the slow-in limit signal, the crushable bumper will "absorb" all the rotational energy to safely stop the hoist at the "full-in" position. The rotational energy at full speed is capable of producing loads in excess of the cable rating. This provides a last line safety device to minimize to the greatest extent possible a hoist cable breakage due to a single point failure. A crushed bumper also provides the operator a visual indication that a failure has occurred.

The cable cutter assembly consists of a pressure cartridge, guillotine and anvil, all of which are contained in the cable guide housing. The specific minimum "all fire" current of the pressure cartridge is five amps per bridge. This cartridge has the highest EMI rejection with a guaranteed "no fire" rating of one watt.

14-1-1.3.2 Crewmember's Hover Grip.

The crewmember's hover grip allows a continuously variable hoist speed from zero to a full speed of 350 feet per minute. The rescue hoist assembly incorporates four guide rollers to guide the cable into the hoist at large fleet angles due to either high winds (30 m/s) or aircraft movement. The bell mouth swivels to follow the angle of the cable at that condition.

14-1-1.3.3 Rescue Hoist/Cargo Hook Pendant.

The rescue hoist/cargo hook pendant allows crew to select the up or down direction of the hoist cable by using the thumbwheel. The rescue hoist/cargo hook pendant operates with the rescue hoist control box and has a thumbwheel potentiometer for controlling the rescue hoist at variable speeds from the cabin. Cable speed is directly proportional to the thumbwheel deflection.

14-1-1.3.4 Pilot's and Copilot's Cyclic Sticks.

Two switches, located on the pilot's and copilot's cyclic sticks, allow the pilot or copilot to completely override any hoist command and operate the hoist at a fixed speed. Provisions for cutting the hoist cable are provided with a guarded switch to avoid accidental firing of the cable cutter.

14-1-1.3.5 Rescue Hoist Control Box.

The rescue hoist includes a rescue hoist control box. The rescue hoist control box has a BACKUP CONTROL POWER switch that allows the hoist operator to select backup electrical power and operate the rescue hoist using the HOIST UP/DOWN switch. Two indicator lights indicate when normal power is selected and hydraulic fluid temperature is within limits. The indicators are labelled NORMAL POWER and HYD OVERHEAT.

14-1-1.3.6 Auxiliary Rescue Hoist Control Panel.

The auxiliary rescue hoist control panel has a HOIST SHEAR guarded switch which can shear the rescue hoist cable. Also included is a RESCUE LIGHT ON/OFF/DIM switch and OFF/BRT variable control.

14-1-1.3.7 Hover Trim Control Panel

The SAFE/EMER/SHEAR on the hover trim control panel provides cable shear from the cabin.

14-1-1.3.8 Rescue Hoist Light.

The rescue hoist light provides variable intensity, IR light illumination for use with the rescue hoist (Figure 14-1-1-3). The rescue hoist light is mounted on the front of the rescue hoist support. The rescue hoist light is controlled by the cabin auxiliary rescue hoist control panel and relays in the hoist relay panel. Electrical power for the rescue hoist light is supplied by three circuit breakers on the mission readiness circuit breaker panel. The RESQ HST LT DIM circuit breaker applies 115 vac from the No. 1 ac primary bus to the primary winding of the RESCUE LIGHT OFF/BRT control, on the auxiliary rescue hoist control

panel, to vary the brightness of the rescue hoist light. The RESQ HST LT CONT circuit breaker applies 28 vdc from the No. 1 dc primary bus to the RESCUE LIGHT ON/OFF/DIM switch. The RESQ HST LT PWR circuit breaker applies 28 vdc from the No. 1 dc primary bus to the normally open contacts of relay K105 in the hoist relay panel. The normally open contacts of relay K107 in the hoist relay panel receive 0 to 115 vac variable voltage from the RESCUE LIGHT OFF/BRT control. When the RESCUE LIGHT switch is placed to the DIM position, the first relay de-energizes and the second relay is energized to supply the variable 0 to 115 vac voltage from the RESCUE LIGHT OFF/BRT control to the rescue hoist light.

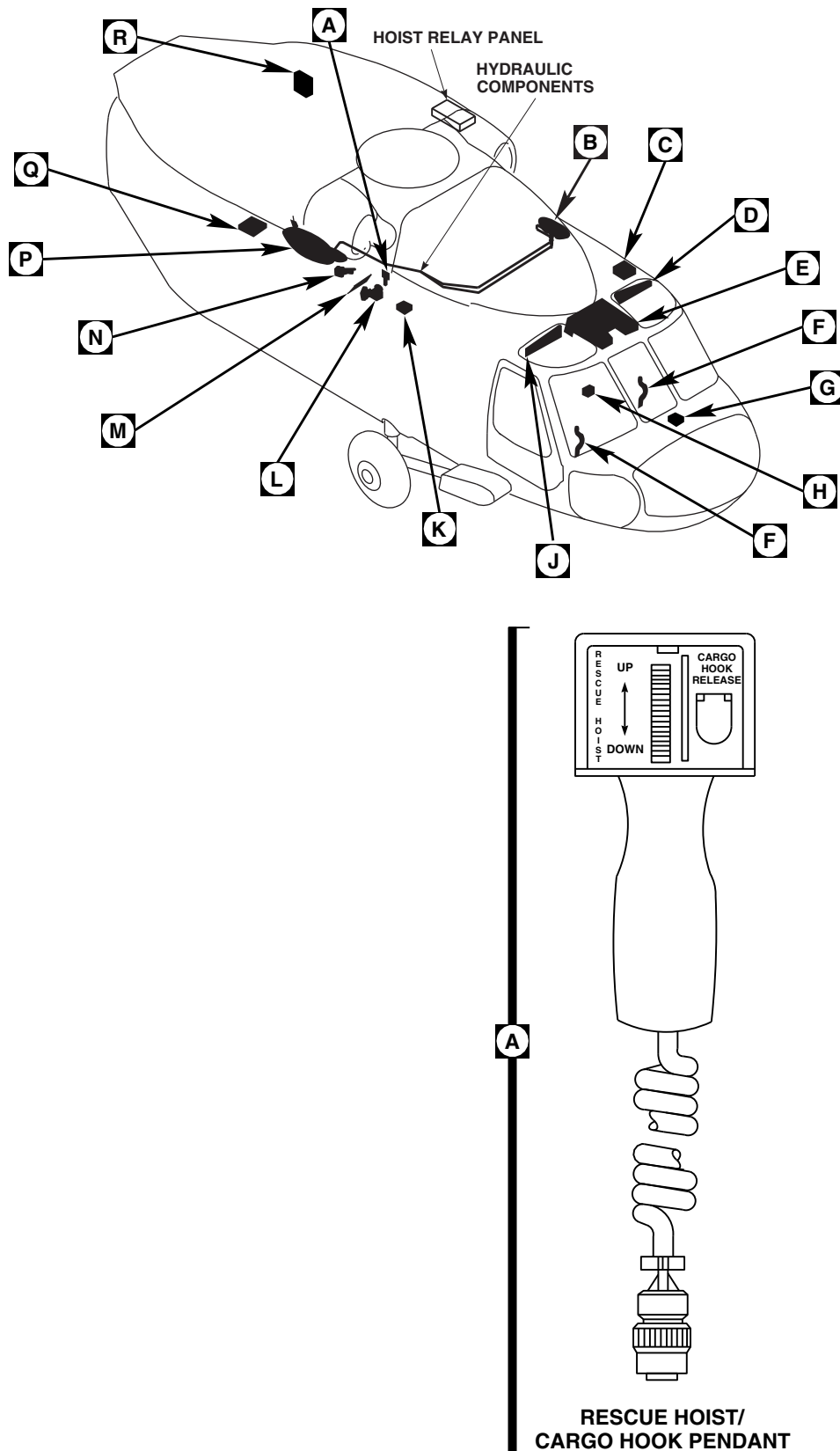


FIGURE 14-1-1-1. RESCUE HOIST SYSTEM LOCATION DIAGRAM (SHEET 1 OF 7)

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TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
J267	CARGO HOOK DISCONNECT
J452	HOVER / TRIM PANEL
P3 / J3	NO. 7 RELAY PANEL ASSEMBLY
P4 / J4	RESCUE HOIST ASSEMBLY
P5 / J5	RESCUE HOIST ASSEMBLY
P6 / J6	RESCUE HOIST ASSEMBLY
P7 / J7	RESCUR HOIST ASSEMBLY
P8 / J8	RESCUE HOIST ASSEMBLY
P9 / J9	RESCUE HOIST ASSEMBLY
P25R	INTERCOMMUNICATION SET SYSTEM
P112 / J112	DISCONNECT
P114 / J114	DISCONNECT
P118 / J118	CAUTION / ADVISORY SYSTEM
P119 / J119	COCKPIT TUB, BL 30 RH, STA 220
P120 / J120	COCKPIT TUB, BL 30 RH, STA 220
P126 / J126	UPPER CONSOLE DISCONNECT
P154R	CREWMEMBER'S HOVER GRIP
P200 / J200	UPPER CONSOLE DISCONNECT
P217 / J217	LEFT MAIN DISCONNECT
P224 / J224	MAIN ROTOR PYLON DECK, BL 4 RH, STA 247
P231 / J231	UPPER CONSOLE DISCONNECT
P236 / J236	DISCONNECT
P246 / J246	UPPER CONSOLE DISCONNECT
P247 / J247	COCKPIT, BL 40 RH, STA 249
P248 / J248	COCKPIT, BL 40 LH, STA 248
P249 / J249	COPILOT'S CIRCUIT BREAKER PANEL

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P262 / J262	DISCONNECT
P263 / J263	DISCONNECT
P264 / J264	DISCONNECT
P352 / J352	AUXILIARY RESCUE HOIST CONTROL PANEL
P353 / J353	HOIST RELAY PANEL
P357 / J357	HOIST RELAY PANEL
P452 / J452	HOVER TRIM PANEL
P902	DISCONNECT
P907 / J907	LEFT MAIN DISCONNECT
P909 / J909	LEFT MAIN DISCONNECT
P922 / J922	DISCONNECT
P962 / J2	RESCUE HOIST ASSEMBLY
P963 / J1	RESCUE HOIST ASSEMBLY
P964 / J3	RESCUE HOIST ASSEMBLY
P965 / J1	RESCUE HOIST CONTROL BOX
P966 / J2	RESCUE HOIST CONTROL BOX
P967 / J967	OVER-TEMPERATURE SW, BL 30 RH, STA 340
P968 / J968	RESCUE HOIST / CARGO HOOK PENDANT
P969 / J969	RESCUE HOIST DISCONNECT, BL 42 RH, STA 339
TB1	RESCUE HOIST ASSEMBLY

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FIGURE 14-1-1-1. RESCUE HOIST SYSTEM LOCATION DIAGRAM (SHEET 2 OF 7)

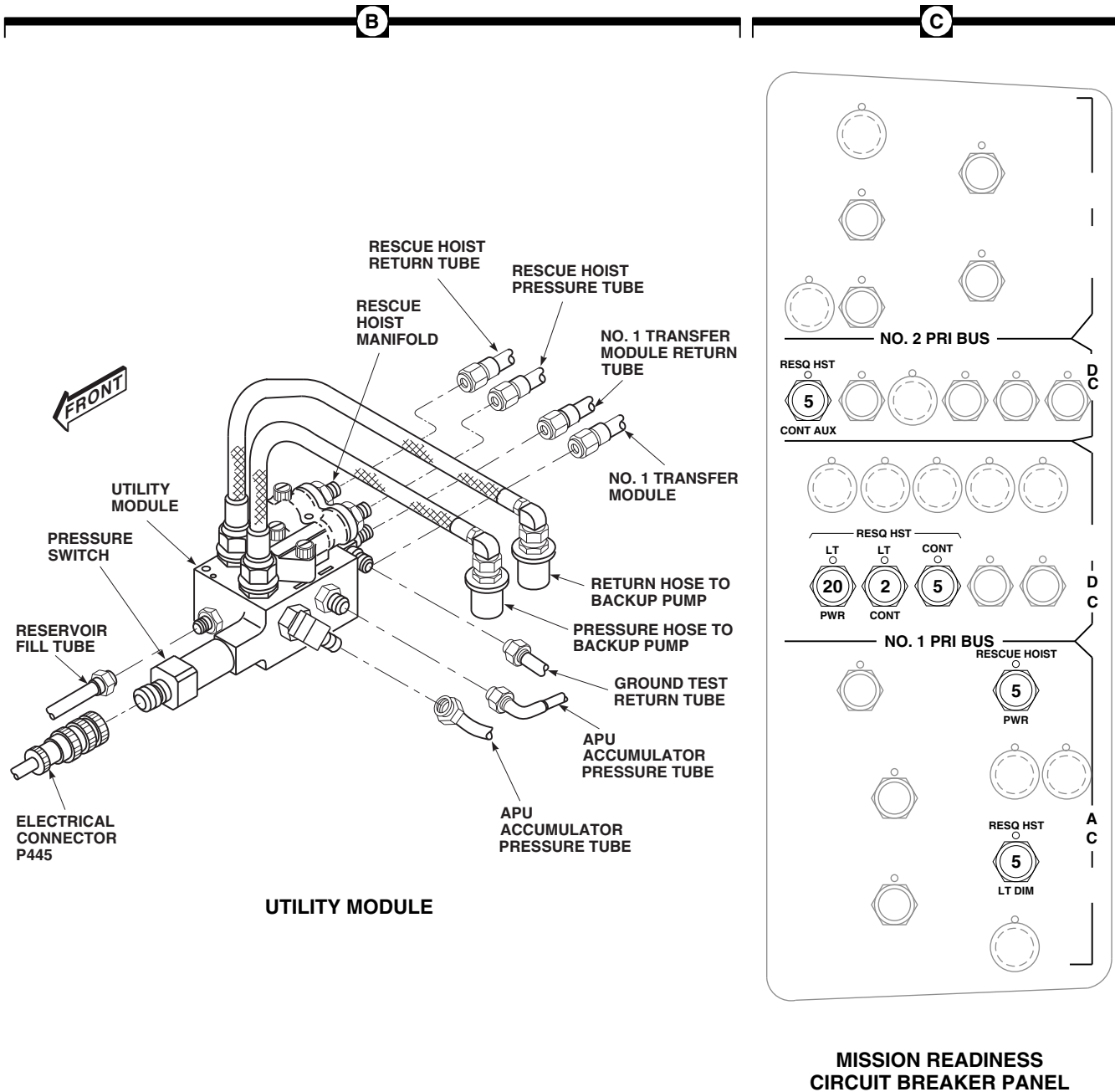
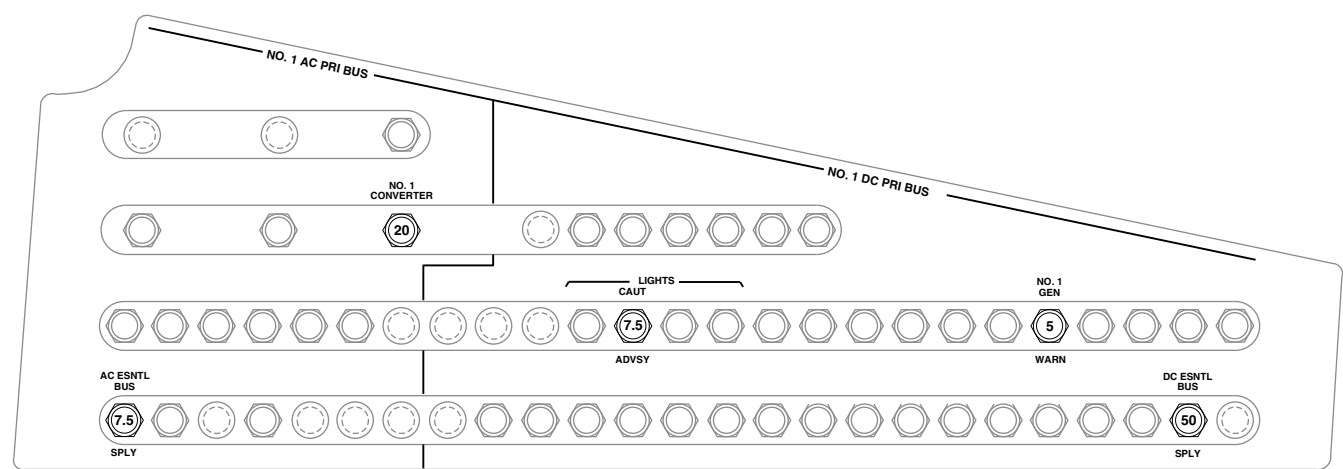


FIGURE 14-1-1-1. RESCUE HOIST SYSTEM LOCATION DIAGRAM (SHEET 3 OF 7)

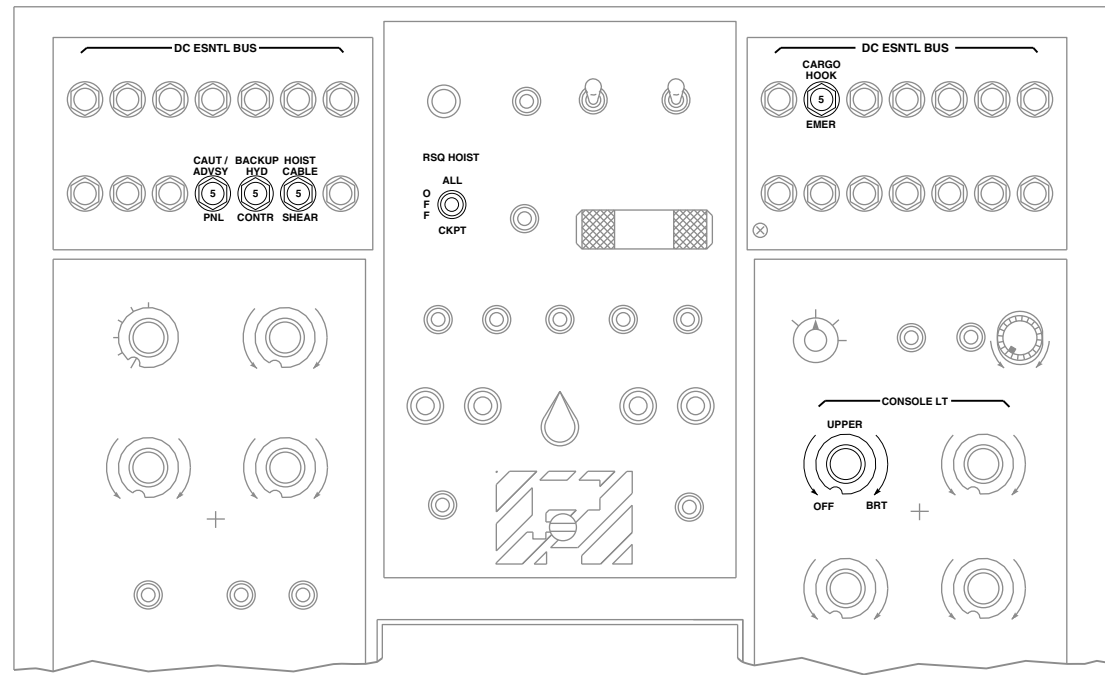
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COPILOT'S CIRCUIT BREAKER PANEL

D

E



UPPER CONSOLE

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FIGURE 14-1-1-1. RESCUE HOIST SYSTEM LOCATION DIAGRAM (SHEET 4 OF 7)

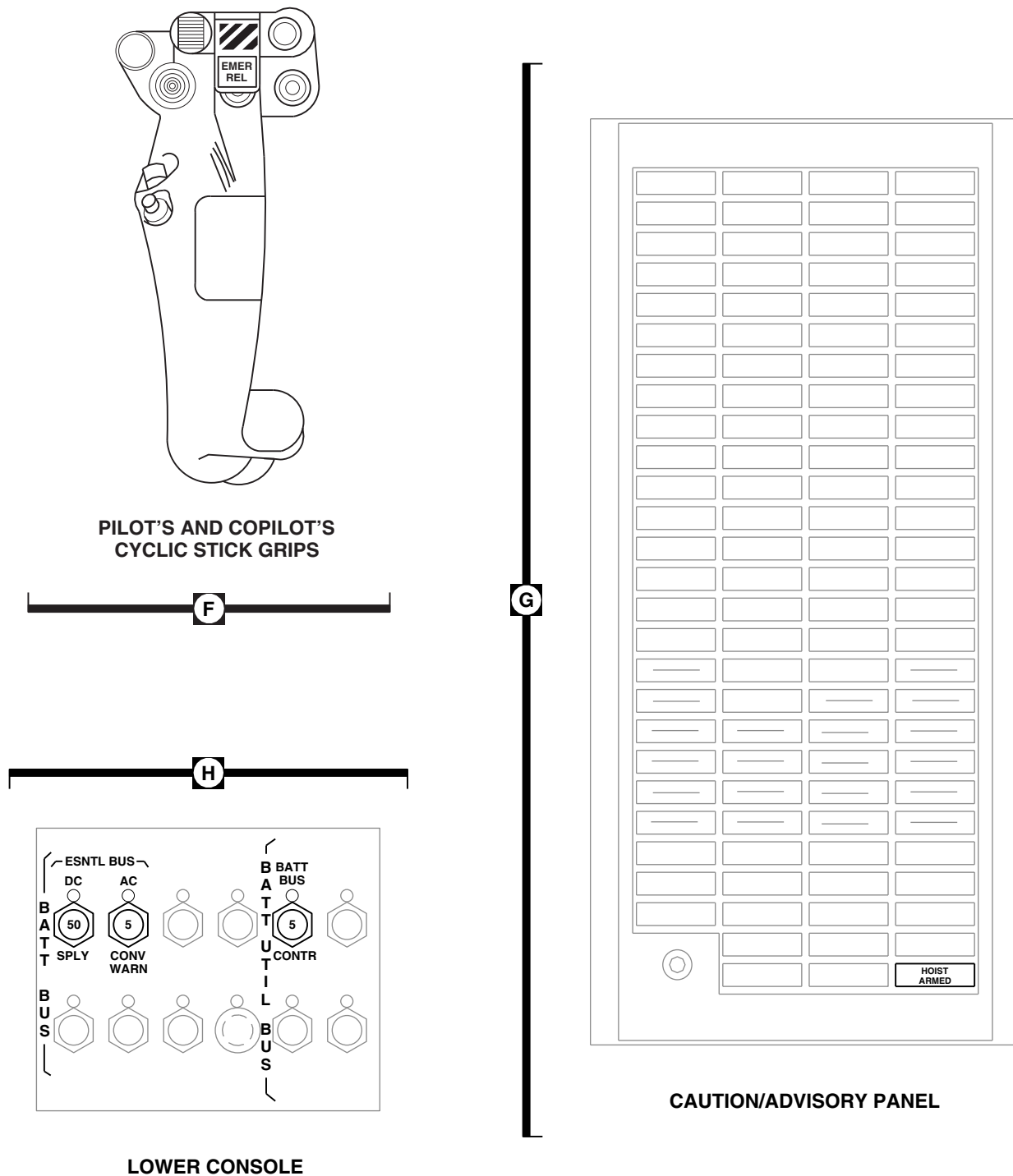
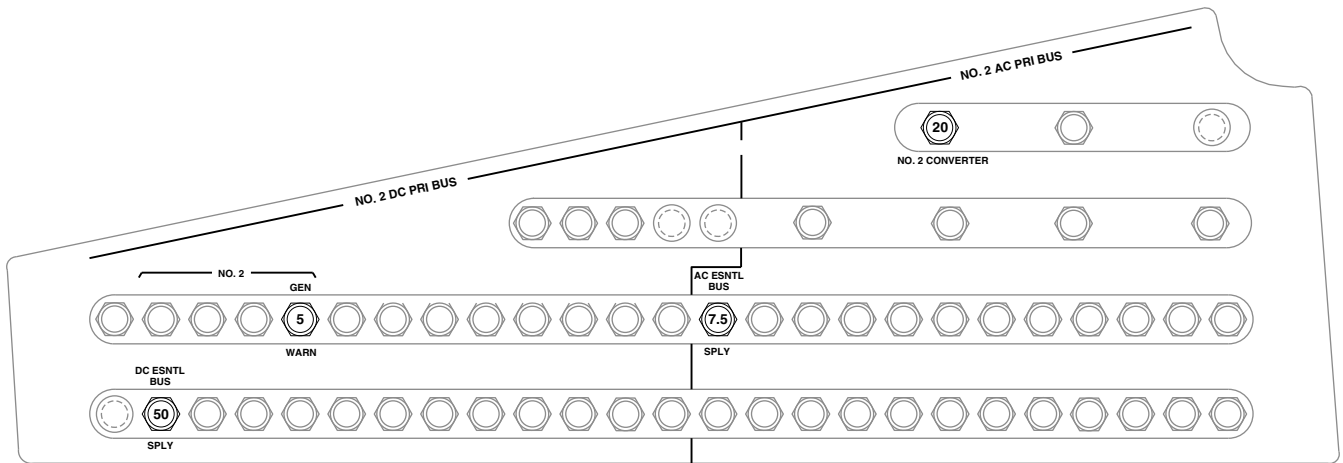


FIGURE 14-1-1-1. RESCUE HOIST SYSTEM LOCATION DIAGRAM (SHEET 5 OF 7)

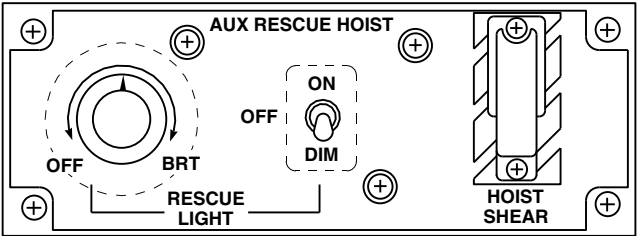
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PILOT'S CIRCUIT BREAKER PANEL

J

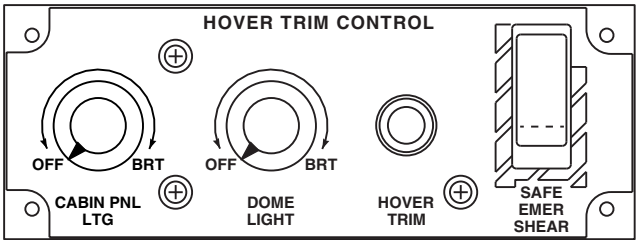
K



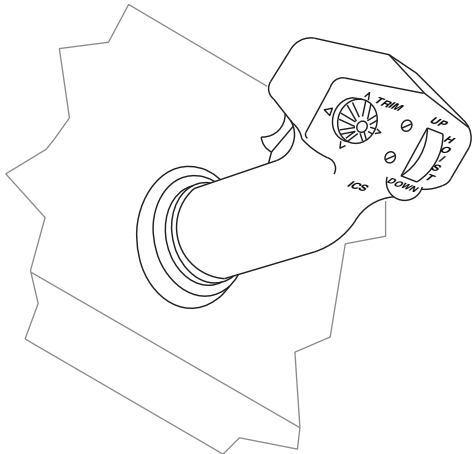
AUXILIARY RESCUE HOIST CONTROL PANEL

M

L



HOVER TRIM CONTROL PANEL



CREWMEMBER'S HOVER GRIP

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FIGURE 14-1-1-1. RESCUE HOIST SYSTEM LOCATION DIAGRAM (SHEET 6 OF 7)

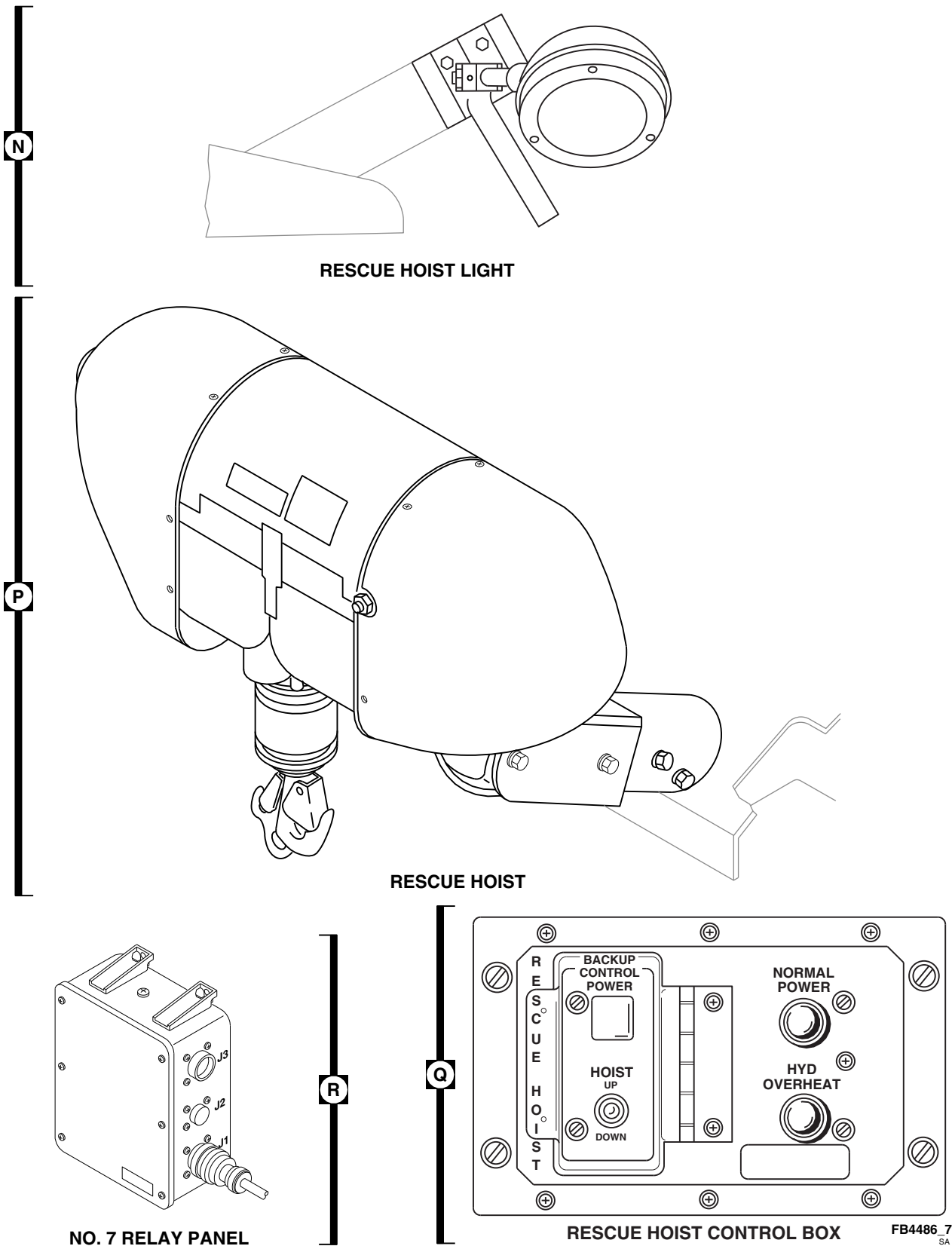


FIGURE 14-1-1-1. RESCUE HOIST SYSTEM LOCATION DIAGRAM (SHEET 7 OF 7)

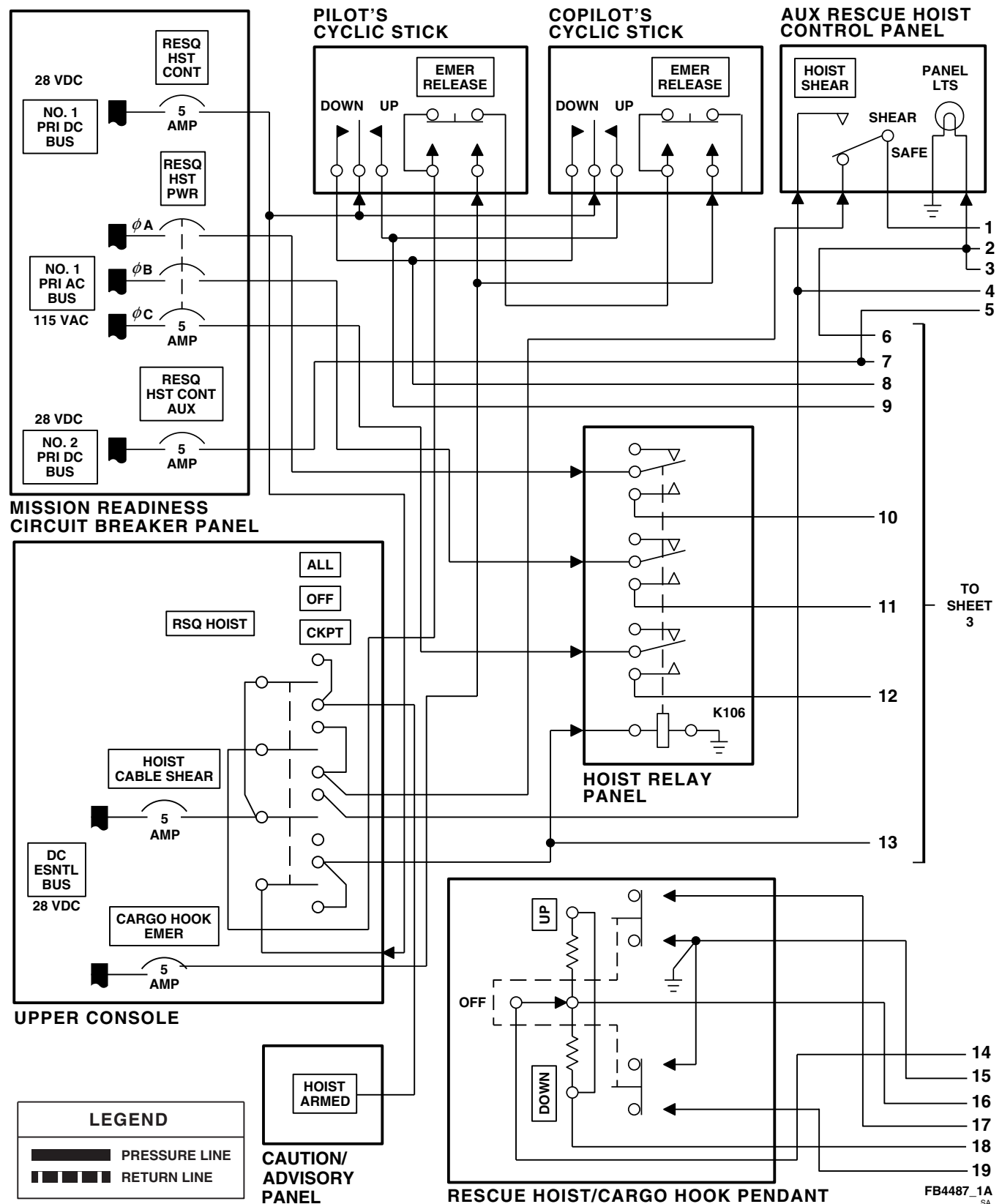


FIGURE 14-1-1-2. RESCUE HOIST SYSTEM BLOCK DIAGRAM (SHEET 1 OF 3)

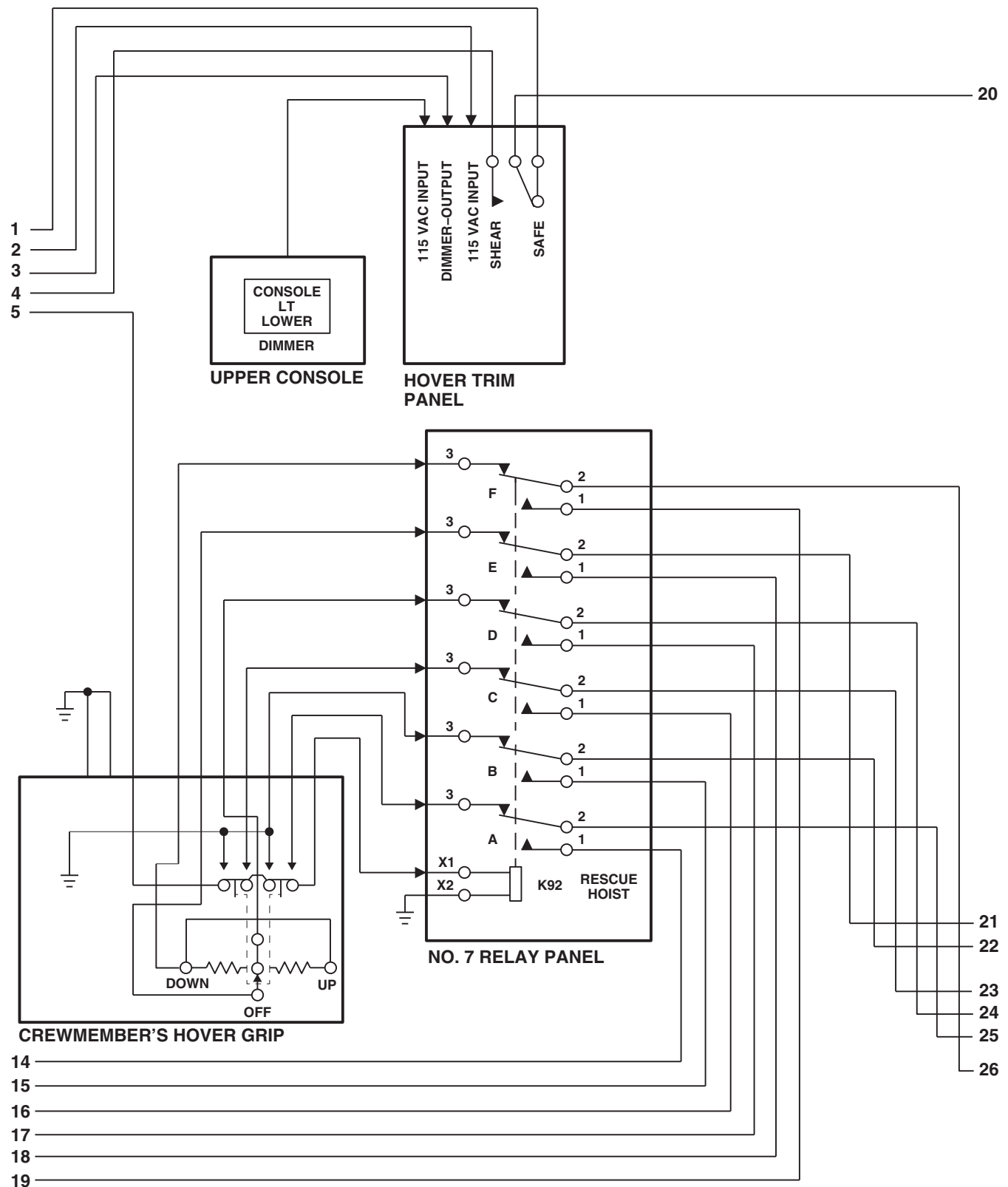
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FIGURE 14-1-1-2. RESCUE HOIST SYSTEM BLOCK DIAGRAM (SHEET 2 OF 3)

14-1-1-12

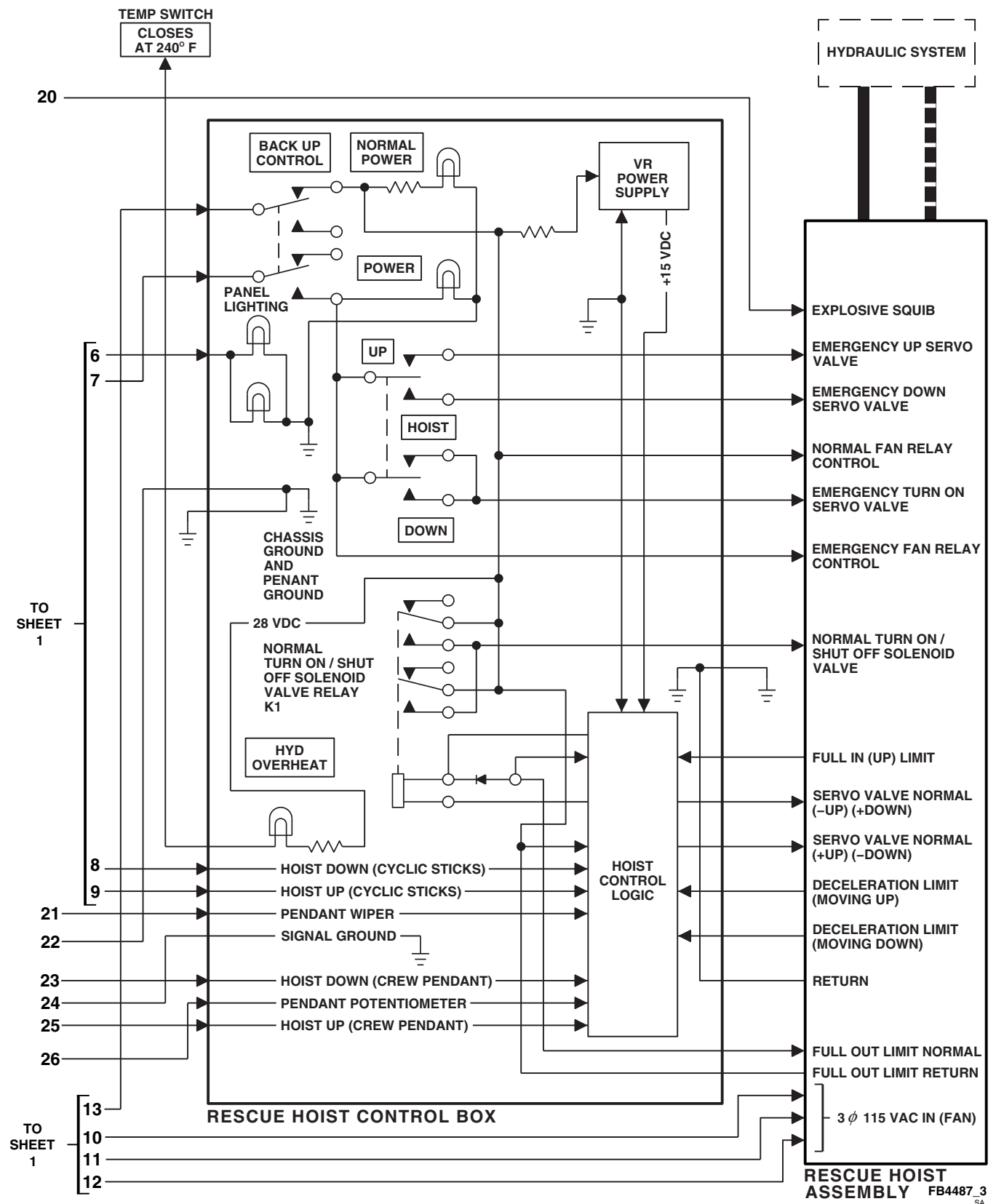


FIGURE 14-1-1-2. RESCUE HOIST SYSTEM BLOCK DIAGRAM (SHEET 3 OF 3)

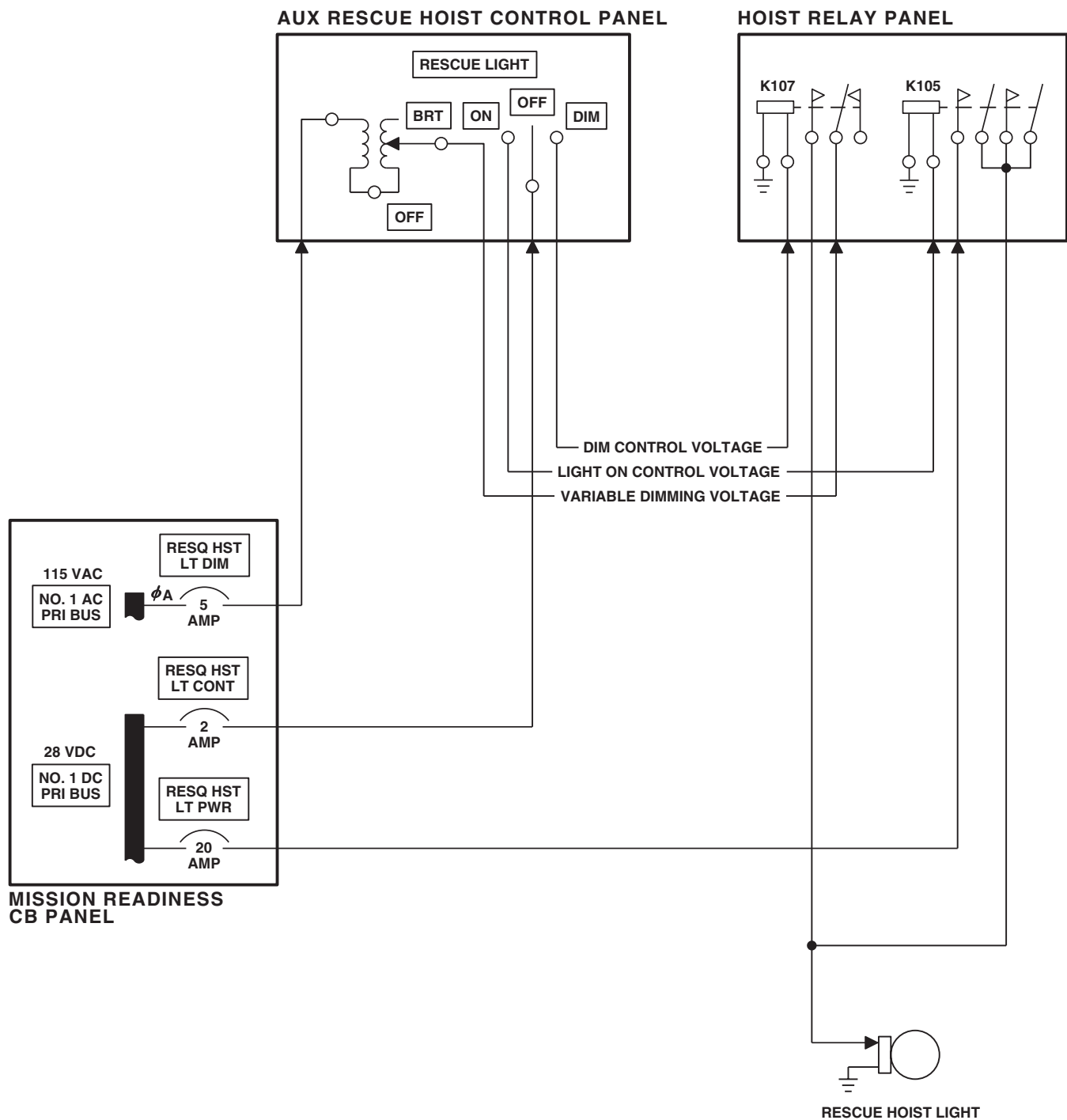
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FIGURE 14-1-1-3. RESCUE HOIST SYSTEM LIGHT BLOCK DIAGRAM

14-1-2 RESCUE HOIST SYSTEM DESCRIPTION AND DATA

HOIST BL-29900-30 .

14-1-2.1 General.

The rescue hoist system consists of a rescue hoist assembly, a rescue hoist pendant, a pilot's hoist control panel, and a crew's control panel. These components are shown in Figure 14-1-2-1. The hoist system requires 200 V, three-phase 400 Hz, 24 amperes main power; 28 vdc, five amperes control power, and a separate 28 vdc, five amperes input for cable cutter power.

14-1-2.2 Description.

14-1-2.2.1 Rescue Hoist Assembly.

The rescue hoist assembly operates with the pilot's hoist control panel or the rescue hoist pendant in this order of precedence. The assembly includes an electric motor, a hoist, a controller section, and aerodynamic fairings. The hoist assembly contains a rotating drum, which is driven by the electric motor through a gear train and a clutch assembly to raise and lower a 290 foot cable terminated in a swivel hook. A brake assembly inside the drum locks the drum when input power is cut off or the load tries to overhaul the motor, and the clutch assembly allows slippage if the load on the hook exceeds 1800 pounds. A levelwind mechanism on the hoist assembly ensures even winding of the cable on the drum. Limit switches sense when the cable approaches the full-in or full-out position, and when the cable reaches these positions. A potentiometer coupled to the drum continuously senses cable position for cable pay out indication.

The hoist uses a tension roller located between the cable guide and the drum. The tension roller is gear driven by the drum through an over-running clutch; such that when reeling out, the tension roller is rotating faster than the tangential speed of the drum, thereby maintaining a "pull" on the cable at no load. In like manner, the tension roller rotates slower than the tangential speed of the drum when reeling in, thus providing drag on the cable which ensures tight cable storage on the drum under all conditions.

The cable assembly for the rescue hoist consists of a wire rope with a swaged stainless steel ball/shank fitting. The wire rope is 3/16 inch diameter, corrosion resistant, preformed spin-resistant type wire. The drum end of the cable assembly is cut off by an inert gas shielded electric arc process such that the cable end is sealed and made suitable for attachment at the drum anchor point. The swivel hook assembly prevents inadvertent load release and is operable with one hand.

The swivel hook assembly incorporates a lightweight, high-strength, honeycomb aluminum crushable bumper assembly. During normal hoist operations after the slow in limit switch has actuated, the crushable bumper triggers the "full-in" limit switch plunger to stop the hoist. In the event of a failure of the slow-in limit signal, the crushable bumper will "absorb" all the rotational energy to safely stop the hoist at the "full-in" position. The rotational energy at full speed is capable of producing loads in excess of the rating of the cable. This provides a last line safety device to minimize to the greatest extent possible a hoist cable breakage due to a single point failure. A crushed bumper also provides the operator a visual indication that a failure has occurred.

The cable cutter assembly consists of a pressure cartridge, guillotine and anvil, all of which are contained in the cable guide housing. The specific minimum "all fire" current of the pressure cartridge is five amperes per bridge. This cartridge has the highest EMI rejection with a guaranteed "no fire" rating of one watt.

The speed of the hoist is continuously variable from zero to a full speed of 350 feet per minute. The rescue hoist assembly incorporates four guide rollers to guide the cable into the hoist at large fleet angles due to either high winds (30 m/s) or helicopter movement. The bell mouth swivels to follow the angle of the cable at that condition. Table 14-1-2-1 lists leading particulars of the rescue hoist.

Table 14-1-2-1. Leading Particulars.

RESCUE HOIST**Functional Characteristics:**

- Min. 350 fpm cable speed with 600 lb load (average)
- Max. Operating Load 600 lb
- Design Limit Load 1800 lb
- Structural Ultimate Load 2700 lb
- 28 vdc, five amperes power for cable cutter (momentary)
- 200 V, three-phase, 400 Hz, 24 amperes ac input power
- 28 vdc, five amperes control power
- Cable angle capability of 30° from the vertical
- Electro-ballistic cable cutter

Physical Characteristics:

- Brushless dc motor with Integral Power electronics
- Built-in heat sink fins on cable drum
- Optimized fail-safe Weston type brake
- Dual levelwind shaft system
- Optimized cable tensioning system
- Integrated hoist control unit
- Safety overload clutch - slip between 1200 lb and 1850 lb.
- Redundant Limit Switches:
 - a. Slow down lowering
 - b. Slow down hoisting
 - c. Full in

14-1-2-2

Table 14-1-2-1. Leading Particulars. (Cont)

RESCUE HOIST

- d. Full out
- e. Emergency full out
- f. Full out and emergency full out override (for cable installation and removal only)

14-1-2.2.2 Rescue Hoist Pendant.

The rescue hoist pendant allows crew selection of the direction and speed of hoist cable travel via thumbwheel activation (Figure 14-1-2-1). Cable speed is directly proportional to thumbwheel deflection. The pendant control handle also provides a Night Vision Goggle (NVG) compatible/Sunlight readable cable pay out display. The display will dim with a 28 vdc signal to the controller. The pendant control handle provides the

hoist operator with three NVG indicators, for one second following application of 28 vdc power all indicators are illuminated as a test feature. The indicators provide the operator with cable "FULL IN," "FULL OUT," and "MTR (motor) HOT" information. A four-way switch is provided for search light control.

Table 14-1-2-2 lists leading particulars of the control pendant.

Table 14-1-2-2. Leading Particulars.**RESCUE HOIST PENDANT****Functional Characteristics:**

Input power - 28 vdc supplied by EC-23000 Rescue Hoist Electronic Controller

Output:

- Signal of variable voltage for speed control
- Up and down directional signals
- NVG compatible/Sunlight readable, digital cable pay out display
- Motor thermal overheat signal
- Full in limit switch, full out limit switch, and search light control

Physical Characteristics:

- Three NVG indicators
- Variable speed thumbwheel
- Ten-foot retractable cord

14-1-2.2.3 Rescue Hoist Pendant Cable Assembly.

The rescue hoist pendant cable assembly interconnects the rescue hoist pendant and crew's hoist control panel.

14-1-2.2.4 Pilot's Hoist Control Panel.

The rescue hoist includes a pilot's hoist control panel. The pilot can completely override any hoist command and operate the hoist at a fixed speed by using the PILOT OVERRIDE toggle switch. Provisions for cutting the hoist cable are provided with a guarded switch to avoid accidental firing of the cable cutter. All indicators are NVG compatible for one second following application of 28 vdc power. All indicators are illuminated as a test feature.

14-1-2.2.5 Crew's Hoist Control Panel.

The rescue hoist includes a crew's hoist control panel. The crew's hoist control panel has a switch for the SEARCH LIGHT, a SQUIB TEST switch, and an ARM TEST switch. Provisions for cutting the hoist cable are provided with a guarded switch to avoid accidental firing of the cable cutter. Also, the crew control panel can be used to activate the NVG lighting by placing the SEARCH LIGHT switch to the lower position.

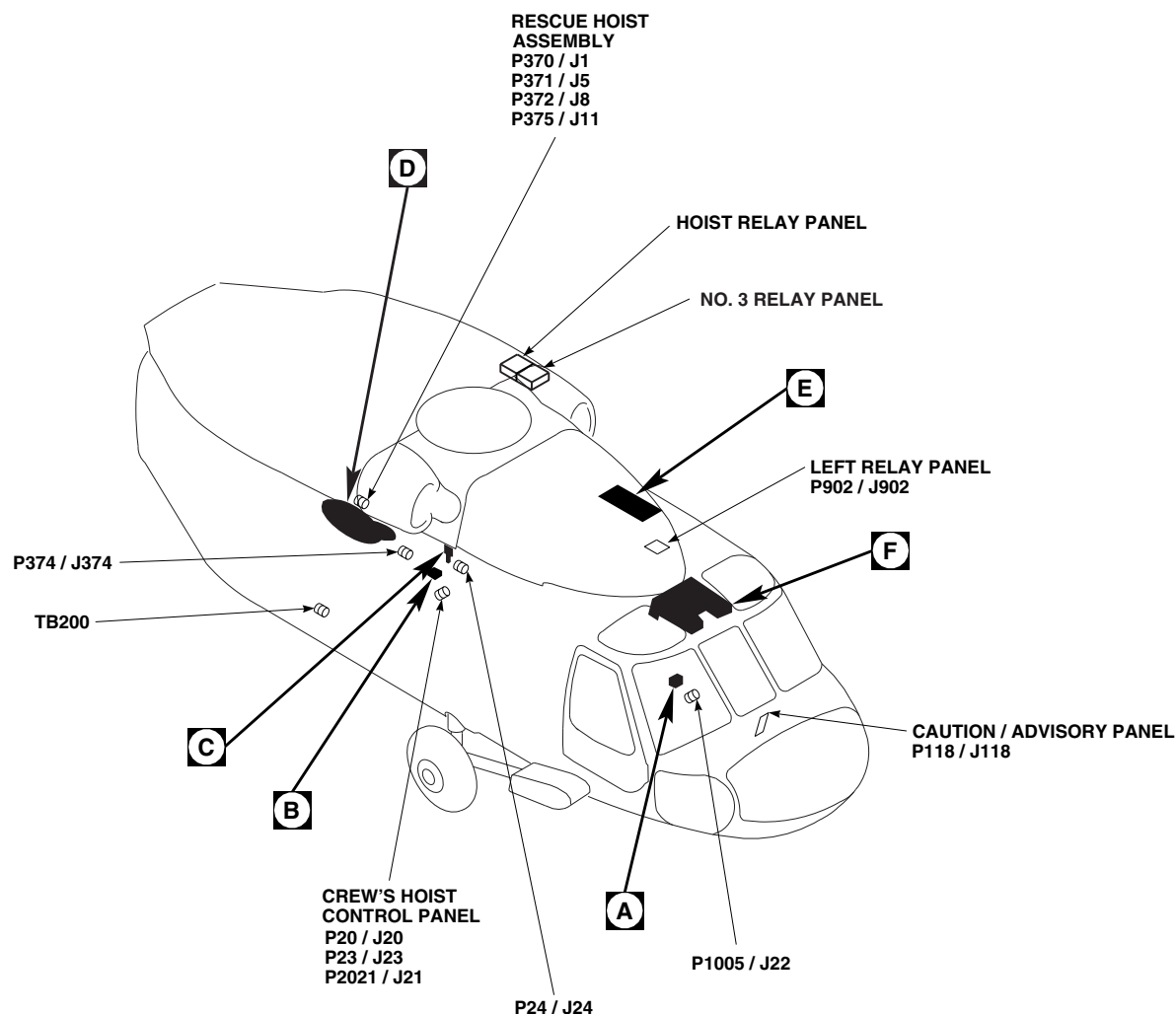
14-1-2.2.6 Theory of Operation.

The control system allows for two modes of control with the pilot having the highest priority and the rescue hoist pendant having the lowest priority. The rescue hoist pendant provides indications of overheating, cable extremes and cable length and

allows control of the hoist operation. The speed command from the pilot's control panel is fixed while variable speed is provided at the crew's hoist control panel and rescue hoist pendant. The pendant controls hoist speed via the deflection of the thumbwheel which is continuously variable and spring loaded to the off position. All controls have

automatic slow speed prior to the automatic stop at the cable extremes.

The pilot's and the crew's controls have a guarded switch for activation of the cable cutter electro-ballistic device. Searchlight capability is provided in the pendant control handle.

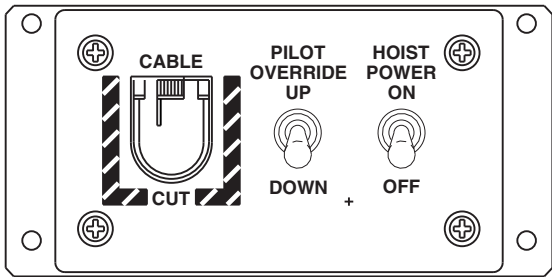


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P6 / J6	RESCUE HOIST ASSEMBLY PRESSURE CARTRIDGE (SQUIB)
P20 / J20	CREW'S HOIST CONTROL PANEL
P23 / J23	CREW'S HOIST CONTROL PANEL
P24 / J24	RESCUE HOIST PENDANT
P115 / J115	COCKPIT, BL 6 RH, STA 210
P118 / J118	CAUTION / ADVISORY PANEL
P168	UPPER CONSOLE DISCONNECT
P200 / J200	UPPER CONSOLE DISCONNECT
P370 / J1	RESCUE HOIST ASSEMBLY
P371 / J5	RESCUE HOIST ASSEMBLY

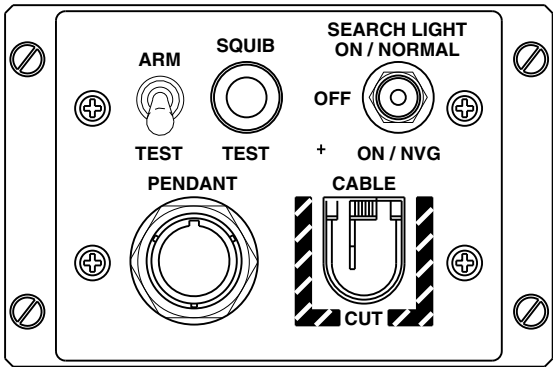
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P372 / J8	RESCUE HOIST ASSEMBLY
P373 / J373	TOP DECK DISCONNECT
P374 / J374	TOP DECK DISCONNECT
P375 / J11	RESCUE HOIST ASSEMBLY
P902 / J902	LEFT RELAY PANEL
P914 / J914	COCKPIT, BL 24 LH, STA 247
P1005 / J22	PILOT'S HOIST CONTROL PANEL
P2020 / J2020	NO. 3 RELAY PANEL
P2021 / J21	CREW'S HOIST CONTROL PANEL
TB200	DUAL-MODE CONTROLLABLE SEARCHLIGHT

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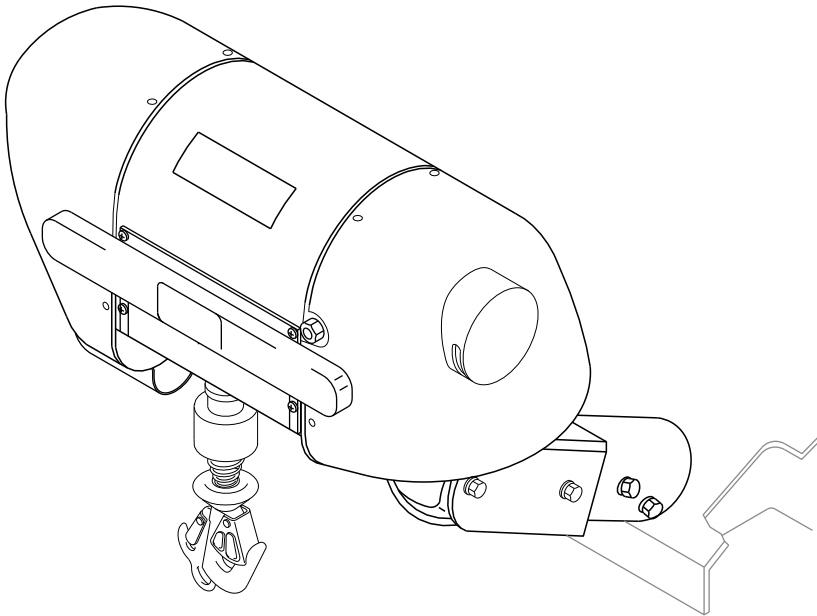
FIGURE 14-1-2-1. RESCUE HOIST SYSTEM LOCATION DIAGRAM (SHEET 1 OF 3)



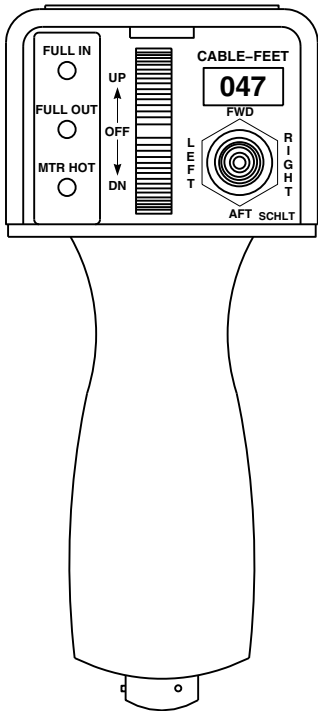
PILOT'S HOIST CONTROL PANEL



CREW'S HOIST CONTROL PANEL



RESCUE HOIST ASSEMBLY



RESCUE HOIST PENDANT

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FIGURE 14-1-2-1. RESCUE HOIST SYSTEM LOCATION DIAGRAM (SHEET 2 OF 3)

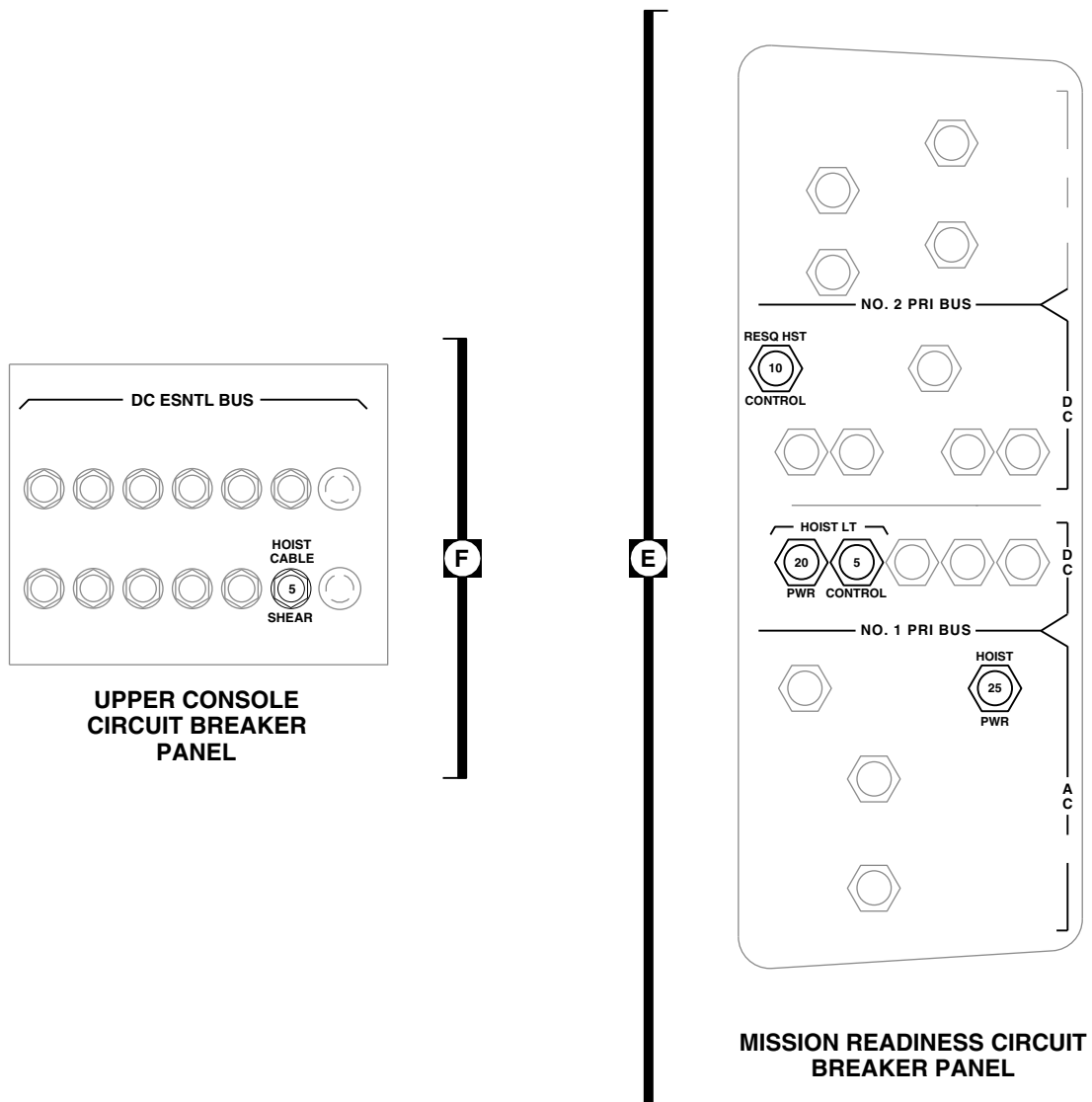
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FIGURE 14-1-2-1. RESCUE HOIST SYSTEM LOCATION DIAGRAM (SHEET 3 OF 3)

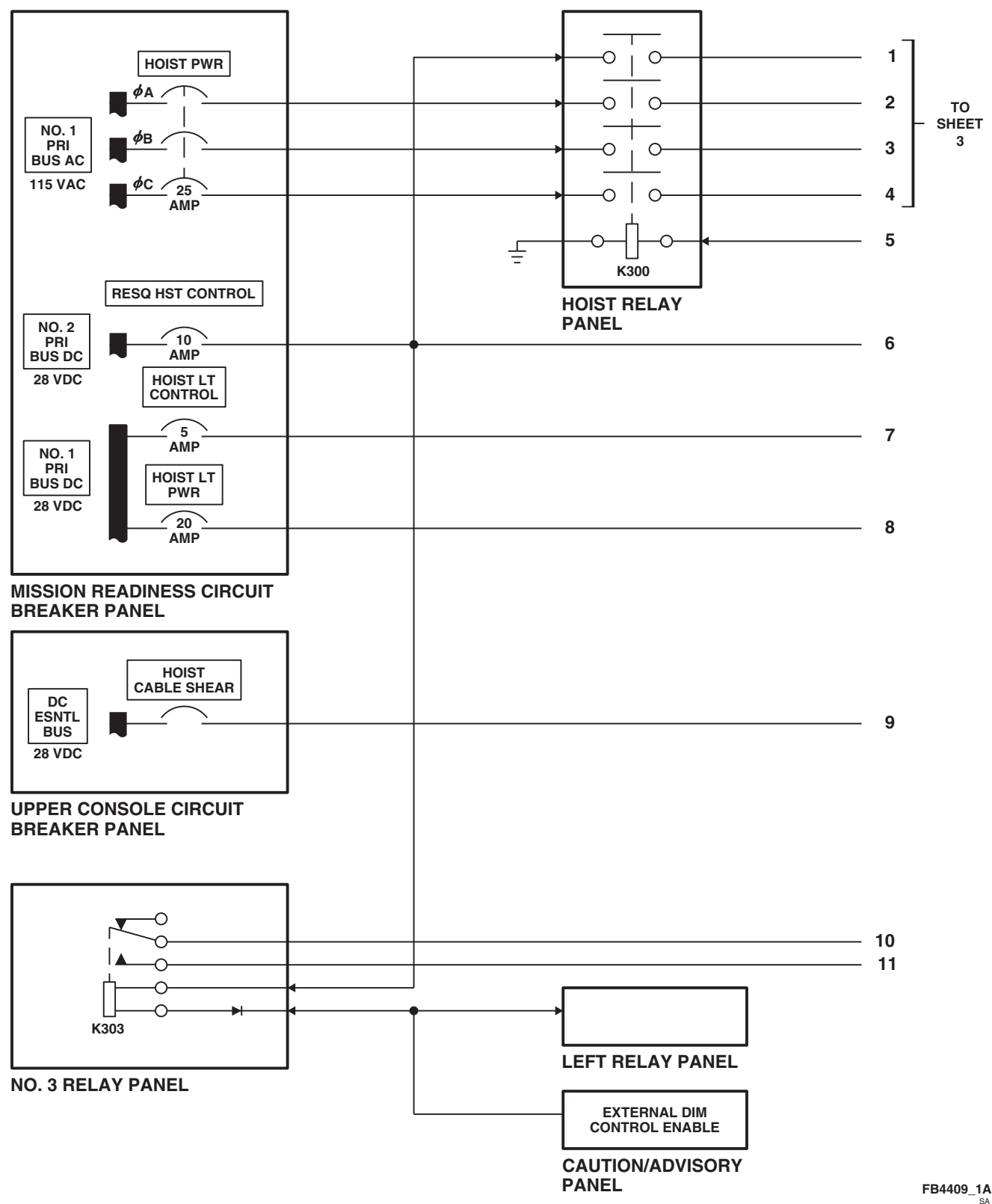


FIGURE 14-1-2-2. RESCUE HOIST SYSTEM BLOCK DIAGRAM (SHEET 1 OF 4)

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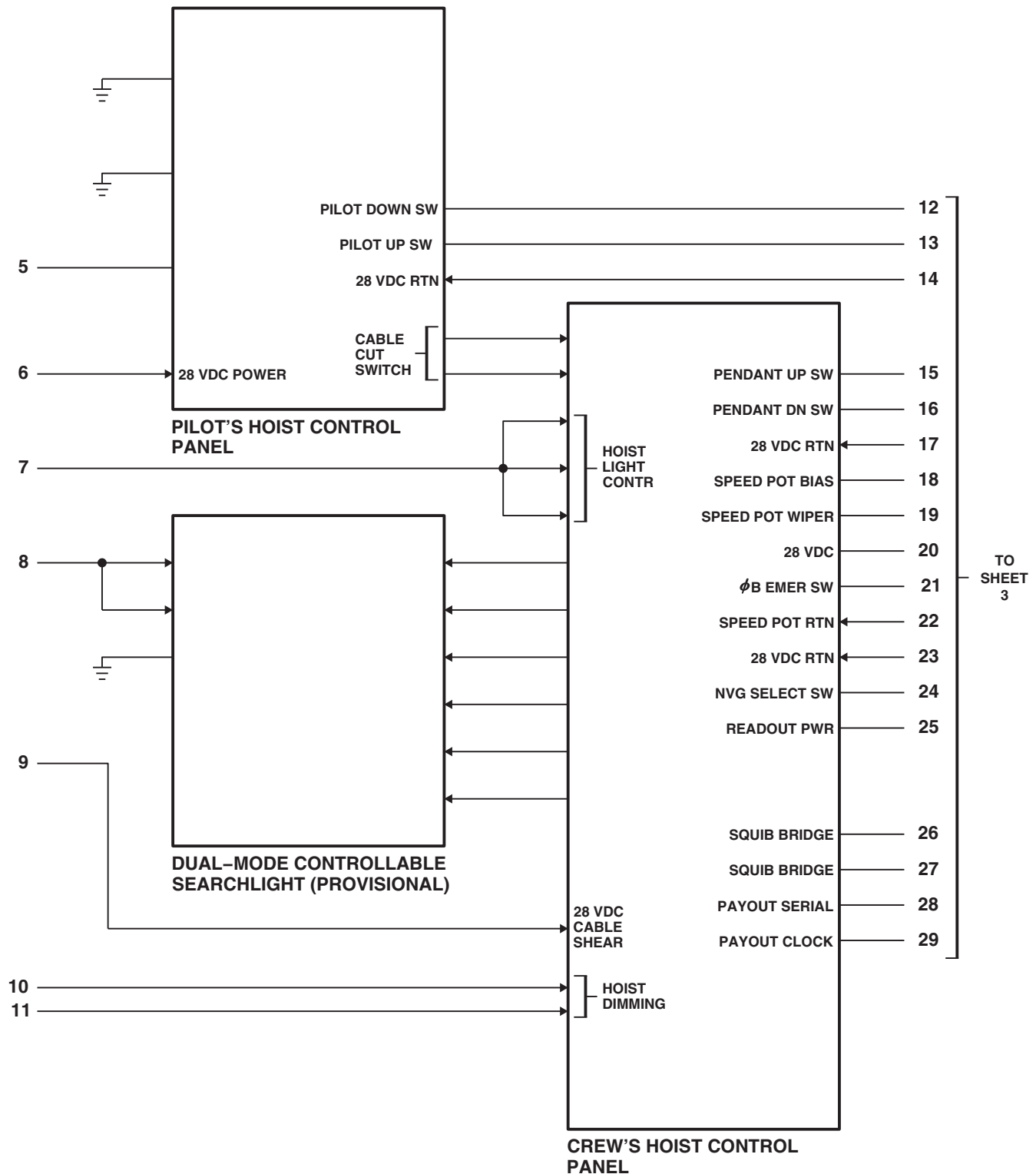
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FIGURE 14-1-2-2. RESCUE HOIST SYSTEM BLOCK DIAGRAM (SHEET 2 OF 4)

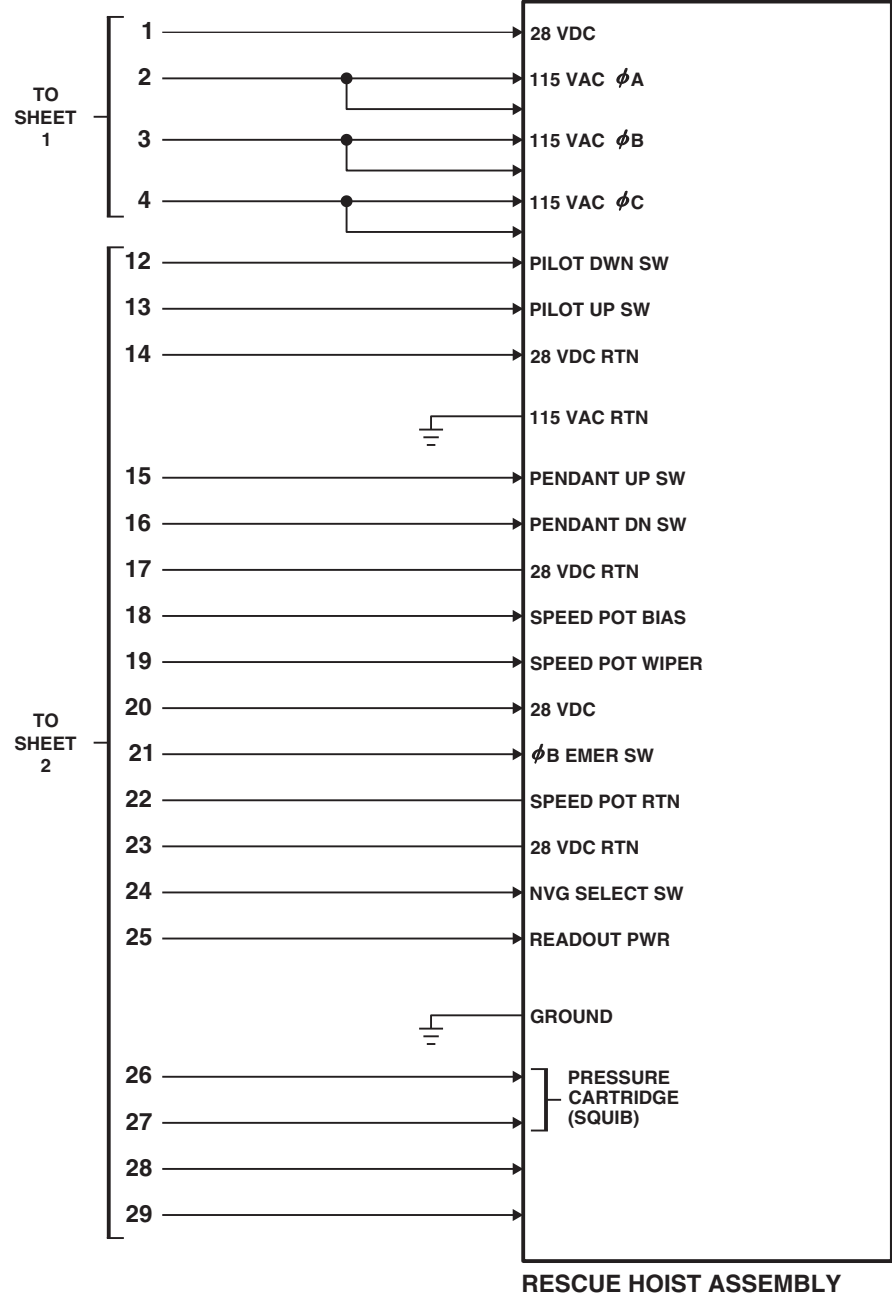
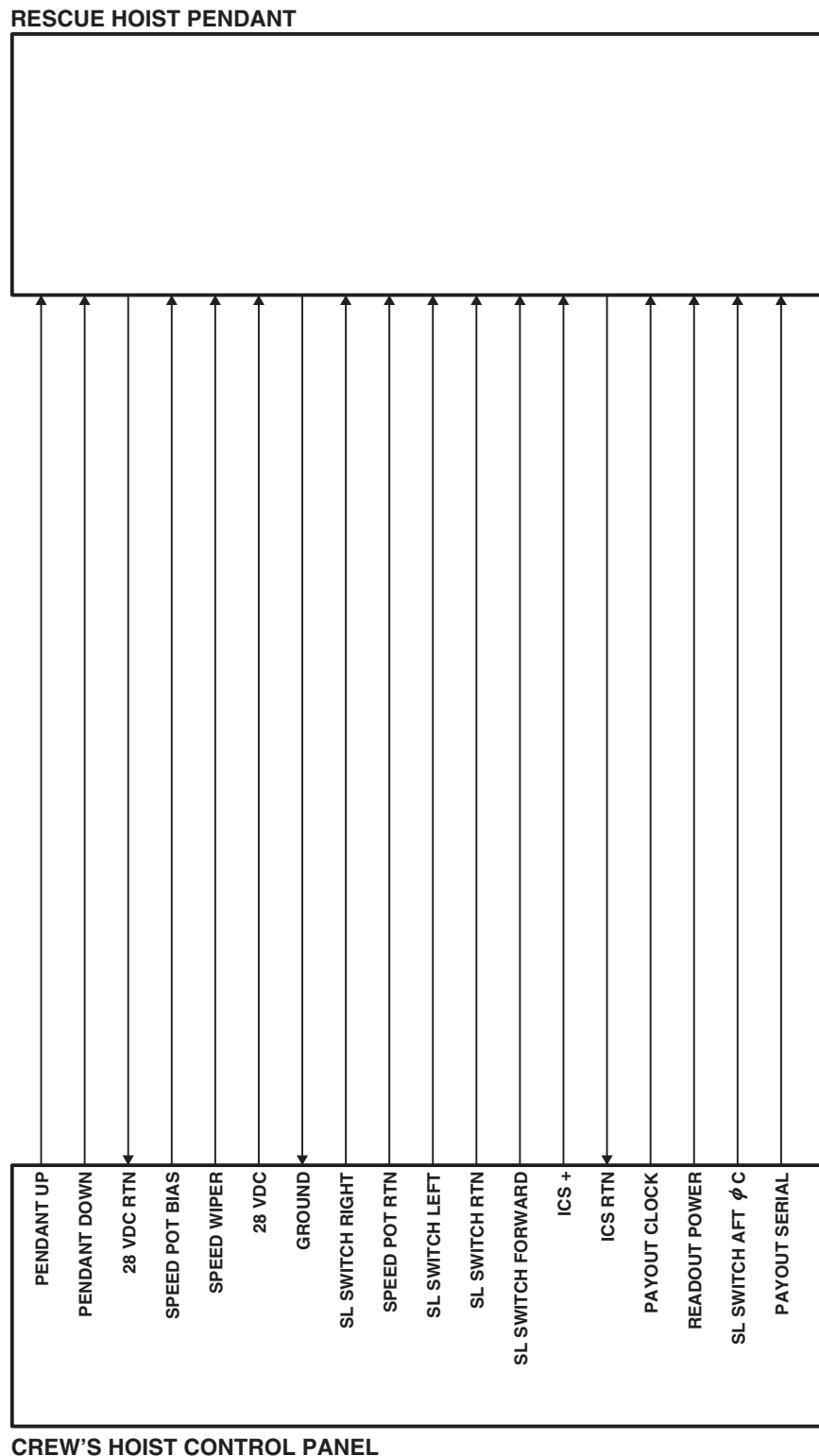


FIGURE 14-1-2-2. RESCUE HOIST SYSTEM BLOCK DIAGRAM (SHEET 3 OF 4)

FB4409_3
 SA



FB4409_4
SA

FIGURE 14-1-2-2. RESCUE HOIST SYSTEM BLOCK DIAGRAM (SHEET 4 OF 4)

SECTION 14-2

TROUBLESHOOTING

SECTION OVERVIEW

PARAGRAPH	TITLE
14-2-1	Rescue Hoist System HOIST, BL-27100-71 HOIST, 42315-10
14-2-2	Rescue Hoist System HOIST BL-29900-30

14-2-1 RESCUE HOIST SYSTEM **HOIST, BL-27100-71** **HOIST, 42315-10** .

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Container Suitable for Holding Approximately 225 Feet of Rescue Hoist Cable
- Electrical Repairer Toolkit
- Insulated Jumper Wire

References

- Appropriate Flight Manual
- PARA 1-6-15
- PARA 1-6-16
- PARA 14-4-4

Equipment Conditions

- External Electrical Power Available and/or APU Operational
- External Hydraulic Power Available or Backup Hydraulic System Operational

- Rescue Hoist Center Access Cover Removed (PARA 14-4-4).

WARNING

Ensure that rescue hoist cable does not contact aircraft during testing or operational checkout. Failure to prevent any contact with aircraft could result in serious injury or death to personnel.

WARNING

The squib is an electrically activated explosive device. Improper checkout of the cable shear function could result in serious injury or death to personnel. The emergency shear function activates the squib. With aircraft on the ground, if cartridge P6 is connected, and EMER REL on either cyclic stick is pressed, or if SHEAR switches on hover trim or auxiliary rescue hoist control panel are set to SHEAR, hoist cable will shear.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

14-2-1.1 Rescue Hoist Operational Check.

- Remove rescue hoist center access panel (PARA 14-4-4).
- Disconnect P6 on rescue hoist from J6 and disconnect P256 from cargo hook. Remove squibs and immediately place squibs in foil envelope.

- Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- Pull out the following circuit breakers on upper console:

BACKUP HYD CONTR

HOIST CABLE SHEAR

- e. Place upper console RSQ HOIST switch to OFF.
- f. Turn on electrical power (PARA 1-6-16).
- g. Place upper console RSQ HOIST switch to ALL. If rescue hoist control box BACKUP CONTROL POWER switch light is on, press and release switch.

RESULT: Rescue hoist control box NORMAL POWER light shall go on.

- If result is not as specified, go to TABLE NO. 14-2-1-1.
- If after pressing BACKUP CONTROL POWER switch, switch light does not go off, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
- h. Place upper console RSQ HOIST switch to CKPT.

RESULT: Rescue hoist control panel NORMAL POWER light shall remain on.

- If result is not as specified, check continuity between RSQ HOIST switch S100-4 and S100-6.
 - (a) If continuity is present, replace RSQ HOIST switch (PARA 9-4-5).
 - (b) If continuity is not present, repair/replace wiring (Appendix F).
- i. Press rescue hoist control box BACKUP CONTROL POWER switch.

RESULT:

- (1) NORMAL POWER light shall go off.
- (2) BACKUP CONTROL POWER switch light shall go on.

- If result (1) is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
- If result (2) is not as specified, go to TABLE NO. 14-2-1-2.

- j. Press rescue hoist control box BACKUP CONTROL POWER switch.

RESULT:

- (1) BACKUP CONTROL POWER switch light shall go off.
- (2) NORMAL POWER light shall go on.
- If results are not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).

WARNING

Rescue hoist cable is stiff and abrasive. Broken cable strands are sharp. Work gloves should be worn whenever handling rescue hoist cable.

CAUTION

Maintaining tension on rescue hoist cable during hoist operation is necessary to make sure that rescue hoist cable feeds properly and does not tangle.

NOTE

Rescue hoist cable pays out very quickly when using pendant.

- k. Place a large sheet of paper or a large clean bucket under rescue hoist. Coil rescue hoist cable on paper or into bucket as cable is reeled out.
- l. Push in BACKUP HYD CONTR circuit breaker.

14-2-1-2

- m. Place BACKUP HYD PUMP switch on upper console to ON.
- n. On rescue hoist/cargo hook pendant, turn RESCUE HOIST thumbwheel switch full DOWN, and hold. At same time, start helicopter elapsed-time clock.
- o. Record time cable takes to pay out.

RESULT:

- (1) Cable shall pay out at 250 feet-per-minute until 3 to 5 feet of cable remains on drum.
- (2) When 3 to 5 feet of cable remains on drum, hoist shall slow to 50 feet-per-minute.
- (3) Hoist shall stop with 3-1/2 to 4-1/2 wraps of cable remaining on hoist drum.
- (4) Rescue hoist elapsed-time indicator shall operate.
- (5) Elapsed time shall not exceed 63 seconds.
- If cable does not pay out, go to TABLE NO. 14-2-1-3.
- If cable pays out at 50 feet-per-minute for entire length, go to TABLE NO. 14-2-1-4.
- If result (2) is not as specified, go to TABLE NO. 14-2-1-5.
- If result (3) is not as specified, go to TABLE NO. 14-2-1-6.
- If result (4) is not as specified, go to TABLE NO. 14-2-1-7.
- If cable does not pay out smoothly, check hoist (PARA 14-4-1).
- p. Reset helicopter elapsed-time clock.
- q. Turn rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch full UP, and hold. At same time, start helicopter elapsed-time clock.

- r. Record time cable takes to reel in.

RESULT:

- (1) Cable shall reel in at 250 feet-per-minute until 8 to 10 feet of cable remains to be reeled in.
- (2) When 8 to 10 feet of cable remains, hoist slows to 50 feet-per-minute.
- (3) Hoist stops when all cable is reeled in.
- (4) Elapsed time shall not exceed 63 seconds.
- If cable does not reel in, go to TABLE NO. 14-2-1-8.
- If cable reels in at 50 feet-per-minute for entire length, go to TABLE NO. 14-2-1-9.
- If result (2) is not as specified, go to TABLE NO. 14-2-1-10.
- If result (3) is not as specified, go to TABLE NO. 14-2-1-11.
- If cable wraps loosely on hoist, check hoist (PARA 14-4-1).
- s. Reset helicopter elapsed-time clock.
- t. On crewmember's hover grip, turn HOIST thumbwheel switch full DOWN, and hold. At same time, start helicopter elapsed-time clock.
- u. Record time cable takes to pay out.

RESULT:

- (1) Cable shall pay out at 250 feet-per-minute until 3 to 5 feet of cable remains on drum.
- (2) When 3 to 5 feet of cable remains on drum, hoist shall slow to 50 feet-per-minute.
- (3) Hoist shall stop with 3-1/2 to 4-1/2 wraps of cable remaining on hoist drum.
- (4) Elapsed time shall not exceed 63 seconds.

- If cable does not pay out, go to TABLE NO. 14-2-1-12.
 - If cable pays out at 50 feet-per-minute for entire length, go to TABLE NO. 14-2-1-4.
 - If result (2) is not as specified, go to TABLE NO. 14-2-1-5.
 - If result (3) is not as specified, go to TABLE NO. 14-2-1-6.
 - If cable does not pay out smoothly, check hoist (PARA 14-4-1).
- v. Reset helicopter elapsed-time clock.
- w. Turn crewmember's hover grip RESCUE HOIST thumbwheel switch full UP, and hold. At same time, start helicopter elapsed-time clock.
- x. Record time cable takes to reel in.

RESULT:

- (1) Cable shall reel in at 250 feet-per-minute until 8 to 10 feet of cable remains to be reeled in.
 - (2) When 8 to 10 feet of cable remains, hoist slows to 50 feet-per-minute.
 - (3) Hoist stops when all cable is reeled in.
 - (4) Elapsed time shall not exceed 63 seconds.
- If cable does not reel in, go to TABLE NO. 14-2-1-13.
 - If cable reels in at 50 feet-per-minute for entire length, go to TABLE NO. 14-2-1-14.
 - If result (2) is not as specified, go to TABLE NO. 14-2-1-10.
 - If result (3) is not as specified, go to TABLE NO. 14-2-1-11.
 - If cable wraps loosely on hoist, check hoist (PARA 14-4-1).

- y. Turn rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch full DOWN, and hold. While cable is paying out, turn crewmember's hover grip HOIST thumbwheel switch to UP.

RESULT: Cable shall stop paying out and shall begin to reel in.

- If result is not as specified, replace crewmember's hover grip (TM 11-70-23).
- z. With crewmember's hover grip thumbwheel HOIST switch turned to UP, turn rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch to DOWN.

RESULT: Cable shall continue to reel in.

- If result is not as specified, replace crewmember's hover grip (TM 11-70-23).
- aa. Reset helicopter elapsed-time clock.
- ab. Place pilot's cyclic stick HOIST switch to DOWN. At same time, start helicopter elapsed-time clock.

RESULT:

- (1) Cable pays out at 100 feet-per-minute.
 - (2) Elapsed time shall not exceed 175 seconds.
- If result (1) is not as specified, go to TABLE NO. 14-2-1-15.
 - If result (2) is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
 - If cable pays out at 50 feet-per-minute for entire length, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
- ac. Reset helicopter elapsed-time clock.

14-2-1-4

- ad. Place pilot's cyclic stick HOIST switch to UP. At the same time, start helicopter elapsed-time clock.

RESULT:

- (1) Cable reels in at 100 feet-per-minute.
 - (2) Time from hoist start to stop shall not exceed 175 seconds.
- If result (1) is not as specified, go to TABLE NO. 14-2-1-16.
 - If result (2) is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
 - If cable reels in at 50 feet-per-minute for entire length, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
- ae. Reset helicopter elapsed-time clock.
- af. Place copilot's cyclic stick HOIST switch to DOWN. At same time, start helicopter elapsed-time clock.

RESULT:

- (1) Cable pays out at 100 feet-per-minute.
 - (2) Time from hoist start to stop shall not be over 175 seconds.
- If result (1) is not as specified, go to TABLE NO. 14-2-1-17.
 - If result (2) is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
 - If cable pays out at 50 feet-per-minute for entire length, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
- ag. Reset helicopter elapsed-time clock.

- ah. Place copilot's cyclic stick HOIST switch to UP. At the same time, start helicopter elapsed-time clock.

RESULT:

- (1) Cable reels in at 100 feet-per-minute.
 - (2) Elapsed time shall not exceed 175 seconds.
- If result (1) is not as specified, go to TABLE NO. 14-2-1-18.
 - If result (2) is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
 - If cable reels in at 50 feet-per-minute for entire length, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
- ai. Reset helicopter elapsed-time clock.
- aj. Turn rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch full DOWN, and hold. While cable is paying out, place pilot's cyclic stick HOIST switch to UP.

RESULT: Hoist shall stop paying out cable and shall reel in cable.

- If result is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).
- ak. Release rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch and pilot's cyclic stick HOIST switch.
- al. Turn rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch full UP, and hold. While cable is reeling in, place pilot's cyclic stick switch to DOWN.

RESULT: Hoist shall stop reeling in cable and pay out cable.

- If result is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).

am. Release rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch and pilot's cyclic stick HOIST switch.

hoist/cargo hook pendant RESCUE HOIST thumbwheel switch and from both cyclic sticks.

an. Turn rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch full UP, and hold. Reel in cable to its limit.

RESULT: Hoist shall not operate.

- If result is not as specified, replace RSQ HOIST switch S100 (PARA 9-4-5).

ao. Turn rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch to DOWN and reel out 100 feet of cable.

at. Place upper console RSQ HOIST switch to ALL. Make sure NORMAL POWER light is on.

ap. Press hoist control panel BACKUP CONTROL POWER switch.

au. Using rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch, reel in cable to full-in limit stop.

RESULT:

(1) BACKUP CONTROL POWER switch light shall go on and NORMAL POWER light shall go off.

av. Set BACKUP HYD PUMP switch on upper console to OFF.

(2) Hoist heat exchanger fan shall run.

aw. Pull out BACKUP HYD CONTR circuit breaker on upper console.

- If result (2) is not as specified, go to TABLE NO. 14-2-1-19.

ax. Turn off electrical (PARA 1-6-16) and hydraulic power (PARA 1-6-15).

aq. Hold rescue hoist control box BACKUP CONTROL HOIST switch to DOWN for 5 to 10 seconds.

ay. Push in the HOIST CABLE SHEAR circuit breaker on the upper console.

RESULT: Hoist cable shall pay out.

az. Pull out CARGO HOOK EMER circuit breaker on upper console.

- If result is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).

ba. Place upper console RSQ HOIST switch to OFF.

ar. Press hoist control panel BACKUP CONTROL POWER switch.

bb. Make sure pilot's and copilot's cyclic stick EMER REL switches are not activated.

RESULT: BACKUP CONTROL POWER switch light shall go off and NORMAL POWER light shall go on.

bc. Make sure auxiliary rescue hoist control panel HOIST SHEAR and hover trim control panel SAFE/SHEAR/EMER switches are set to SAFE.

- If result is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27).

bd. Check continuity between:

P6-C	and	P6-A
P6-B	and	P6-D

as. Place upper console RSQ HOIST switch to OFF. Attempt to operate hoist from rescue

14-2-1-6

RESULT: Continuity shall be present.

- If result is not as specified, repair/replace wiring, as required.

be. Turn on electrical power (PARA 1-6-16).

bf. Connect multimeter between P6-A and P6-B.

bg. Place upper console RSQ HOIST switch to ALL.

RESULT: HOIST ARMED advisory capsule shall illuminate.

- If result is not as specified, troubleshoot caution/advisory system (PARA 8-2-4).

bh. Hold auxiliary rescue hoist control panel HOIST SHEAR switch to SHEAR.

RESULT: Multimeter shall indicate 28 vdc.

- If result is not as specified, go to TABLE NO. 14-2-1-20.

bi. Release auxiliary rescue hoist control panel HOIST SHEAR switch to SAFE.

bj. Press and hold pilot's cyclic stick EMER REL switch.

RESULT: Multimeter shall indicate 28 vdc.

- If result is not as specified, go to TABLE NO. 14-2-1-21.

bk. Release pilot's cyclic stick EMER REL switch.

bl. Press and hold copilot's cyclic stick EMER REL switch.

RESULT: Multimeter shall indicate 28 vdc.

- If result is not as specified, go to TABLE NO. 14-2-1-22.

bm. Release copilot's cyclic stick EMER REL switch.

bn. Place upper console RSQ HOIST switch to CKPT.

RESULT: HOIST ARMED advisory capsule shall illuminate.

- If result is not as specified, troubleshoot caution/advisory system (PARA 8-2-4).

bo. Hold auxiliary rescue hoist control panel HOIST SHEAR switch to SHEAR.

RESULT: Multimeter shall indicate 0 vdc.

- If result is not as specified, replace RSQ HOIST switch S100 (PARA 9-4-5).

bp. Release auxiliary rescue hoist control panel HOIST SHEAR switch to SAFE.

bq. Place upper console RSQ HOIST switch to OFF.

RESULT: HOIST ARMED advisory capsule shall go off.

- If result is not as specified, troubleshoot caution/advisory system (PARA 8-2-4).

br. On upper console, set RSQ HOIST switch to ALL.

bs. On hover trim control panel, hold switch to SHEAR.

bt. Check for 28 vdc between P6-A and P6-B.

RESULT: Multimeter shall indicate 28 vdc.

- If result is not as specified, replace hover trim panel (TM 11-70-23).

bu. On hover trim control panel, release SHEAR switch.

bv. On pilot's cyclic stick, press EMER REL.

RESULT: Multimeter shall indicate 28 vdc.

- If result is not as specified, troubleshoot pilot's cyclic stick (PARA 11-2-1).

- bw. Release EMER REL switch.
- bx. On copilot's cyclic stick, press EMER REL switch.

RESULT: Multimeter shall indicate 28 vdc.

- If result is not as specified, troubleshoot copilot's cyclic stick (PARA 11-2-1).

- by. Release EMER REL switch.
- bz. Disconnect P967 from temperature switch on rescue hoist hydraulic return line.
- ca. Connect an insulated jumper wire from P967-1 to ground.
- cb. Place upper console RSQ HOIST switch to ALL.

RESULT: Rescue hoist control box HYD OVERHEAT lamp shall go on.

- If result is not as specified, go to TABLE NO. 14-2-1-23.

- cc. Place upper console RSQ HOIST switch to OFF.
- cd. Remove jumper wire from P967 and connect P967 to temperature switch on rescue hoist hydraulic return line.
- ce. Remove IR filter from rescue hoist light (PARA 14-4-23).

- cf. Place auxiliary rescue hoist control panel RESCUE LIGHT switch to ON.

RESULT: Rescue hoist light shall go on full bright.

- If result is not as specified, go to TABLE NO. 14-2-1-24.

- cg. Place auxiliary rescue hoist control panel RESCUE LIGHT switch to DIM.
- ch. Rotate the RESCUE LIGHT dimmer control from OFF to BRT.

RESULT: The intensity of rescue hoist light illumination shall go from off to full bright.

- If result is not as specified, go to TABLE NO. 14-2-1-25.

- ci. Place auxiliary rescue hoist control panel RESCUE LIGHT switch to OFF.
- cj. Turn off electrical power (PARA 1-6-16).
- ck. Remove squibs from foil envelope and install in rescue hoist and cargo hook. Connect P6 and P256.
- cl. Install rescue hoist center access panel (PARA 14-4-4).
- cm. Install IR filter on rescue hoist light (PARA 14-4-23).

TABLE NO. 14-2-1-1. NORMAL POWER Light Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check that RESQ HST CONT circuit breaker is pushed in.	Step 1. If circuit breaker is in, go to 2. Step 2. If circuit breaker is not in, push in circuit breaker. Go to 7.
2. Check for 28 vdc between P965-2 and ground.	

TABLE NO. 14-2-1-1. NORMAL POWER Light Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If voltage is as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 7 .	Step 2. If voltage is not as specified, go to 3 .
3. Check for 28 vdc between RSQ HOIST switch S100-4 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between S100-4 and P965-2. Go to 7 .	Step 2. If voltage is not as specified, go to 4 .
4. Check for 28 vdc between S100-5 and ground.	
Step 1. If voltage is as specified, replace RSQ HOIST switch S100 (PARA 9-4-5). Go to 7 .	Step 2. If voltage is not as specified, go to 5 .
5. Check for 28 vdc between P200-d and ground.	
Step 1. If voltage is as specified, repair/replace wiring between S100-5 and J200-d. Go to 7 .	Step 2. If voltage is not as specified, go to 6 .
6. Check for 28 vdc between RESQ HST CONT circuit breaker terminal 1 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P200-d. Go to 7 .	Step 2. If voltage is not as specified, replace RESQ HST CONT circuit breaker (PARA 9-4-1). Go to 7 .
7. Procedure completed.	

TABLE NO. 14-2-1-2. After Pressing BACKUP CONTROL POWER Switch, Switch Light Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check that RESQ HST CONT AUX circuit breaker is pushed in.	

TABLE NO. 14-2-1-2. After Pressing BACKUP CONTROL POWER Switch, Switch Light Does Not Go On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
Step 1. If circuit breaker is in, go to 2. Step 2. If circuit breaker is not in, push in circuit breaker. Go to 4.
2. Check for 28 vdc between P965-3 and ground.
Step 1. If voltage is as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 4. Step 2. If voltage is not as specified, go to 3.
3. Check for 28 vdc between RESQ HST CONT AUX circuit breaker terminal 1 and ground.
Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P965-3. Go to 4. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 4.
4. Procedure completed.

TABLE NO. 14-2-1-3. Cable Does Not Pay Out When Using Rescue Hoist/Cargo Hook Pendant.

TEST OR INSPECTION CORRECTIVE ACTION
1. Does cable pay out when crewmember's hover grip HOIST thumbwheel switch is rotated to maximum DOWN?
Step 1. If cable pays out, go to 2. Step 2. If cable does not pay out, go to 9.
2. Check for 28 vdc between P3-V and P3-W.
Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 3.
3. Check continuity between P3-W and ground.
Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring. Go to 31.

TABLE NO. 14-2-1-3. Cable Does Not Pay Out When Using Rescue Hoist/Cargo Hook Pendant. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
4. Check continuity between P3-V and P154R-19.	Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, repair/replace wiring. Go to 31.
5. Check for 28 vdc between P154R-16 and ground.	Step 1. If voltage is as specified, replace crewman's hover grip (TM 11-70-23). Go to 31. Step 2. If voltage is not as specified, repair/replace wiring between P154R-16 and terminal 1 of RESQ HST CONT AUX circuit breaker. Go to 31.
6. Turn rescue hoist/cargo hook pendant RESCUE HOIST thumbwheel switch to full DOWN. Go to 7.	
7. Check continuity between: P968-B and P968-D P968-C and P968-F P968-E and P968-F	Step 1. If continuity is present, go to 8. Step 2. If continuity is not present, replace rescue hoist/cargo hook pendant (PARA 14-4-21). Go to 31.
8. Check continuity between: J968-B and P3-D J968-C and P3-K J968-D and P3-G J968-E and P3-S J968-F and P3-N	Step 1. If continuity is present, replace No. 7 relay panel (PARA 9-4-108). Go to 31. Step 2. If continuity is not present, repair/replace wiring, as required. Go to 31.
9. Press rescue hoist control box BACKUP CONTROL POWER switch and place HOIST switch to DOWN. Check that cable pays out.	Step 1. If cable pays out, go to 10. Step 2. If cable does not pay out, go to 29.

TABLE NO. 14-2-1-3. Cable Does Not Pay Out When Using Rescue Hoist/Cargo Hook Pendant. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
10. Press rescue hoist control box BACKUP CONTROL POWER switch to turn on normal power and turn rescue hoist/cargo hook pendant UP/DOWN thumb-wheel switch full down. Go to 11.	
11. Check continuity between: P965-8 and P3-E P965-6 and P3-H P965-7 and P3-L P965-10 and P3-P P965-9 and P3-T	Step 1. If continuity is present, go to 12. Step 2. If continuity is not present, repair/replace wiring, as required. Go to 31.
12. Disconnect P9 from rescue hoist electrical valve control. Go to 13.	
13. Check for 28 vdc between J9-A and J9-B of electrical valve control.	Step 1. If voltage is as specified, go to 14. Step 2. If voltage is not as specified, go to 24.
14. Check continuity between TB1-5 and TB1-6.	Step 1. If continuity is present, go to 15. Step 2. If continuity is not present, replace full-out (down) limit switch (PARA 14-4-40). Go to 31.
15. Check continuity between: P963-5 and P966-5 P963-6 and P966-6 P963-7 and P966-7 P963-14 and P966-14 P963-15 and P966-15	Step 1. If continuity is present, go to 16. Step 2. If continuity is not present, replace wiring harness (PARA 14-4-22), backup control panel harness (PARA 14-4-26), or rescue hoist control box harness (PARA 14-4-27), as required. Go to 31.
16. Disconnect connector P7 from rescue hoist control valve. Go to 17.	
17. Check for 171 to 209 ohms between:	

TABLE NO. 14-2-1-3. Cable Does Not Pay Out When Using Rescue Hoist/Cargo Hook Pendant. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
J7-A and J7-B J7-C and J7-D	
Step 1. If resistance is as specified, go to 18 .	
Step 2. If resistance is not as specified, replace control valve (PARA 14-4-10).	
Go to 31 .	
18. Check continuity between J1-14 and J1-15 on rescue hoist assembly.	
Step 1. If continuity is present, go to 19 .	
Step 2. If continuity is not present, replace rescue hoist assembly (PARA 14-4-1).	
Go to 31 .	
19. Check for 40 to 80 ohms between P9-A and P9-B.	
Step 1. If resistance is as specified, go to 20 .	
Step 2. If resistance is not as specified, replace solenoid valve (PARA 14-4-18).	
Go to 31 .	
20. Check continuity between:	
P4-F and P7-B	
P4-F and P7-D	
Step 1. If continuity is present, go to 21 .	
Step 2. If continuity is not present, replace electrical valve control assembly (PARA	
14-4-11). Go to 31 .	
21. Check resistance between the following pins of electrical valve control assembly and record meter readings, then go to 22:	
P4-E and P7-A	
P4-E and P7-C	
22. Reverse meter leads and check resistance again. Go to 23.	
23. Check that one meter reading is at least 10X as large as the other.	
Step 1. If one reading is at least 10X the other reading, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 31 .	
Step 2. If trouble remains, go to 27 .	
Step 3. If one reading is not at least 10X the other reading, replace electrical valve control assembly (PARA 14-4-11). Go to 31 .	

TABLE NO. 14-2-1-3. Cable Does Not Pay Out When Using Rescue Hoist/Cargo Hook Pendant. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
24. Check continuity between TB1-7 and TB1-8.	Step 1. If continuity is present, go to 25. Step 2. If continuity is not present, replace emergency full-out (down) limit switch (PARA 14-4-40). Go to 31.
25. Disconnect connector J4 from rescue hoist electrical valve control assembly. Go to 26.	
26. Check for 28 vdc between J4-A and ground.	Step 1. If voltage is as specified, replace electrical valve control (PARA 14-4-11). Go to 31. Step 2. If voltage is not as specified, replace rescue hoist assembly (PARA 14-4-1). Go to 31.
27. Connect a direct reading pressure gauge at rescue hoist hydraulic inlet. Go to 28.	
28. Start back-up hydraulic pump and check for 3000-3100 psi.	Step 1. If 3000-3100 psi is present, replace rescue hoist assembly (PARA 14-4-1). Go to 31. Step 2. If 3000-3100 psi is not present, troubleshoot hydraulic systems (PARA 7-2-1). Go to 31.
29. Check for 28 vdc between the following on rescue hoist control box: J2-2 and ground J2-3 and ground	Step 1. If voltage is as specified, go to 30. Step 2. If voltage is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 31.
30. Check continuity between: P963-2 and P966-2 P963-3 and P966-3	Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-1). Go to 31.

TABLE NO. 14-2-1-3. Cable Does Not Pay Out When Using Rescue Hoist/Cargo Hook Pendant. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If continuity is not present, replace wiring harness (PARA 14-4-22), backup control panel harness (PARA 14-4-26), or rescue hoist control box harness (PARA 14-4-27), as required. Go to 31. 31. Procedure completed.

TABLE NO. 14-2-1-4. Cable Pays Out At 50 Feet-Per-Minute.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Stop hoist with at least 5 feet of cable remaining on drum. Go to 2. 2. Check continuity between P966-12 and P966-13. Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, go to 4. 3. Check continuity between: P966-17 and ground P966-18 and ground Step 1. If continuity is present, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 5. Step 2. If continuity is not present, replace backup control panel (PARA 14-4-26) harness or rescue hoist control box harness (PARA 14-4-27). Go to 5. 4. Check continuity between: P966-12 and P963-12 P966-13 and P963-13 Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-1). Go to 5. Step 2. If continuity is not present, replace wiring harness (PARA 14-4-22), backup control panel harness (PARA 14-4-26), or rescue hoist control box harness (PARA 14-4-27), as required. Go to 5. 5. Procedure completed.

TABLE NO. 14-2-1-5. Cable Pays Out At 250 Feet-Per-Minute But Does Not Slow To 50 Feet-Per-Minute.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Stop hoist with 3 feet of cable remaining on drum. Go to 2.	
2. Check continuity between P966-12 and P966-13.	Step 1. If continuity is present, replace deceleration limit switch (PARA 14-4-14 or PARA 14-4-17). Go to 3. Step 2. If trouble remains, replace rescue hoist assembly (PARA 14-4-1). Go to 3. Step 3. If continuity is not present, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 3.
3. Procedure completed.	

TABLE NO. 14-2-1-6. Rescue Hoist Does Not Stop With 3-1/2 To 4-1/2 Wraps Of Cable On Drum.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Stop hoist with less than 3-1/2 to 4-1/2 wraps on cable drum. Go to 2.	
2. Check continuity between P966-14 and P966-15.	Step 1. If continuity is present, replace emergency full-out (down) limit switch (PARA 14-4-14 or PARA 14-4-17). Go to 3. Step 2. If continuity is not present, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 3.
3. Procedure completed.	

TABLE NO. 14-2-1-7. Elapsed-Time Clock Does Not Operate.

TEST OR INSPECTION	CORRECTIVE ACTION
1. While operating hoist, check for 28 vdc between J8-A and J8-B on electrical control valve.	

TABLE NO. 14-2-1-7. Elapsed-Time Clock Does Not Operate. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, replace elapsed-time clock (PARA 14-4-8). Go to 2. Step 2. If voltage is not as specified, replace rescue hoist assembly (PARA 14-4-1). Go to 2.
2. Procedure completed.	

TABLE NO. 14-2-1-8. Cable Does Not Reel In.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Turn rescue hoist/cargo hook pendant UP/DOWN thumbwheel switch to maximum UP position. Go to 2.	
2. Check continuity between P968-A and P968-B.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, replace rescue hoist/cargo hook pendant (PARA 14-4-21). Go to 14.
3. Check continuity between: J968-A and P3-A J968-B and P3-D P3-B and P965-5 P3-E and P965-8	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring. Go to 14.
4. Connect 28 vdc between J3-V (positive) and J3-W (negative). Check continuity between J3-A and J3-B.	Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, replace No. 7 relay panel (PARA 9-4-108). Go to 14.
5. Place pilot's cyclic stick HOIST switch to UP. Check that cable reels in.	

TABLE NO. 14-2-1-8. Cable Does Not Reel In. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If cable reels in, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 14. Step 2. If cable does not reel in, go to 6.
6. Press rescue hoist control box BACKUP CONTROL POWER switch and place HOIST switch to UP. Check that cable reels in.	Step 1. If cable reels in, go to 7. Step 2. If cable does not reel in, go to 13.
7. Press rescue hoist control box BACKUP CONTROL POWER switch to turn on normal power and turn rescue hoist/cargo hook pendant UP/DOWN thumbwheel switch full UP. Go to 8.	
8. Install an insulated jumper wire between J2-8 and J2-9 on rescue hoist control box. Go to 9.	
9. Check for 28 vdc between J2-7 and ground on rescue hoist control box.	Step 1. If voltage is as specified, go to 10. Step 2. If voltage is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 14.
10. Check for 15 vdc between J2-6 and J2-5 on rescue hoist control box.	Step 1. If voltage is as specified, go to 11. Step 2. If voltage is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 14.
11. Remove jumper wire from J2. Go to 12.	
12. Check continuity between:	
	P963-8 and P966-8 P963-9 and P966-9
	Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-1). Go to 14. Step 2. If continuity is not present, replace wiring harness (PARA 14-4-22), backup control panel harness (PARA 14-4-26), or rescue hoist control box harness (PARA 14-4-27), as required. Go to 14.
13. Check continuity between P963-1 and P966-1.	

TABLE NO. 14-2-1-8. Cable Does Not Reel In. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-1). Go to 14. Step 2. If continuity is not present, repair wiring harness (PARA 14-4-22), backup control panel harness (PARA 14-4-26), or rescue hoist control box harness (PARA 14-4-27), as required. Go to 14.
14. Procedure completed.	

TABLE NO. 14-2-1-9. Cable Reels In At 50 Feet-Per-Minute.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Stop hoist with at least 10 feet of cable payed out. Go to 2.	
2. Check continuity between P966-10 and P966-11.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, go to 4.
3. Check continuity between:	
	P966-17 and ground
	P966-18 and ground
	Step 1. If continuity is present, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 5. Step 2. If continuity is not present, replace wiring harness (PARA 14-4-22), backup control panel harness (PARA 14-4-26), or rescue hoist control box harness (PARA 14-4-27), as required. Go to 5.
4. Check continuity between:	
	P963-10 and P966-10
	P963-11 and P966-11
	Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-1). Go to 5. Step 2. If continuity is not present, replace wiring harness (PARA 14-4-22), backup control panel harness (PARA 14-4-26), or rescue hoist control box harness (PARA 14-4-27), as required. Go to 5.

TABLE NO. 14-2-1-9. Cable Reels In At 50 Feet-Per-Minute. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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|-------------------------|--|
| 5. Procedure completed. | |
|-------------------------|--|

TABLE NO. 14-2-1-10. Cable Reels In At 250 Feet-Per-Minute, But Does Not Slow To 50 Feet-Per-Minute.

TEST OR INSPECTION	CORRECTIVE ACTION
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- | | |
|--|--|
| 1. Stop hoist with 8 feet of cable payed out. Go to 2. | |
| 2. Check continuity between P966-10 and P966-11. | |

Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-1).
Go to 3.

Step 2. If continuity is not present, replace backup control panel (PARA 14-4-26) or
rescue hoist control box (PARA 14-4-27). Go to 3.

- | | |
|-------------------------|--|
| 3. Procedure completed. | |
|-------------------------|--|

TABLE NO. 14-2-1-11. Hoist Does Not Stop With Cable Fully In.

TEST OR INSPECTION	CORRECTIVE ACTION
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- | | |
|--|--|
| 1. Check continuity between P966-8 and P966-9. | |
|--|--|

Step 1. If continuity is present, go to 2.

Step 2. If continuity is not present, replace backup control panel (PARA 14-4-26) or
rescue hoist control box (PARA 14-4-27). Go to 3.

- | | |
|---|--|
| 2. Check continuity between J1-8 and J1-9 on rescue hoist assembly. | |
|---|--|

Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-1).
Go to 3.

Step 2. If continuity is not present, replace backup control panel (PARA 14-4-26) or
rescue hoist control box (PARA 14-4-27). Go to 3.

TABLE NO. 14-2-1-11. Hoist Does Not Stop With Cable Fully In. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Procedure completed.	

TABLE NO. 14-2-1-12. Cable Does Not Pay Out When Using Crewmember's Hover Grip.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Turn hover grip UP/DOWN thumbwheel switch to full DOWN. Go to 2.	
2. Check continuity between J154R-15 and J154R-17.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, replace crewman's hover grip (TM 11-70-23). Go to 18.
3. Check continuity between: P154R-15 and P3-J P154R-17 and P3-F P3-E and P965-8 P3-H and P965-6	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring. Go to 18.
4. Check continuity between: J3-F and J3-E J3-J and J3-H	Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, replace No. 7 relay panel (PARA 9-4-108). Go to 18.
5. Place pilot's cyclic stick HOIST thumbwheel switch to DOWN position. Check that cable pays out.	Step 1. If cable pays out, go to 6. Step 2. If cable does not pay out, go to 7.
6. Check continuity between J154R-21 and J154R-22.	

TABLE NO. 14-2-1-12. Cable Does Not Pay Out When Using Crewmember's Hover Grip. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 18. Step 2. If continuity is not present, replace crewman's hover grip (TM 11-70-23). Go to 18.
7. Press rescue hoist control box BACKUP CONTROL POWER switch and place HOIST switch to DOWN . Check that cable pays out.	Step 1. If cable pays out, go to 8. Step 2. If cable does not pay out, go to 15.
8. Press rescue hoist control box BACKUP CONTROL POWER switch to turn on normal power and turn hover grip UP/DOWN thumbwheel switch full DOWN . Go to 9.	
9. Install an insulated jumper wire between J2-14 and J2-15 on rescue hoist control box. Go to 10.	
10. Check for 28 vdc between J2-7 and ground.	Step 1. If voltage is as specified, go to 11. Step 2. If voltage is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 18.
11. Check for 15 vdc between J2-5 and J2-6 .	Step 1. If voltage is as specified, go to 12. Step 2. If voltage is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 18.
12. Check continuity between:	P963-5 and P966-5 P963-6 and P966-6 P963-7 and P966-7 P963-14 and P966-14 P963-15 and P966-15
	Step 1. If continuity is present, remove jumper from J2, then replace rescue hoist assembly (PARA 14-4-1). Go to 18. Step 2. If continuity is not present, remove jumper wire from J2, then replace wiring harness (PARA 14-4-22), backup control panel harness (PARA 14-4-26), or rescue hoist control box harness (PARA 14-4-27), as required. Go to 18.

TABLE NO. 14-2-1-12. Cable Does Not Pay Out When Using Crewmember's Hover Grip. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
13. Connect a direct reading pressure gauge at rescue hoist hydraulic inlet. Go to 14.	
14. Start back-up hydraulic pump and check for 3000-3100 psi.	
	Step 1. If 3000-3100 psi is present, replace rescue hoist assembly (PARA 14-4-1). Go to 18.
	Step 2. If 3000-3100 psi is not present, troubleshoot hydraulic systems (PARA 7-2-1). Go to 18.
15. Check for 28 vdc between: J2-2 and ground J2-3 and ground	
	Step 1. If voltage is as specified, go to 16.
	Step 2. If voltage is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 18.
16. Check continuity between: P966-2 and P963-2 P966-3 and P963-3	
	Step 1. If continuity is present, go to 17.
	Step 2. If continuity is not present, replace wiring harness (PARA 14-4-22). Go to 18.
17. Check continuity between J1-3 on rescue hoist assembly and J9-D on electrical valve control.	
	Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-1). Go to 18.
	Step 2. If continuity is not present, repair/replace wiring. Go to 18.
18. Procedure completed.	

TABLE NO. 14-2-1-13. Cable Does Not Reel In Using Crewman's Hover Grip.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Turn crewman's hover grip UP/DOWN thumbwheel switch to maximum UP position. Go to 2.	
2. Check continuity between J154R-17 and J154R-18.	
	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, replace crewmember's hover grip (TM 11-70-23). Go to 14.
3. Check continuity between: P154R-17 and P3-F P154R-18 and P3-C P3-E and P965-8 P3-B and P965-5	
	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring. Go to 14.
4. Check continuity between: J3-E and J3-F J3-C and J3-B	
	Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, replace No. 7 relay panel (PARA 9-4-108). Go to 14.
5. Place pilot's cyclic stick HOIST switch to UP. Check that cable reels in.	
	Step 1. If cable reels in, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 14. Step 2. If cable does not reel in, go to 6.
6. Press rescue hoist control box BACKUP CONTROL POWER switch and place HOIST switch to UP. Check that cable reels in.	
	Step 1. If cable reels in, go to 7. Step 2. If cable does not reel in, go to 13.
7. Press rescue hoist control box BACKUP CONTROL POWER switch to turn on normal power and turn rescue hoist/cargo hook pendant UP/DOWN thumbwheel switch full UP. Go to 8.	

TABLE NO. 14-2-1-13. Cable Does Not Reel In Using Crewman's Hover Grip. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
8. Install an insulated jumper wire between J2-8 and J2-9 on rescue hoist control box. Go to 9.	
9. Check for 28 vdc between J2-7 and ground on rescue hoist control box.	
	Step 1. If voltage is as specified, go to 10. Step 2. If voltage is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 14.
10. Check for 15 vdc between J2-6 and J2-5 on rescue hoist control box.	
	Step 1. If voltage is as specified, go to 11. Step 2. If voltage is not as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 14.
11. Remove jumper wire from J2. Go to 12.	
12. Check continuity between: P963-8 and P966-8 P963-9 and P966-9	
	Step 1. If continuity is present, replace wiring harness (PARA 14-4-22), backup control panel harness (PARA 14-4-26), or rescue hoist control box harness (PARA 14-4-27), as required. Go to 14. Step 2. If continuity is not present, replace rescue hoist assembly (PARA 14-4-1). Go to 14.
13. Check continuity between P963-1 and P966-1.	
	Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-1). Go to 14. Step 2. If continuity is not present, replace wiring harness (PARA 14-4-22), backup control panel harness (PARA 14-4-26), or rescue hoist control box harness (PARA 14-4-27), as required. Go to 14.
14. Procedure completed.	

TABLE NO. 14-2-1-14. Cable Reels In At 50 Feet-Per-Minute.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Stop hoist with at least 10 feet of cable payed out. Go to 2.	
2. Check continuity between P966-10 and P966-11.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, go to 4.
3. Check continuity between: P966-17 and ground P966-18 and ground	Step 1. If continuity is present, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 5. Step 2. If continuity is not present, replace backup control panel harness (PARA 14-4-26) or rescue hoist control box harness (PARA 14-4-27). Go to 5.
4. Check continuity between: P963-10 and P966-10 P963-11 and P966-11	Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-1). Go to 5. Step 2. If continuity is not present, replace wiring harness (PARA 14-4-22). Go to 5.
5. Procedure completed.	

TABLE NO. 14-2-1-15. Cable Does Not Pay Out Using Pilot's Cyclic Stick HOIST Switch.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between J120-S and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, repair/replace wiring between J120-S and RESQ HST CONT circuit breaker terminal 1. Go to 5.
2. Set pilot's cyclic stick HOIST switch to DOWN. Go to 3.	

TABLE NO. 14-2-1-15. Cable Does Not Pay Out Using Pilot's Cyclic Stick HOIST Switch. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Check continuity between P120-S and P120-c.	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, troubleshoot pilot's cyclic stick (PARA 11-2-1). Go to 5.
4. Check continuity between J120-c and P965-12.	Step 1. If continuity is present, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 5. Step 2. If continuity is not present, repair/replace wiring. Go to 5.
5. Procedure completed.	

TABLE NO. 14-2-1-16. Cable Does Not Reel In Using Pilot's Cyclic Stick HOIST Switch.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With pilot's cyclic stick HOIST switch in UP, check continuity between P120-R and P120-S.	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, troubleshoot pilot's cyclic stick (PARA 11-2-1). Go to 3.
2. With pilot's cyclic stick HOIST switch in UP, check for 28 vdc between P965-11 and ground.	Step 1. If voltage is as specified, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 3. Step 2. If voltage is not as specified, repair/replace wiring between J120-R and P965-11. Go to 3.
3. Procedure completed.	

TABLE NO. 14-2-1-17. Cable Does Not Pay Out Using Copilot's Cyclic Stick HOIST Switch.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between J119-c and ground.	Step 1. If voltage is as specified, repair/replace wiring between J119-c and P247-G. Go to 2. Step 2. If voltage is not as specified, repair/replace wiring between J119-S and P248-B. Go to 2.
2. Procedure completed.	

TABLE NO. 14-2-1-18. Cable Does Not Reel In Using Copilot's Cyclic Stick HOIST Switch.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With copilot's cyclic stick HOIST switch in UP, check continuity between P119-R and P119-S.	Step 1. If continuity is present, repair/replace wiring between J119-R and P247-Z. Go to 2. Step 2. If continuity is not present, troubleshoot pilot's cyclic stick (PARA 11-2-1). Go to 2.
2. Procedure completed.	

TABLE NO. 14-2-1-19. Hoist Heat Exchanger Fan Motor Does Not Run.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 vac between: P962-1 and P962-4 P962-2 and P962-4 P962-3 and P962-4	Step 1. If voltage is as specified, replace rescue hoist assembly (PARA 14-4-1). Go to 7. Step 2. If voltage is not as specified, go to 2.
2. Check continuity between:	

TABLE NO. 14-2-1-19. Hoist Heat Exchanger Fan Motor Does Not Run. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	P353-B and P962-1 P353-D and P962-2 P353-F and P962-3 P962-4 and ground Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring or wiring harness (PARA 14-4-22), as required. Go to 7.
3. Check for 115 vac between:	P353-C and ground P353-E and ground P353-G and ground Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, go to 6.
4. Check for 28 vdc between P353-A and ground.	Step 1. If voltage is as specified, go to 5. Step 2. If voltage is not as specified, repair/replace wiring between P353-A and P246-V. Go to 7.
5. Check continuity between:	P353-N and ground P353-P and ground P353-T and ground Step 1. If continuity is present, replace hoist relay panel (PARA 14-4-31). Go to 7. Step 2. If continuity is not present, repair/replace wiring, as required. Go to 7.
6. Check for 115 vac between RESQ HST PWR circuit breaker:	Terminal A1 and ground Terminal B1 and ground Terminal C1 and ground Step 1. If voltage is as specified, repair/replace wiring, as required between, then go to 7: Terminal A1 and P353-C Terminal B1 and P353-E Terminal C1 and P353-G

TABLE NO. 14-2-1-19. Hoist Heat Exchanger Fan Motor Does Not Run. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, replace RESQ HST PWR circuit breaker (PARA 9-4-1). Go to 7 .
7. Procedure completed.	

TABLE NO. 14-2-1-20. Multimeter Does Not Indicate 28 Vdc Across P6-A And P6-B With RSQ HOIST Switch To ALL And Auxiliary Rescue Hoist Control Panel HOIST SHEAR Switch Held To SHEAR.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between J352-B and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 6.
2. With auxiliary rescue hoist shear switch held to SHEAR, check continuity between P352-B and P352-D.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, replace auxiliary rescue hoist control panel (PARA 14-4-25). Go to 9.
3. Reconnect J352. With HOIST SHEAR switch held to SHEAR, check for 28 vdc between P452-C and ground.	Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, repair/replace wiring. Go to 9.
4. With hover trim panel SHEAR/SAFE/EMER switch in SAFE position, check continuity between J452-C and J452-P.	Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, replace hover trim control panel (TM 11-70-23). Go to 9.
5. Check for 28 vdc between P6-A and ground.	Step 1. If voltage is as specified, repair/replace wiring between P6-B and GND119-1. Go to 9.

**TABLE NO. 14-2-1-20. Multimeter Does Not Indicate 28 Vdc Across P6-A And P6-B With RSQ HOIST Switch To ALL And Auxiliary Rescue Hoist Control Panel HOIST SHEAR Switch Held To SHEAR.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
Step 2. If voltage is not as specified, repair/replace wiring between P6-A and P452-P. Go to 9 .	
6. Check for 28 vdc between RSQ HOIST switch terminal 1 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between J352-B and terminal 1. Go to 9 .	
Step 2. If voltage is not as specified, go to 7 .	
7. Check for 28 vdc between RSQ HOIST switch terminal 2 and ground.	
Step 1. If voltage is as specified, replace RSQ HOIST switch S100 (PARA 9-4-5). Go to 9 .	
Step 2. If voltage is not as specified, go to 8 .	
8. Check for 28 vdc between HOIST CABLE SHEAR circuit breaker terminal 1 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and RSQ HOIST switch terminal 2. Go to 9 .	
Step 2. If voltage is not as specified, replace HOIST CABLE SHEAR circuit breaker (PARA 9-4-1). Go to 9 .	
9. Procedure completed.	

TABLE NO. 14-2-1-21. Multimeter Does Not Indicate 28 Vdc With Pilot's Cyclic Stick EMER REL Switch Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between J120-e and ground.	
Step 1. If voltage is as specified, go to 2 .	
Step 2. If voltage is not as specified, go to 3 .	

TABLE NO. 14-2-1-21. Multimeter Does Not Indicate 28 Vdc With Pilot's Cyclic Stick EMER REL Switch Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
2. While pressing and holding pilot's cyclic stick EMER REL switch, check continuity between P120-e and P120-f.	Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, troubleshoot pilot's cyclic stick (PARA 11-2-1). Go to 8.
3. Check for 28 vdc between J200-J and ground.	Step 1. If voltage is as specified, repair/replace wiring between P200-J and J120-e. Go to 8. Step 2. If voltage is not as specified, go to 4.
4. Check for 28 vdc between CARGO HOOK EMER circuit breaker terminal 1 and ground.	Step 1. If voltage is as specified, repair/replace wiring. Go to 8. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 8.
5. Check continuity between: J120-f and P247-q J247-q and P246-c P246-X and J352-M J352-D and P452-C J969-23 and P452-P P969-23 and P964-3 P6-A and J3-1	Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, repair/replace wiring or wiring harness (PARA 14-4-22), as required. Go to 8.
6. Check continuity between J246-X and J246-c.	Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, repair/replace RSQ HOIST switch S100 (PARA 9-4-5). Go to 8.
7. Check continuity between P352-M and P352-D.	

2. While pressing and holding pilot's cyclic stick EMER REL switch, check continuity between P120-e and P120-f.

Step 1. If continuity is present, go to 5.

Step 2. If continuity is not present, troubleshoot pilot's cyclic stick (PARA 11-2-1).

Go to 8.

3. Check for 28 vdc between J200-J and ground.

Step 1. If voltage is as specified, repair/replace wiring between P200-J and J120-e.

Go to 8.

Step 2. If voltage is not as specified, go to 4.

4. Check for 28 vdc between CARGO HOOK EMER circuit breaker terminal 1 and ground.

Step 1. If voltage is as specified, repair/replace wiring. Go to 8.

Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 8.

5. Check continuity between:

J120-f and P247-q

J247-q and P246-c

P246-X and J352-M

J352-D and P452-C

J969-23 and P452-P

P969-23 and P964-3

P6-A and J3-1

Step 1. If continuity is present, go to 6.

Step 2. If continuity is not present, repair/replace wiring or wiring harness (PARA 14-4-22), as required. Go to 8.

6. Check continuity between J246-X and J246-c.

Step 1. If continuity is present, go to 7.

Step 2. If continuity is not present, repair/replace RSQ HOIST switch S100 (PARA 9-4-5). Go to 8.

7. Check continuity between P352-M and P352-D.

TABLE NO. 14-2-1-21. Multimeter Does Not Indicate 28 Vdc With Pilot's Cyclic Stick EMER REL Switch Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, replace hover trim control panel (TM 11-70-23). Go to 8. Step 2. If continuity is not present, replace auxiliary rescue hoist control panel (PARA 14-4-25). Go to 8.
8. Procedure completed.	

TABLE NO. 14-2-1-22. Multimeter Does Not Indicate 28 Vdc With Copilot's Cyclic Stick EMER REL Switch Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between J119-e and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, repair/replace wiring between J119-e and P247-e. Go to 3.
2. Check continuity between J119-e and J119-f.	Step 1. If continuity is present, repair/replace wiring between J119-f and J120-d. Go to 3. Step 2. If continuity is not present, troubleshoot copilot's cyclic stick (PARA 11-2-1). Go to 3.
3. Procedure completed.	

TABLE NO. 14-2-1-23. HYD OVERHEAT Lamp Does Not Light.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between P965-4 and J224-B.	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, repair/replace wiring. Go to 3.

TABLE NO. 14-2-1-23. HYD OVERHEAT Lamp Does Not Light. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

2. Check continuity between P967-2 and ground.	
--	--

Step 1. If continuity is present, replace temperature switch (PARA 7-4-45). Go to 3.	
--	--

Step 2. If trouble remains, replace backup control panel (PARA 14-4-26) or rescue hoist control box (PARA 14-4-27). Go to 3.	
--	--

Step 3. If continuity is not present, repair/replace wiring. Go to 3.	
---	--

3. Procedure completed.	
-------------------------	--

TABLE NO. 14-2-1-24. Rescue Hoist Light Does Not Go On Full Bright.

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

1. Check that RESQ HST LT CONT and RESQ HST LT PWR circuit breakers are pushed in.	
--	--

Step 1. If breakers are in, go to 2.	
--------------------------------------	--

Step 2. If breakers are not in, push in circuit breakers. Go to 6.	
--	--

2. Check for 28 vdc between P969-1 and P969-4.	
--	--

Step 1. If voltage is as specified, go to 3.	
--	--

Step 2. If voltage is not as specified, go to 4.	
--	--

3. Check continuity between:	
------------------------------	--

P969-1 and rescue hoist light terminal +	
--	--

P969-4 and rescue hoist light terminal -	
--	--

Step 1. If continuity is present, replace rescue hoist light (PARA 14-4-23). Go to 6.	
---	--

Step 2. If continuity is not present, repair/replace wiring. Go to 6.	
---	--

4. Check continuity between:	
------------------------------	--

J969-1 and P357-B	
-------------------	--

J969-4 and ground	
-------------------	--

P357-A and RESQ HST LT PWR circuit breaker terminal 1	
---	--

J352-J and RESQ HST LT CONT circuit breaker terminal 1	
--	--

J352-K and P353-H	
-------------------	--

TABLE NO. 14-2-1-24. Rescue Hoist Light Does Not Go On Full Bright. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, repair/replace wiring. Go to 6.
5. Check continuity between P352-J and P352-K.	Step 1. If continuity is present, replace hoist relay panel (PARA 14-4-31). Go to 6. Step 2. If continuity is not present, replace auxiliary rescue hoist control panel (PARA 14-4-25). Go to 6.
6. Procedure completed.	

TABLE NO. 14-2-1-25. Rescue Hoist Light Intensity Does Not Go From Off To Full Bright.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check that RESQ HST LT DIM circuit breaker is pushed in.	Step 1. If breaker is in, go to 2. Step 2. If breaker is not in, push in circuit breaker. Go to 5.
2. Check continuity between:	J352-F and RESQ HST LT DIM circuit breaker terminal 1 J352-G and ground J352-H and ground J352-V and ground J352-S and P353-U J352-T and P353-U J352-U and P353-R P353-S and J969-1 Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring. Go to 5.
3. Check continuity between P352-J and P352-U.	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, replace auxiliary rescue hoist control panel (PARA 14-4-25). Go to 5.

TABLE NO. 14-2-1-25. Rescue Hoist Light Intensity Does Not Go From Off To Full Bright. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
4. Check continuity between P352-S and P352-F.	Step 1. If continuity is present, replace hoist relay panel (PARA 14-4-31). Go to 5 . Step 2. If continuity is not present, replace auxiliary rescue hoist control panel (PARA 14-4-25). Go to 5 .
5. Procedure completed.	

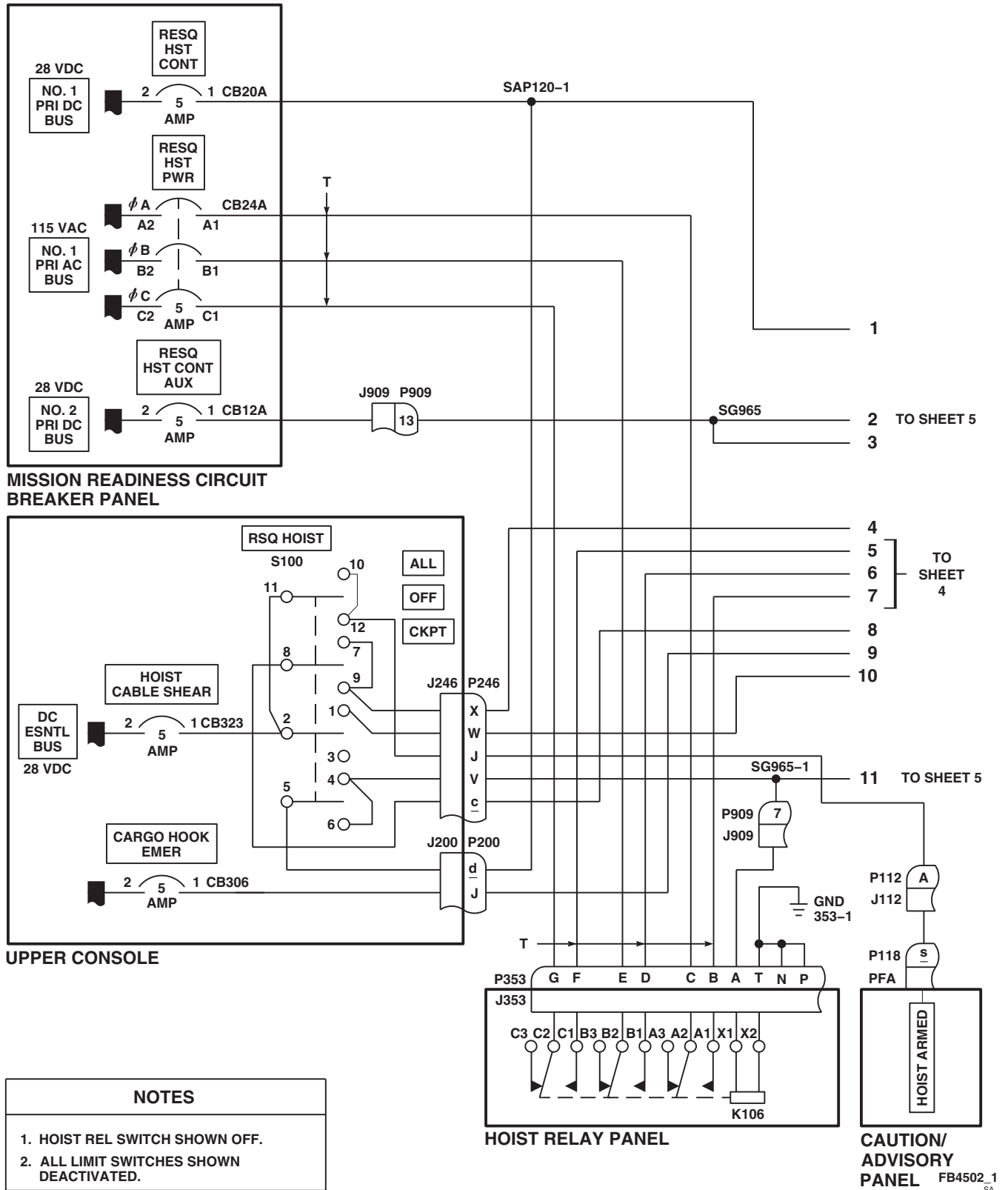


FIGURE 14-2-1-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 8)

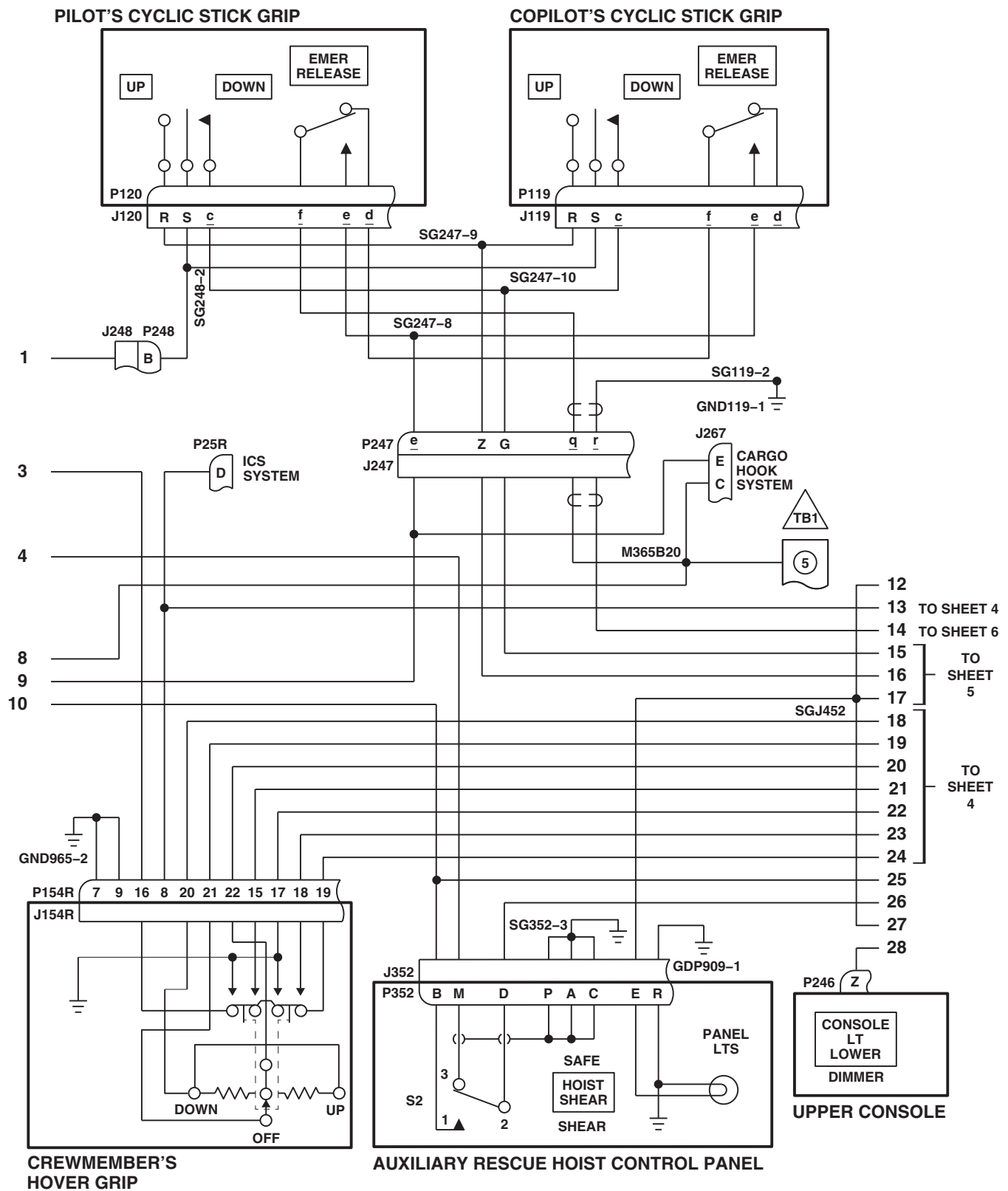
FB4502_2
SA

FIGURE 14-2-1-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 8)

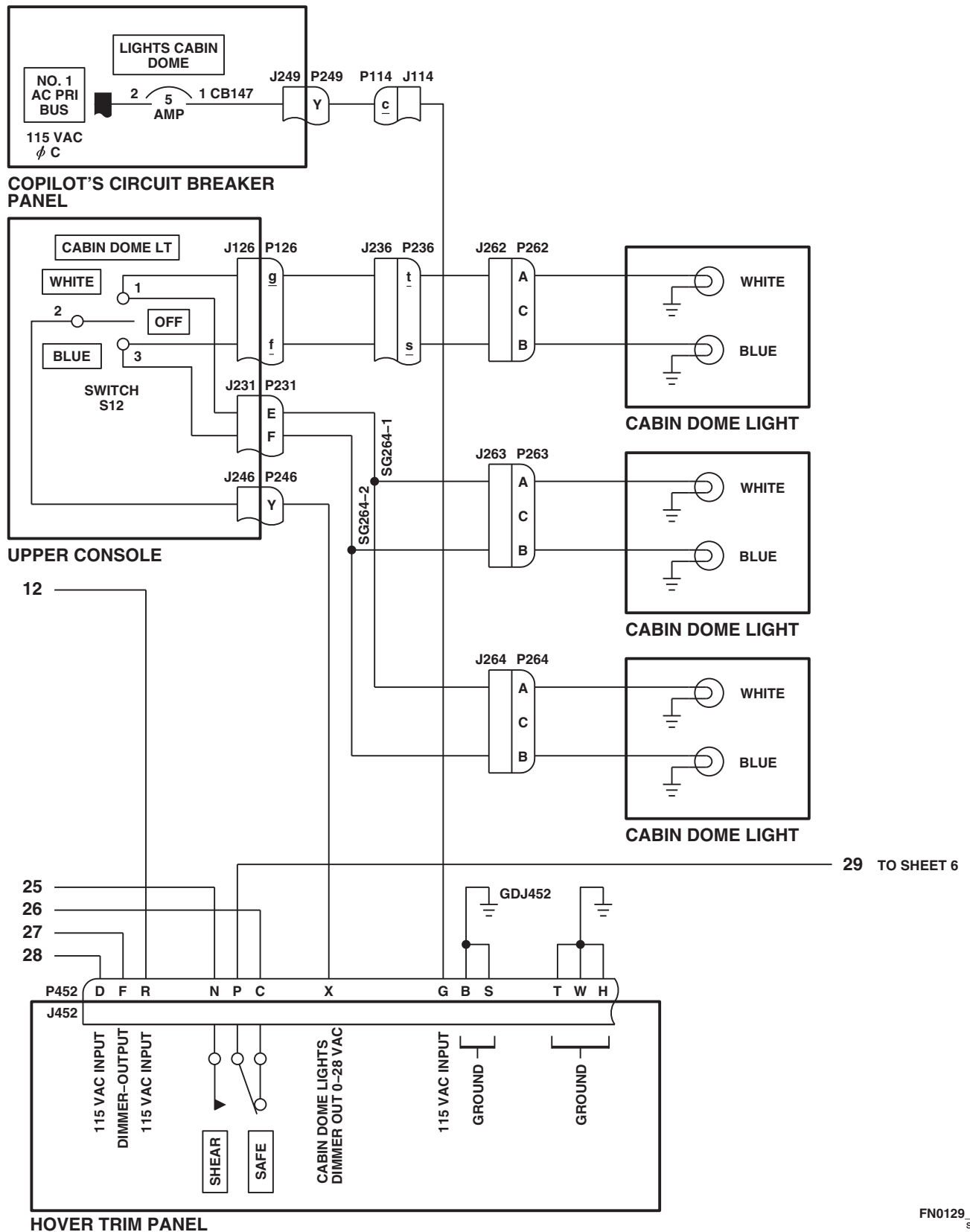


FIGURE 14-2-1-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 8)

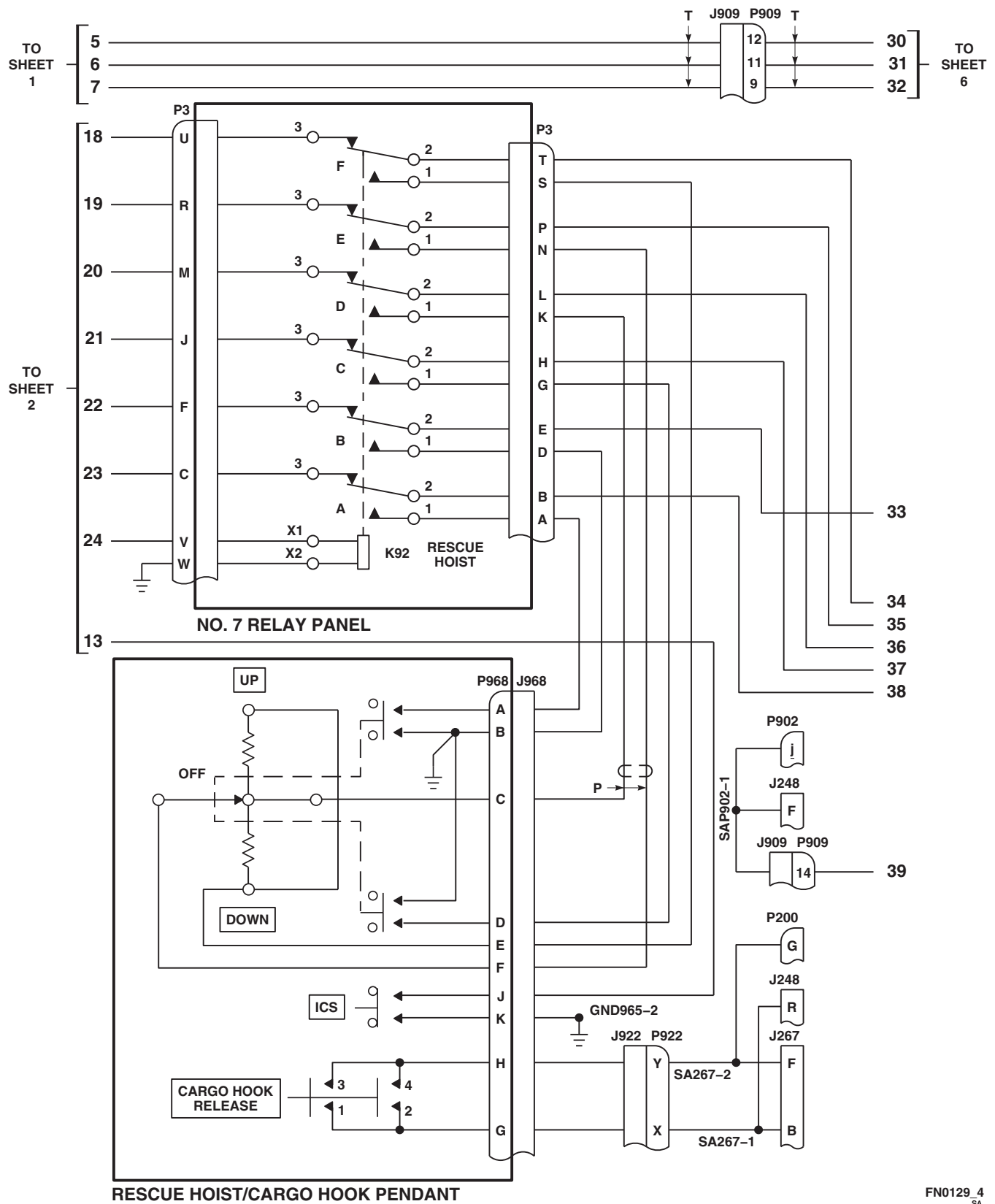
FN0129_4
SA

FIGURE 14-2-1-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 8)

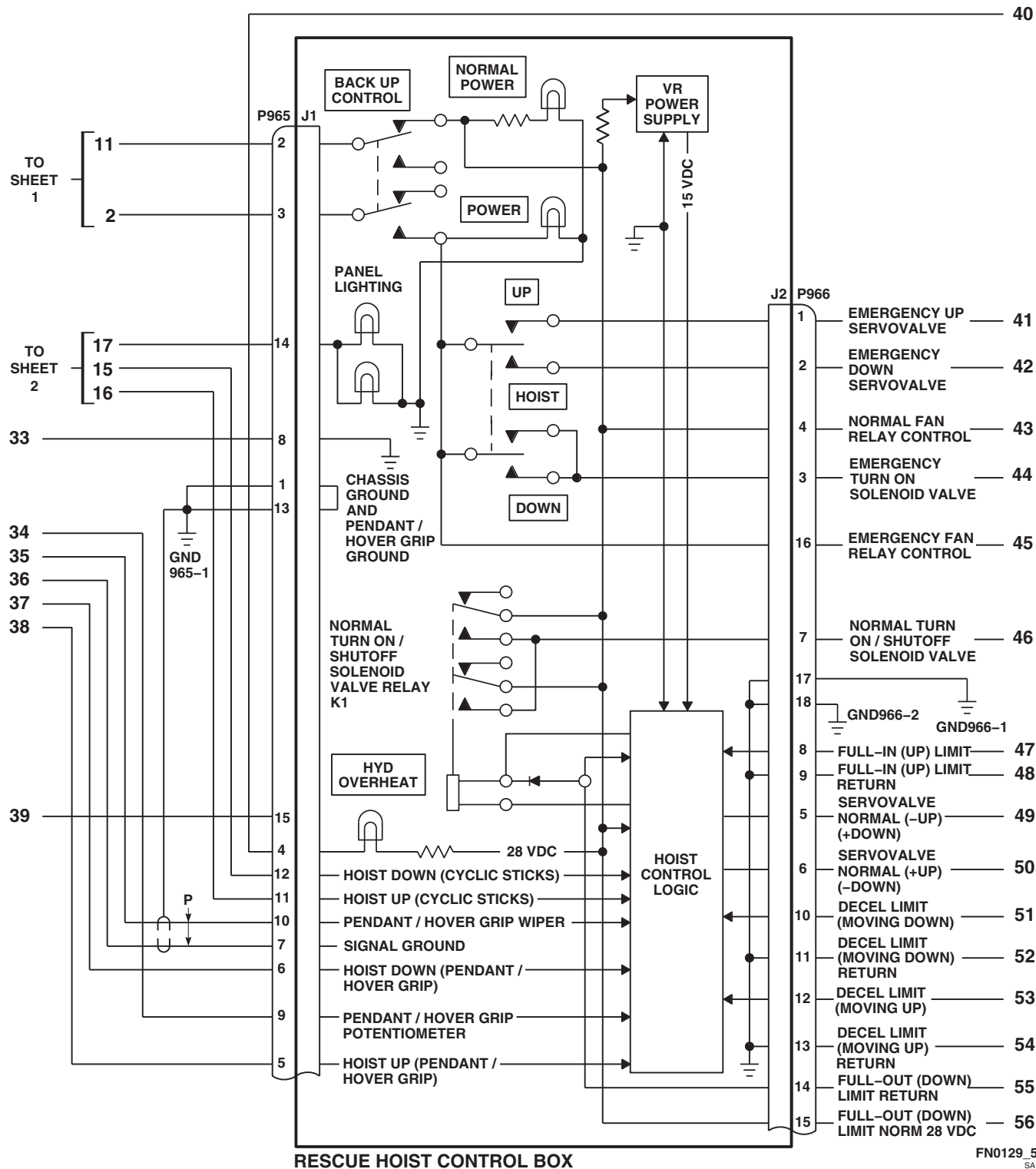


FIGURE 14-2-1-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 8)

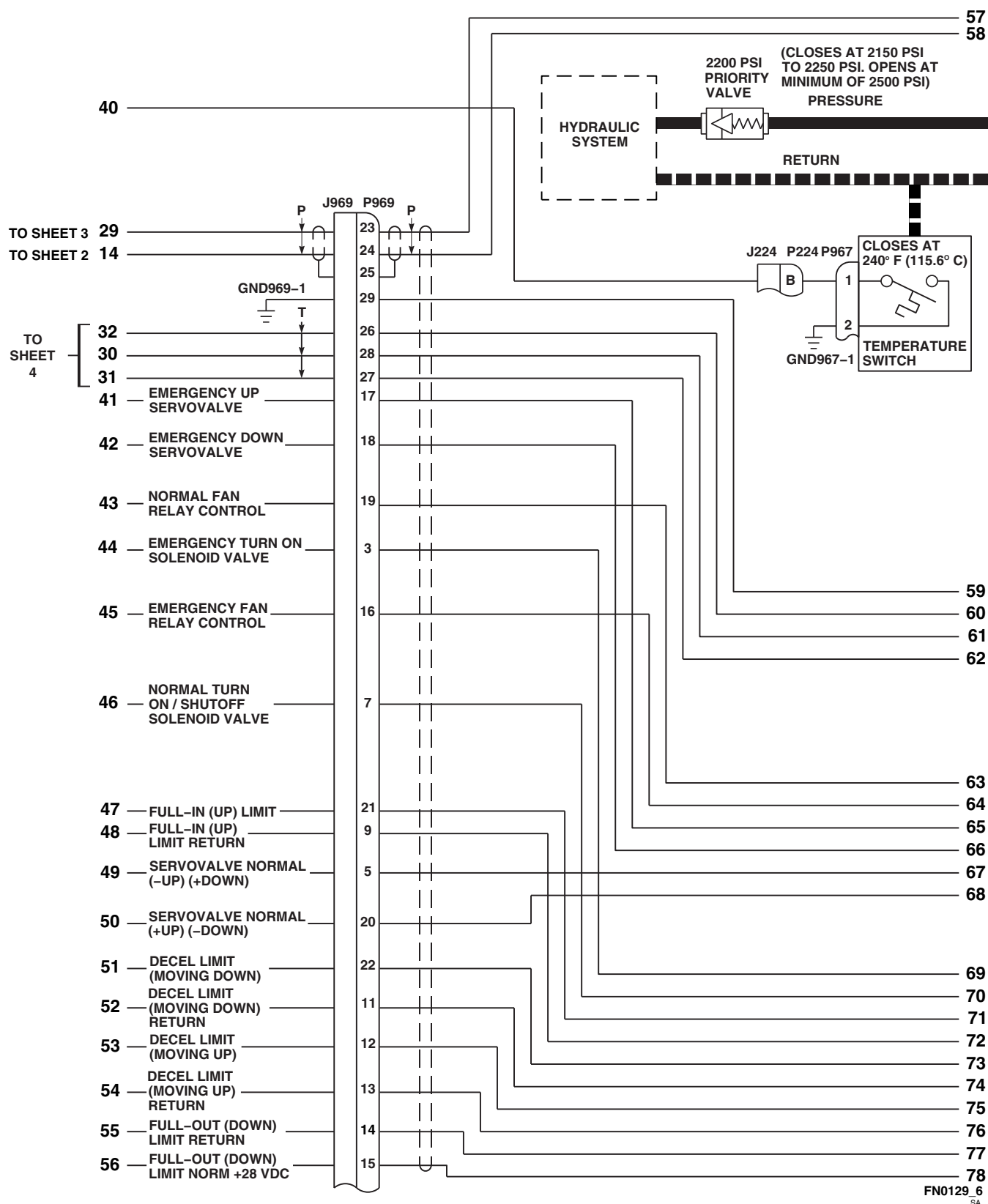


FIGURE 14-2-1-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 6 OF 8)

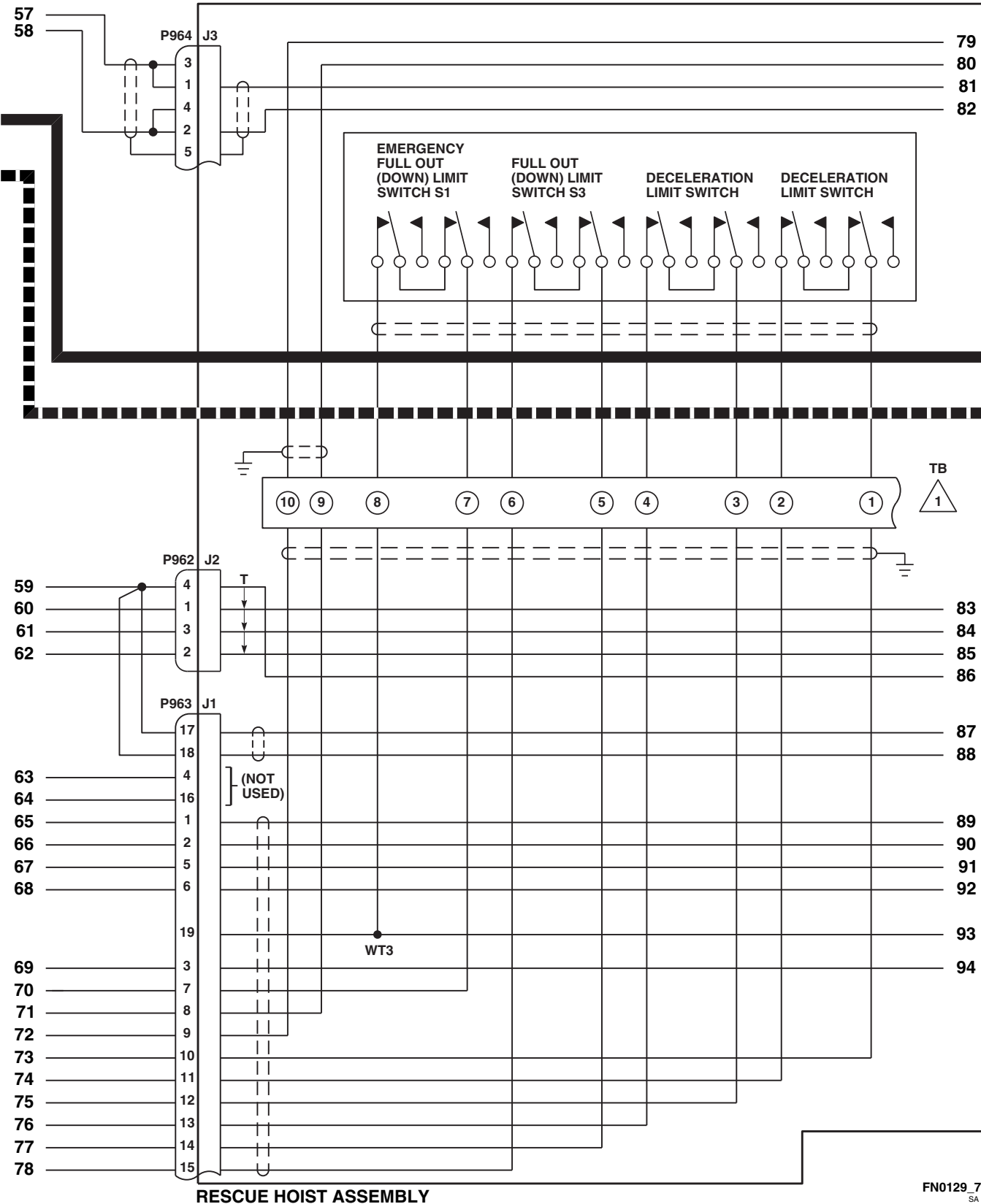


FIGURE 14-2-1-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 7 OF 8)

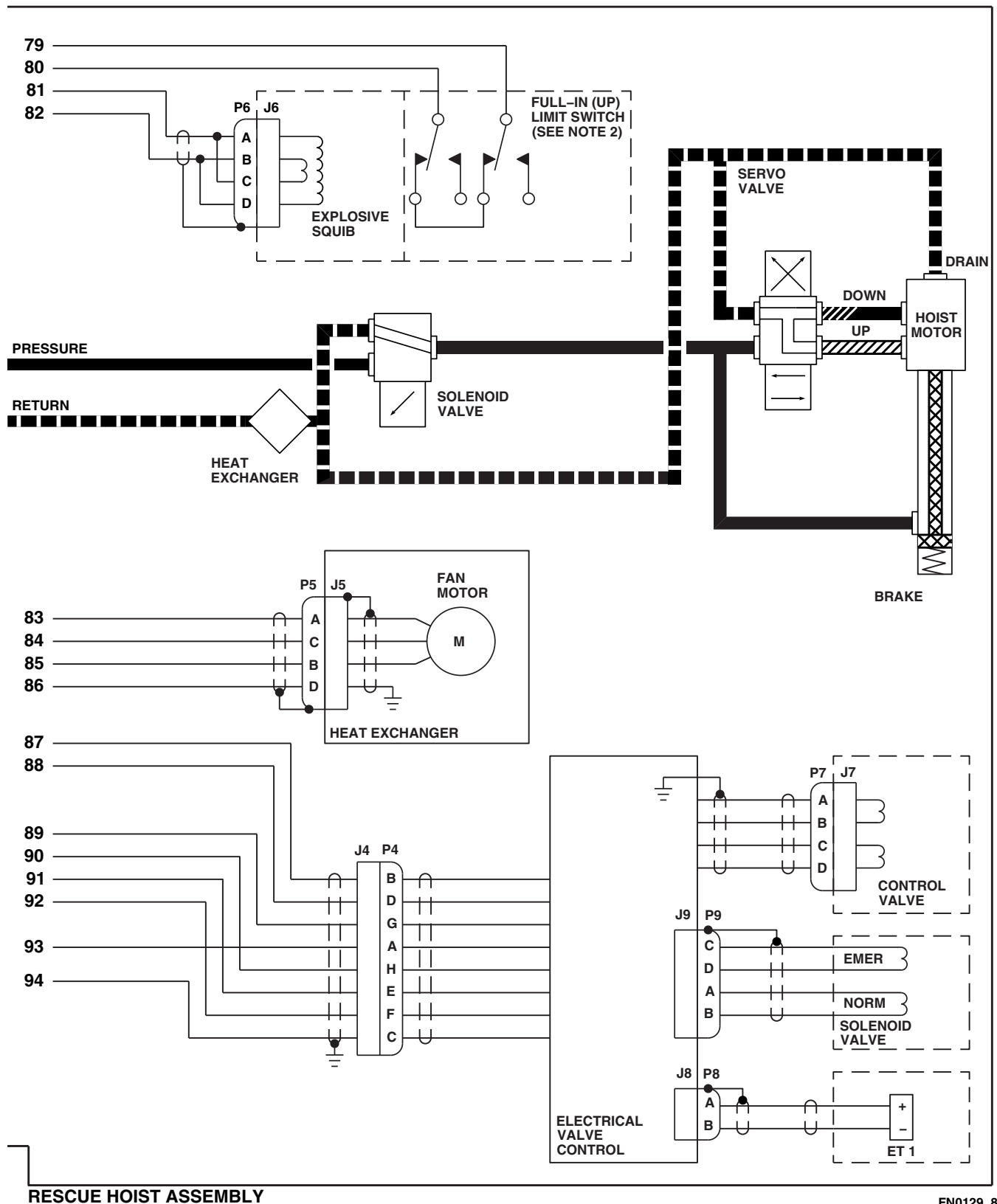
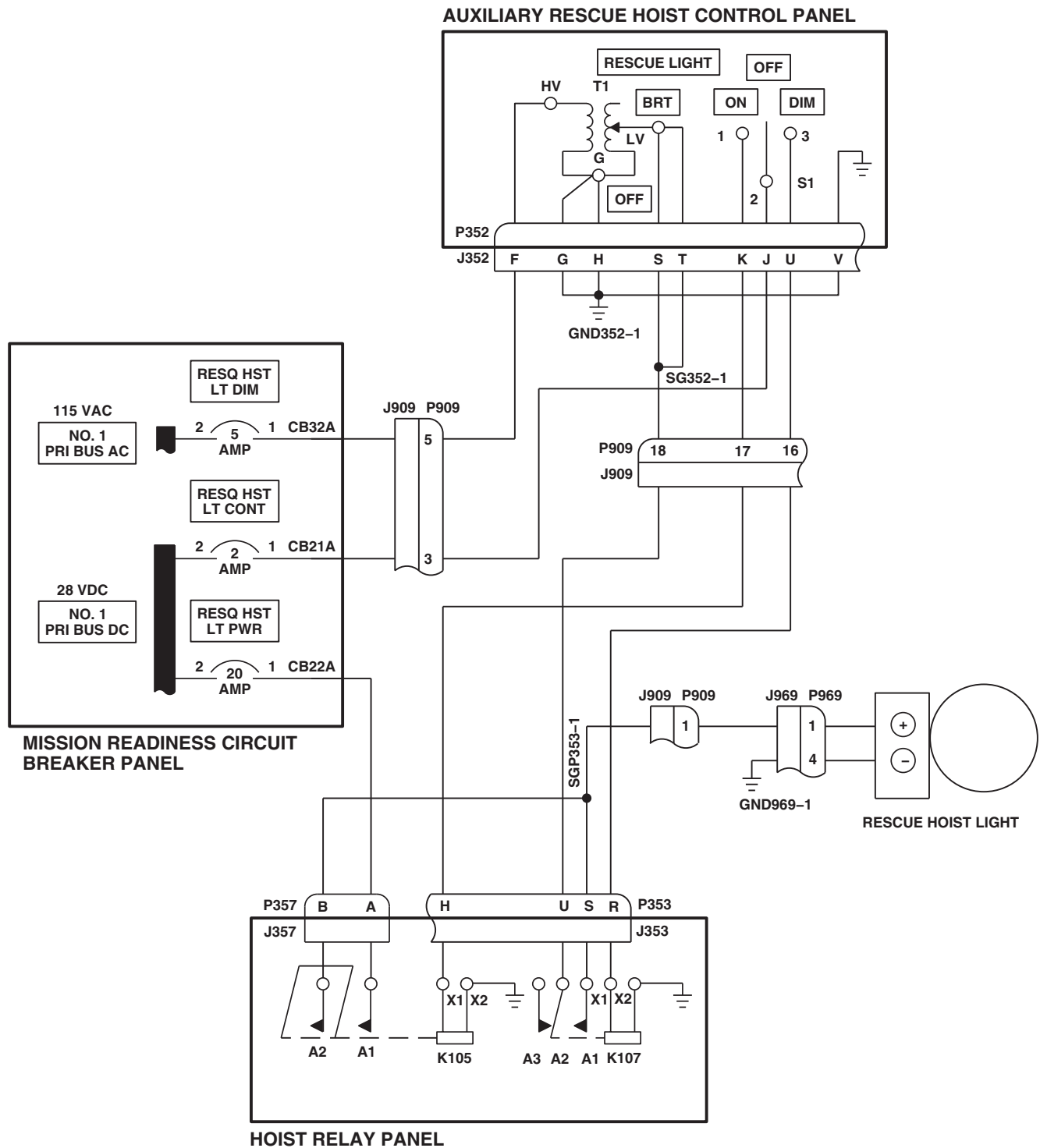
FN0129_8
SA

FIGURE 14-2-1-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 8 OF 8)



FB4503
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FIGURE 14-2-1-2. RESCUE HOIST LIGHT SCHEMATIC DIAGRAM

14-2-1-45/(14-2-1-46 Blank)

14-2-2 RESCUE HOIST SYSTEM **HOIST BL-29900-30** .

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45
- Large Container
- Infrared Light Test Set, Figure H-149, Appendix H
- 28 Vdc Test Lamp
- Jumper Wires

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit
- Gloves

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The rescue hoist kit operational/troubleshooting procedure is subdivided as follows:

- (1) Cable Shear
- (2) Rescue Hoist Assembly
- (3) Dual-Mode Controllable Searchlight (Provisional)

14-2-2.1 Cable Shear.

WARNING

The squib is an electrically activated explosive device. Improper checkout of the cable shear function could result in serious injury or death to personnel.

- Disconnect connector P6 from squib on rescue hoist assembly.

WARNING

All squibs and squib actuated devices shall be handled as live ammunition. Squibs or squib actuated devices which

have been fired may retain an explosive residue capable of presenting a hazardous condition.

- Carefully install a piece of aluminum foil around squib pins and install shipping cap.
- Make sure HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel is pushed in.
- Turn on external electrical power (PARA 1-6-16).
- On pilot’s hoist control panel, place HOIST POWER-ON-OFF switch to ON.
- Connect 28 vdc test lamp between pins A and B on connector P6.
- Place crew’s hoist control panel ARM-TEST switch to ARM.

- h. On crew's hoist control panel, lift CABLE CUT switch guard, press and hold CABLE CUT switch.

RESULT: Test lamp shall go on.

- If result is not as specified, go to TABLE NO. 14-2-2-1.

- i. Release crew's hoist control panel CABLE CUT switch and lower guard.

RESULT: Test lamp shall go off.

- If test lamp does not go off, replace crew's hoist control panel (PARA 14-4-39).

- j. On pilot's hoist control panel, lift CABLE CUT switch guard and press and hold CABLE CUT switch.

RESULT: Test lamp shall go on.

- If test lamp does not go on, go to TABLE NO. 14-2-2-2.

- k. Release pilot's hoist control panel CABLE CUT switch and lower guard.

RESULT: Test lamp shall go off.

- If test lamp does not go off, replace pilot's hoist control panel (PARA 14-4-38).

- l. Place crew's hoist control panel ARM-TEST switch to TEST.

- m. Turn off external electrical power (PARA 1-6-16).

- n. Disconnect test lamp from connector P6.

- o. Install jumper wires between pins A and B, and between C and D on connector P6.

- p. Turn on external electrical power (PARA 1-6-16).

- q. On pilot's hoist control panel, place HOIST POWER-ON-OFF switch to ON.

- r. On crew's hoist control panel, lift CABLE CUT switch guard, press and hold CABLE CUT switch.

RESULT: SQUIB TEST indicator shall go on.

- If result is not as specified, go to TABLE NO. 14-2-2-3.

- s. Release crew's hoist control panel CABLE CUT switch and lower guard.

RESULT: SQUIB TEST indicator shall go off.

- If SQUIB TEST does not go off, replace crew's hoist control panel (PARA 14-4-39).

- t. On pilot's hoist control panel, lift CABLE CUT switch guard and press and hold CABLE CUT switch.

RESULT: SQUIB TEST indicator shall go on.

- If SQUIB TEST indicator does not go on, replace crew's hoist control panel (PARA 14-4-39).

- u. Release pilot's hoist control panel CABLE CUT switch and lower guard.

RESULT: SQUIB TEST indicator shall go off.

- If SQUIB TEST indicator does not go off, replace crew's hoist control panel (PARA 14-4-39).

- v. On pilot's hoist control panel, place HOIST POWER-ON-OFF switch to OFF.

14-2-2-2

- w. Turn off external electrical power (PARA 1-6-16).
- x. Disconnect jumper wires from connector P6.

WARNING

To prevent injury to personnel, do not connect electrical harness if voltage is found at connector, until problem is repaired.

- y. Using multimeter, check for 0 vdc at pins A and B of connector P6.
- z. Remove shipping cap and aluminum foil from around squib pins and connect P6 to J6 to squib on rescue hoist assembly.

14-2-2.2 Rescue Hoist Assembly.

- a. Make sure HOIST PWR and RESQ HST CONTROL circuit breakers on mission readiness circuit breaker panel are pushed in.
- b. Pull out HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel.
- c. Connect rescue hoist pendant to crew's hoist control panel.
- d. Turn on external electrical power (PARA 1-6-16).

WARNING

Ensure that rescue hoist cable does not contact aircraft during testing or operational checkout. Failure to prevent any contact with aircraft could result in serious injury or death to personnel.

- e. On pilot's hoist control panel, place HOIST POWER-ON-OFF switch to ON.

RESULT: Hoist fan motor shall go on.

- If fan motor does not go on, go to TABLE NO. 14-2-2-4.

WARNING

Rescue hoist cable is stiff and abrasive. Broken cable strands are sharp. Work gloves should be worn whenever handling rescue hoist cable.

CAUTION

To prevent damage to hook or cable, rescue hoist cable must be kept clear of all parts of the aircraft and free from other external obstacles when operating the rescue hoist. Cable abrasion during rescue hoist operations can lead to cable failure. If cable contact or snagging occurs, suspend rescue hoist operations and inspect the cable for damage in accordance with applicable procedures.

- f. Place large container under rescue hoist. Coil rescue hoist cable in container as cable is reeled out.
- g. Place rescue hoist pendant UP-OFF-DN thumbwheel to full DN position and observe the following:

RESULT:

- (1) Hoist cable should reel out at approximately 300 feet-per-minute until 8 to 12 feet remains from full-out limit switch.
- (2) CABLE-FEET indicator on rescue hoist pendant shall show increasing total.
- (3) When 8 to 12 feet of cable remains from full-out limit switch, hoist cable slows to 50 feet per minute.

(4) Hoist stops with 3-1/2 to 4-1/2 wraps of cable remaining on hoist drum.

(5) FULL OUT indicator on rescue hoist pendant is illuminated.

- If rescue hoist cable does not reel out, go to TABLE NO. 14-2-2-5.

- If rescue hoist cable reels out, but does not operate correctly, go to TABLE NO. 14-2-2-6.

- If rescue hoist pendant CABLE-FEET indicator does not show increasing total, go to TABLE NO. 14-2-2-7.

- If rescue hoist pendant FULL OUT indicator does not illuminate, replace rescue hoist pendant (PARA 14-4-36).

h. Place rescue hoist pendant UP-OFF-DN thumbwheel to full UP position and observe the following:

RESULT:

(1) Hoist cable shall reel in at approximately 300 feet-per-minute until 8 to 12 feet remains from full-in limit switch.

(2) CABLE-FEET indicator on rescue hoist pendant shall show decreasing total.

(3) When 8 to 12 feet of cable remains from full-in limit switch, hoist cable slows to 50 feet per minute.

(4) Hoist stops when cable is fully reeled in.

(5) FULL IN indicator on rescue hoist pendant is illuminated.

- If rescue hoist cable does not reel in, go to TABLE NO. 14-2-2-8.

- If rescue hoist cable reels in, but does not operate correctly, go to TABLE NO. 14-2-2-6.

- If rescue hoist pendant CABLE-FEET indicator does not show decreasing total, go to TABLE NO. 14-2-2-7.

- If rescue hoist pendant FULL IN indicator does not illuminate, replace rescue hoist pendant (PARA 14-4-36).

i. Release rescue hoist pendant UP-OFF-DN thumbwheel.

RESULT: UP-OFF-DN thumbwheel shall return to OFF position.

- If thumbwheel does not return to OFF position, replace rescue hoist pendant (PARA 14-4-36).

j. Place pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch to DOWN.

RESULT:

(1) Hoist cable shall reel out at approximately 200 feet-per-minute until 8 to 12 feet remains from full-out limit switch.

(2) When 8 to 12 feet of cable remains from full-out limit switch, hoist slows to 50 feet per minute.

(3) Hoist stops with 3-1/2 to 4-1/2 wraps of cable remaining on hoist drum.

- If hoist cable does not reel out, go to TABLE NO. 14-2-2-9.

- If hoist cable reels out, but does not operate correctly, replace rescue hoist assembly (PARA 14-4-32).

k. Place pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch to UP.

RESULT:

(1) Hoist cable shall reel in at approximately 200 feet-per-minute until 8 to 12 feet remains from full-in limit switch.

14-2-2-4

- (2) When 8 to 12 feet of cable remains from full-in limit switch, hoist slows to 50 feet per minute.
- (3) Hoist stops when cable is reeled in.
 - If hoist cable does not reel in, go to TABLE NO. 14-2-2-10.
 - If hoist cable reels in, but does not operate correctly, replace rescue hoist assembly (PARA 14-4-32).
- l. Place rescue hoist pendant UP-OFF-DN thumbwheel to DN for 5 seconds to reel off approximately 25 feet of cable.
- m. Set and hold rescue hoist pendant UP-OFF-DN thumbwheel to DN and place pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch to UP.

RESULT: Hoist cable shall reel in.

- If hoist cable does not reel in, replace rescue hoist assembly (PARA 14-4-32).
- n. Set and hold rescue hoist pendant UP-OFF-DN thumbwheel to UP and place pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch to DOWN.

RESULT: Hoist cable shall reel out.

- If hoist cable does not reel out, replace rescue hoist assembly (PARA 14-4-32).
- o. Release rescue hoist pendant UP-OFF-DN thumbwheel and pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch.
- p. Reel in hoist cable by setting rescue hoist pendant UP-OFF-DN thumbwheel to UP.
- q. On pilot's hoist control panel, place HOIST POWER ON/OFF switch to OFF.
- r. Turn off electrical power (PARA 1-6-16).

- s. Push in HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel.

14-2-2.3 Dual-Mode Controllable Searchlight (Provisional).

- a. Make certain HOIST LT PWR and HOIST LT CONTROL circuit breakers on mission readiness circuit breaker panel are pushed in.
- b. Pull out HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel.
- c. Turn on electrical power (PARA 1-6-16).

NOTE

If light is already fully extended, no movement will be seen.

- d. Set and hold rescue hoist pendant SCHLT switch to FWD position for 5 seconds.

RESULT: Dual-mode controllable searchlight shall extend from stored position.

- If dual-mode controllable searchlight does not extend, go to TABLE NO. 14-2-2-11.
- e. Set and hold rescue hoist pendant SCHLT switch to RIGHT for 5 seconds.

RESULT: Dual-mode controllable searchlight shall turn to the right.

- If dual-mode controllable searchlight does not turn right, go to TABLE NO. 14-2-2-12.
- f. Set and hold rescue hoist pendant SCHLT switch to LEFT for 5 seconds.

RESULT: Dual-mode controllable searchlight shall turn to the left.

- If dual-mode controllable searchlight does not turn left, go to TABLE NO. 14-2-2-13.
- g. Set and hold rescue hoist pendant SCHLT switch to AFT position.

RESULT: Dual-mode controllable searchlight shall retract to the stored position.

- If dual-mode controllable searchlight does not retract, go to TABLE NO. 14-2-2-14.

WARNING

Looking directly into searchlight when it is turned on could cause damage to eyes.

- h. Set crew's hoist control panel SEARCH LIGHT ON/NORMAL-OFF-ON/NVG switch to ON/NORMAL.

RESULT: Dual-mode controllable searchlight normal lamps shall go on.

- If dual-mode controllable searchlight does not go on, go to TABLE NO. 14-2-2-15.
- i. Set crew's hoist control panel SEARCH LIGHT ON/NORMAL-OFF-ON/NVG switch to OFF.

RESULT: Dual-mode controllable searchlight shall go off.

- If dual-mode controllable searchlight does not go off, replace crew's hoist control panel (PARA 14-4-39).
- j. Place IR light test set (test set) POWER switch to ON.

RESULT: Test set POWER ON lamp shall go on.

- If test set POWER ON lamp does not go on, replace lamp, battery, or test set.

- k. Place test set IR detector lead over POWER ON lamp.

RESULT: Test set IR LIGHT ON lamp shall go on.

- If IR/DET LEDs on test set do not go on, replace test set.

WARNING

Looking directly into searchlight when it is turned on could cause damage to eyes.

- l. Place crew's hoist control panel SEARCH LIGHT ON/NORMAL-OFF-ON/NVG switch to ON/NVG.

- m. Place test set IR detector lead over dual-mode controllable searchlight IR lamp.

RESULT: Test set IR LIGHT ON lamp shall go on steady.

- If test set IR LIGHT does not go on, go to TABLE NO. 14-2-2-16.
- n. Set SEARCH LIGHT ON/NORMAL-OFF-ON/NVG switch to OFF.
- o. Turn off electrical power (PARA 1-6-16).
- p. Push in HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel.

TABLE NO. 14-2-2-1. Test Lamp Does Not Go On When Crew's Hoist Control Panel CABLE CUT Switch Is Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P2021-U and ground.	

TABLE NO. 14-2-2-1. Test Lamp Does Not Go On When Crew's Hoist Control Panel CABLE CUT Switch Is Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.	2. Check continuity between: P23-A and P6-A P23-A and P6-C P23-B and P6-B P23-B and P6-D
Step 1. If continuity is present, replace crew's hoist control panel (PARA 14-4-39). Go to 4. Step 2. If continuity is not present, replace harness, as required. Go to 4.	3. Check continuity between HOIST CABLE SHEAR circuit breaker terminal 1 and P2021-U.
Step 1. If continuity is present, replace HOIST CABLE SHEAR circuit breaker (PARA 9-4-1). Go to 4. Step 2. If continuity is not present, repair/replace wiring. Go to 4.	4. Procedure completed.

TABLE NO. 14-2-2-2. Test Lamp Does Not Go On When Pilot's Hoist Control Panel CABLE CUT Switch Is Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between: P1005-N and P2021-S P1005-M and P2021-q	Step 1. If continuity is present, replace pilot's hoist control panel (PARA 14-4-38). Go to 2. Step 2. If trouble remains, replace crew's hoist control panel (PARA 14-4-39). Go to 2. Step 3. If continuity is not present, repair/replace wiring, as required. Go to 2.

TABLE NO. 14-2-2-2. Test Lamp Does Not Go On When Pilot's Hoist Control Panel CABLE CUT Switch Is Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	2. Procedure completed.

TABLE NO. 14-2-2-3. SQUIB TEST Indicator Does Not Go On When Crew's Hoist Control Panel CABLE CUT Switch Is Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Is the backup pump operating?	Step 1. If backup pump is operating, turn off pump. Go to 8. Step 2. If backup pump is not operating, go to 2.
2. With HOIST POWER ON-OFF switch ON, check for 28 vdc between P371-A and ground.	Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 4.
3. Check continuity between:	P371-B and ground P371-C and ground P372-F and P2021-F P372-L and P2021-L Step 1. If continuity is present, replace rescue hoist assembly (PARA 14-4-32). Go to 8. Step 2. If trouble remains, replace crew's hoist control panel (PARA 14-4-39). Go to 8. Step 3. If continuity is not present, repair/replace wiring or harness, as required. Go to 8.
4. Check for 28 vdc between K300-X1 and ground.	Step 1. If voltage is as specified, go to 5. Step 2. If voltage is not as specified, go to 6.
5. Check continuity between:	

TABLE NO. 14-2-2-3. SQUIB TEST Indicator Does Not Go On When Crew's Hoist Control Panel CABLE CUT Switch Is Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
K300-X2 and ground K300-12 and P371-A K300-11 and terminal 1 (RESQ HST CONTROL circuit breaker) Step 1. If continuity is present, replace hoist relay panel. Go to 8. Step 2. If trouble remains, replace No. 3 relay panel (PARA 9-4-4). Go to 8. Step 3. If continuity is not present, repair/replace wiring, as required. Go to 8. 6. Check for 28 vdc between RESQ HST CONTROL circuit breaker terminal 1 and ground. Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, replace RESQ HST CONTROL circuit breaker (PARA 9-4-1). Go to 8. 7. Check continuity between: K300-X1 and P1005-E P1005-F and terminal 1 (RESQ HST CONTROL circuit breaker) Step 1. If continuity is present, replace pilot's hoist control panel. Go to 8. Step 2. If continuity is not present, repair/replace wiring, as required. Go to 8. 8. Procedure completed.	

TABLE NO. 14-2-2-4. Rescue Hoist Fan Motor Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With HOIST POWER ON-OFF switch ON, check for 115 vac between: P370-A and P370-G P370-B and P370-G P370-C and P370-G P370-D and P370-G P370-E and P370-G P370-F and P370-G Step 1. If voltage is as specified, replace rescue hoist assembly (PARA 14-4-32). Go to 5.	

TABLE NO. 14-2-2-4. Rescue Hoist Fan Motor Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
---------------------------	--------------------------

Step 2. If voltage is not as specified, go to **2**.

2. Check continuity between:

K300-A2 and P370-A
K300-A2 and P370-B
K300-B2 and P370-C
K300-B2 and P370-D
K300-C2 and P370-E
K300-C2 and P370-F
P370-G and ground

Step 1. If continuity is present, go to **3**.

Step 2. If continuity is not present, repair/replace wiring, as required. Go to **5**.

3. Check for 115 vac between:

K300-A1 and ground
K300-B1 and ground
K300-C1 and ground

Step 1. If voltage is as specified, replace hoist relay panel. Go to **5**.

Step 2. If voltage is not as specified, go to **4**.

4. Check continuity between:

K300-A1 and terminal A2 (HOIST PWR circuit breaker)
K300-B1 and terminal B2 (HOIST PWR circuit breaker)
K300-C1 and terminal C2 (HOIST PWR circuit breaker)

Step 1. If continuity is present, replace HOIST PWR circuit breaker (PARA 9-4-1).
 Go to **5**.

Step 2. If continuity is not present, repair/replace wiring, as required. Go to **5**.

5. Procedure completed.

TABLE NO. 14-2-2-5. Rescue Hoist Cable Does Not Reel Out With Rescue Hoist Pendant Thumbwheel In DN Position.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With rescue hoist pendant UP-OFF-DN thumbwheel in DN position, check for 0 ohms between P20-B and P20-C.	Step 1. If resistance is as specified, go to 2. Step 2. If resistance is not as specified, go to 4.
2. Check continuity between: P2021-B and P372-B P2021-C and P372-C	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring or harness, as required. Go to 5.
3. Check limit switches for proper adjustment (PARA 14-4-40).	Step 1. If limit switches are not properly adjusted, adjust limit switches (PARA 14-4-40). Go to 5. Step 2. If limit switches are properly adjusted, replace crew's hoist control panel (PARA 14-4-39). Go to 5. Step 3. If trouble remains, replace rescue hoist assembly (PARA 14-4-32). Go to 5.
4. Check continuity between: P20-B and P24-B P20-C and P24-C	Step 1. If continuity is present, replace rescue hoist pendant (PARA 14-4-36). Go to 5. Step 2. If continuity is not present, replace pendant control cable (PARA 14-4-36). Go to 5.
5. Procedure completed.	

TABLE NO. 14-2-2-6. Rescue Hoist Cable Reels In/Out Incorrectly.

TEST OR INSPECTION	CORRECTIVE ACTION
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1. Check for 235 to 265 ohms between P20-D and P20-J.

TABLE NO. 14-2-2-6. Rescue Hoist Cable Reels In/Out Incorrectly. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
	Step 1. If resistance is as specified, go to 2. Step 2. If resistance is not as specified, go to 3.
2. Check for 40 to 50 ohms between P20-E and P20-J.	Step 1. If resistance is as specified, go to 4. Step 2. If resistance is not as specified, go to 7.
3. Check continuity between: P20-D and P24-D P20-J and P24-J	Step 1. If continuity is present, replace rescue hoist pendant (PARA 14-4-36). Go to 8. Step 2. If continuity is not present, replace pendant control cable (PARA 14-4-36). Go to 8.
4. With hoist pendant thumbwheel in full UP then in full DN position, check for 40 to 50 ohms between P20-D and P20-E.	Step 1. If resistance is as specified, go to 5. Step 2. If resistance is not as specified, go to 7.
5. Check continuity between: P2021-D and P372-D P2021-E and P372-E P2021-J and P372-J	Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, repair/replace wiring or harness, as required. Go to 8.
6. Check limit switches for proper adjustment (PARA 14-4-40).	Step 1. If limit switches are not properly adjusted, adjust limit switches (PARA 14-4-40). Go to 8. Step 2. If limit switches are properly adjusted, replace crew's hoist control panel (PARA 14-4-39). Go to 8. Step 3. If trouble remains, replace rescue hoist assembly (PARA 14-4-32). Go to 8.

TABLE NO. 14-2-2-6. Rescue Hoist Cable Reels In/Out Incorrectly. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
7. Check continuity between: P20-D and P24-D P20-E and P24-E	Step 1. If continuity is present, replace rescue hoist pendant (PARA 14-4-36). Go to 8. Step 2. If continuity is not present, replace pendant control cable (PARA 14-4-36). Go to 8.
8. Procedure completed.	

TABLE NO. 14-2-2-7. Rescue Hoist Pendant CABLE-FEET Indicator Does Not Show Increasing/Decreasing Total.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between: P20-U and P24-U P20-S and P24-S P20-R and P24-R	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, replace pendant control cable (PARA 14-4-36). Go to 3.
2. Check continuity between: P372-R and P2021-R P372-P and P2021-P P372-N and P2021-N	Step 1. If continuity is present, replace rescue hoist pendant (PARA 14-4-36). Go to 3. Step 2. If trouble remains, replace crew's hoist control panel (PARA 14-4-39). Go to 3. Step 3. If trouble still remains, replace rescue hoist assembly (PARA 14-4-32). Go to 3. Step 4. If continuity is not present, repair/replace wiring, as required. Go to 3.
3. Procedure completed.	

TABLE NO. 14-2-2-8. Rescue Hoist Cable Does Not Reel In With Rescue Hoist Pendant Thumbwheel In UP Position.

TEST OR INSPECTION CORRECTIVE ACTION
1. With rescue hoist pendant UP-OFF-DN thumbwheel in UP position, check for 0 ohms between P20-A and P20-C. Step 1. If resistance is as specified, go to 2. Step 2. If resistance is not as specified, go to 4. 2. Check continuity between P2021-A and P372-A Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring. Go to 5. 3. Check limit switches for proper adjustment (PARA 14-4-40). Step 1. If limit switches are not properly adjusted, adjust limit switches (PARA 14-4-40). Go to 5. Step 2. If limit switches are properly adjusted, replace crew's hoist control panel (PARA 14-4-39). Go to 5. Step 3. If trouble remains, replace rescue hoist assembly (PARA 14-4-32). Go to 5. 4. Check continuity between P24-A and P20-A. Step 1. If continuity is present, replace rescue hoist pendant (PARA 14-4-36). Go to 5. Step 2. If continuity is not present, replace pendant control cable (PARA 14-4-36). Go to 5. 5. Procedure completed.

TABLE NO. 14-2-2-9. Hoist Cable Does Not Reel Out With Pilot's Hoist Control Panel PILOT OVER-RIDE UP-DOWN Switch In DOWN.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check continuity between:

TABLE NO. 14-2-2-9. Hoist Cable Does Not Reel Out With Pilot's Hoist Control Panel PILOT OVERRIDE UP-DOWN Switch In DOWN. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	P1005-H and P372-X P1005-K and P372-a P1005-D and ground Step 1. If continuity is present, replace pilot's hoist control panel (PARA 14-4-38). Go to 2. Step 2. If trouble remains, replace rescue hoist assembly (PARA 14-4-32). Go to 2. Step 3. If continuity is not present, repair/replace wiring, as required. Go to 2. 2. Procedure completed.

TABLE NO. 14-2-2-10. Rescue Hoist Cable Does Not Reel In With Pilot's Hoist Control Panel PILOT OVERRIDE UP-DOWN Switch In UP.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check continuity between: P1005-G and P372-W P1005-D and ground Step 1. If continuity is present, replace pilot's hoist control panel (PARA 14-4-38). Go to 2. Step 2. If trouble remains, replace rescue hoist assembly (PARA 14-4-32). Go to 2. Step 3. If continuity is not present, repair/replace wiring or replace harness, as required. Go to 2. 2. Procedure completed.

TABLE NO. 14-2-2-11. Dual-Mode Controllable Searchlight Does Not Extend.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check for 28 vdc between: P2021-c and ground P2021-j and ground P2021-n and ground

TABLE NO. 14-2-2-11. Dual-Mode Controllable Searchlight Does Not Extend. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

Step 1. If voltage is as specified, go to **2**.
 Step 2. If voltage is not as specified, go to **4**.

2. With SCHLT switch in FWD position, check for 0 ohms between P20-L and P20-M.

Step 1. If resistance is as specified, go to **3**.
 Step 2. If resistance is not as specified, go to **5**.

3. Check continuity between:
TB200-6 and P2021-a
TB200-1 and ground

Step 1. If continuity is present, replace crew's hoist control panel (PARA 14-4-39).
 Go to **6**.
 Step 2. If trouble remains, replace dual-mode controllable searchlight. Go to **6**.
 Step 3. If continuity is not present, repair/replace wiring, as required. Go to **6**.

4. Check continuity between:
P2021-c and terminal 1 (HOIST LT CONTROL circuit breaker)
P2021-j and terminal 1 (HOIST LT CONTROL circuit breaker)
P2021-n and terminal 1 (HOIST LT CONTROL circuit breaker)

Step 1. If continuity is present, replace HOIST LT CONTROL circuit breaker (PARA 9-4-1). Go to **6**.
 Step 2. If continuity is not present, repair/replace wiring, as required. Go to **6**.

5. Check continuity between:
P20-M and P24-M
P20-L and P24-L

Step 1. If continuity is present, replace rescue hoist pendant (PARA 14-4-36).
 Go to **6**.
 Step 2. If continuity is not present, replace pendant control cable (PARA 14-4-36).
 Go to **6**.

6. Procedure completed.

TABLE NO. 14-2-2-12. Dual-Mode Controllable Searchlight Does Not Turn Right.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With SCHLT switch in RIGHT position, check for 0 ohms between P20-H and P20-L.	Step 1. If resistance is as specified, go to 2. Step 2. If resistance is not as specified, go to 3.
2. Check continuity between P2021-Y and TB200-4.	Step 1. If continuity is present, replace crew's hoist control panel (PARA 14-4-39). Go to 4. Step 2. If trouble remains, replace dual-mode controllable searchlight. Go to 4. Step 3. If continuity is not present, repair/replace wiring. Go to 4.
3. Check continuity between P20-H and P24-H.	Step 1. If continuity is present, replace rescue hoist pendant (PARA 14-4-36). Go to 4. Step 2. If continuity is not present, replace pendant control cable (PARA 14-4-36). Go to 4.
4. Procedure completed.	

TABLE NO. 14-2-2-13. Dual-Mode Controllable Searchlight Does Not Turn Left.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With SCHLT switch in LEFT position, check for 0 ohms between P20-K and P20-L.	Step 1. If resistance is as specified, go to 2. Step 2. If resistance is not as specified, go to 3.
2. Check continuity between P2021-Z and TB200-5.	Step 1. If continuity is present, replace crew's hoist control panel (PARA 14-4-39). Go to 4. Step 2. If trouble remains, replace dual-mode controllable searchlight. Go to 4. Step 3. If continuity is not present, repair/replace wiring. Go to 4.

TABLE NO. 14-2-2-13. Dual-Mode Controllable Searchlight Does Not Turn Left. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Check continuity between P20-K and P24-K.	Step 1. If continuity is present, replace rescue hoist pendant (PARA 14-4-36). Go to 4. Step 2. If continuity is not present, replace pendant control cable (PARA 14-4-36). Go to 4.
4. Procedure completed.	

TABLE NO. 14-2-2-14. Dual-Mode Controllable Searchlight Does Not Retract.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With SCHLT switch in AFT position, check for 0 ohms between P20-T and P20-L.	Step 1. If resistance is as specified, go to 2. Step 2. If resistance is not as specified, go to 3.
2. Check continuity between P2021-b and TB200-7.	Step 1. If continuity is present, replace crew's hoist control panel (PARA 14-4-39). Go to 4. Step 2. If trouble remains, replace dual-mode controllable searchlight. Go to 4. Step 3. If continuity is not present, repair/replace wiring. Go to 4.
3. Check continuity between P20-K and P24-K.	Step 1. If continuity is present, replace rescue hoist pendant (PARA 14-4-36). Go to 4. Step 2. If continuity is not present, replace pendant control cable (PARA 14-4-36). Go to 4.
4. Procedure completed.	

TABLE NO. 14-2-2-15. Dual-Mode Controllable Searchlight Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for burnt-out normal search light bulb.	Step 1. If lamp is good, go to 2. Step 2. If lamp is burnt out, replace lamp. Go to 5.
2. Check for 28 vdc between TB200-3 and ground.	Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 4.
3. Check continuity between P2021-k and TB200-2.	Step 1. If continuity is present, replace crew's hoist control panel (PARA 14-4-39). Go to 5. Step 2. If trouble remains, replace dual-mode controllable searchlight. Go to 5. Step 3. If continuity is not present, repair/replace wiring. Go to 5.
4. Check continuity between TB200-3 and terminal 1 (HOIST LT PWR circuit breaker).	Step 1. If continuity is present, replace HOIST LT PWR circuit breaker (PARA 9-4-1). Go to 5. Step 2. If continuity is not present, repair/replace wiring. Go to 5.
5. Procedure completed.	

TABLE NO. 14-2-2-16. Test Set IR LIGHT Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for burnt-out NVG search light bulb.	Step 1. If lamp is good, go to 2. Step 2. If lamp is burnt out, replace lamp. Go to 5.
2. Check for 28 vdc between TB200-9 and ground.	Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 4.

TABLE NO. 14-2-2-16. Test Set IR LIGHT Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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3. Check continuity between P2021-p and TB200-8.

Step 1. If continuity is present, replace crew's hoist control panel (PARA 14-4-39).

Go to **5**.

Step 2. If trouble remains, replace dual-mode controllable searchlight. Go to **5**.

Step 3. If continuity is not present, repair/replace wiring. Go to **5**.

4. Check continuity between TB200-9 and terminal 1 (HOIST LT PWR circuit breaker).

Step 1. If continuity is present, replace HOIST LT PWR circuit breaker (PARA 9-4-1). Go to **5**.

Step 2. If continuity is not present, repair/replace wiring. Go to **5**.

5. Procedure completed.

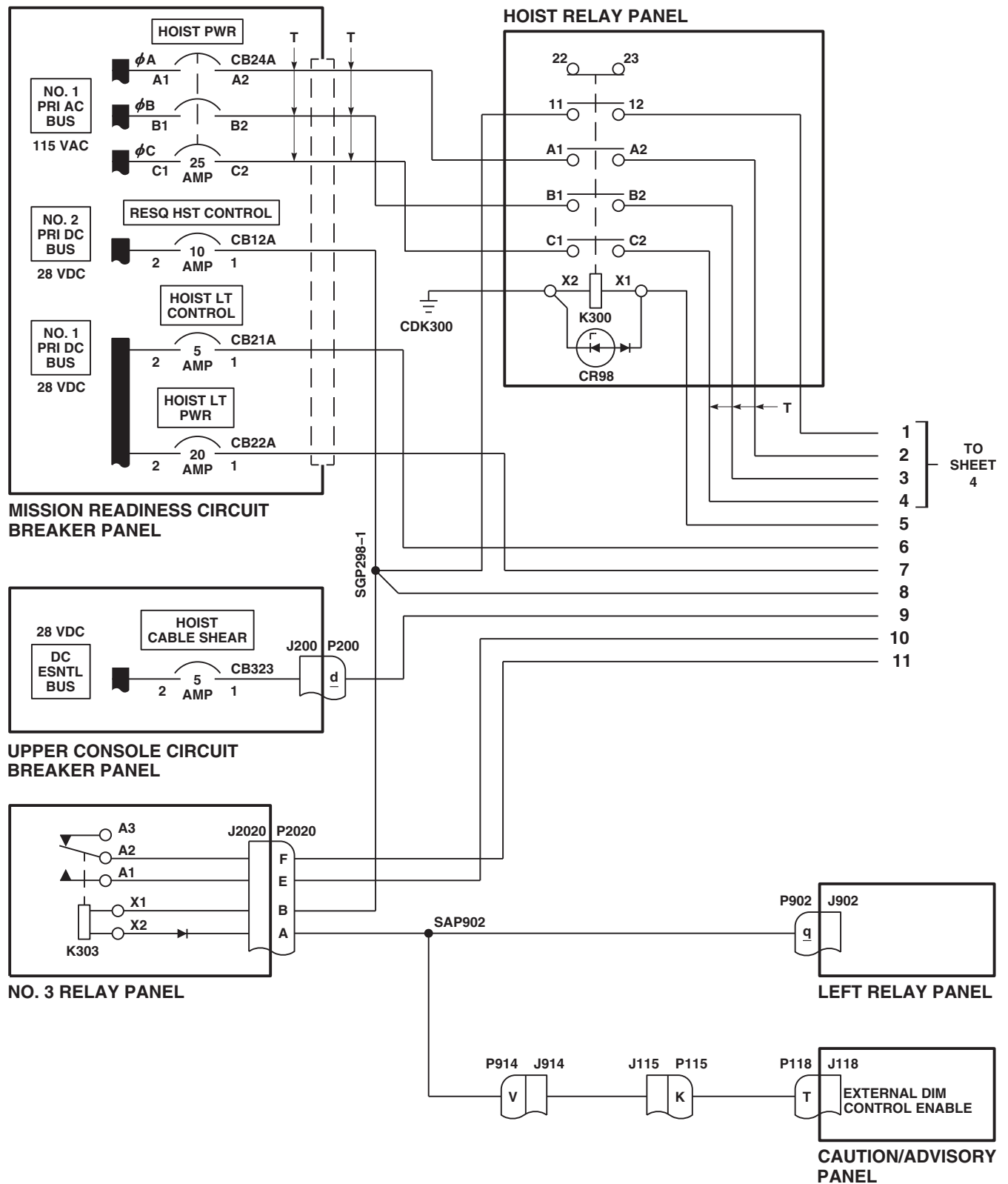
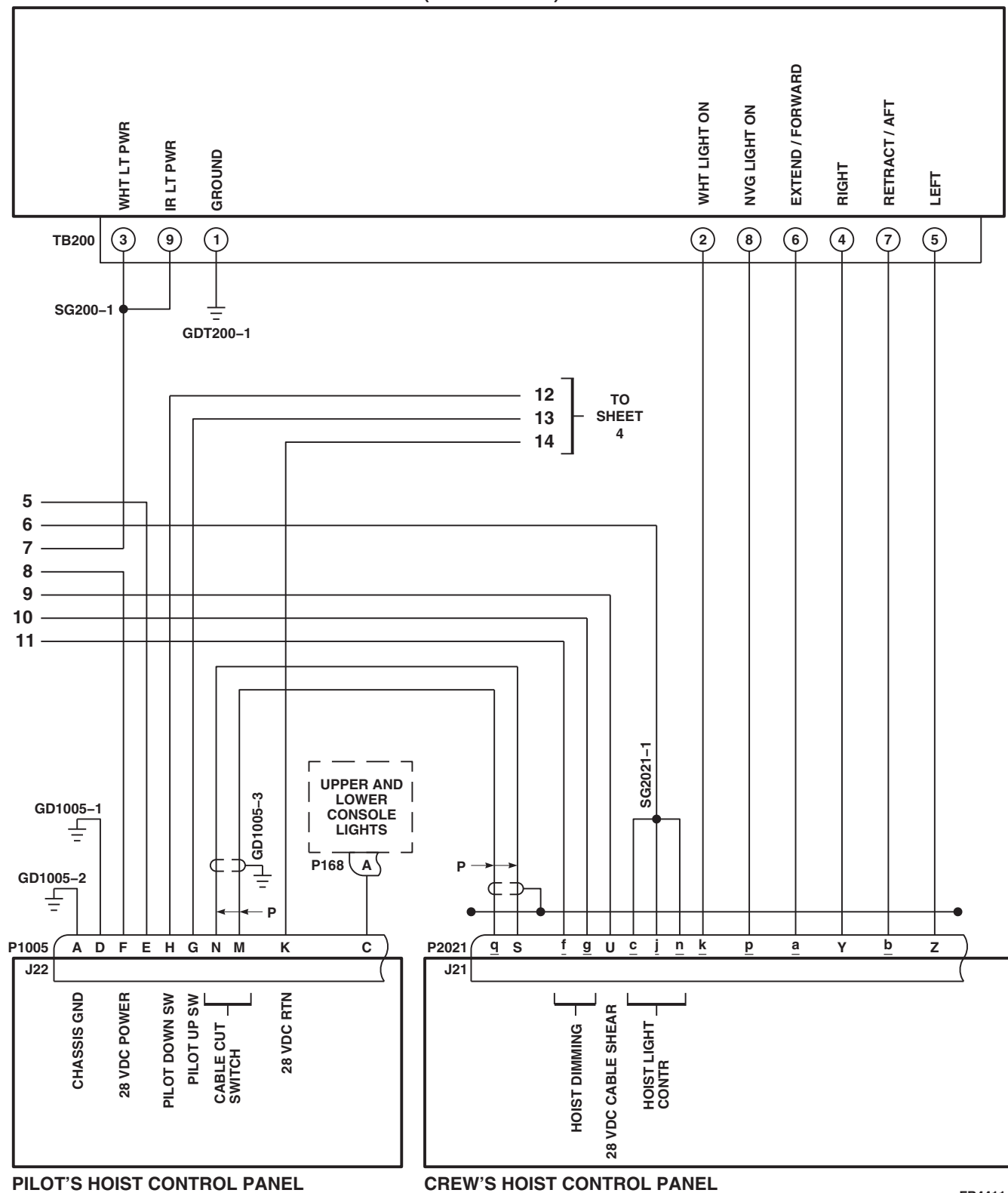


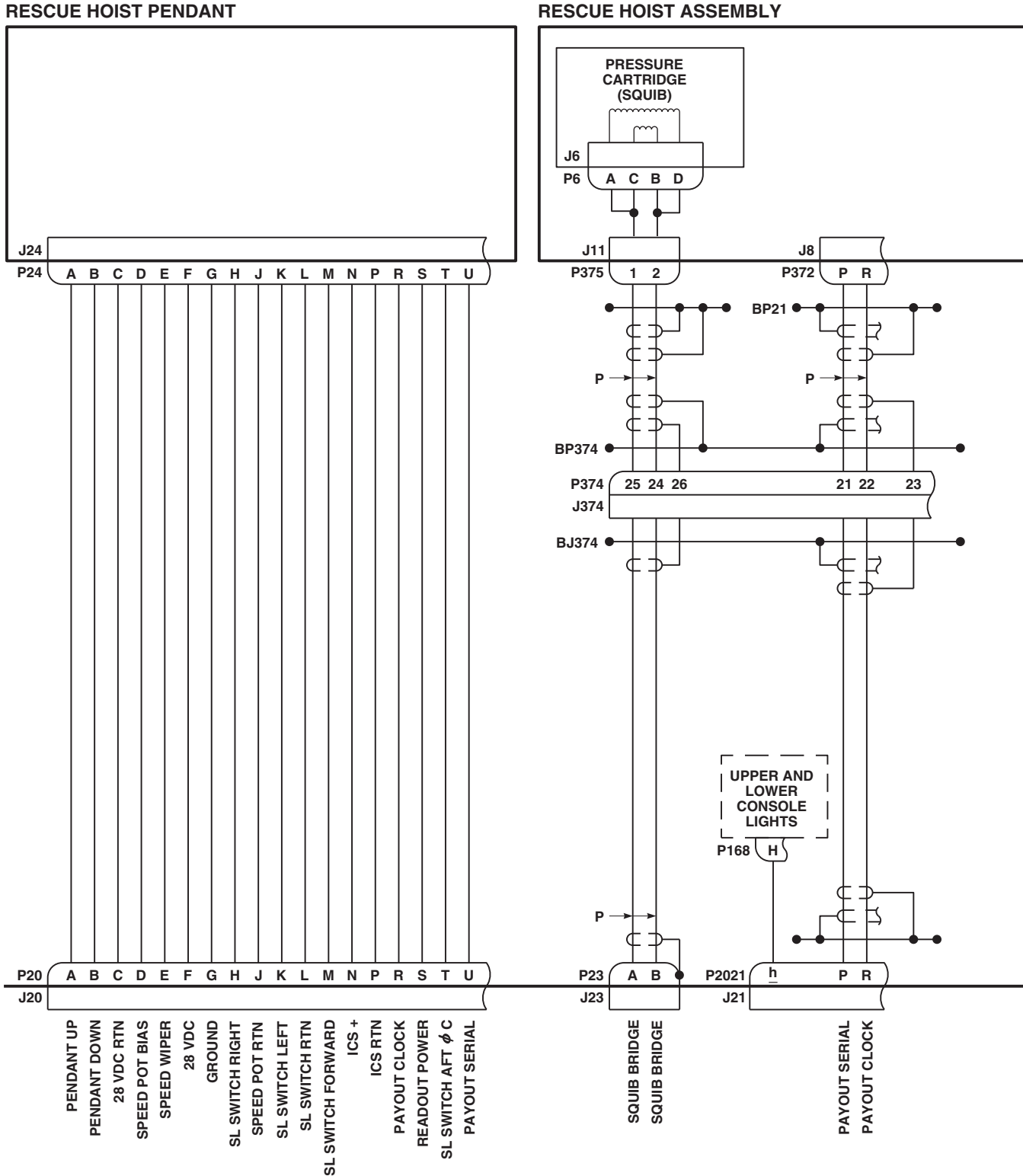
FIGURE 14-2-2-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 4)

DUAL-MODE CONTROLLABLE SEARCHLIGHT (PROVISIONAL)



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FIGURE 14-2-2-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 4)

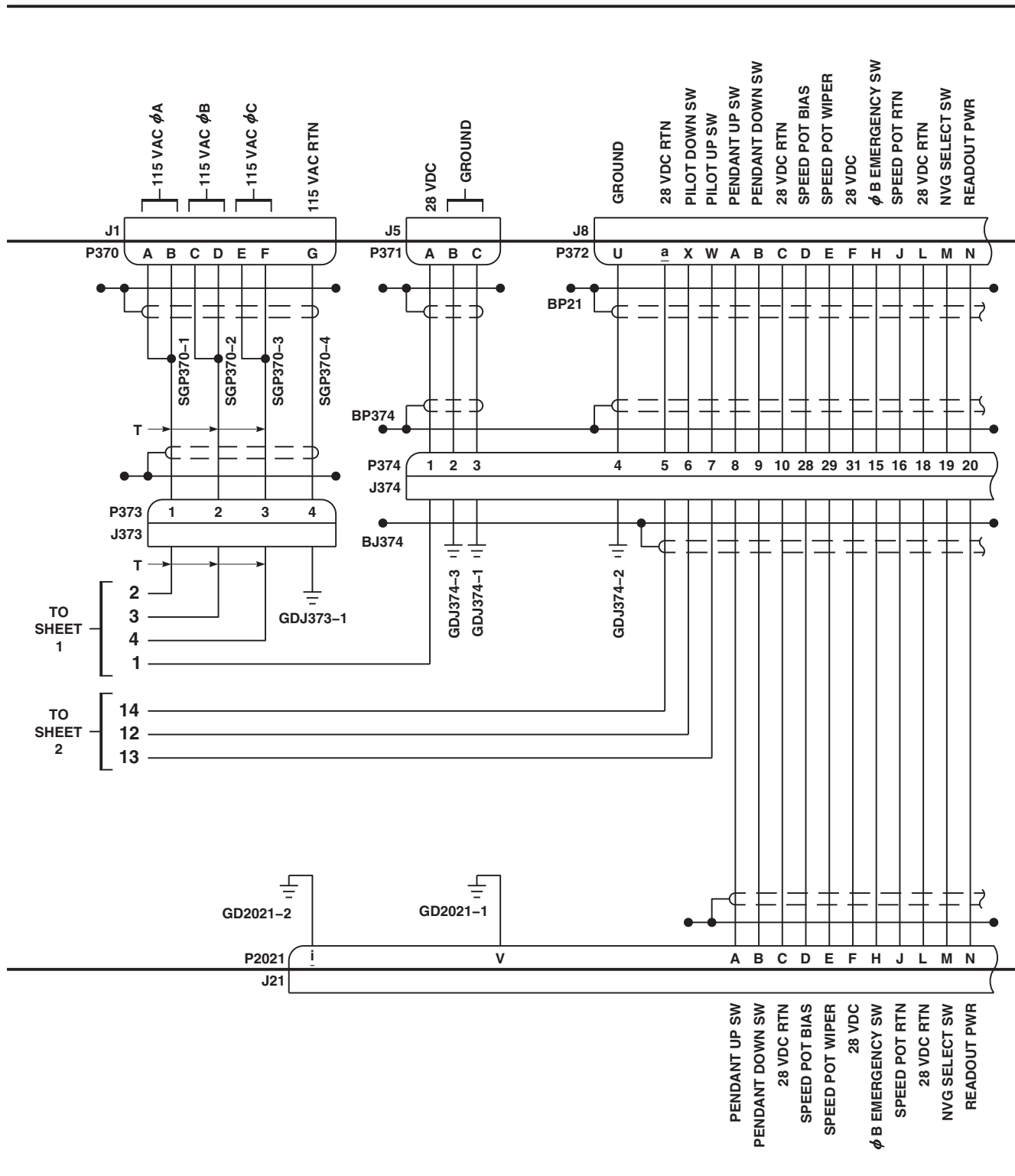


CREW'S HOIST CONTROL PANEL

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FIGURE 14-2-2-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 4)

RESCUE HOIST ASSEMBLY



CREW'S HOIST CONTROL PANEL

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FIGURE 14-2-2-1. RESCUE HOIST SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 4)

SECTION 14-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

PARAGRAPH	TITLE
14-3-1	Rescue Hoist Inspection HOIST, BL-27100-71 HOIST, 42315-10
14-3-2	Rescue Hoist Inspections HOIST BL-29900-30

14-3-1

RESCUE HOIST INSPECTION

HOIST, BL-27100-71

HOIST, 42315-10

.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-3-1.1	Inspect Rescue Hoist Cable	14-3-1-1	14-3-1.2	Inspect Rescue Hoist Cable	14-3-1-3
	HOIST, BL-27100-71			HOIST, 42315-10	

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Light
- Magnifying Glass, 12X
- Protective Gloves

Materials/Parts

- Lockwire, Item 195, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 14-4-2
- PARA 14-4-4

14-3-1.1

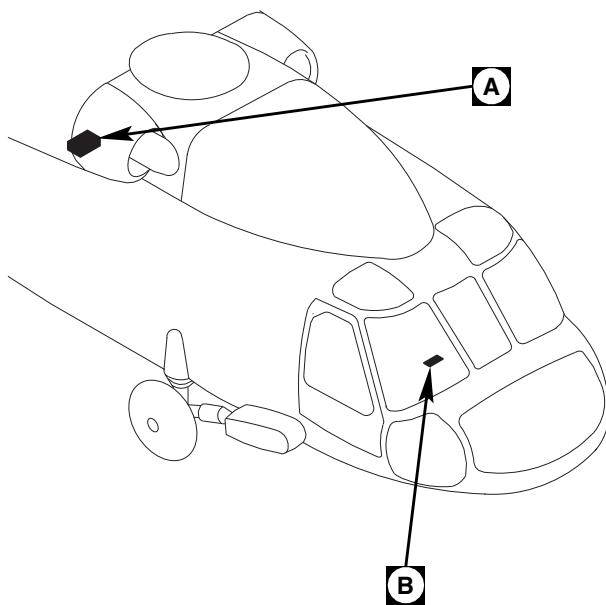
Inspect Rescue Hoist Cable

HOIST, BL-27100-71

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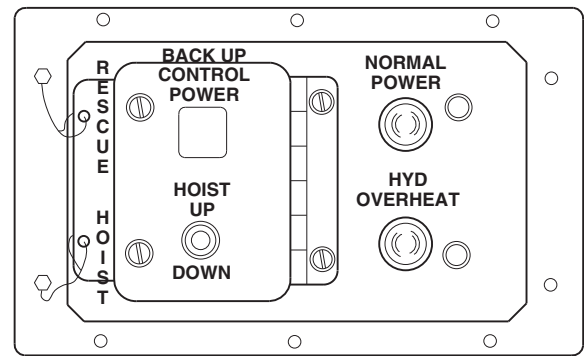
- Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- On lower console, set RESCUE HOIST switch to ALL (Figure 14-3-1-1, Detail B).
- In cabin do the following:

- Check that the NORMAL POWER light on the cabin control panel is on (Figure 14-3-1-1, Detail A).
- If BACKUP CONTROL POWER switch light is on, open switch guard assembly, and press switch. BACKUP CONTROL POWER switch light should go off and NORMAL POWER light should go on.

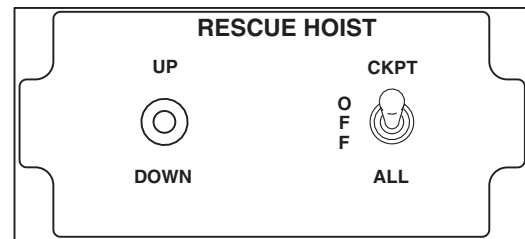


A

B



CABIN CONTROL PANEL



LOWER CONSOLE
RESCUE HOIST CONTROL PANEL

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FIGURE 14-3-1-1. INSPECT RESCUE HOIST CABLE **HOIST, BL-27100-71**

- (3) Close switch guard assembly and lock-wire, Item 195, Appendix D, switch guard to screws.

WARNING

- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of the helicopter and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.
- Injury to personnel will occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.

CAUTION

Damage to hoist and loose cable wraps will result if cable is restrained from motion while reeling out. Do not restrain cable while reeling out.

- d. In cabin, operate rescue hoist and cargo hook pendant to reel out at least 3 feet of cable. When 3 feet of cable is reached, stop hoist.
- e. Slide bumper and spring up the cable to permit inspection of cable.
- f. Using 12X magnifying glass and a light, inspect the first four inches of exposed cable as follows:
 - (1) Inspect cable for necking down. **NECKING DOWN OF CABLE NOT ALLOWED.**
 - (2) Inspect for nicks, broken, loose, or deformed wires. **NO NICKS, BROKEN, LOOSE, OR DEFORMED WIRES ALLOWED.**

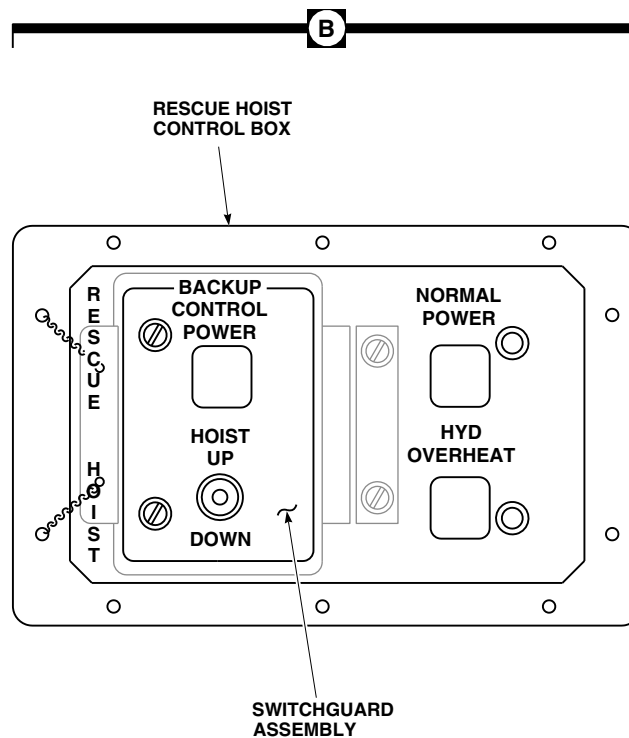
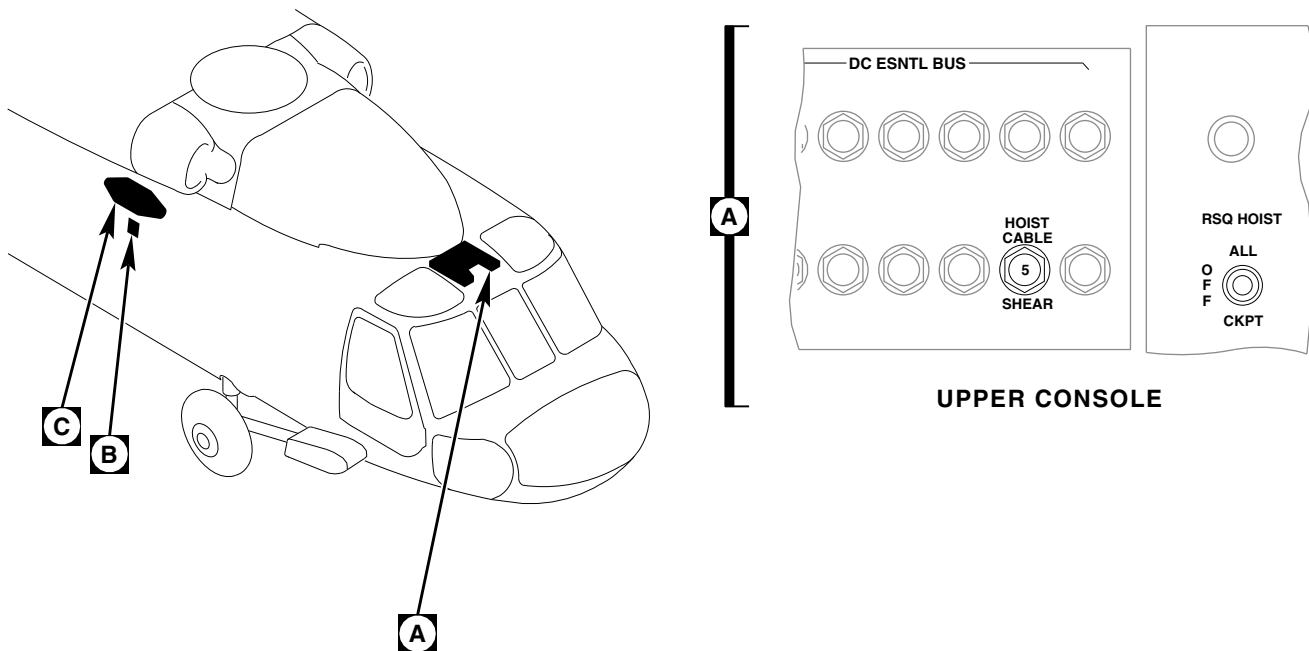
- (3) Inspect for corrosion and damage to cable ball shank. **NO CORROSION OR DAMAGE ALLOWED.**

- g. Slowly reel cable onto cable drum, while assistant applies firm drag on cable with a gloved hand and checks for damage. Observe the cable closely when winding back onto the drum. **NO DAMAGE IS ALLOWED.**
- h. Replace cable if damaged (PARA 14-4-4).
- i. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- j. Make sure area is clean and free of foreign material.

14-3-1.2 Inspect Rescue Hoist

Cable **HOIST, 42315-10** .

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Remove explosive cartridge (PARA 14-4-2).
- c. Remove screws and washers holding outboard fairing and top access panel to hoist (Figure 14-3-1-2, Sheet 2, Detail C). Remove outboard fairing and top access panel.
- d. Inspect cable drum gears, cable guide, and level wind shafts for damage. If damage exists, remove hoist for repair or replacement of damaged parts.
- e. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- f. On upper console, set RSQ HOIST switch to ALL (Figure 14-3-1-2, Sheet 1, Detail A).
- g. In cabin do the following:
 - (1) Check that cabin RESCUE HOIST control box NORMAL POWER light is on.
 - (2) If BACKUP CONTROL POWER switch light is on, open switch guard assembly, and press switch. BACKUP CONTROL POWER switch light should go off and NORMAL POWER light should go on.



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FIGURE 14-3-1-2. INSPECT CABLE **HOIST, 42315-10** (SHEET 1 OF 2)

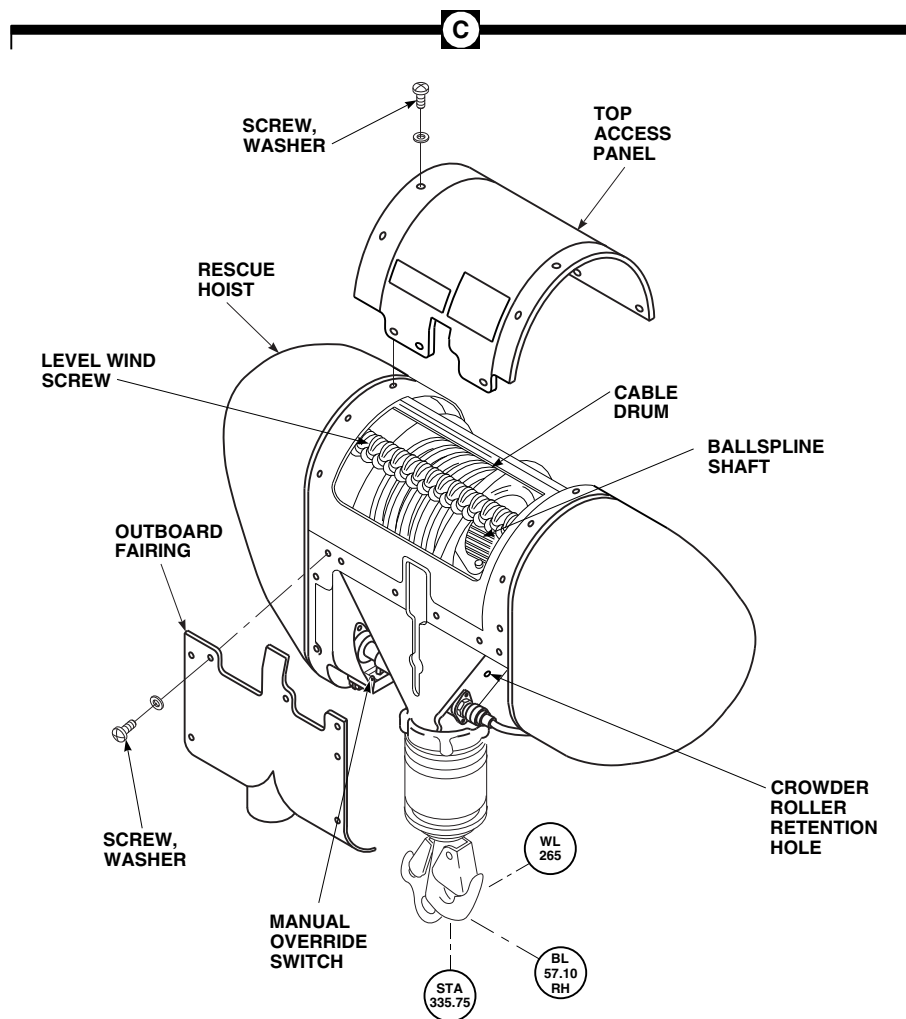


FIGURE 14-3-1-2. INSPECT CABLE **HOIST, 42315-10** (SHEET 2 OF 2)

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- (3) Close switch guard assembly and lock-wire, Item 195, Appendix D, switch guard

to screws (Figure 14-3-1-2, Sheet 1, Detail B).

14-3-2 RESCUE HOIST INSPECTIONS **HOIST BL-29900-30**

PARAGRAPH BREAKDOWN

		PAGE		PAGE
14-3-2.1	In-Flight Load Test	14-3-2-1	14-3-2.2	Dimensional Check
				14-3-2-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Outside Micrometer, 0 - 1 inch
- Protective Gloves

Materials/Parts

- Antiseize Compound, Item 75, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-58
- PARA 14-4-1
- PARA 14-4-40

14-3-2.1 In-Flight Load Test.

NOTE

Hoist operator should put on the safety harness, and attach and secure before opening the sliding door.

- The hoist operator’s safety harness should be put on, adjusted, and secured.
- Open right side cabin door.
- On pilot’s hoist control panel, turn HOIST POWER switch ON.
- With helicopter in hover at an altitude of 300 feet, using control pendant, lower hoist cable.

WARNING

Injury to personnel and damage to equipment will occur if static electricity

is not dissipated before ground personnel touch hook or cable.

- Perform load test as follows:
 - Operate rescue hoist for one duty cycle as specified in Figure 14-3-2-1. For each cycle, check cable speed and time required to travel full cable stroke. Verify proper limit switch operation.

WARNING

Injury to personnel and damage to equipment will occur if hoist operator is not wearing protective gloves to protect operator’s hand from broken cable strands.

NOTE

- Hoist operator should control hoist load in forward flight to minimize cable

The following test cycle is used for hoist testing:
BL-29900-30 TEST REQUIREMENTS

		DATE _____			
		OPERATOR _____			
		SERIAL NUMBER _____			
TEST CONDITIONS		START TEMPERATURE PH TO PH AC VOLTAGE		200 $\pm$ 10 VRMS	
CYCLE	CYCLE CONDITION		LOAD [LBS]	SPEED (top cable layer) [FT/MIN.]	PHASE CURRENT [ARMS]
1 FULL CABLE TRAVEL	PILOT	RAISE	300	205 $\pm$ 40	25 Max.
		LOWER	300	205 $\pm$ 40	5 Max.
2 FULL CABLE TRAVEL	PENDANT	RAISE	300	300 Min.	25 Max.
		LOWER	300	300 Min.	5 Max.
3 FULL CABLE TRAVEL	PENDANT	RAISE	600	350 Min.	25 Max.
		LOWER	600	300 Min.	5 Max.
4 FULL CABLE TRAVEL	PENDANT	RAISE	600	350 Min.	25 Max.
		LOWER	600	300 Min.	5 Max.
5 FULL CABLE TRAVEL	PENDANT	RAISE	600	350 Min.	25 Max.
		LOWER	600	300 Min.	5 Max.
6 FULL CABLE TRAVEL	PENDANT	RAISE	600	350 Min.	25 Max.
		LOWER	600	300 Min.	5 Max.
7 TOP LAYER ONLY	SLOW LIMIT FROM STOP	RAISE	600	37-63	10 Max.
START DRUM FLANGE TEMPERATURE				deg F	
FINAL DRUM FLANGE TEMPERATURE				deg F	200°F Max.
VERIFY LIMIT SWITCH OPERATION				Pass/Fail	
VERIFY MOTOR FAN OPERATION & DIRECTION				Pass/Fail	
CYCLE COUNTER OPERATES				Pass/Fail	
CYCLE COUNTER READING					
SLIP CLUTCH VALUE		LBS.	1200 LBS. Min, 1850 LBS. Max		
CABLE PAYOUT	FULL IN	_____	FULL OUT	_____	

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FIGURE 14-3-2-1. DUTY CYCLE (REPRODUCE LOCALLY)

contact with side of helicopter. When practical, extension or retraction of load should be restricted to a hover or as slow a speed as possible.

- Swinging loads may cause erratic radar altimeter readings.
- (2) During cycle 3 (variable speed operation), check for proper variable speed operation and smoothness of control. There shall be no excessive non-linearity control.

14-3-2.2 Dimensional Check.

14-3-2.2.1 Tension Roller Replacement Criteria.

- a. Before each mission use, inspect tension roller as follows:
 - (1) Turn off all electrical power (PARA 1-6-16).
 - (2) Remove screws and washers securing access cover and backup plate to rescue hoist. Remove access cover and backup plate (Figure 14-3-2-2, Sheet 1, Detail A).
 - (3) Inspect tension roller for abnormal wear or obvious defects such as cracks, tears, abrasion, or degradation of material (Figure 14-3-2-2, Sheet 2, Detail B). If damage exceeds criteria, replace hoist (PARA 14-4-1). If roller wear does not exceed limit, proceed to step a. (4).
 - (4) Lubricate threads of screws with antiseize compound, Item 75, Appendix D.
 - (5) Install access cover and backup plate on rescue hoist and secure with screws and washers (Figure 14-3-2-2, Sheet 1, Detail A).
 - (6) Using lockwire, Item 196, Appendix D, lockwire screws.
- b. Every month, inspect outside diameter of tension roller as follows:
 - (1) Turn off all electrical power (PARA 1-6-16).

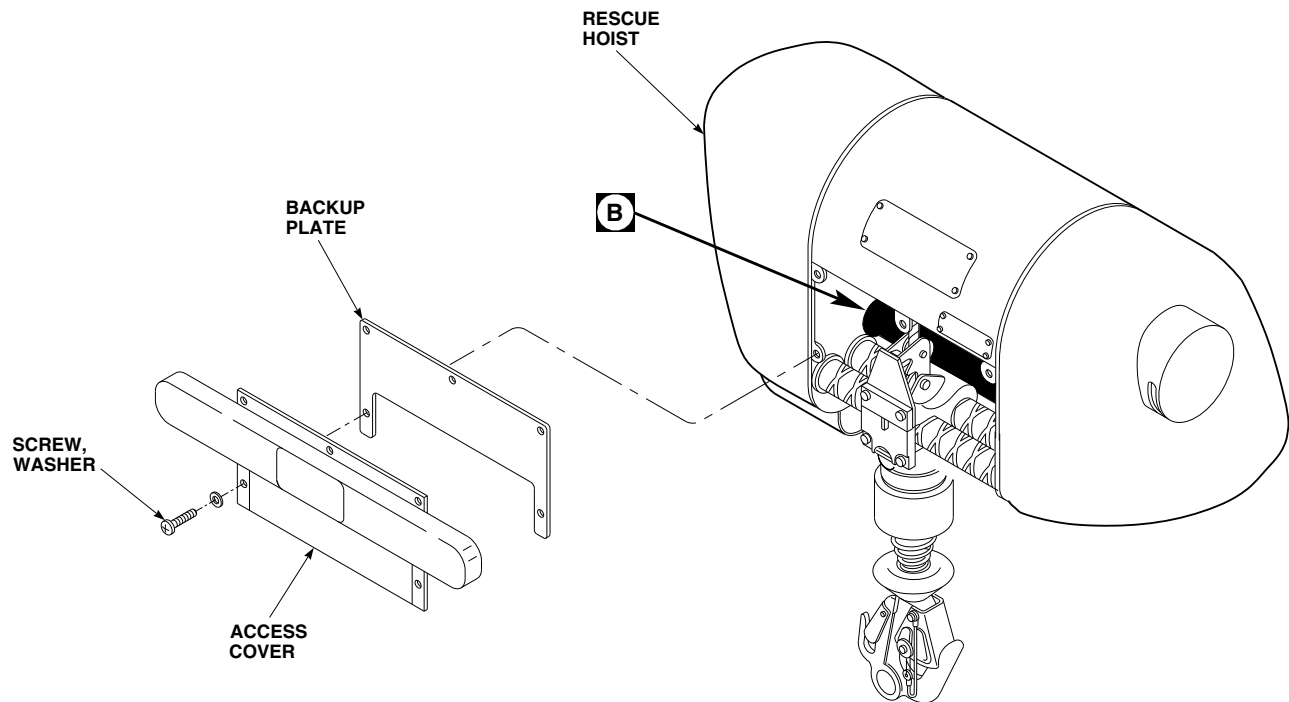
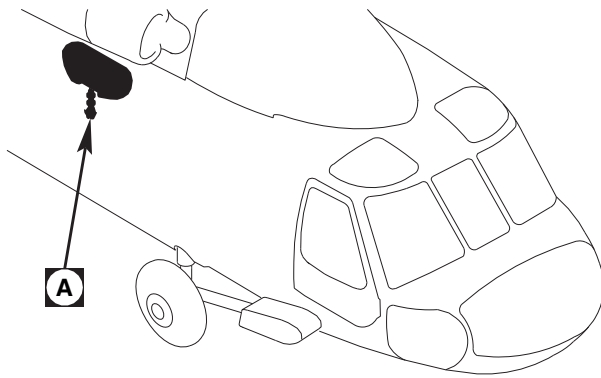
- (2) Remove screws and washers securing access cover and backup plate to rescue hoist. Remove access cover and backup plate (Figure 14-3-2-2, Sheet 1, Detail A).
- (3) Using outside micrometer, measure outside diameter of tension roller (Figure 14-3-2-2, Sheet 2, Detail B). If roller wear is below 0.801-inch at any cable contact point, replace hoist (PARA 14-4-1). If roller wear does not exceed limit, proceed to step b. (4).
- (4) Lubricate threads of screws with antiseize compound, Item 75, Appendix D.
- (5) Install access cover and backup plate on rescue hoist and secure with screws and washers (Figure 14-3-2-2, Sheet 1, Detail A).
- (6) Using lockwire, Item 196, Appendix D, lockwire screws.

14-3-2.2.2 Crushable Bumper and Spring Replacement Criteria.

NOTE

Rescue hoist uses crushable bumper mounted above hook to dissipate kinetic energy within hoist moving parts. This is accomplished by controlled crushing of sacrificial element within bumper when bumper contacts bellmouth of cable guide and full-in switch. Therefore, it is necessary the bumper be visually and dimensionally inspected before each mission use.

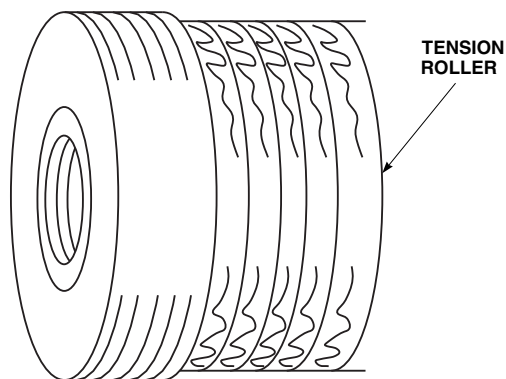
- a. Before each mission use, inspect crushable bumper and spring as follows:
 - (1) When hook is at full-in position, measure height from top of crushable bumper to bottom of spring retainer (Figure 14-3-2-3, Sheet 2, Detail B). **HEIGHT SHALL BE 3.15 INCHES.** If not within limit, check operation of full-in limit switch (PARA 14-4-40).
 - (2) When hook is at full-in position, check concavity in two sections of crushable



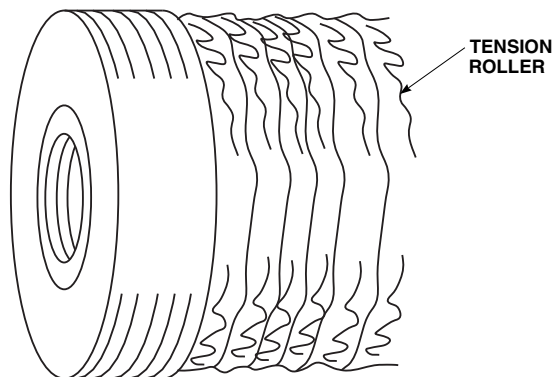
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FIGURE 14-3-2-2. TENSION ROLLER REPLACEMENT CRITERIA (SHEET 1 OF 2)

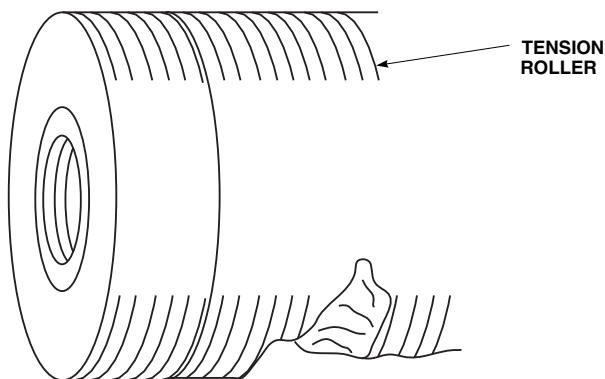
B



(A) NORMAL SLIGHT WEAR, CONTINUE USE.



(B) NORMAL EXCESSIVE WEAR. PERFORM DIMENSIONAL CHECK. IF DIAMETER MEASURES BELOW 0.801-INCH, REPLACE RESCUE HOIST.



(C) ABNORMAL WEAR. REPLACE RESCUE HOIST.

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FIGURE 14-3-2-2. TENSION ROLLER REPLACEMENT CRITERIA (SHEET 2 OF 2)

bumper by measuring depth of crushable bumper at its deepest point. **DEPTH SHALL NOT EXCEED 0.375 INCHES.** If not within limits, replace bumper assembly and check operation of full-in limit switch (PARA 14-4-40).

- (3) When hook is at full-in position, measure height of bumper assembly from top of

bumper retainer to bottom of spring (Figure 14-3-2-3, Sheet 2, Detail C). **HEIGHT SHALL BE 1.5 - 2.0 INCHES.** If not within limits, check operation of full-in limit switch (PARA 14-4-40).

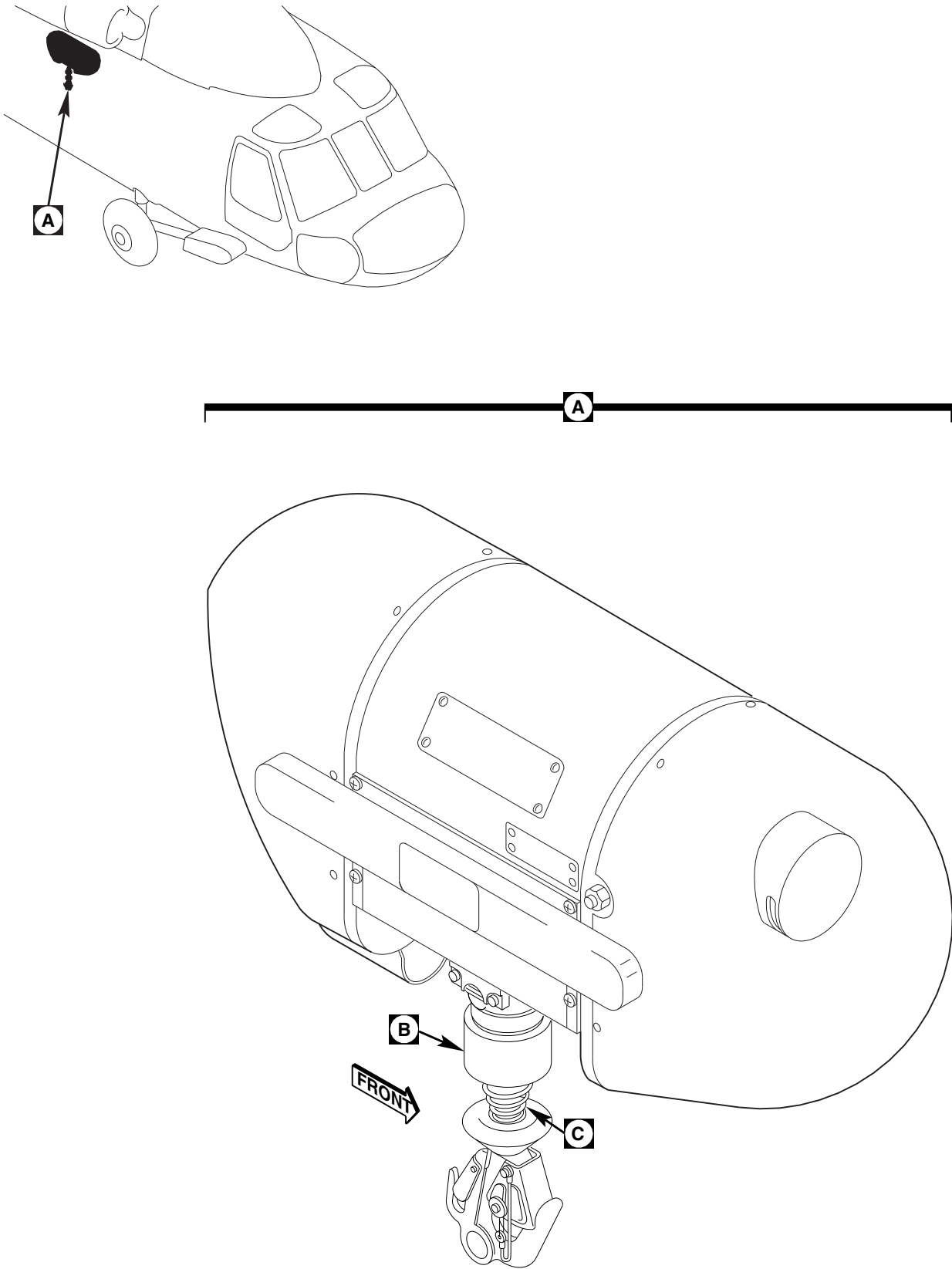
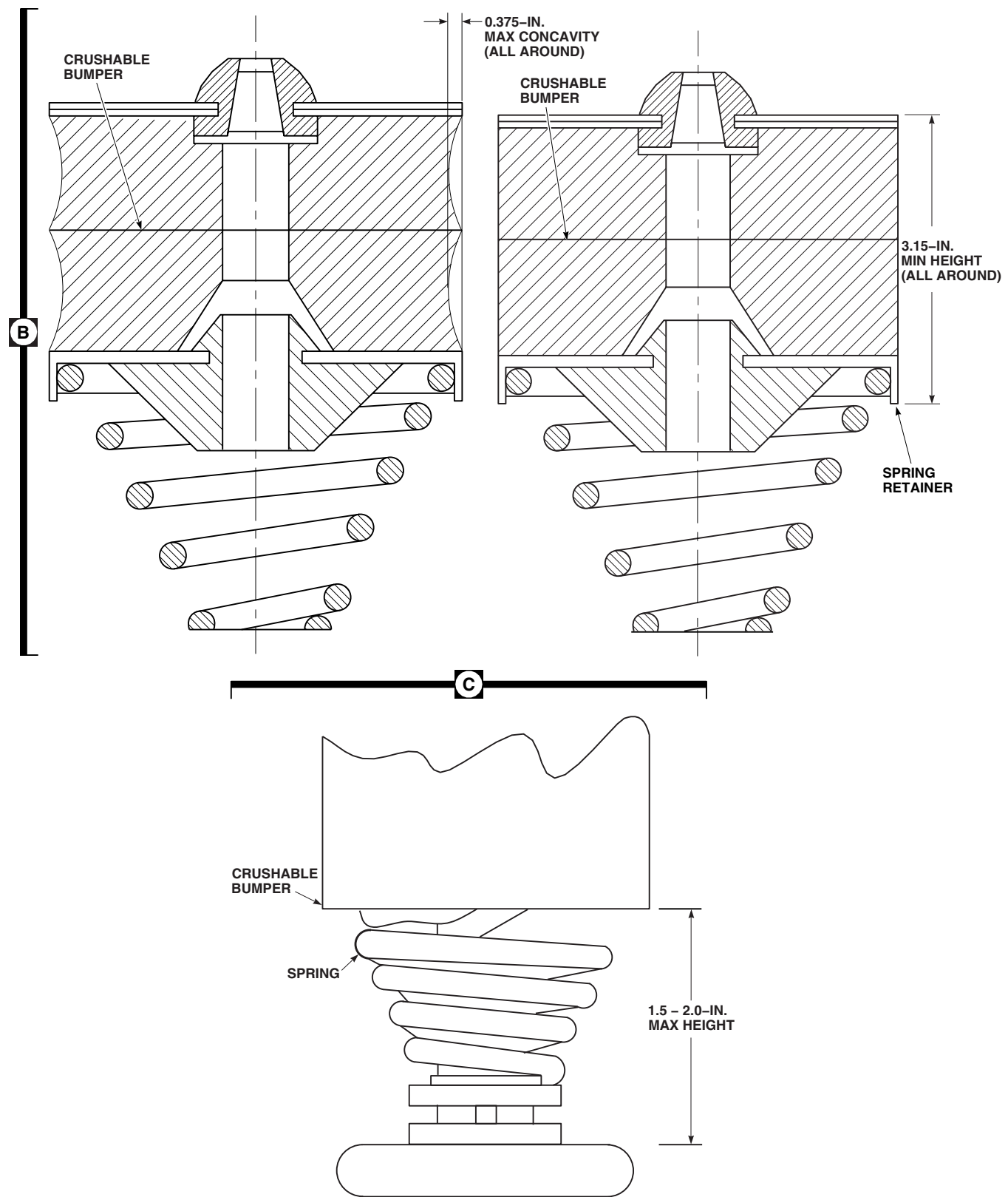


FIGURE 14-3-2-3. CRUSHABLE BUMPER AND SPRING REPLACEMENT CRITERIA
 (SHEET 1 OF 2)

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**FIGURE 14-3-2-3. CRUSHABLE BUMPER AND SPRING REPLACEMENT CRITERIA
(SHEET 2 OF 2)**

SECTION 14-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

PARAGRAPH	TITLE
14-4-1	Rescue Hoist HOIST, BL-27100-71 HOIST, 42315-10
14-4-2	Explosive Cartridge HOIST, BL-27100-71 HOIST, 42315-10
14-4-3	Cable Cutter HOIST, BL-27100-71 HOIST, 42315-10
14-4-4	Rescue Hoist Cable HOIST, BL-27100-71 HOIST, 42315-10
14-4-5	Rescue Hoist Hook HOIST, BL-27100-71 HOIST, 42315-10
14-4-6	Hydraulic Motor HOIST, BL-27100-71 HOIST, 42315-10
14-4-7	Heat Exchanger HOIST, BL-27100-71 HOIST, 42315-10
14-4-8	Elapsed Time Indicator HOIST, BL-27100-71 HOIST, 42315-10
14-4-9	Front and Rear Cowlings HOIST, BL-27100-71 HOIST, 42315-10
14-4-10	Control Valve HOIST, BL-27100-71
14-4-11	Electrical Valve Control HOIST, BL-27100-71
14-4-12	Electronic Control Box HOIST, 42315-10
14-4-13	Hoist Valve Module Assembly HOIST, 42315-10
14-4-14	Hook Housing Limit Switches HOIST, BL-27100-71
14-4-15	Hydraulic Control Assembly HOIST, BL-27100-71
14-4-16	Rescue Hoist Fan HOIST, 42315-10
14-4-17	Shaft Housing Limit Switches HOIST, BL-27100-71
14-4-18	Solenoid Valve HOIST, BL-27100-71
14-4-19	Support Assembly HOIST, 42315-10
14-4-20	Rescue Hoist Fairings

PARAGRAPH	TITLE
14-4-21	Rescue Hoist/Cargo Hook Pendant HOIST, BL-27100-71 HOIST, 42315-10
14-4-22	Wiring Harness HOIST, BL-27100-71 HOIST, 42315-10
14-4-23	Rescue Hoist Light Assembly HOIST, BL-27100-71 HOIST, 42315-10
14-4-24	Rescue Hoist Light Lamp HOIST, BL-27100-71 HOIST, 42315-10
14-4-25	Aux Rescue Hoist Control Panel HOIST, BL-27100-71 HOIST, 42315-10
14-4-26	Backup Control Panel HOIST, BL-27100-71
14-4-27	Rescue Hoist Control Box HOIST, 42315-10
14-4-28	Switch Guard Cover HOIST, BL-27100-71
14-4-29	Indicator and Backup Control Power Lamps HOIST, BL-27100-71
14-4-30	Backup Control Panel Information Plate HOIST, BL-27100-71
14-4-31	Auxiliary Hoist Relay Panel HOIST, BL-27100-71
14-4-32	Rescue Hoist HOIST BL-29900-30
14-4-33	Load Cable HOIST BL-29900-30
14-4-34	Hook Assembly HOIST BL-29900-30
14-4-35	Cable Cutter and Anvil HOIST BL-29900-30
14-4-36	Rescue Hoist Pendant HOIST BL-29900-30
14-4-37	Support Fairings and Cowlings HOIST BL-29900-30
14-4-38	Pilot's Hoist Control Panel HOIST BL-29900-30
14-4-39	Crew Hoist Control Panel HOIST BL-29900-30
14-4-40	Limit Switch HOIST BL-29900-30
14-4-41	Explosive Cartridge HOIST BL-29900-30

14-4-1 RESCUE HOIST **HOIST, BL-27100-71** **HOIST, 42315-10** .

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-1.1	Replace Rescue Hoist HOIST, BL-27100-71	14-4-1-2	14-4-1.2	Replace Rescue Hoist HOIST, 42315-10	14-4-1-12

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Drain Tube, Locally-Made
- Drill
- Electrical Repairer’s Toolkit
- Flashlight
- Fluorescent Penetrant Inspection Kit
- Inspection Mirror
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 231, Appendix D
- Plastic Tubing, 3-inches long (0.133 or 0.186-inch inside diameter) (for Drain Tube, Locally-Made)
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 296, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-3-30
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 1-7-22
- PARA 2-6-1
- PARA 2-6-5
- PARA 7-4-13
- PARA 7-4-43
- PARA 7-4-65

• PARA 14-2-1

• PARA 14-4-20

14-4-1.1 Replace Rescue Hoist**HOIST, BL-27100-71 .****14-4-1.1.1 Remove.**

- a. Remove support arm fairings and cuff fairing (PARA 14-4-20).
- b. Disconnect electrical connectors, P962, P963, and P964, from rescue hoist (Figure 14-4-1-1, Sheet 1, Detail B).
- c. Install protective caps and plugs on electrical connectors.
- d. Disconnect hydraulic pressure and return line quick-disconnects on backup hydraulic pump module (PARA 7-4-13).
- e. Using 1-quart container and machinery towel, Item 211, Appendix D, to catch hydraulic fluid, disconnect hydraulic pressure and return lines from rescue hoist (Figure 14-4-1-1, Sheet 1, Detail A).
- f. Install protective caps on fittings and plugs in lines.
- g. Remove screws, washers, clamps, and electrical harness from support arm (Figure 14-4-1-1, Sheet 2, Detail D).
- h. Remove screw, washer, clamp, channel, spacer, and block securing hydraulic lines and electrical harness to support arm.

WARNING

Injury to personnel and damage to equipment will result if rescue hoist is not supported during removal. Rescue hoist weighs 90 pounds. Use assistant to support rescue hoist during removal.

- i. Remove bolts, washers, and nuts securing support arm to base support (Figure 14-4-1-1,

Sheet 2, Detail C). Remove hoist and support arm from helicopter.

14-4-1.1.2 Disassemble.

- a. Remove return and pressure unions from rescue hoist hydraulic pressure and return ports (Figure 14-4-1-1, Sheet 1, Detail A). Discard packings.
- b. Install protective caps on fittings and plugs in ports.
- c. Remove bolts, washers, and nuts securing support arm to rescue hoist support. Remove support arm.
- d. Remove bolts, washers, and nuts securing base support to airframe fitting. Remove base support (Figure 14-4-1-1, Sheet 2, Detail C).
- e. Using nonmetallic scraper, remove sealing compound from base support and airframe fitting mating surface.
- f. Disconnect electrical connector, P969 from J969, on airframe fitting. Remove electrical harness.
- g. Install protective cap and plug on electrical connectors.
- h. Disconnect hydraulic pressure and return lines from hydraulic system (PARA 7-4-43).

14-4-1.1.3 Inspect.**NOTE**

Viewed from below, level wind shafts are two crosscut steel shafts parallel with each other with cable guide between them.

- a. Visually inspect rescue hoist to determine if one or more shims are present between front ends of level wind shafts and shaft drive gears (Figure 14-4-1-1, Sheet 2, Detail F).

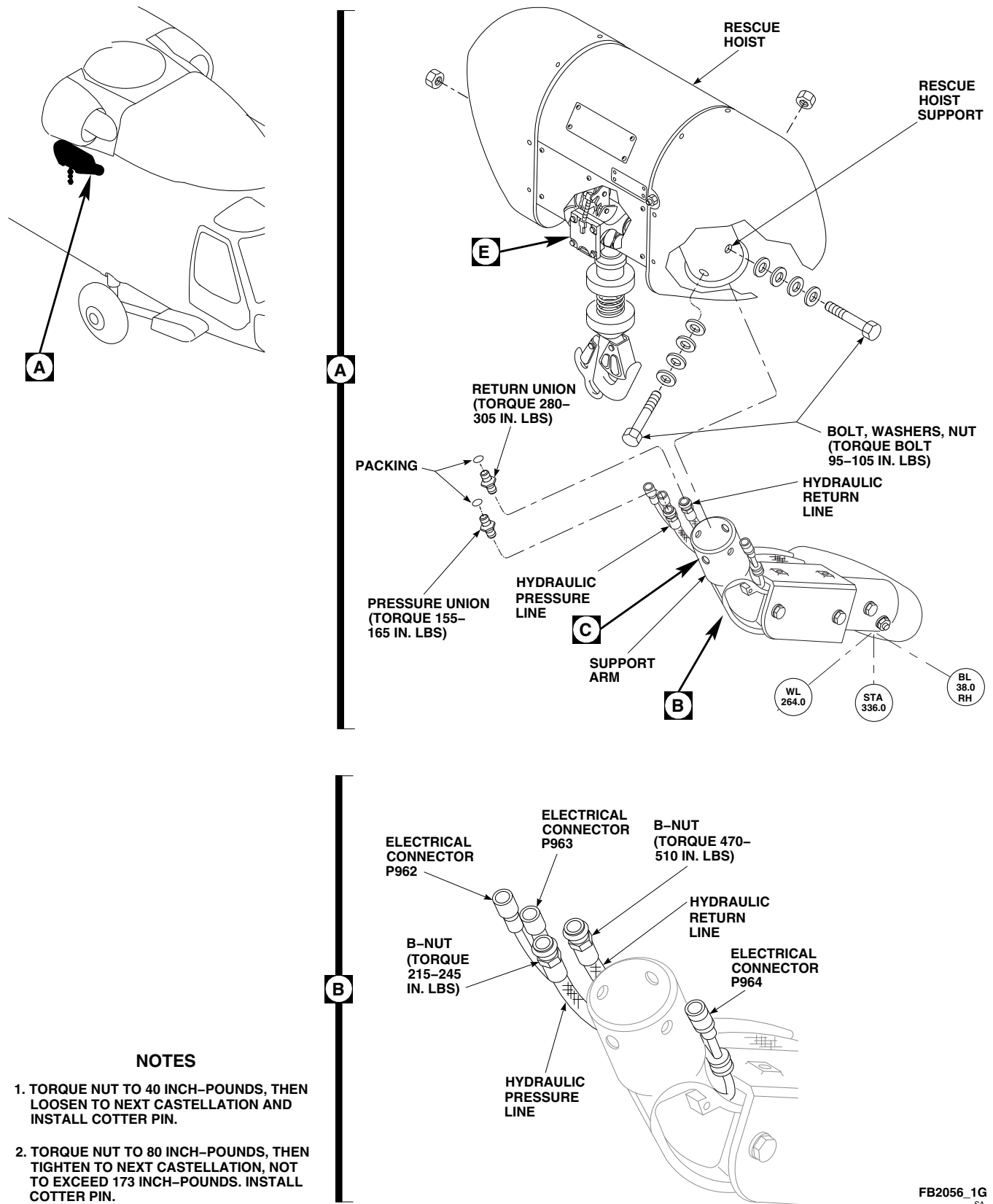


FIGURE 14-4-1-1. REPLACE RESCUE HOIST HOIST, BL-27100-71 (SHEET 1 OF 2)

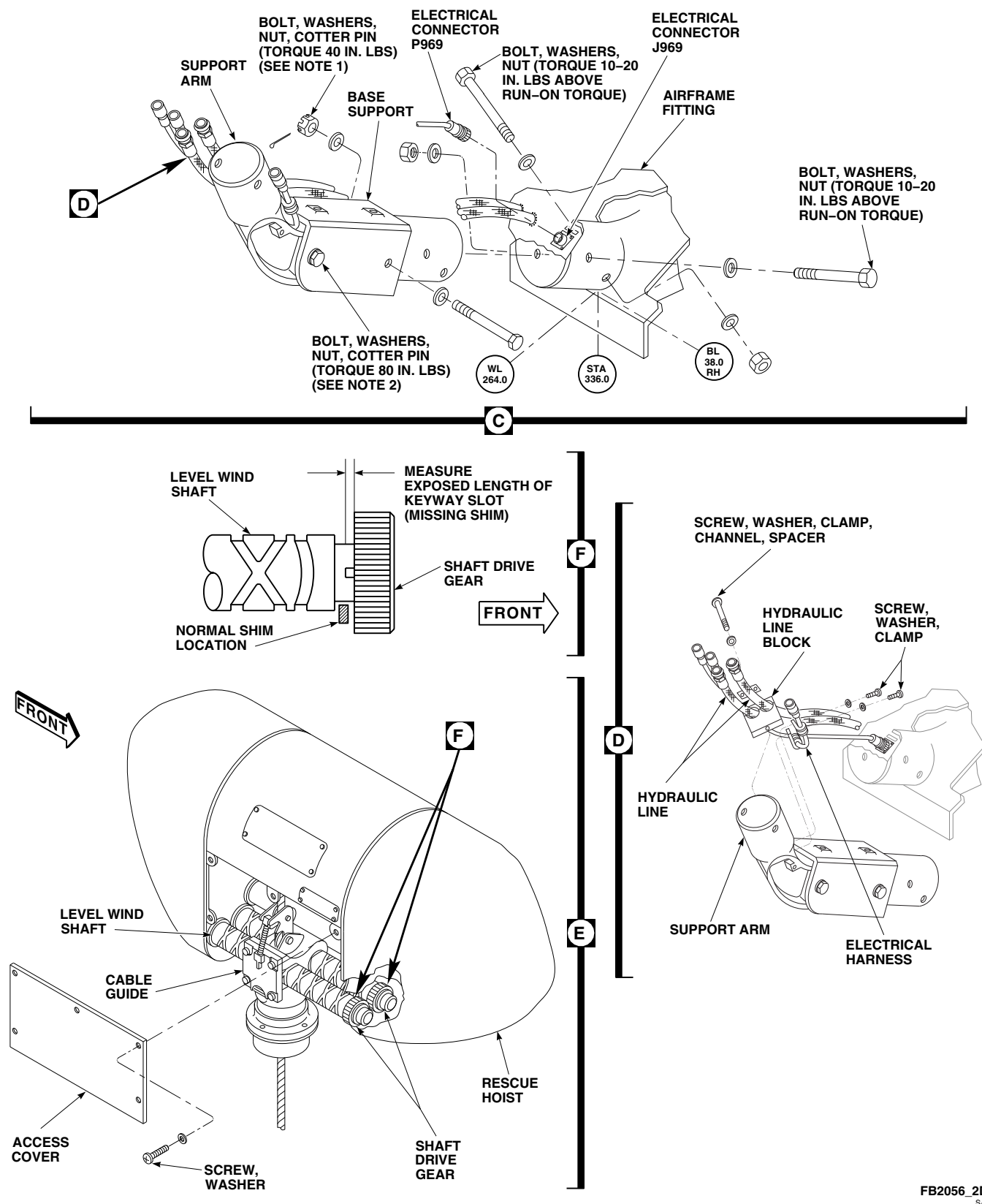
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FIGURE 14-4-1-1. REPLACE RESCUE HOIST HOIST, BL-27100-71 (SHEET 2 OF 2)

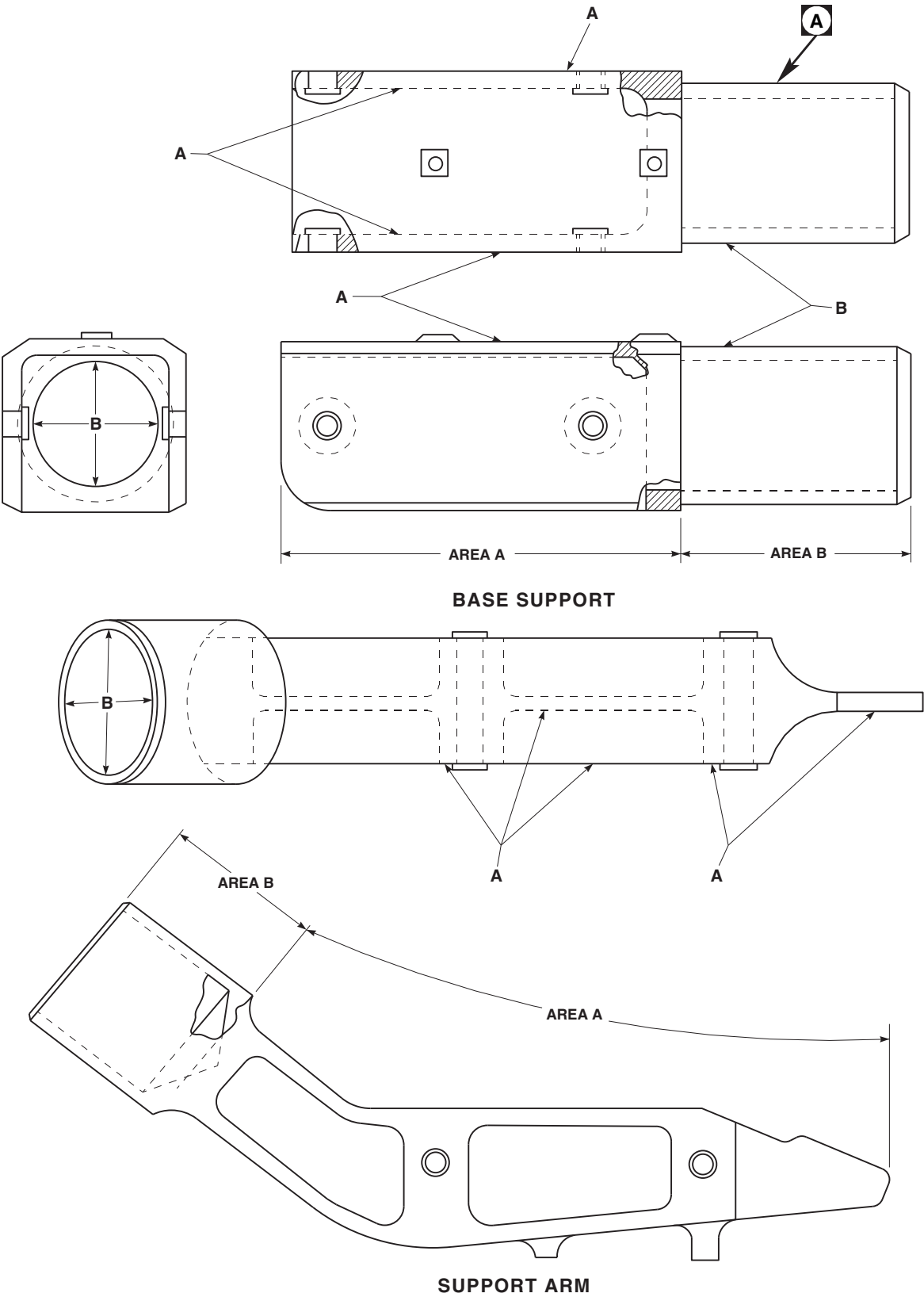
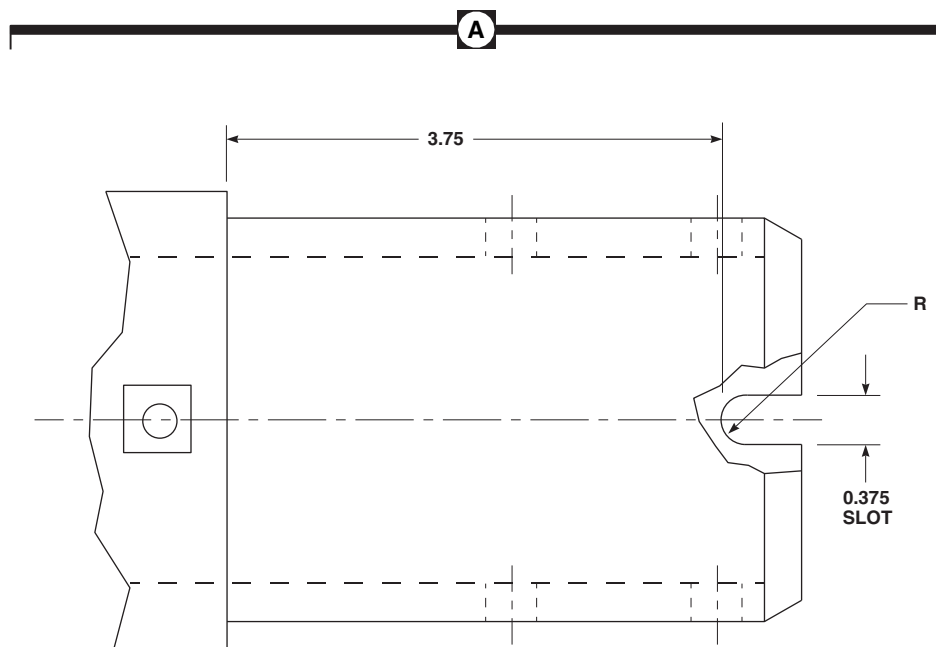


FIGURE 14-4-1-2. INSPECT RESCUE HOIST HOIST, BL-27100-71 (SHEET 1 OF 2)

AK3231_1
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AK3231_2
SA**FIGURE 14-4-1-2. INSPECT RESCUE HOIST** **HOIST, BL-27100-71 (SHEET 2 OF 2)**

- b. If no shims are present, proceed as follows:
- (1) Remove screws and washers securing access cover to rescue hoist (Figure 14-4-1-1, Sheet 2, Detail E). Remove cover.
 - (2) Measure keyway slots in front ends of level wind shafts (Figure 14-4-1-1, Sheet 2, Detail F).

NOTE

It may be necessary to use an inspection mirror and flashlight if rescue hoist is stopped with keyways facing inboard.

- (3) If measurement of exposed length of keyway is $\frac{1}{8}$ -inch or less, install access cover on rescue hoist with screws and washers (Figure 14-4-1-1, Sheet 2, Detail E). Using lockwire, Item 196, Appendix D, lockwire screws.
- (4) If measurement of exposed length of keyway is greater than $\frac{1}{8}$ -inch, replace rescue hoist.

- c. Inspect base support for cracks as follows (Figure 14-4-1-2, Sheet 1):

- (1) Using paint remover, Item 231, Appendix D, remove paint, sealing compound, and primer from inside and outside surfaces of cylindrical section mating with fitting and channel section wall, AREA B and AREA A, respectively.

NOTE

During fluorescent penetrant inspection, pay particular attention to areas around boltholes.

- (2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type 1, Method B, C, or D, Sensitivity Level 3 or 4, on cylindrical areas for cracks (ASTM E1417). **NONE ALLOWED.**
 - (3) Visually inspect all other areas for cracks. **NONE ALLOWED.**
- d. Inspect base support for nicks, dents, scratches, pits, and gouges. **NONE ALLOWED.** If found,

using abrasive cloth, Item 7, Appendix D, repair as follows:

(1) Repair AREA A as follows:

(a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.010-INCH DEEP. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.**

(b) **DAMAGE SHALL NOT BE CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLES.**

(c) If damage or repairs exceed limits, replace base support.

(2) Repair AREA B as follows:

(a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.002-INCH DEEP AND 0.750-INCH DIAMETER.**

(b) If damage or repairs exceed limits, replace base support.

e. Inspect base support for corrosion. **NONE ALLOWED.** If corrosion is found, using abrasive cloth, Item 7, Appendix D, repair as follows:

(1) Repair AREA A as follows:

(a) **BLEND OUT CORROSION UP TO 0.005-INCH DEEP OVER 100% OF SURFACE AREA.**

(b) If damage or repairs exceed limits, replace base support.

(2) Repair AREA B as follows:

(a) **BLEND OUT CORROSION UP TO 0.010-INCH DEEP OVER 100% OF INNER DIAMETER, AND UP TO 0.002-INCH DEEP INTO THE WALL THICKNESS TO MINIMUM OUTSIDE DIAMETER OF 2.966-INCHES.**

(b) If damage or repairs exceed limits, replace base support.

f. Visually inspect base support for cracks, nicks, dents, scratches, gouges, and corrosion. **NONE ALLOWED.**

g. Inspect base support for incorporation of 0.375-inch slot at bottom of base support. If slot is not present, perform the following:

(1) Add slot to base support as shown in Figure 14-4-1-2, Sheet 2, Detail A.

(2) Treat repaired area with corrosion resistant coating, Item 118, Appendix D.

(3) Using epoxy primer coating kit, Item 135, Appendix D, prime stripped and repaired area.

h. Inspect support arm for cracks as follows (Figure 14-4-1-2, Sheet 1):

(1) Using paint remover, Item 231, Appendix D, remove paint, sealing compound, and primer from inside and outside surfaces of socket section that mates with fitting and channel section wall, AREA B and AREA A, respectively.

NOTE

Pay particular attention to areas around boltholes during fluorescent penetrant inspection.

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type 1, Method B, C, or D, Sensitivity Level 3 or 4, on cylindrical areas for cracks (ASTM E1417). **NONE ALLOWED.**

(3) Visually inspect all other areas for cracks. **NONE ALLOWED.**

(4) If cracks are found, replace support arm.

i. Inspect support arm for nicks, dents, scratches, pits, or gouges. **NONE ALLOWED.** If found, using abrasive cloth, Item 7, Appendix D, repair as follows:

(1) Repair AREA A as follows:

(a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.010-INCH DEEP. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.**

(b) **DAMAGE SHALL NOT BE CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLES.**

(c) If damage or repairs exceed limits, replace support arm.

(2) Repair AREA B as follows:

(a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.020-INCH DEEP OVER 100 % OF SURFACE AREA. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.**

(b) If damage or repairs exceed limits, replace support arm.

j. Inspect support arm for corrosion. **NONE ALLOWED.** If corrosion is found, using abrasive cloth, Item 7, Appendix D, repair as follows:

(1) Repair AREA A as follows:

(a) **BLEND OUT CORROSION UP TO 0.005-INCH DEEP OVER 100% OF SURFACE AREA.**

(b) If damage or repairs exceed limits, replace support arm.

(2) Repair AREA B as follows:

(a) **BLEND OUT CORROSION UP TO 0.010-INCH DEEP OVER 100% OF INNER AND OUTER DIAMETER.**

(b) If damage or repairs exceed limits, replace support arm.

k. Inspect airframe fitting for cracks, corrosion, or deformation (PARA 2-6-1).

l. Inspect airframe fitting for plugged drain hole at bottom of fitting at 6 o'clock position as follows:

(1) Inspect drain hole to make sure hole is located at 45° angle at bottom of airframe fitting (Figure 14-4-1-4).

(2) If no hole exists or is located at 90° angle, proceed to step n.

(3) If hole is located at 45° angle and is blocked or clogged, remove drain tube and clean hole.

(4) Install new drain tube as follows:

NOTE

If drain hole diameter is 0.188-inch, use 0.133-inch diameter drain tube. If drain hole diameter is 0.250-inch, use 0.186-inch diameter drain tube.

(a) Locally make drain tube from 0.133 or 0.186-inch inside diameter plastic tubing, 3-inch long.

(b) Install drain tubing into fitting with sealing compound, Item 296, Appendix D. Make sure drain tube is 0.010 - 0.040-inch below inner surface of fitting (Figure 14-4-1-3).

(c) Seal drain tube to fitting with sealing compound, Item 296, Appendix D.

(d) Make sure drain tube is clear of sealing compound.

m. If airframe fitting drain hole is missing or in wrong location, rework as follows:

(1) If drain hole exists at 90° angle at bottom of fitting at 6 o'clock position, plug hole with sealing compound, Item 296, Appendix D (Figure 14-4-1-4).

(2) Drill hole to 0.250-inch diameter at 45° angle (Figure 14-4-1-4, Detail A).

- (3) Apply corrosion resistant coating, Item 118, Appendix D, to drilled hole.
- (4) Using epoxy primer coating kit, Item 135, Appendix D, prime drilled hole.
- (5) Locally make drain tube from 0.186-inch inside diameter plastic tubing, 3-inch long.
- (6) Install drain tubing into fitting with sealing compound, Item 296, Appendix D. Make sure drain tube is 0.010 - 0.040-inch below inner surface of fitting (Figure 14-4-1-3).
- (7) Seal drain tube to fitting with sealing compound, Item 296, Appendix D.
- (8) Make sure drain tube is clear of sealing compound.
- n. Using corrosion resistant coating, Item 118, Appendix D, treat stripped and repaired areas.
- o. Using epoxy primer coating kit, Item 135, Appendix D, prime stripped and repaired areas.

NOTE

Do not paint outside diameter of AREA B, holes, bushings, and inserts of support base assembly or outside diameter of AREA B, holes, bushings, and inserts of support arm.

- p. Paint stripped and repaired areas (PARA 2-6-5).

14-4-1.1.4 Assemble.

- a. Remove protective caps and plugs from ports.
- b. Connect hydraulic pressure and return lines to hydraulic system (PARA 7-4-43).
- c. Remove protective cap and plug from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).

- e. Connect electrical connector, P969 to J969, on airframe fitting (Figure 14-4-1-1, Sheet 2, Detail C).

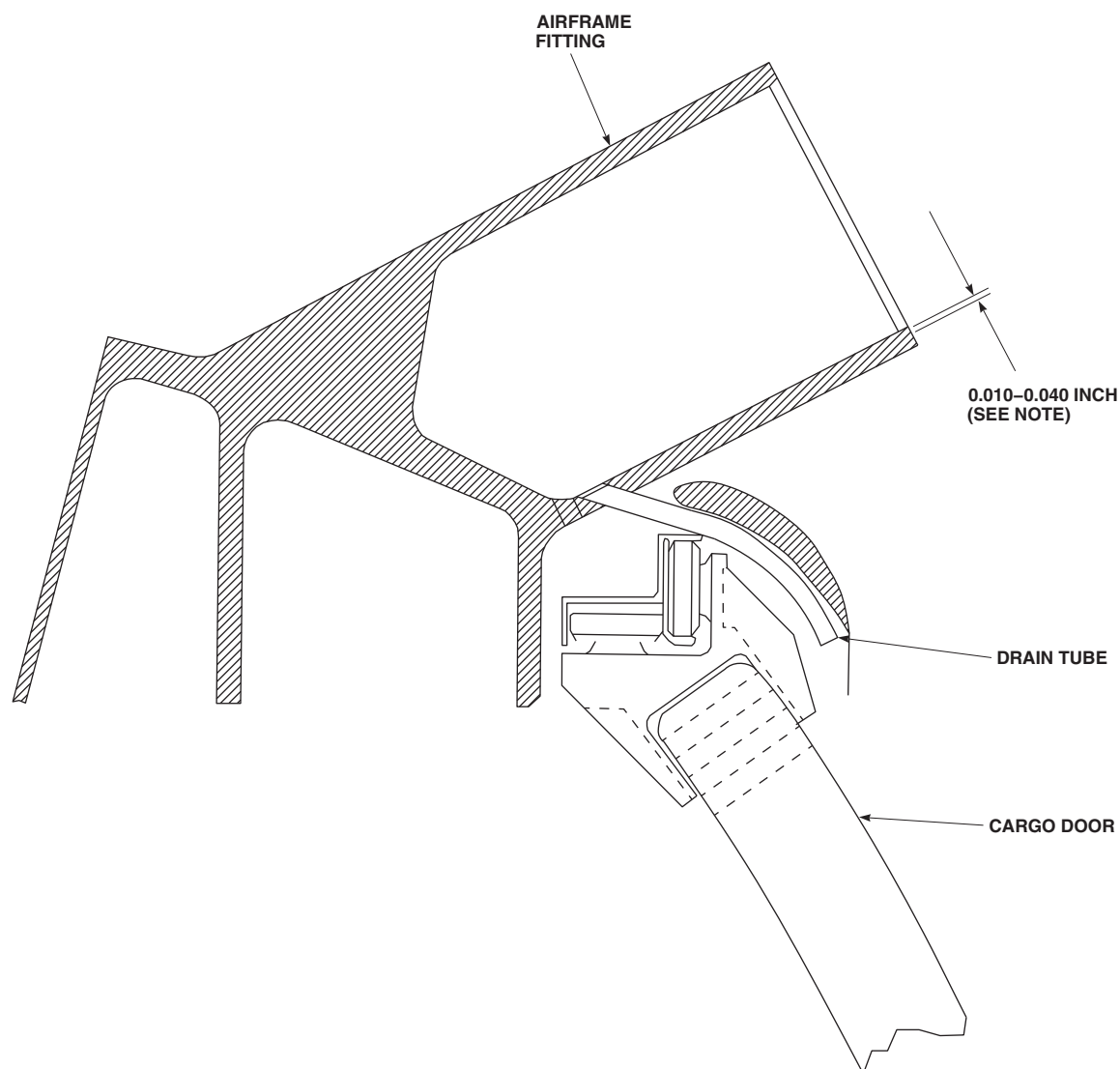
NOTE

Avoid blocking drain hole of airframe fitting (cup area) with sealing compound.

- f. Apply sealing compound, Item 296, Appendix D, to base support and airframe fitting mating surface.
- g. Install base support in airframe fitting with bolts, washers, and nuts. **TORQUE NUTS TO 10 - 20 INCH-POUNDS ABOVE RUN-ON TORQUE.**
- h. Apply sealing compound, Item 296, Appendix D, around base support/airframe fitting mating edges and fastener locations.
- i. Apply sealing compound, Item 296, Appendix D, onto support arm and rescue hoist mating surface.
- j. Install support arm into rescue hoist support with bolts, washers, and nuts (Figure 14-4-1-1, Sheet 1, Detail A). **TORQUE BOLTS TO 95 - 105 INCH-POUNDS.**
- k. Coat packings and union threads with hydraulic fluid, Item 155, Appendix D. Install packings on unions.
- l. Install unions in rescue hoist pressure and return ports. **TORQUE PRESSURE LINE UNION TO 155 - 165 INCH-POUNDS. TORQUE RETURN LINE UNION TO 280 - 305 INCH-POUNDS.**

14-4-1.1.5 Install.

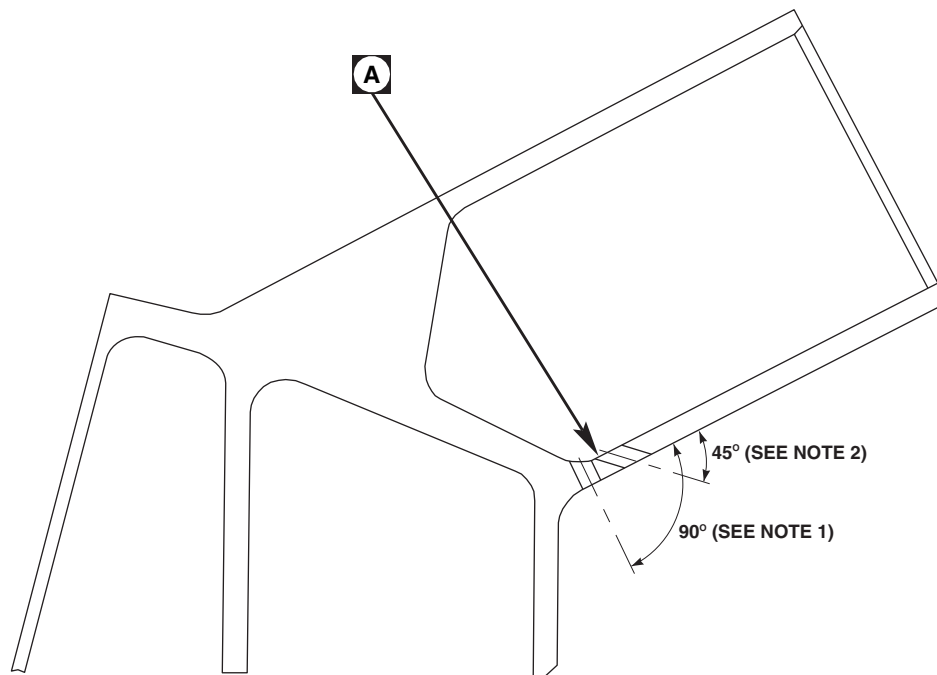
- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Apply sealing compound, Item 296, Appendix D, onto support arm and base support mating surface.

**NOTE**

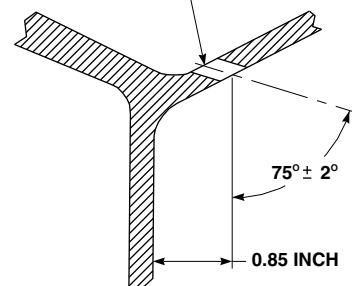
MAKE SURE DRAIN TUBE IS BELOW
INNER SURFACE OF FITTING.

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FIGURE 14-4-1-3. INSPECT RESCUE HOIST DRAIN TUBE **HOIST, BL-27100-71**



0.250 INCH
(SEE NOTE 3)



NOTES

1. PLUG 90° HOLE WITH SEALING COMPOUND.
2. CLEAN EXISTING HOLE (0.188 INCH DIA). ADD 0.250 INCH DIA HOLE IF NO HOLE EXISTS.
3. AVOID GOUGING INNER WALL OF FITTING DURING REWORK.

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FIGURE 14-4-1-4. INSPECT RESCUE HOIST DRAIN HOLE **HOIST, BL-27100-71**

WARNING

Injury to personnel and damage to equipment will result if rescue hoist is not supported during installation. Rescue hoist weighs 90 pounds. Use assistant to support rescue hoist during installation.

- c. Install rescue hoist and support arm to base support with bolts, washers, and nuts (Figure 14-4-1-1, Sheet 2, Detail C).
- d. **TORQUE UPPER NUT TO 80 INCH-POUNDS. THEN TIGHTEN NUT TO NEXT CASTELLATION. DO NOT EXCEED 173 INCH-POUNDS TORQUE.** Install cotter pin.
- e. **TORQUE LOWER NUT TO 40 INCH-POUNDS.** If required, loosen nut to next castellation. Install cotter pin.
- f. Apply sealing compound, Item 292, Appendix D, around boltheads and nuts.
- g. Secure electrical harness and hydraulic lines to support arm with screw, washer, clamp, channel, spacer, and block (Figure 14-4-1-1, Sheet 2, Detail D).
- h. Secure electrical harness to support arm with screws, washers, and clamps.
- i. Remove protective caps and plugs.
- j. Connect hydraulic pressure and return lines to rescue hoist (Figure 14-4-1-1, Sheet 1, Detail B). **TORQUE PRESSURE LINE B-NUT TO 215 - 245 INCH-POUNDS. TORQUE RETURN LINE B-NUT TO 470 - 510 INCH-POUNDS.**
- k. Connect hydraulic pressure and return line quick-disconnects on backup hydraulic pump module (PARA 7-4-13).
- l. Remove protective caps and plugs from electrical connectors.
- m. Inspect and treat electrical connectors (PARA 1-7-20).

- n. Connect electrical connectors, P962, to J1, P963, to J2, and P964 to J3 on rescue hoist. Using lockwire, Item 196, Appendix D, lockwire electrical connectors.
- o. Service rescue hoist (PARA 1-3-30).
- p. Bleed hydraulic system (PARA 7-4-65).
- q. Perform operational check of rescue hoist system (PARA 14-2-1).
- r. Install support arm fairings and cuff fairing (PARA 14-4-20).

14-4-1.2 Replace Rescue Hoist

HOIST, 42315-10 .

NOTE

- Any hoist kept in storage must be tested in accordance with overload clutch margin validation load check before being returned to service.
- Keep gear box filled with oil during long-term storage. Prepare tag that reads: "Gearbox filled. Drain and service to proper level before use." Attach tag to rescue hoist.

14-4-1.2.1 Remove.

- a. If rescue hoist is being removed from storage, test rescue hoist in accordance with overload clutch margin validation load check (PARA 1-7-22).
- b. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- c. Remove support arm fairings and cuff fairing (PARA 14-4-20).
- d. Disconnect electrical connectors, P962, P963, and P964, from rescue hoist (Figure 14-4-1-5, Detail A).
- e. Install protective caps and plugs on electrical connectors.
- f. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124,

Appendix D, clean area around rescue hoist hydraulic line connections.

- g. Using 1-quart container and machinery towel, Item 211, Appendix D, to catch hydraulic fluid, disconnect hydraulic pressure and return lines from rescue hoist. Install protective caps on fittings and plugs in lines.

WARNING

Injury to personnel and damage to equipment will result if rescue hoist is not supported during removal. Rescue hoist weighs 96 pounds. Use assistant to support rescue hoist during removal.

- h. Remove bolts, washers, and nuts securing rescue hoist to support. Using an assistant, remove rescue hoist from support.
- i. Remove unions from rescue hoist hydraulic pressure and return ports. Install protective caps on fittings and plugs in ports.

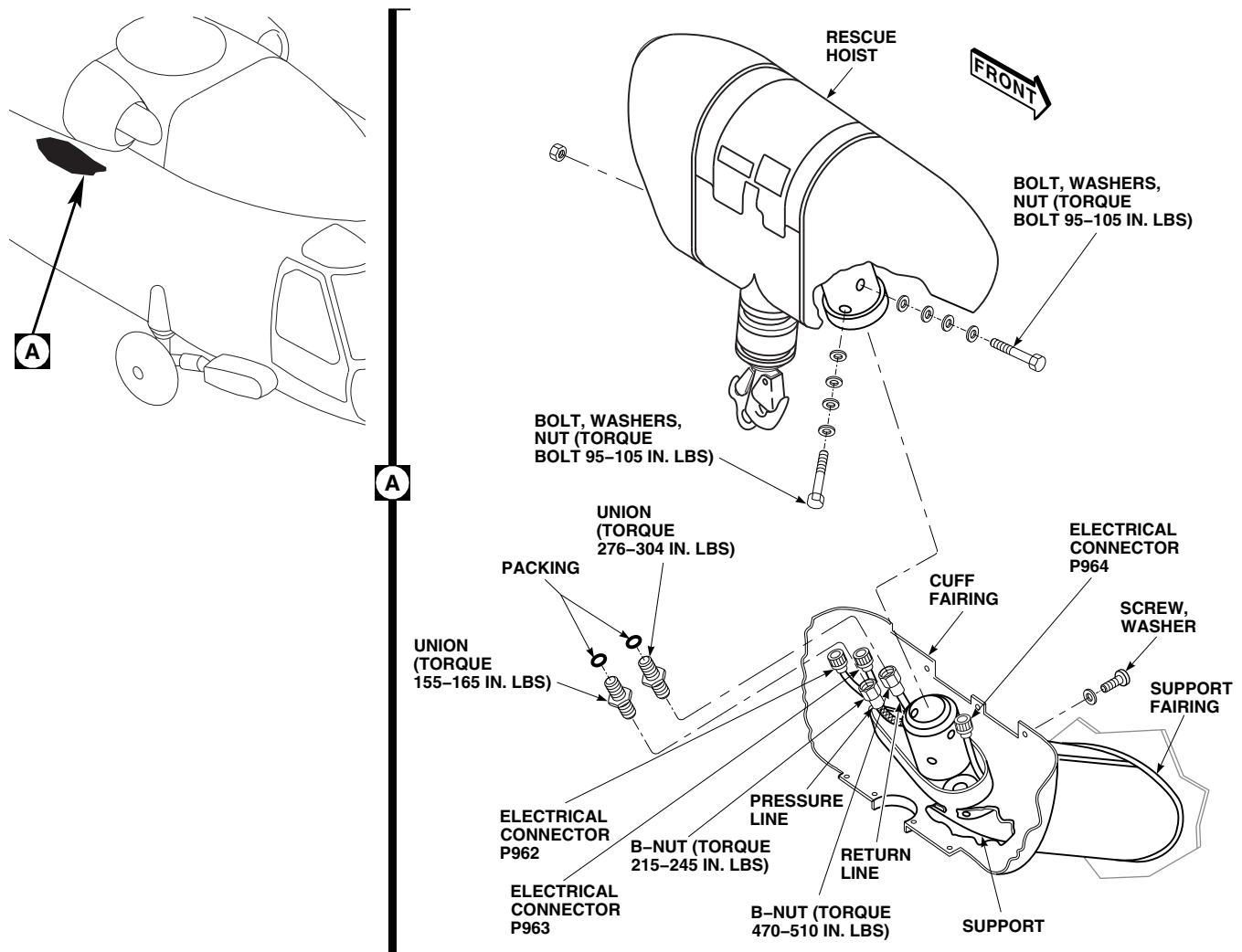
14-4-1.2.2 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Remove protective caps and plugs.
- c. Coat packings and union threads with hydraulic fluid, Item 155, Appendix D. Install packings on unions (Figure 14-4-1-5, Detail A).
- d. Install unions in rescue hoist pressure and return ports.
- e. **TORQUE PRESSURE UNION TO 155 - 165 INCH-POUNDS.**
- f. **TORQUE RETURN UNION TO 280 - 305 INCH-POUNDS.**
- g. Apply sealing compound, Item 296, Appendix D, onto support arm and base support mating surface.

WARNING

Injury to personnel and damage to equipment will result if rescue hoist is not supported during installation. Rescue hoist weighs 96 pounds. Use assistant to support rescue hoist during installation.

- h. Using an assistant, install rescue hoist on support while aligning boltholes on rescue hoist and support arm.
- i. Install bolts, washers, and nuts. Make sure lower bolthead faces rear. **TORQUE BOLTS TO 95 - 105 INCH-POUNDS.**
- j. Apply sealing compound, Item 292, Appendix D, around boltheads and nuts.
- k. Remove protective caps and plugs.
- l. Connect hydraulic pressure and return lines to rescue hoist.
- m. **TORQUE PRESSURE LINE B-NUT TO 215 - 245 INCH-POUNDS.**
- n. **TORQUE RETURN LINE B-NUT TO 470 - 510 INCH-POUNDS.**
- o. Remove protective caps and plugs from electrical connectors.
- p. Inspect and treat electrical connectors (PARA 1-7-20).
- q. Connect electrical connectors, P962, P963, and P964, to rescue hoist.
- r. Using lockwire, Item 196, Appendix D, lockwire electrical connectors.
- s. Service rescue hoist (PARA 1-3-30).
- t. Bleed hydraulic system (PARA 7-4-65).
- u. Perform operational check of rescue hoist system (PARA 14-2-1).

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SAFIGURE 14-4-1-5. REPLACE RESCUE HOIST **HOIST, 42315-10**

- v. Install support arm fairings and cuff fairing (PARA 14-4-20).
- w. Perform overload clutch margin validation load check (PARA 1-7-22).

14-4-2
 EXPLOSIVE CARTRIDGE
 HOIST, BL-27100-71
 HOIST, 42315-10
 .

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-2.1	Replace Explosive Cartridge	14-4-2-1	14-4-2.2	Replace Explosive Cartridge	14-4-2-5
		HOIST, BL-27100-71			HOIST, 42315-10

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Shorting Jumper
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Pressure Cartridge Installation Kit, KT-444

- Protective Cap and Plug

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 1-8-1
- PARA 9-4-34
- PARA 9-4-106
- PARA 14-2-1

14-4-2.1
 Replace Explosive Cartridge
 HOIST, BL-27100-71
 .

NOTE

Make sure service life date (month and year) is vibroetched on each side (four places) of explosive cartridge. Refer to PARA 1-8-1 for service life date.

14-4-2.1.1 Remove.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Disconnect battery connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

WARNING

- Injury to personnel will result if explosive cartridge accidentally fires. Explosive cartridges may be fired by radio or radar frequencies and static electricity. Short-circuit explosive cartridge connector pins to prevent accidental firing.

- Injury to personnel and damage to equipment will result if cartridges are inadvertently activated. All cartridges and cartridge actuated devices shall be handled as live ammunition. Explosive cartridges which have been fired may retain an explosive residue capable of presenting a hazardous condition.

- c. Disconnect electrical connector, P6, from explosive cartridge (Figure 14-4-2-1, Detail A).
- d. Install suitable shorting jumper on explosive cartridge receptacle and electrical connector.
- e. Install protective cap and plug on explosive cartridge receptacle and electrical connector.
- f. Remove explosive cartridge and shim(s) (if installed) from cartridge housing. Remove and discard packing. If same cartridge is to be installed, retain shims. If cartridge is to be replaced, discard shims.
- g. Clean inside of cartridge housing using dry-cleaning solvent, Item 124, Appendix D.

14-4-2.1.2 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. On replacement explosive cartridge, perform the following:
 - (1) Remove and discard packing.
 - (2) Remove and discard collar.
- c. Install explosive cartridge in cartridge housing to determine initial key orientation (Figure 14-

4-2-1, Detail A and Detail B). **NOTE POSITION OF KEY.** Evaluate as follows:

- (1) If key is located in desired position "A" (10:30 o'clock position, $\pm$ 30 minutes tolerance), then no shims are required, perform the following:
 - (a) Remove cartridge from cartridge housing (Figure 14-4-2-1, Detail A).
 - (b) Install packing on cartridge.
 - (c) Proceed to step d.
- (2) If key is not located in desired position "A" (10:30 o'clock position, $\pm$ 30 minutes tolerance), shim(s) are required. Refer to Table 14-4-2-1 and (Figure 14-4-2-1, Detail B) to determine the type and number of shims required to move key orientation to position "A". Perform the following:
 - (a) Obtain shim(s) from pressure cartridge installation kit, KT-444.
 - (b) Remove cartridge from cartridge housing (Figure 14-4-2-1, Detail A).
 - (c) Install shims on cartridge determined in step c. (2).
 - (d) Install packing.

- d. Install cartridge in cartridge housing. **TORQUE CARTRIDGE TO 80 - 90 INCH-POUNDS.**

- e. Using lockwire, Item 197, Appendix D, lockwire cartridge.
- f. Remove protective cap and plug from cartridge receptacle and electrical connector.
- g. Remove shorting jumper from cartridge receptacle and electrical connector.
- h. Inspect and treat electrical connector, P6 (PARA 1-7-20).

14-4-2-2 Change 10

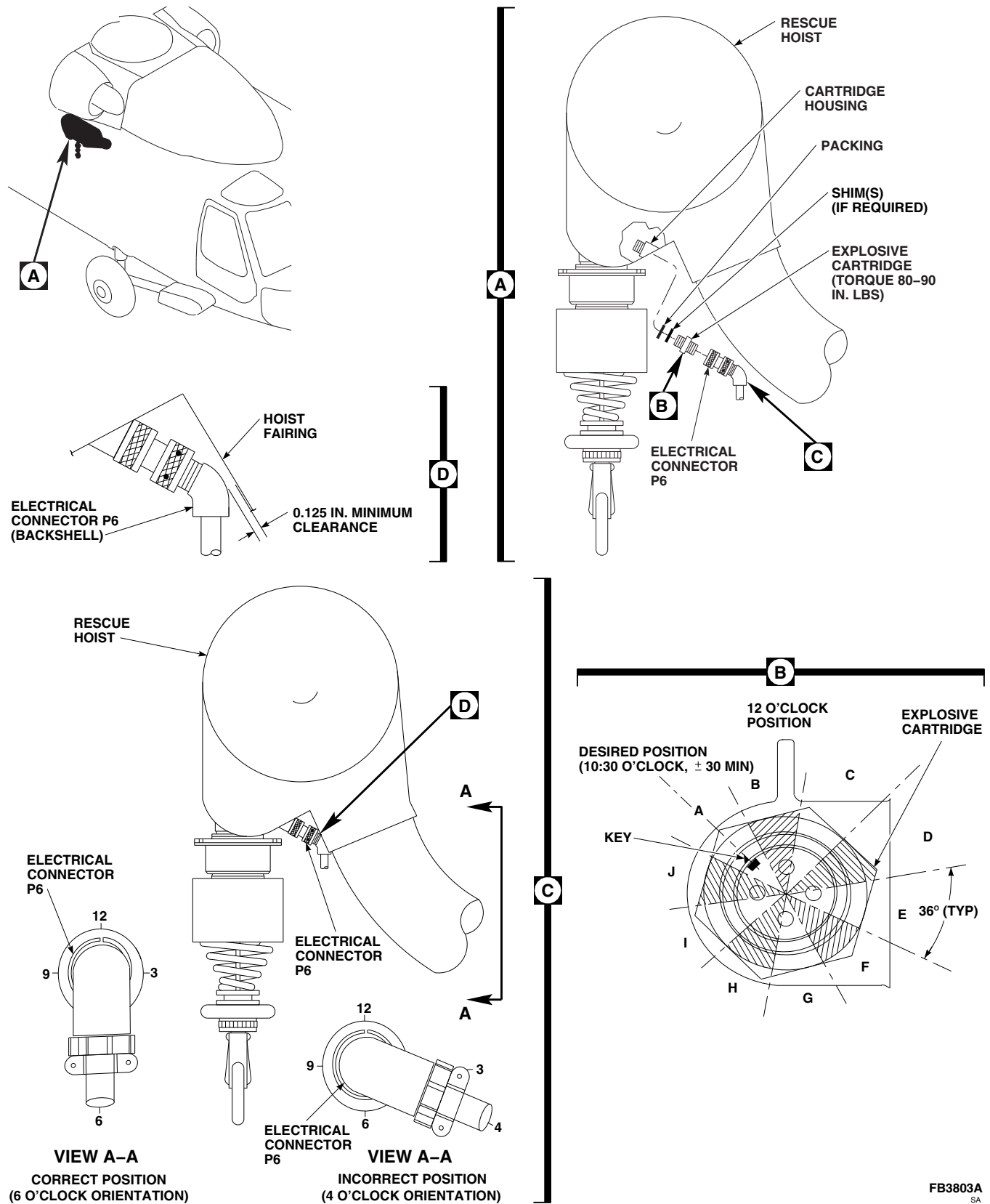


FIGURE 14-4-2-1. EXPLOSIVE CARTRIDGE ELECTRICAL CONNECTOR, P6 **HOIST, BL-27100-71**

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WARNING

Injury to personnel and damage to equipment will result if cartridge is fired by stray voltage. When installing cartridge, do not connect electrical connector until stray voltage check has been performed.

- i. Perform stray voltage check (PARA 14-2-1).
- j. Connect electrical connector, P6, to explosive cartridge. Make sure backshell of connector is orientated to 6 o'clock position (± 30 minutes tolerance) (Figure 14-4-2-1, Detail C).
- k. Using lockwire, Item 197, Appendix D, lock-wire electrical connector to cartridge.

- l. Connect battery connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

CAUTION

To prevent damage to backshell of electrical connector and hoist fairing during operational check, clearance check must be performed.

- m. Measure clearance between backshell of electrical connector and hoist fairing (Figure 14-4-2-1, Detail D). **CLEARANCE SHALL NOT BE LESS THAN 0.125-INCH.** If clearance is not within limit, repeat step c.
- n. Perform operational check of rescue hoist system (PARA 14-2-1).

Table 14-4-2-1. Shim Requirements

Key Position (See Note)	Shim (0.005 in.) Qty	Shim (0.010 in.) Qty	Shim (0.031 in.) Qty
A (Desired Position)	-	-	-
B	1	-	-
C	-	1	-
D	1	1	-
E	-	2	-
F	1	2	-
G	-	-	1
H	1	-	1
I	-	1	1
J	1	1	1

Note: Key positions
are $\pm 18^\circ$.

14-4-2.2 Replace Explosive Cartridge

HOIST, 42315-10 .

14-4-2.2.1 Remove.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Connect static ground cable to lower stage air valve on shock strut.
- c. Disconnect battery connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- d. Remove screws and washers securing outboard fairing to hoist and remove fairing (Figure 14-4-2-2, Detail A).

WARNING

- Injury to personnel will result if explosive cartridge accidentally fires. Explosive cartridges may be fired by radio or radar frequencies and static electricity. Short-circuit explosive cartridge connector pins to prevent accidental firing.
 - Injury to personnel and damage to equipment will result if cartridges are inadvertently activated. All cartridges and cartridge actuated devices shall be handled as live ammunition. Explosive cartridges which have been fired may retain an explosive residue capable of presenting a hazardous condition.
- e. Disconnect electrical connector, P6, from explosive cartridge.
 - f. Install suitable shorting jumper on explosive cartridge receptacle and electrical connector.
 - g. Remove explosive cartridge, packing, and spacer from cable cutter assembly. Remove packing from spacer.
 - h. If cartridge housing is dirty, clean using dry-cleaning solvent, Item 124, Appendix D.

14-4-2.2.2 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

NOTE

Make sure service life date (month and year) is vibroetched on each side (four places) of explosive cartridge. Refer to PARA 1-8-1 for service life date.

- b. On replacement explosive cartridge, perform the following:
 - (1) Remove and discard packing.
 - (2) Remove and discard collar.
 - (3) Install packing (Figure 14-4-2-2, Detail A).
- c. Install suitable shorting jumper on new explosive cartridge.
- d. Install packing onto spacer and install spacer into cable cutter assembly.

WARNING

Injury to personnel and damage to equipment will result if explosive cartridge is fired by stray voltage. When installing explosive cartridge, do not connect electrical connector until stray voltage check has been performed.

- e. Install explosive cartridge into cable cutter assembly. **TORQUE CARTRIDGE TO 80 - 90 INCH-POUNDS.**
- f. Using lockwire, Item 197, Appendix D, lockwire explosive cartridge.
- g. Connect battery connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- h. Connect external electrical power (PARA 1-6-16).

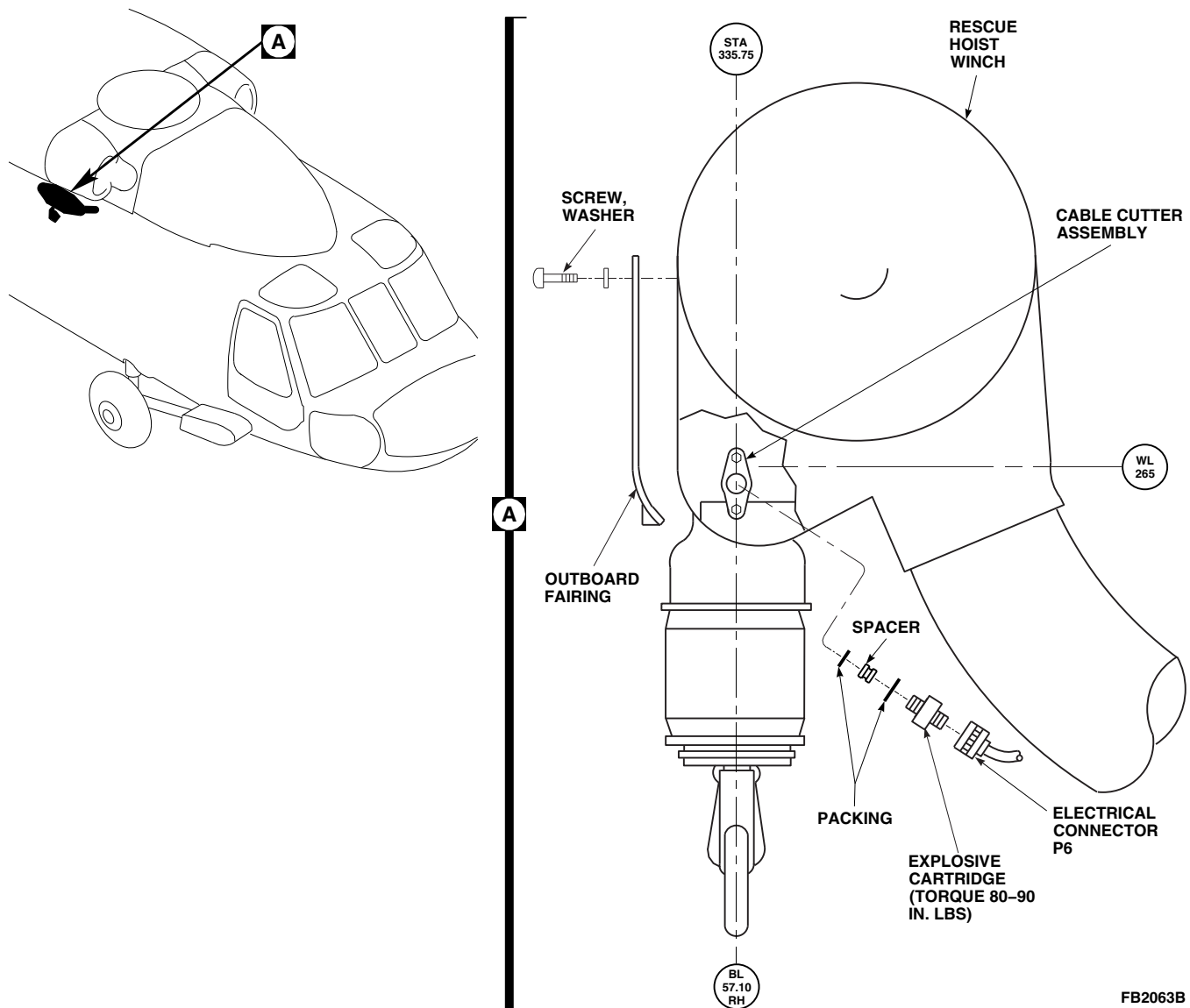


FIGURE 14-4-2-2. REPLACE EXPLOSIVE CARTRIDGE **HOIST, 42315-10**

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- i. Perform stray voltage check (PARA 14-2-1).
- j. Disconnect electrical power (PARA 1-6-16).
- k. Disconnect battery connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- l. Remove shorting jumper from explosive cartridge.
- m. Inspect and treat electrical connectors (PARA 1-7-20).
- n. Connect electrical connector, P6, to explosive cartridge.
- o. Using lockwire, Item 196, Appendix D, lockwire electrical connector, P6.
- p. Make sure area is free and clear of foreign objects.
- q. Install outboard fairing to hoist with screws and washers.
- r. Connect battery connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- s. Disconnect and remove static ground cable.

- t. Perform operational check of rescue hoist system (PARA 14-2-1).
- u. Record explosive cartridge replacement in helicopter logbook.

14-4-3 CABLE CUTTER **HOIST, BL-27100-71** **HOIST, 42315-10** .

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-3.1	Replace Cable Cutter HOIST, BL-27100-71	14-4-3-1	14-4-3.2	Replace Cable Cutter HOIST, 42315-10	14-4-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 14-4-2

14-4-3.1 Replace Cable Cutter **HOIST, BL-27100-71** .

14-4-3.1.1 Remove.

- Remove explosive cartridge (PARA 14-4-2).
- Turn cable cutter anvil counterclockwise and remove from cartridge housing (Figure 14-4-3-1, Detail A).
- Remove cotter pin from cable cutter.
- Remove cable cutter and packing. Discard packing.
- Clean inside of cartridge housing with dry-cleaning solvent, Item 124, Appendix D.

14-4-3.1.2 Install.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

- Install new cable cutter into cartridge housing (Figure 14-4-3-1, Detail A).
- Align cotter pin holes and install cotter pin.
- Install packing on cable cutter anvil.
- Thread cable cutter anvil into cartridge housing. Lockwire, Item 197, Appendix D, cable cutter anvil.
- Install explosive cartridge (PARA 14-4-2).

14-4-3.2 Replace Cable Cutter **HOIST, 42315-10** .

14-4-3.2.1 Remove.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Remove explosive cartridge (PARA 14-4-2).

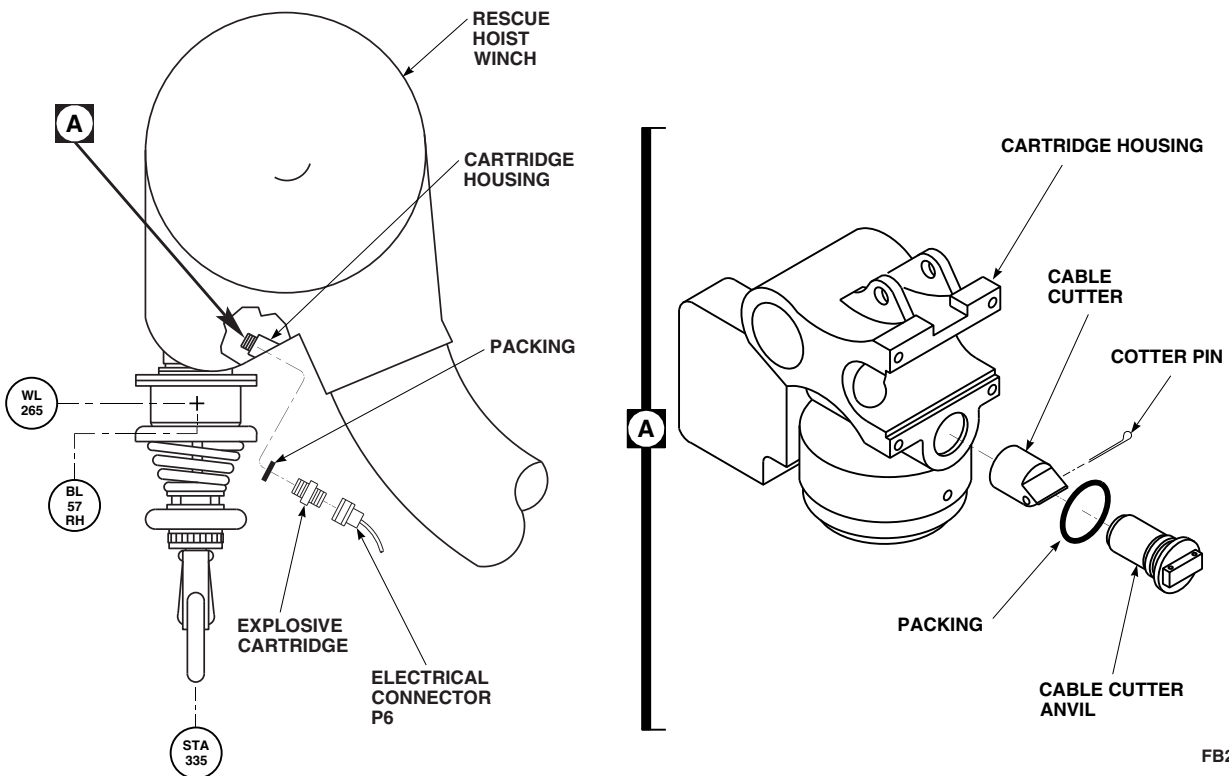


FIGURE 14-4-3-1. REPLACE CABLE CUTTER **HOIST, BL-27100-71**

- c. Remove bolt and washer holding full up limit switch and switch bracket to hoist (Figure 14-4-3-2, Sheet 2, Detail D). Move switch out of way.
- d. Remove cable cutter cap and cable cutter anvil (Figure 14-4-3-2, Sheet 2, Detail C). Discard anvil if cartridge has been fired.
- e. Remove bolts and washers holding cable cutter barrel to hoist. Remove cable cutter barrel.
- f. Remove cable cutter and packing from cable cutter barrel. Discard packing.
- g. Remove remains of shear screw which retained cable cutter, if cartridge has been fired.

14-4-3.2.2 Inspect.

Inspect all components for corrosion. **NONE ALLOWED.** Replace corroded components.

14-4-3.2.3 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Thoroughly clean inside of cartridge housing with dry-cleaning solvent, Item 124, Appendix D.
- c. Install packing on cable cutter and install cable cutter into cable cutter barrel (Figure 14-4-3-2, Sheet 2, Detail C). Make sure hole in cable cut-

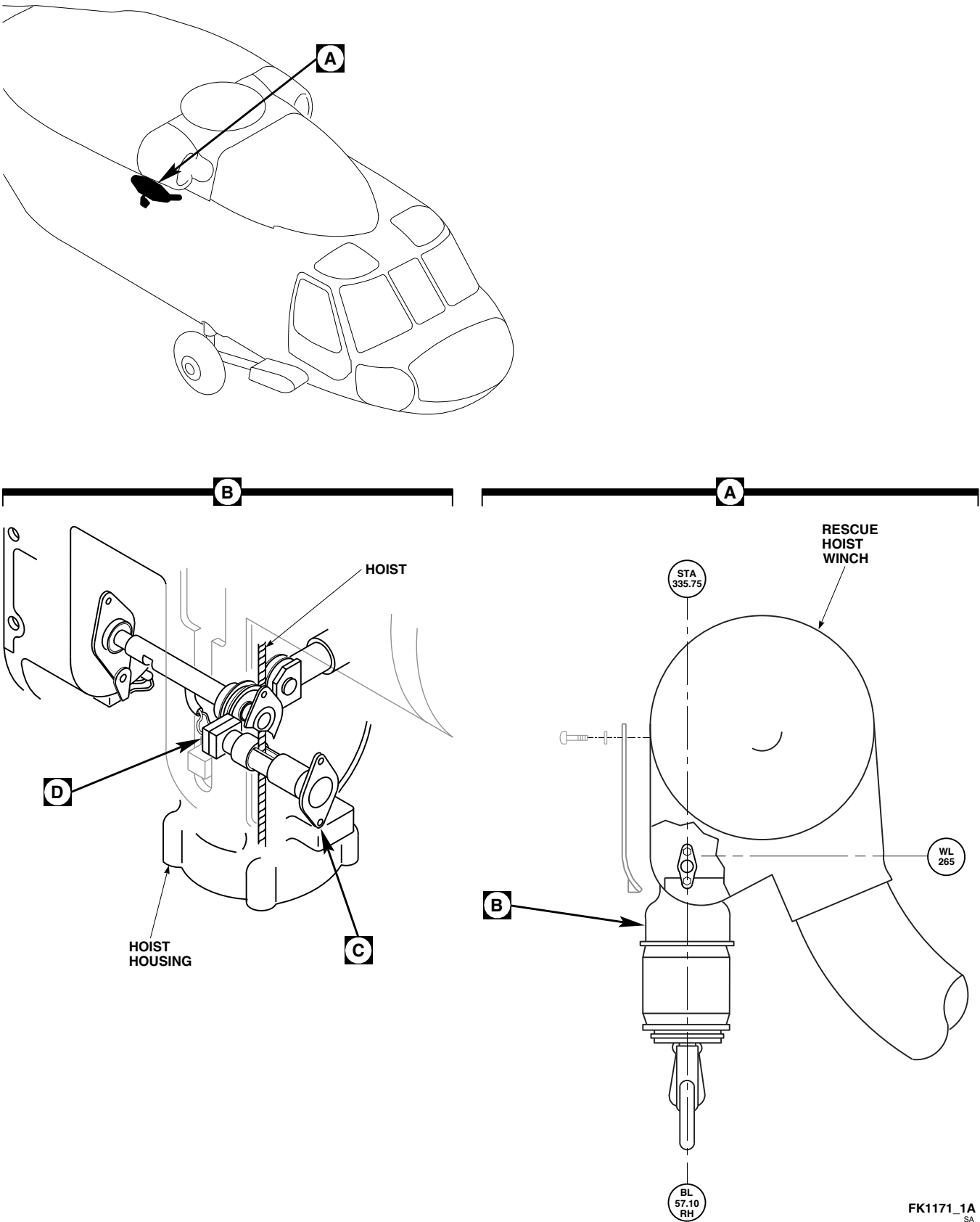


FIGURE 14-4-3-2. REPLACE CABLE CUTTER **HOIST, 42315-10** (SHEET 1 OF 2)

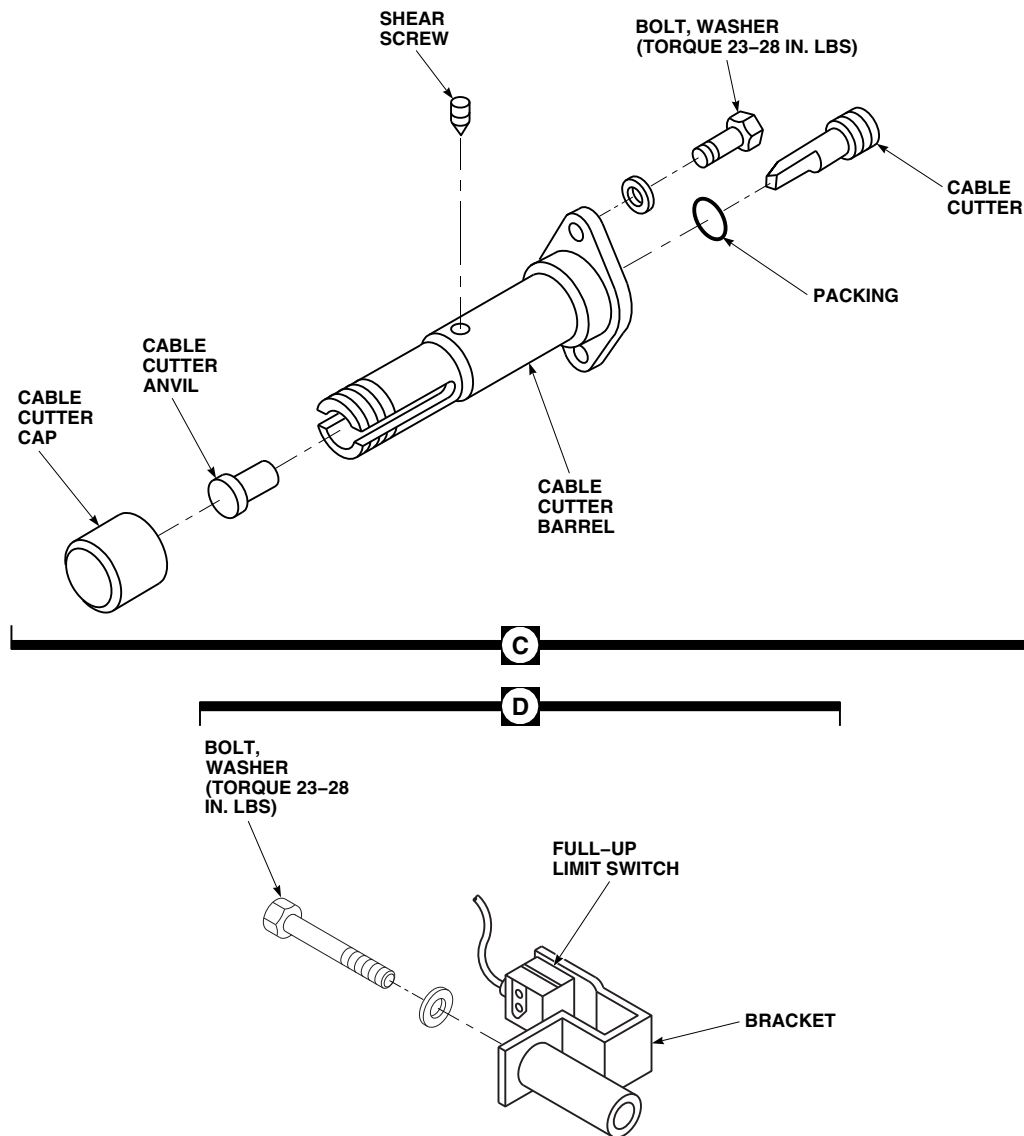
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FIGURE 14-4-3-2. REPLACE CABLE CUTTER **HOIST, 42315-10 (SHEET 2 OF 2)**

- ter lines up with set screw hole in cable cutter barrel.
- d. Install new shear screw into cable cutter barrel. Tighten shear screw.
- e. Slide cable cutter barrel into hoist housing and install with bolts and washers. **TORQUE BOLTS 23 - 28 INCH-POUNDS.**
- f. Install new anvil into cable cutter cap and install cap onto cable cutter barrel.
- g. Install full-up limit switch and bracket to hoist using bolt and washer (Figure 14-4-3-2, Sheet 2, Detail D). **TORQUE BOLTS 23 - 28 INCH-POUNDS.**
- h. Install explosive cartridge (PARA 14-4-2).

14-4-4 RESCUE HOIST CABLE **HOIST, BL-27100-71**
HOIST, 42315-10 .

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-4.1	Replace Rescue Hoist Cable	14-4-4-2	14-4-4.2	Replace Rescue Hoist Cable	14-4-4-6
	HOIST, BL-27100-71			HOIST, 42315-10	

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Rod, 7/64-inch (or equivalent)
- Electrical Repairer’s Toolkit
- Open-End Wrench, 7/16-inch
- Protective Gloves
- Slide Caliper
- Torque Wrench, 0 - 30 in. lbs
- Vernier Caliper

Materials/Parts

- Barrier Material, Item 80, Appendix D
- Grease, Item 145, Appendix D

- Lockwire, Item 195, Appendix D
- Lubricating Oil, Item 208, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-22
- PARA 14-2-1
- PARA 14-4-1
- PARA 14-4-2
- PARA 14-4-5
- PARA 14-4-9
- PARA 14-4-14
- PARA 14-4-17

14-4-4.1 Replace Rescue Hoist Cable**HOIST, BL-27100-71 .****14-4-4.1.1 Remove.**

- a. Remove explosive cartridge (PARA 14-4-2).
- b. Remove screws and washers securing access cover to rescue hoist. Remove access cover (Figure 14-4-4-1, Sheet 1, Detail B).
- c. On lower console, check that RESCUE HOIST switch is OFF (center position) (Figure 14-4-4-1, Sheet 2, Detail E).
- d. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- e. On lower console, set RESCUE HOIST switch to ALL.
- f. Place barrier material, Item 80, Appendix D, on deck under rescue hoist.

WARNING

- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of helicopter and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.
- Injury to personnel will occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.

CAUTION

Damage to hoist and loose cable wraps will result if cable is restrained from motion while reeling out. Do not restrain cable while reeling out.

NOTE

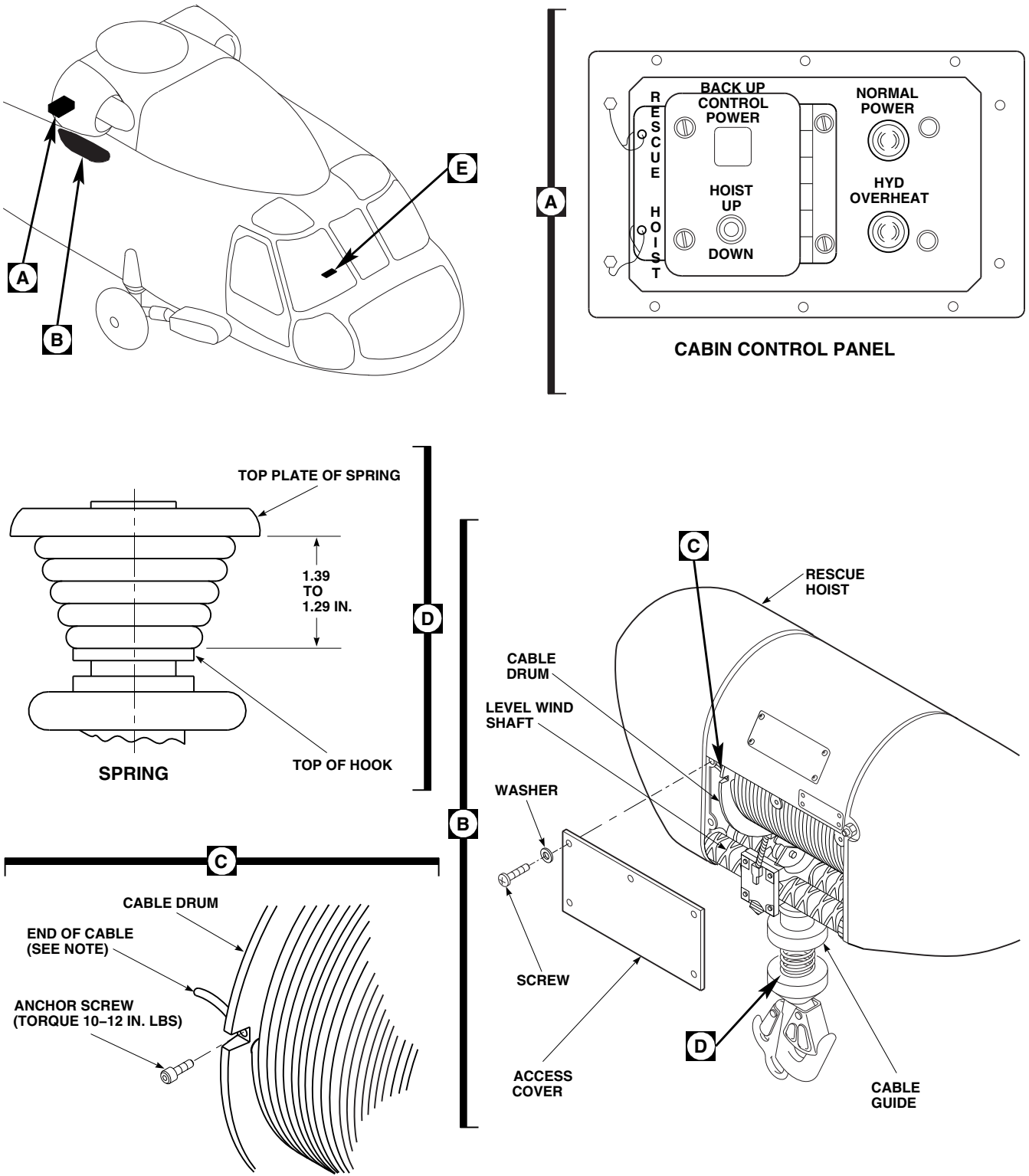
Hoist contains 225 usable feet of cable. Initial and final 20 feet of cable are colored orange.

- g. In cabin, operate rescue hoist and cargo hook pendant and reel out hoist cable while assistant walks end of cable as far from hoist as space permits. When extreme position is reached, stop hoist.
- h. Have assistant grasp cable just under exit point from hoist. Reel cable out again while assistant walks cable out as far as possible.
- i. Repeat step h. until full-out limit switch is reached.
- j. In cabin, open switch guard assembly and press BACK UP CONTROL POWER switch on cabin control panel. Switch should light (Figure 14-4-4-1, Sheet 1, Detail A).

CAUTION

Damage to hoist will result if anchor screw securing cable to drum is allowed to pass beyond scrub roller when reeling out last cable wrap on hoist drum. Do not allow anchor screw to pass beyond scrub roller.

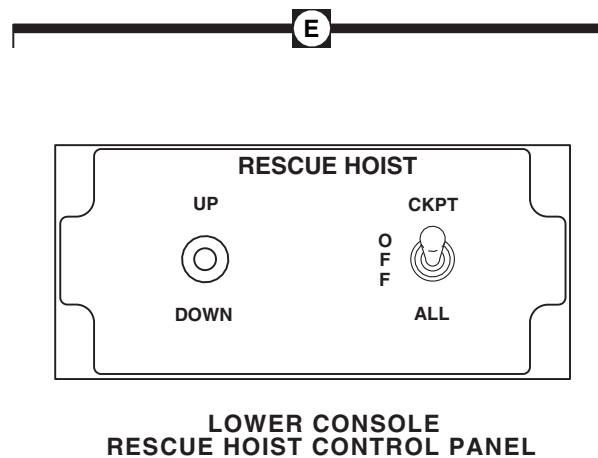
- k. With assistant observing cable drum, alternately press and release HOIST switch to DOWN on cabin control panel until last cable wrap is off drum and anchor screw is accessible (Figure 14-4-4-1, Sheet 1, Detail C).
- l. Inspect cable drum gears, cable guide, and level wind shafts for damage. **NONE ALLOWED.** If damage is found, replace rescue hoist (PARA 14-4-1).
- m. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- n. Remove anchor screw securing cable to drum and remove cable from drum.



NOTE
INSTALL CABLE WITH END
BOTTOMED OUT IN BLIND HOLE.

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FIGURE 14-4-4.1. REPLACE RESCUE HOIST CABLE **HOIST, BL-27100-71** (SHEET 1 OF 2)

FB2236 2A
SA**FIGURE 14-4-4-1. REPLACE RESCUE HOIST CABLE** **HOIST, BL-27100-71 (SHEET 2 OF 2)**

- o. Pull cable through cable guide (Figure 14-4-4-1, Sheet 1, Detail B).
- p. Remove rescue hoist hook (PARA 14-4-5).
- g. On lower console, set RESCUE HOIST switch to ALL.
- h. In cabin, open switch guard cover and press BACK UP CONTROL POWER switch on cabin control panel (Figure 14-4-4-1, Sheet 1, Detail A). Switch should light.

14-4-4.1.2 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Push cable through cable guide (Figure 14-4-4-1, Sheet 1, Detail B).
- c. Insert cable on cable drum until cable end bottoms out in blind hole (Figure 14-4-4-1, Sheet 1, Detail C).
- d. Insert anchor screw into strands of cable behind welded area. **TORQUE ANCHOR SCREW TO 10 - 12 INCH-POUNDS.**
- e. On lower console, set RESCUE HOIST switch to OFF (Figure 14-4-4-1, Sheet 2, Detail E).
- f. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

WARNING

- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of helicopter and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.
- Injury to personnel will occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.



- Backwinding of cable will result if backup control is run in down position when cable is fully extended.
 - To prevent damage to hoist, at any sign of hoist or cable trouble, stop hoist and investigate problem.
- i. With assistant observing cable drum and applying firm drag with gloved hand, reel in cable by pressing HOIST switch on cabin control panel to UP.
 - j. Continue to reel in cable. When cable is out of orange zone, release switch.
 - k. Press BACK UP CONTROL POWER switch and check that NORMAL POWER light goes on, indicating system is in normal power mode. Close switch guard cover.
 - l. Using lockwire, Item 195, Appendix D, lockwire switch guard cover to bolts.
 - m. Using rescue hoist and cargo hook pendant, slowly reel out hoist cable, while assistant applies firm drag on cable with gloved hand. Hoist should stop at full-out limit switch with 3 ½ - 4 ½ wraps of cable on drum.
 - n. If cable does not stop at full-out position, replace shaft housing limit switches (PARA 14-4-17).
 - o. Continue using pendant and slowly reel cable onto drum, while assistant applies firm drag on cable with gloved hand. Have assistant make sure cable feeds on drum correctly and that cable is kept clean and is not damaged in any way. Stop reeling cable when approximately three feet of cable remains to be loaded.
 - p. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
 - q. Install rescue hoist hook (PARA 14-4-5).
 - r. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
 - s. On lower console, set RESCUE HOIST switch to ALL. In cabin, operate pendant and continue to reel in cable to full-in of system limit switch (Figure 14-4-4-1, Sheet 2, Detail E).
 - t. With hook making contact with full-in limit switch, check compression of spring by measuring from top of hook to top plate of spring using slide caliper. **SPRING COMPRESSION SHALL BE 1.29 - 1.39-INCH** (Figure 14-4-4-1, Sheet 1, Detail D).
 - u. If spring compression is not 1.29 - 1.39-inch, replace hook housing limit switch (PARA 14-4-14).
 - v. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
 - w. Install access cover on rescue hoist and secure with screws and washers (Figure 14-4-4-1, Sheet 1, Detail B). Lockwire, Item 195, Appendix D.
 - x. Install explosive cartridge (PARA 14-4-2).
 - y. Make sure area is clean and free of foreign material.

- z. Perform operational check of rescue hoist system (PARA 14-2-1).

14-4-4.2 Replace Rescue Hoist Cable

HOIST, 42315-10 .

14-4-4.2.1 Remove.

WARNING

- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of helicopter and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.
- Injury to personnel will occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.

CAUTION

Damage to cable will occur if cable is kinked. Use extreme care when reeling cable in or out.

NOTE

- Hoist contains 200 usable feet of cable. Initial and final 20 feet of cable is colored orange.
 - Clean barrier material must be placed under hoist area to lay cable on during hoist operation.
- a. Place barrier material, Item 80, Appendix D, under hoist area to lay cable on during hoist operation.
 - b. Operate rescue hoist/cargo hook pendant and reel out hoist cable while assistant walks end of cable as far from hoist as space permits. When extreme position is reached, stop hoist.
 - c. Have assistant grasp cable just under exit point from hoist. Reel cable out again while assistant walks cable out as far as possible.
 - d. Repeat step c. until full-out limit switch is reached.

CAUTION

To prevent damage to cable reeling system, do not activate hoist in either direction without cable on drum.

NOTE

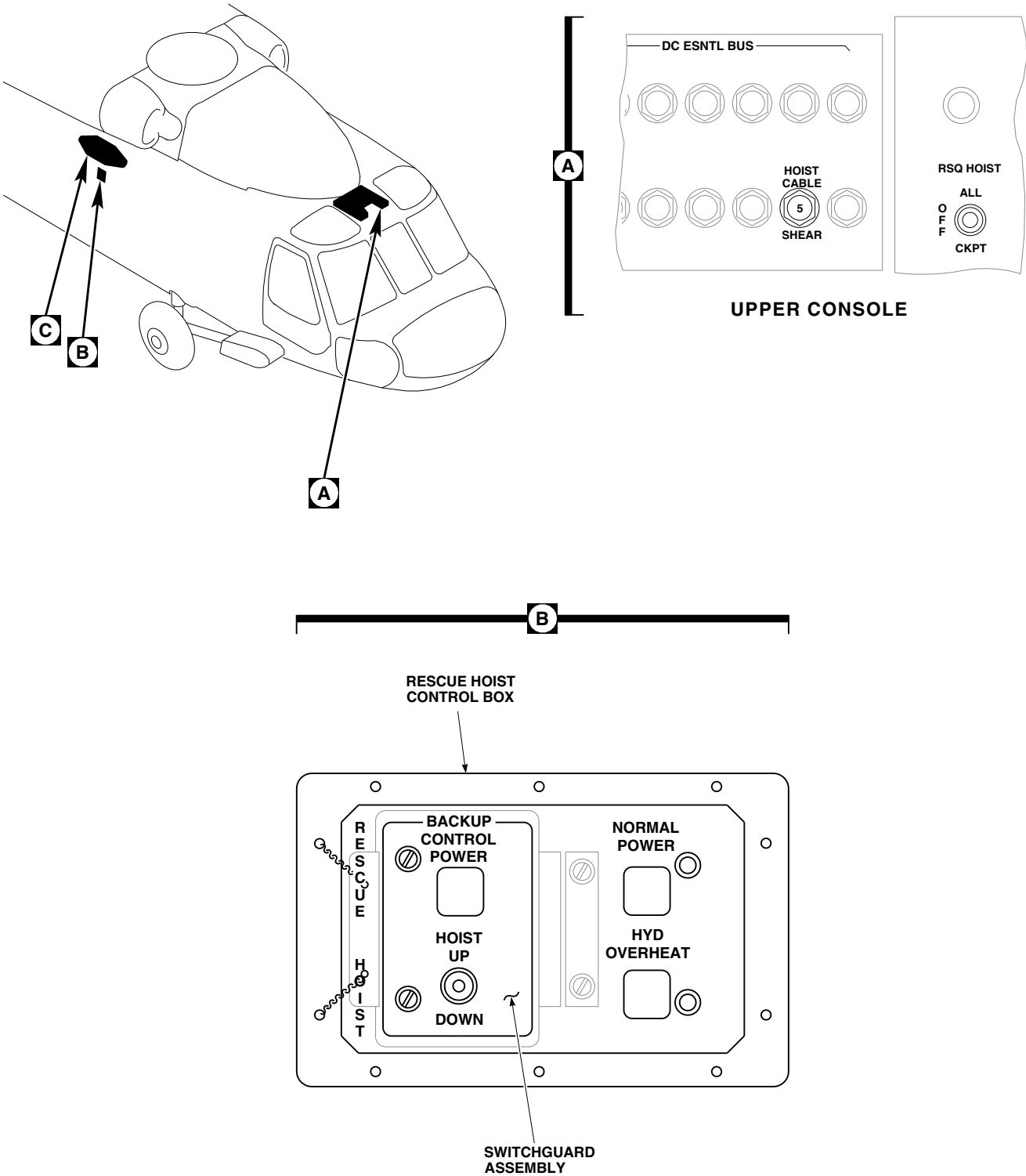
When cable retention screws are centered in main housing inspection window, there should not be any cable remaining on drum.

- e. Depress two manual override switches at rear corner of exposed hoist area, and using rescue hoist/cargo hook pendant, slowly reel out remaining cable until drum returns from bottoming on rear end of hoist and cable retention screws are directly in center of main housing inspection window (Figure 14-4-4-2, Sheet 2, Detail D).
- f. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- g. Remove rescue hoist hook (PARA 14-4-5).

NOTE

If crowder roller assembly is not retained in its compressed position before cable is removed, it may be difficult to reinsert cable.

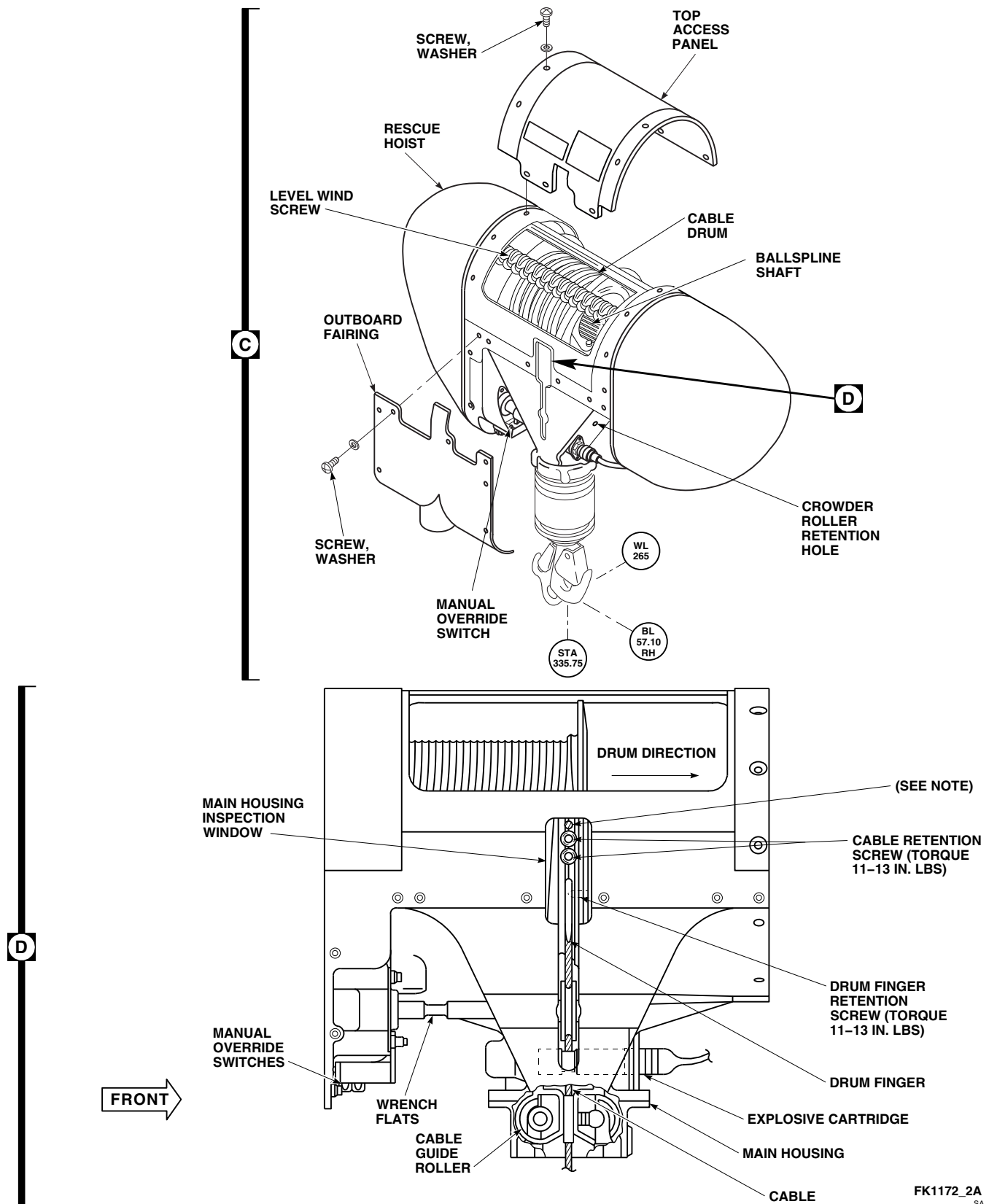
- h. Insert 7/64-inch drill rod or equivalent into crowder roller retention hole located next to explosive cartridge in main housing (Figure 14-4-4-2, Sheet 2, Detail D).



NOTE
BEFORE TORQUING CABLE RETENTION SET
SCREWS MAKE SURE CABLE IS FULLY INSERTED
IN THE DRUM AND VISIBLE IN THIS INSPECTION HOLE.

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FIGURE 14-4-4-2. REPLACE RESCUE HOIST CABLE **HOIST, 42315-10 (SHEET 1 OF 2)**

FIGURE 14-4-4-2. REPLACE RESCUE HOIST CABLE **HOIST, 42315-10** (SHEET 2 OF 2)

NOTE

- Make sure to loosen two cable retention screws and drum finger retention screw on drum flange before removing cable from hoist.
 - Rotating drum after cable is removed from hoist may misposition drum and limit switches, which could require adjustment after new cable is installed. Do not rotate drum after cable is removed.
 - Install new cable immediately after old cable has been removed.
- i. Loosen cable retention screws securing hoist cable to drum and remove hoist cable from drum (Figure 14-4-4-2, Sheet 2, Detail D). Pull cable through cable guide.

14-4-4.2.2 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Do not apply grease to ballspline shaft.

- b. Lubricate ballspline shaft by applying light film of lubricating oil, Item 208, Appendix D (Figure 14-4-4-2, Sheet 2, Detail C).
- c. Lubricate level wind screw and drive gears with grease, Item 145, Appendix D.
- d. If during cable removal original drum position is lost, perform the following:
 - (1) Rotate drum in downward direction until drum bottoms to left (toward rear of helicopter) and begins moving to right (toward front of helicopter).
 - (2) Just as drum begins moving to right, cable installation hole in drum will align in

center of housing inspection window. Cable should be installed in this position.

CAUTION

To prevent damage to cable, make sure hole in drum is aligned in center of housing inspection window.

- e. Have assistant push hoist cable end through cable guide rollers, cable cutter and traction drive sheave while rotating traction sheave drive shaft with $\frac{7}{16}$ -inch open-end wrench (Figure 14-4-4-2, Sheet 2, Detail D).
- f. Insert cable end into drum retention hole. Through inspection hole, check to make sure cable end is fully bottomed.
- g. Install cable with cable retention screws. **TORQUE RETENTION SCREWS AND DRUM FINGER RETENTION SCREW TO 11 - 13 INCH-POUNDS.**
- h. Remove drill rod from crowder roller retention hole.
- i. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

WARNING

- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of helicopter and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.
- Injury to personnel will occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.

CAUTION

- Backwinding of cable will result if backup control is run in downward position when cable is fully extended.
 - To prevent damage to hoist, at any sign of hoist or cable trouble, stop hoist and investigate problem.
- j. Have assistant observe cable drum and apply firm drag with gloved hand to cable.
 - k. Depress two manual override switches at rear corner of exposed area, using rescue hoist and cargo hook pendant, reel in cable until four wraps are on hoist drum. Release override switches.
 - l. Press BACKUP CONTROL POWER switch and check that NORMAL POWER light goes on, indicating system is in normal power mode. Close switch guard assembly.
 - m. Using lockwire, Item 195, Appendix D, lockwire switch guard to screws.
 - n. Using rescue hoist and cargo hook pendant, slowly reel hoist cable onto cable drum, while assistant applies firm drag on cable with gloved hand. Have assistant make sure cable feeds on drum correctly and does not tangle. Stop reeling when cable end is within three feet of hoist.
 - o. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
 - p. Install rescue hoist hook (PARA 14-4-5).
 - q. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
 - r. Operate rescue hoist pendant and continue to reel in cable until full-in limit switch is contacted.
 - s. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
 - t. Install explosive cartridge (PARA 14-4-2).

- u. Make sure area is clean and free of foreign material.
- v. Install top access cover and outboard fairing using screws and washers.
- w. Perform operational check of rescue hoist system (PARA 14-2-1).

14-4-4.2.3 Adjust Limit Switches.

WARNING

- Improperly adjusted or failed full-in limit switch may damage overload clutch if hoist motor continues to operate when cable hook is in homed position, resulting in possible damage and degradation of load hold capability. If improperly adjusted or failed full-in limit switch is found, perform overload clutch margin validation load check (PARA 1-7-22).
- During course of adjusting full-in limit switch, or in event of improperly adjusted or failed full-in limit switch, overload clutch may be cycled if motor operates with hook in homed position. Excessive cycling of overload clutch when driven by motor can rapidly accumulate overload clutch cycles and degrade overload clutch's margin to hold load. This degraded overload clutch margin could result in reduced hoist overload capacity and possible loss of load event. If overload clutch is repeatedly cycled by hoist motor, rescue hoist should be returned to authorized Goodrich service center for inspection.

CAUTION

Changing type or brand of gear box lubricant will affect performance of overload clutch. Do not change type or brand of gear box lubricant unless, otherwise directed to do so.

- a. Remove screws and washers securing top access panel and outboard fairing (Figure 14-4-4-2, Sheet 2, Detail C).
- b. Remove front and rear cowlings (PARA 14-4-9).
- c. Remove explosive cartridge (PARA 14-4-2).
- d. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- e. In cabin, open switch guard assembly and press BACK UP CONTROL POWER switch on control panel. Switch should light (Figure 14-4-4-1, Sheet 1, Detail A).
- f. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

WARNING

Injury to personnel will occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.

CAUTION

- Tension shall always be applied to cable during rescue hoist operation.
- Extreme care shall be taken not to allow cable to kink, twist, or ball up.
- Care should be taken to keep cable clean for proper rescue hoist operations.
- Backwinding of cable will result if backup control is run in downward position when cable is fully extended.

NOTE

Cams are numbered 1 through 4, left to right as hoist is viewed outboard looking in. During cam setting, if no click is

heard from switch, switch is either too tight against cam or not tight enough. Tighten or loosen adjustment screw as necessary.

- g. Verify there are at least three cable wraps on drum. Set No. 1 cam (emergency full out), to actuate limit switch (S1 and S2) at this cable position. Tighten setscrew on cam (Figure 14-4-4-4).

CAUTION

To prevent damage to hoist, at any sign of hoist or cable trouble, stop rescue hoist and investigate problem.

- h. While pressing override switch (S11 and S12), and using rescue hoist pendant, slowly reel in 0.5 wrap of cable (there should be 3.5 wraps on drum). Set No. 2 cam (normal mode full out) to actuate switch (S9 and S10) at this cable position. Tighten setscrew on cam.

NOTE

Hoist contains 200 usable feet of cable. Initial and final 20 feet of cable is colored orange.

- i. Using pendant, slowly reel in cable until painted portion is just disappearing into bellmouth of hoist. Set No. 3 cam (Deceleration Down) to actuate switch (S7 and S8) at this cable position. Tighten setscrew on cam.
- j. Measure 9 - 11 feet of cable from bottom of carrier assembly of hook. Using pendant, slowly reel in cable, while assistant applies firm drag on cable with gloved hand and makes sure cable is winding properly onto drum. At any sign of hoist or cable trouble, stop hoist and determine problem.
- k. Stop rescue hoist when measured point of cable is at bellmouth point of hoist. Set No. 4 cam (deceleration up), to actuate switch (S5 and S6) at this cable position. Tighten setscrew on cam.

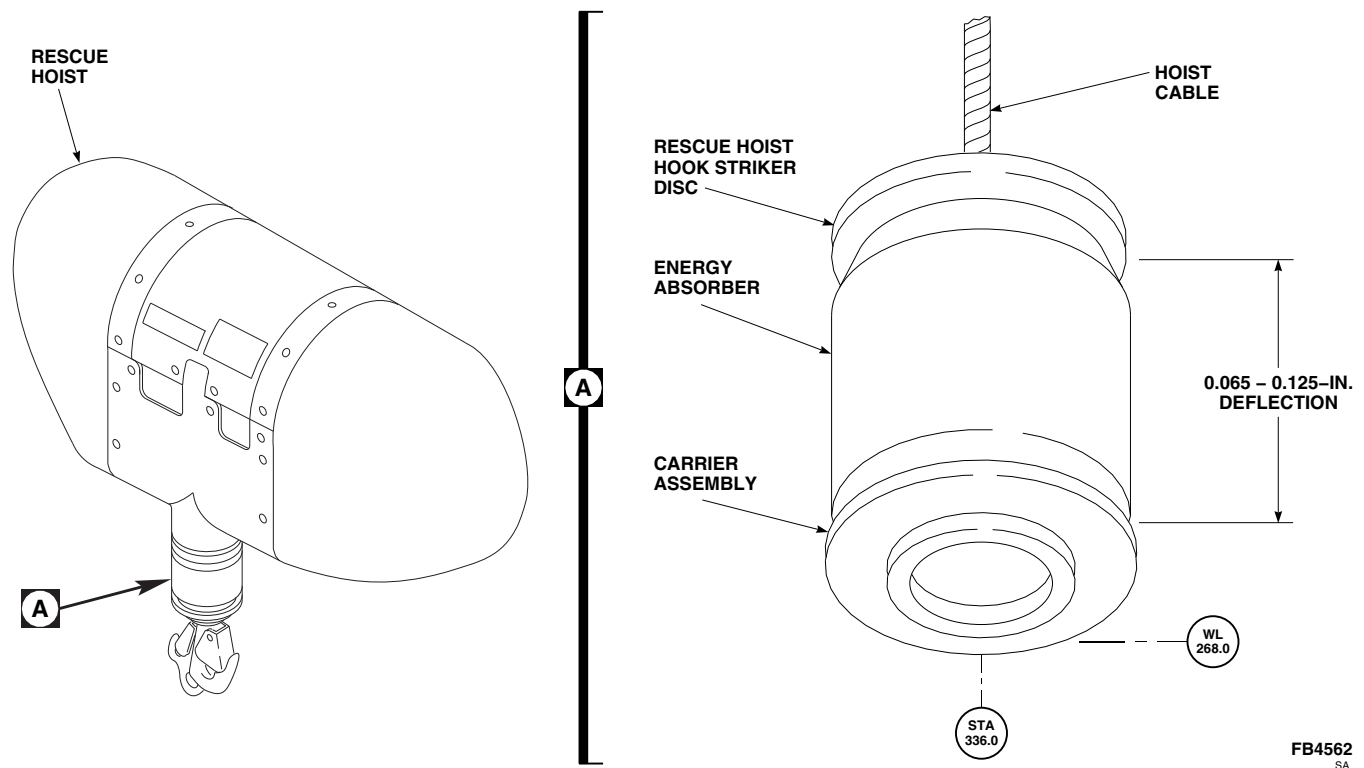
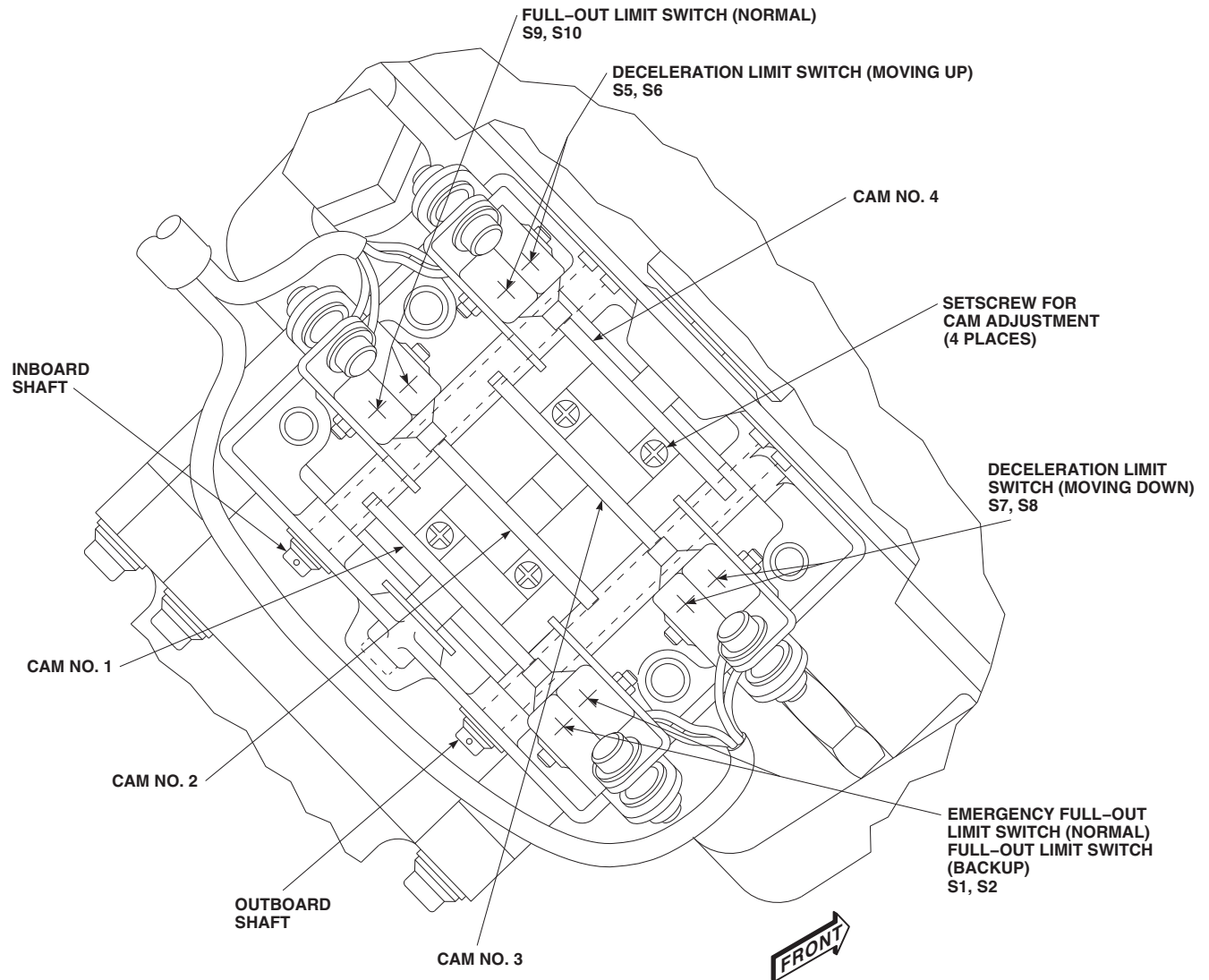


FIGURE 14-4-4-3. ADJUST LIMIT SWITCHES **HOIST, 42315-10**

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- l. Measure distance between bottom of hook striker disc and bottom of carrier assembly (Figure 14-4-4-3, Detail A).
- m. Slowly reel in hoist cable until hook striker disc contacts roller housing and hoist stops.
- n. Using vernier caliper, measure distance between hook striker disc and carrier assembly. **DEFLECTION OF ENERGY ABSORBER SHOULD BE 0.065 - 0.125-INCH.** If deflection is less than limit, perform the following:
 - (1) Reel out hook approximately 1 foot.
 - (2) Loosen full-in limit switch (S3 and S4) bolt, move limit switch up, and tighten bolt.
 - (3) Using pendant, reel in hook slowly until hook stops.
 - (4) Take measurement again. Repeat until deflection is within limits.
- o. Close switch guard assembly.
- p. Using lockwire, Item 195, Appendix D, lockwire switch guard cover to bolts.
- q. Install explosive cartridge (PARA 14-4-2).
- r. Install front and rear cowlings (PARA 14-4-9).
- s. Install access cover on rescue hoist and secure with screws and washers (Figure 14-4-4-2, Sheet 2, Detail C).
- t. Make sure area is clean and free of foreign material.
- u. Perform operational check of rescue hoist system (PARA 14-2-1).



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FIGURE 14-4-4-4. ADJUST UP LIMIT SWITCHES **HOIST, 42315-10**

14-4-5 RESCUE HOIST HOOK **HOIST, BL-27100-71** **HOIST, 42315-10**

.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-5.1	Replace Rescue Hoist Hook	14-4-5-1	14-4-5.2	Replace Rescue Hoist Hook	14-4-5-4
	HOIST, BL-27100-71			HOIST, 42315-10	

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Protective Gloves
- Pin Punch
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Petrolatum, Item 234, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-21
- PARA 14-4-2

14-4-5.1 Replace Rescue Hoist Hook **HOIST, BL-27100-71** .

14-4-5.1.1 Remove.

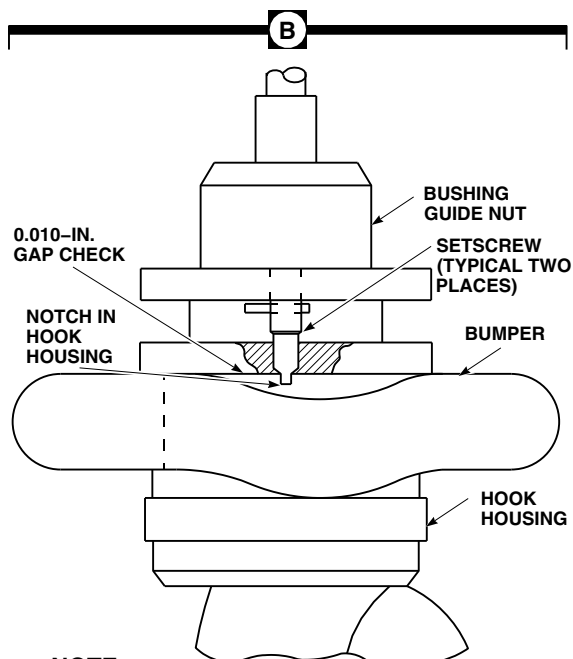
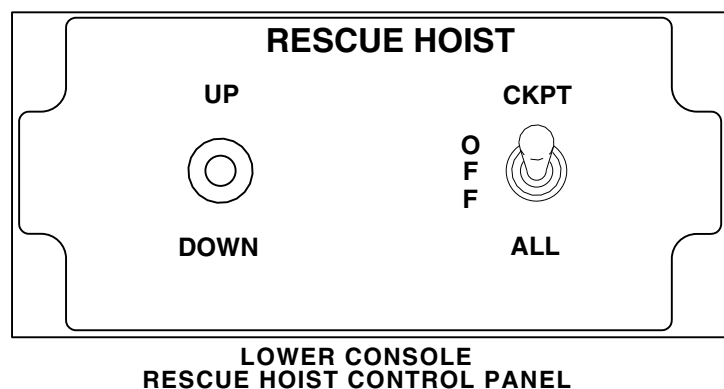
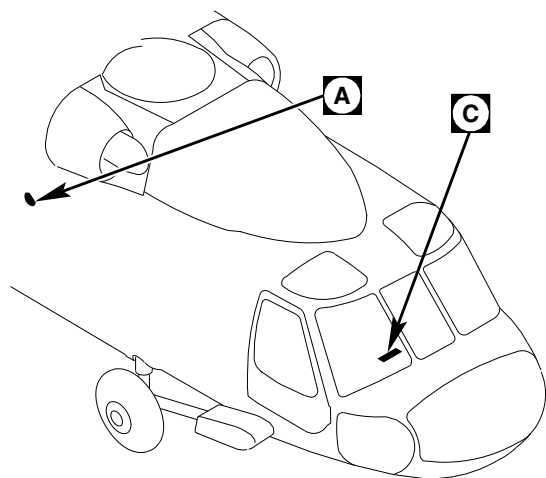
- Remove explosive cartridge (PARA 14-4-2).
- Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- On lower console, set RESCUE HOIST switch to ALL (Figure 14-4-5-1, Detail C).

WARNING

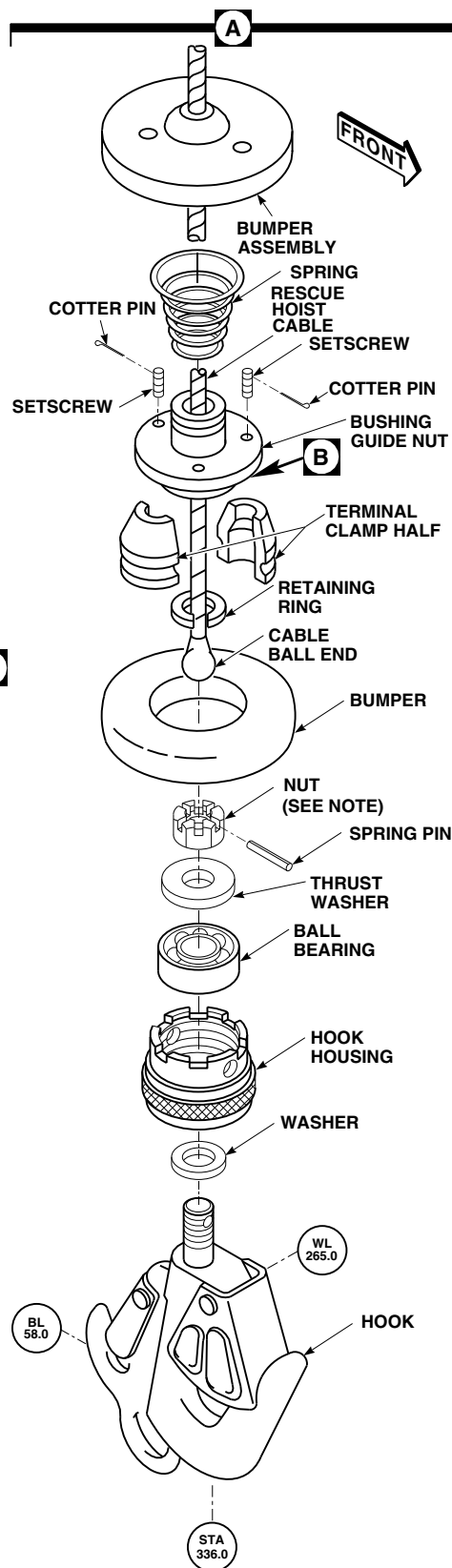
- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of heli-

copter and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.

- Injury to personnel will occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.
- In cabin, operate rescue hoist pendant and lower rescue hoist cable about three feet from hoist.
- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Pull spring off bushing guide nut (Figure 14-4-5-1, Detail A).



NOTE
TIGHTEN NUT ON HOOK UNTIL NUT
BOTTOMS. BACK NUT OFF 1/2 TURN
AND ALIGN WITH HOLE IN HOOK.



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FIGURE 14-4-5-1. REPLACE RESCUE HOIST HOOK **HOIST, BL-27100-71**

- g. Remove cotter pins from setscrews.
- h. Remove setscrews from bushing guide nut.
- i. Unscrew hook housing from bushing guide nut and remove hook and hook housing. Remove bumper from hook housing.
- j. Using pin punch, remove spring pin securing nut to hook and remove nut.
- k. Remove thrust washer, ball bearing, hook housing, and washer from hook.
- l. Slide bushing guide nut up to expose terminal clamp halves. Remove retaining ring and terminal clamp halves. Slide bushing guide nut and spring off hoist cable.

14-4-5.1.2 Inspect.

- a. Check all components for wear or damage. **NONE ALLOWED.** If worn or damaged, replace components as necessary.
- b. Inspect ball bearing for looseness, roughness, or stiffness caused by wear, deterioration of lubricants, or corrosion. **NONE ALLOWED.** If necessary, replace ball bearing.
- c. Inspect spring pin for corrosion, cracks, and correct part number (PARA 1-7-21).

14-4-5.1.3 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Push bumper over cable ball end onto hoist cable (Figure 14-4-5-1, Detail A).

NOTE

Terminal clamps must be replaced as a matched set.

- c. Slide spring and bushing guide nut over cable ball end. Position terminal clamp halves over

cable and secure with retaining ring. Slide bushing guide nut down over terminal clamp halves.

- d. Place washer, hook housing, ball bearing, and thrust washer on hook and secure with nut.
- e. **TIGHTEN NUT ON HOOK UNTIL NUT BOTTOMS.** Back nut off 1/2 turn and align it with hole in hook.

NOTE

Spring pin is to be replaced each time rescue hoist hook is reassembled.

- f. Check hook assembly for free rotation. **NO BINDING ALLOWED.** If binding is felt, proceed as follows:
 - (1) Loosen nut until binding stops.
 - (2) Align nut with hole in hook.
 - (3) Install new spring pin in hole in hook.
- g. Install bumper on hook housing.
- h. Screw hook tightly onto bushing guide nut.

CAUTION

Make sure hook housing is screwed in all the way and, after contact is made, back off to nearest castellation only. Setscrew must bottom into castellation of hook housing to lock.

- i. Align setscrew holes with notches in hook housing (Figure 14-4-5-1, Detail A and Detail B). Install setscrews in bushing guide nut. **TIGHTEN SETSCREWS UNTIL THEY CONTACT AND LOCK INTO HOOK HOUSING.** Install cotter pins through slot in setscrews (Figure 14-4-5-1, Detail A).

WARNING

Injury to personnel and damage to equipment will result if gap between faces of hook housing and bushing guide nut exceeds limits or hook housing can be backed off bushing guide nut. Hook may separate from cable during operation. Make sure gap is within limit and that hook housing cannot be backed off bushing guide nut.

- j. Check gap between faces of hook housing and bushing guide nut as follows:
 - (1) Pull back bumper and check between faces of hook housing and bushing guide nut (Figure 14-4-5-1, Detail B).
 - (2) **IF GAP IS GREATER THAN 0.010-INCH OR HOOK HOUSING CAN BE BACKED OFF BUSHING GUIDE NUT, CHECK SETSCREWS FOR PROPER INSTALLATION.**
- k. Push spring on bushing guide nut (Figure 14-4-5-1, Detail A).
- l. Check for component security and make sure area is clean and free of foreign material.
- m. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

WARNING

- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of helicopter and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.
- Injury to personnel will occur if protective gloves are not worn when reeling in

cable. Use protective gloves when reeling in cable.

- n. On lower console, place RESCUE HOIST switch to ALL (Figure 14-4-5-1, Detail C). In cabin, operate rescue hoist/cargo pendant, to reel rescue hoist cable back into hoist.
- o. Install explosive cartridge (PARA 14-4-2).

14-4-5.2 Replace Rescue Hoist

Hook **HOIST, 42315-10** .

14-4-5.2.1 Remove.

- a. Remove explosive cartridge (PARA 14-4-2).
- b. Turn on hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

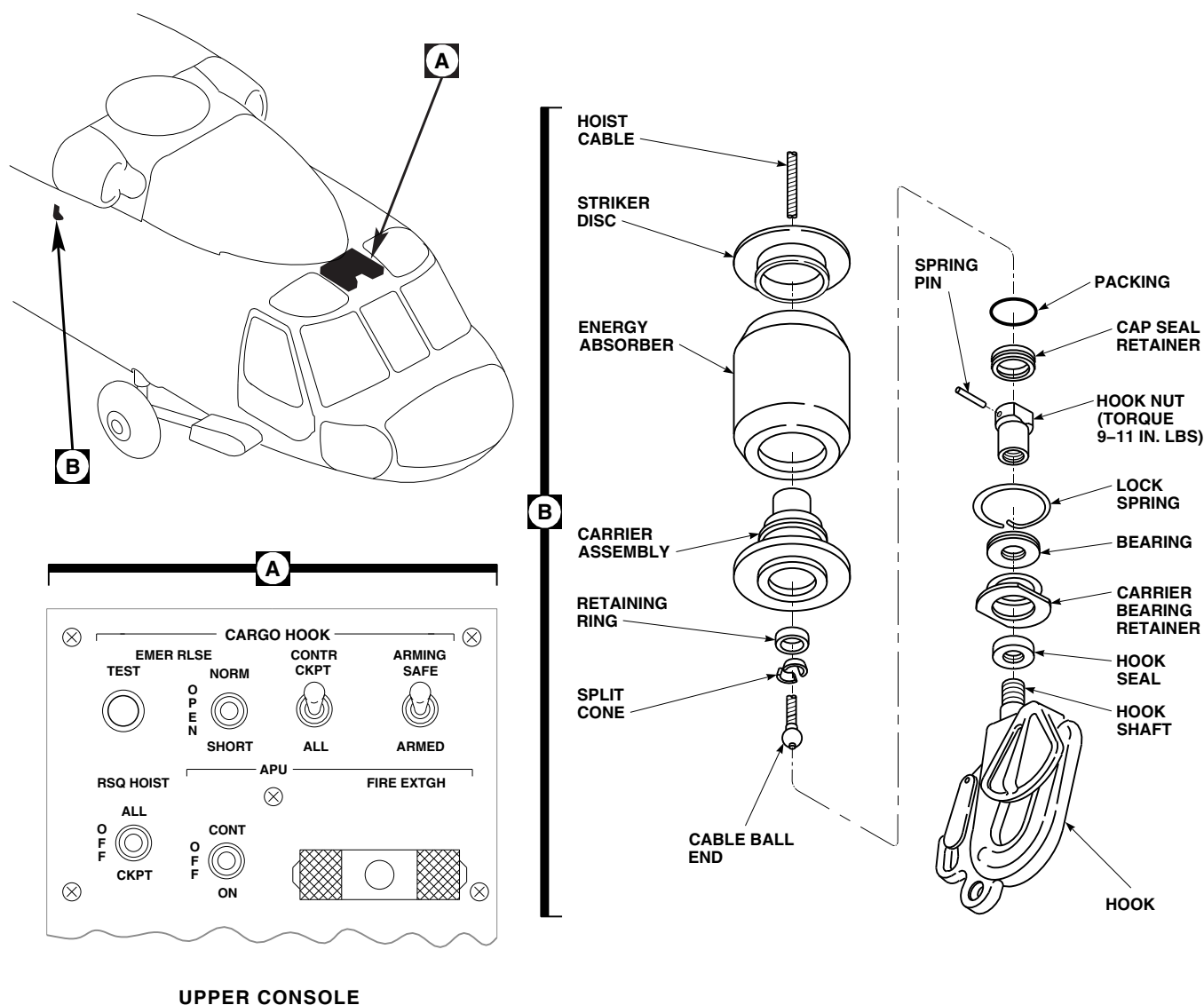
WARNING

- To prevent injury to personnel or damage to equipment, rescue hoist cable must be kept clear of all parts of helicopter and free from other external obstacles. Cable abrasion during hoist operation can lead to cable failure. If cable contact or snagging occurs, inspect cable for damage immediately.
- Injury to personnel will occur if protective gloves are not worn when reeling cable in or out. Use protective gloves when reeling cable in or out.

CAUTION

When reeling out hoist cable from any control input, cable must not be restrained from motion. Loose cable wraps and hoist damage will result.

- c. If questionable operation is suspected, perform the following:
 - (1) Reel out hoist cable to full-out limit.



UPPER CONSOLE

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FIGURE 14-4-5-2. REPLACE RESCUE HOIST HOOK **HOIST, 42315-10**

- (2) Give firm tug on cable to set remaining wraps on drum.
- (3) Reel cable onto drum while applying firm drag on cable with gloved hand.
- d. On upper console, set RSQ HOIST switch to ALL (Figure 14-4-5-2, Detail A). In cabin, operate rescue hoist/cargo hook pendant and lower rescue hoist cable about one foot from hoist.
- e. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- f. Remove lock spring from carrier assembly (Figure 14-4-5-2, Detail B).

WARNING

Injury to personnel will occur if serialized matched carrier assembly, consisting of carrier bearing retainer and carrier assembly, is not saved and installed as a matched set. Save carrier bearing retainer and carrier assembly as a matched set for installation.

NOTE

Make sure to note how holes are referenced on carrier assembly.

- g. Unscrew carrier bearing retainer/hook assembly from carrier. Note how holes are referenced on carrier assembly.
- h. Push cap seal retainer out of carrier assembly and remove packing from retainer. Discard packing.

WARNING

Injury to personnel will occur if split cones are not saved as matched set. Save split cones as matched set for installation.

- i. Remove and retain split cones and retaining ring from cable ball end of hoist cable.
- j. Remove carrier assembly from energy absorber.
- k. Remove spring pin from hook nut and remove nut, bearing, hook, and hook seal from carrier bearing retainer.
- l. Remove striker disc from energy absorber.

14-4-5.2.2 Inspect.

- a. Check all components for wear or damage. **NONE ALLOWED.** If worn or damaged, replace components as necessary.
- b. Inspect spring pin for corrosion, cracks, and correct part number (PARA 1-7-21).

14-4-5.2.3 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

WARNING

Injury to personnel will occur if serialized matched carrier assembly, consist-

ing of carrier bearing retainer and carrier assembly, is not used. Install serialized matched carrier bearing retainer and carrier assembly.

- b. Install hook seal on carrier bearing retainer and install retainer on hook (Figure 14-4-5-2, Detail B).

NOTE

Spring pin is to be replaced each time hoist hook is reassembled.

- c. Place bearing onto hook shaft and install with hook nut. **TORQUE NUT TO 9 - 11 INCH-POUNDS WHILE ALIGNING PIN HOLES IN NUT WITH HOLES ON HOOK SHAFT.** Install new spring pin.
- d. Install striker disc into top of energy absorber.

WARNING

Injury to personnel will occur if serialized matched carrier assembly, consisting of carrier bearing retainer and carrier assembly, is not used. Install serialized matched carrier bearing retainer and carrier assembly.

- e. Insert carrier assembly into bottom of energy absorber.
- f. Insert hoist cable through striker disc, energy absorber, and carrier assembly until it extends out bottom of carrier assembly.
- g. Place retaining ring on cable with bevel facing cable ball end.

WARNING

Injury to personnel will occur if matched split cones are not used. Use matched split cones for installation.

- h. Coat cable ball end with petrolatum, Item 234, Appendix D. Place matched split cones on ball

with tapered end resting in bevel of retaining ring.

- i. Pull energy absorber down to cable ball end and split cones, making sure cones and retaining ring rest correctly in carrier.

WARNING

Injury to personnel will occur if serialized matched carrier assembly, consisting of carrier bearing retainer and carrier assembly, is not used. Install serialized matched carrier bearing retainer and carrier assembly.

- j. Place packing onto cap seal retainer. Insert retainer into carrier assembly with crown of retainer facing cable ball end.

NOTE

There shall be no measurable gap at interface of carrier assembly and carrier bearing retainer after assembly.

- k. Screw carrier bearing retainer/hook assembly into carrier assembly. Align hole in carrier assembly as noted in step g. of removal procedure.

NOTE

Do not overexpand lock spring during installation.

- l. Insert lock spring with one tang in smaller hole, then spiral spring slightly, feeding it up to and around groove and then seat other tang in larger hole. Verify lock spring is securely installed.
- m. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- n. On upper console, set RSQ HOIST switch to ALL (Figure 14-4-5-2, Detail A). In cabin, operate rescue hoist/cargo hook pendant and reel cable back into hoist.
- o. Install explosive cartridge (PARA 14-4-2).
- p. Make sure area is free and clear of foreign material.

14-4-6 HYDRAULIC MOTOR **HOIST, BL-27100-71** **HOIST, 42315-10** .

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-6.1	Replace Hydraulic Motor HOIST, BL-27100-71	14-4-6-1	14-4-6.2	Replace Hydraulic Motor HOIST, 42315-10	14-4-6-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 75, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 14-4-13
- PARA 14-4-15

14-4-6.1 Replace Hydraulic Motor **HOIST, BL-27100-71** .

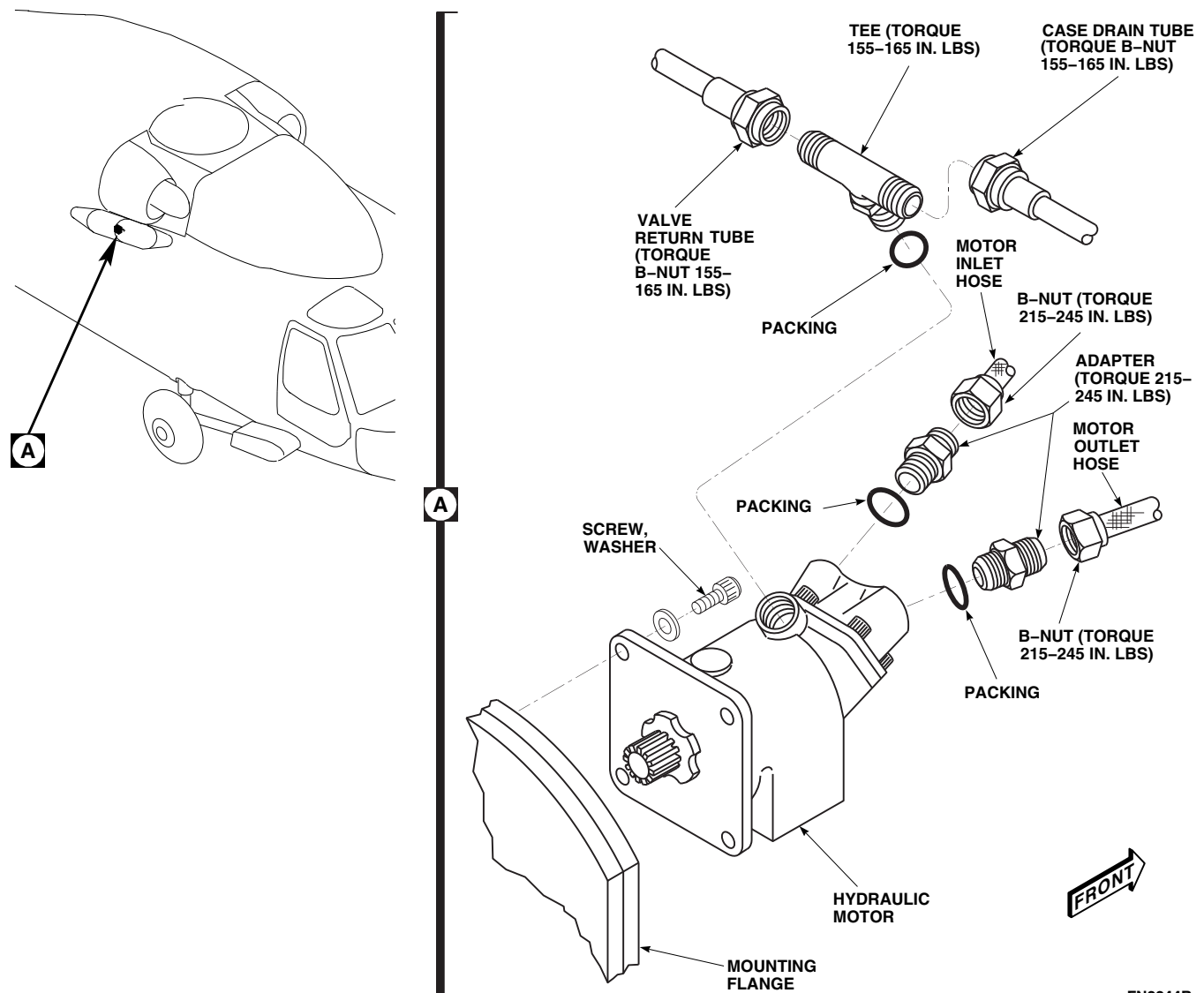
14-4-6.1.1. Remove.

- Remove hydraulic control assembly (PARA 14-4-15).
- Disconnect motor outlet hose from adapter (Figure 14-4-6-1, Detail A).
- Disconnect motor inlet hose from adapter.
- Disconnect valve return tube from tee.
- Disconnect case drain tube from tee.
- Remove adapters and tee from hydraulic motor. Remove and discard packings.
- Install protective caps and plugs.

- Remove screws and washers and carefully remove hydraulic motor from mounting flange.

14-4-6.1.2. Install.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Lubricate threads of screws with antiseize compound, Item 75, Appendix D.
- Position hydraulic motor, serviced with hydraulic fluid, Item 155, Appendix D, on mounting flange, and secure with screws and washers (Figure 14-4-6-1, Detail A).
- Remove protective caps and plugs.
- Lubricate packings with hydraulic fluid, Item 155, Appendix D.
- Install packings on adapters.

FN2244B
SAFIGURE 14-4-6-1. REPLACE HYDRAULIC MOTOR **HOIST, BL-27100-71**

- g. Install adapters in hydraulic motor ports. **TORQUE ADAPTERS TO 215 - 245 INCH-POUNDS.**
- h. Install packing on tee.
- i. Install tee into hydraulic motor. **TORQUE TEE TO 155 - 165 INCH-POUNDS.**
- j. Connect valve return and case drain tubes to tee. **TORQUE B-NUTS TO 155 - 165 INCH-POUNDS.**
- k. Connect motor inlet and outlet hoses to adapters. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**

- l. Install hydraulic control assembly (PARA 14-4-15).

14-4-6.2 Replace Hydraulic Motor **HOIST, 42315-10** .

14-4-6.2.1. Remove.

- a. Remove hoist valve module assembly (PARA 14-4-13).
- b. Remove pressure and return lines from unions (Figure 14-4-6-2, Detail A).
- c. Remove motor case drain line from elbow on top of hydraulic motor.
- d. Remove unions and packings from hydraulic motor. Discard packings.
- e. Install protective caps and plugs to components as they are removed.
- f. Remove screws and washers securing hydraulic motor to mounting flange. Remove hydraulic motor.
- g. Remove elbow and packing from top of hydraulic motor. Discard packing.

14-4-6.2.2. Install.

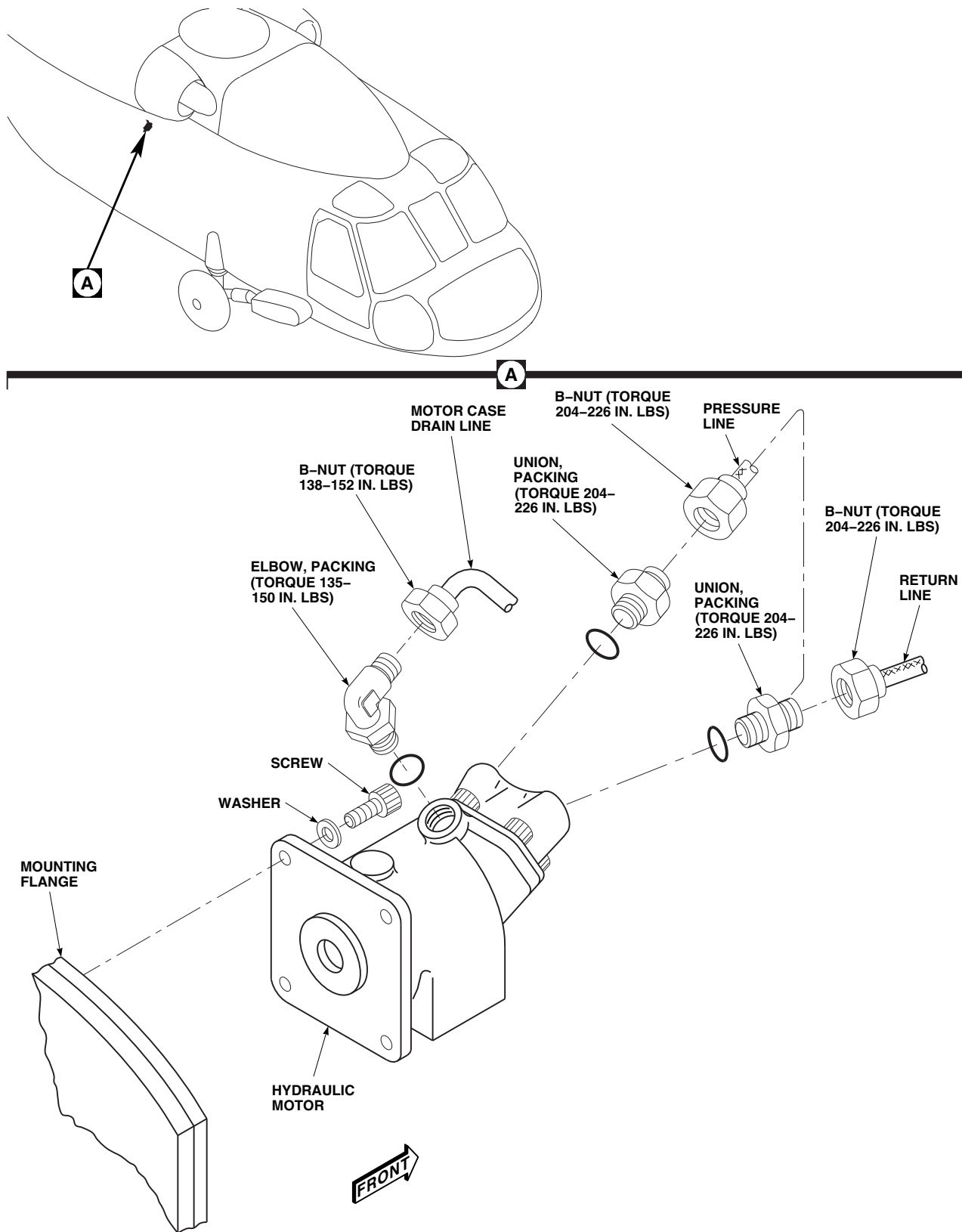
- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Lubricate packings with hydraulic fluid, Item 155, Appendix D.

- c. Install packings on unions and packing on elbow.
- d. Remove protective caps and plugs from components prior to installing them.
- e. Install elbow in hydraulic port on top of hydraulic motor (Figure 14-4-6-2, Detail A).

NOTE

When positioning elbow to connect motor case drain line make sure torque does not fall below 135 inch-pounds or exceed 150 inch-pounds.

- f. **TORQUE ELBOW TO 135 - 150 INCH-POUNDS.** Position elbow and connect motor case drain line to elbow. Do not torque motor case drain line at this time.
- g. Position hydraulic motor, serviced with hydraulic fluid, Item 155, Appendix D, on mounting flange and install with screws and washers.
- h. Install unions in hydraulic motor pressure and return ports. **TORQUE UNIONS TO 204 - 226 INCH-POUNDS.**
- i. **TORQUE MOTOR CASE DRAIN LINE B-NUT TO 138 - 152 INCH-POUNDS.**
- j. Connect motor pressure and return lines. **TORQUE B-NUTS TO 204 - 226 INCH-POUNDS.**
- k. Install hoist valve module assembly (PARA 14-4-13).

FB2067A
SA**FIGURE 14-4-6-2. REPLACE HYDRAULIC MOTOR** **HOIST, 42315-10**

14-4-7 HEAT EXCHANGER **HOIST, BL-27100-71** **HOIST, 42315-10** .

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-7.1	Replace Heat Exchanger	14-4-7-1	14-4-7.2	Replace Heat Exchanger	14-4-7-4
		HOIST, BL-27100-71			HOIST, 42315-10

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Allen Key
- Container, 1-Quart
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Cap and Plug

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-4-65
- PARA 14-2-1
- PARA 14-4-6
- PARA 14-4-9
- PARA 14-4-13
- PARA 14-4-20

14-4-7.1 Replace Heat Exchanger **HOIST, BL-27100-71** .

14-4-7.1.1 Remove.

- Remove rear cowling (PARA 14-4-9).
- Disconnect electrical connector, P5, from heat exchanger (Figure 14-4-7-1, Sheet 1, Detail B).
- Install protective cap and plug on receptacle and electrical connector.
- Position 1-quart container under hoist to catch hydraulic fluid.

- Disconnect main hydraulic input line from input block and rear side of front wall (Figure 14-4-7-1, Sheet 2, Detail C).
- Disconnect main inlet tube from main inlet elbow (Figure 14-4-7-1, Sheet 1, Detail B).
- Disconnect return tube from return tube elbow.
- Remove cuff fairing (PARA 14-4-20).
- Loosen opposite end of return tube and swing tube free from heat exchanger.
- Remove main inlet and return tube elbows from manifold. Discard packings.

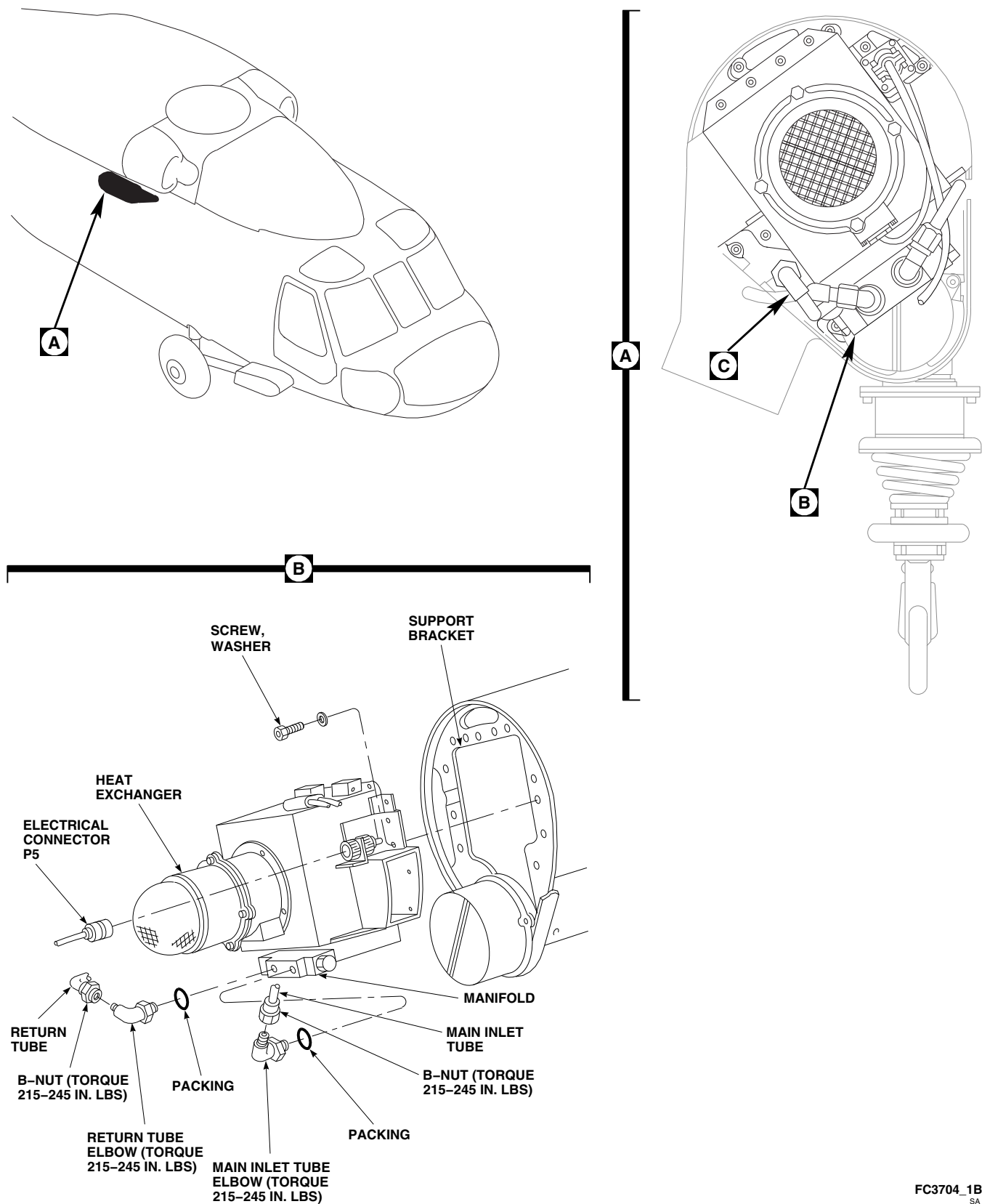
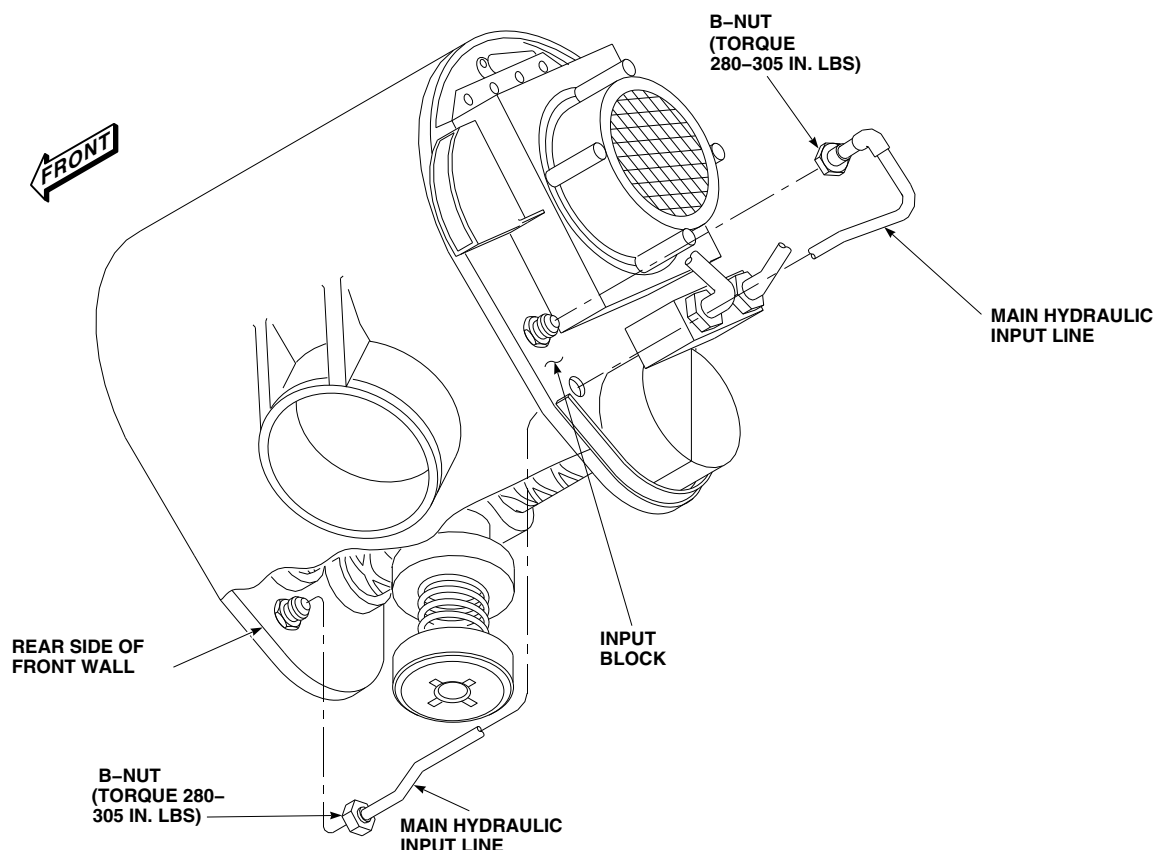


FIGURE 14-4-7-1. REPLACE HEAT EXCHANGER (SHEET 1 OF 2) HOIST, BL-27100-71

14-4-7-2 Change 11

C



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FIGURE 14-4-7-1. REPLACE HEAT EXCHANGER (SHEET 2 OF 2) HOIST, BL-27100-71

CAUTION

To keep limit switch synchronization, do not remove limit switch housing while removing screws.

NOTE

A slight bend to Allen key may be required to gain access to lower screw.

- k. Remove screws and washers and carefully remove heat exchanger from support bracket.

14-4-7.1.2 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Position heat exchanger on support bracket and secure with screws and washers (Figure 14-4-7-1, Sheet 1, Detail B).
- c. Lubricate packings with hydraulic fluid, Item 155, Appendix D.
- d. Install packings on elbows.
- e. Install main inlet and return tube elbows in manifold.

- f. Position elbow and connect heat exchanger return tube. **TORQUE B-NUT AND ELBOW TO 215 - 245 INCH-POUNDS.**
- g. Position main inlet tube elbow and connect main inlet tube. **TORQUE B-NUT AND ELBOW TO 215 - 245 INCH-POUNDS.**
- h. Connect main hydraulic input line to input block and rear side of front wall (Figure 14-4-7-1, Sheet 2, Detail C). **TORQUE B-NUTS TO 280 - 305 INCH-POUNDS.**
- i. Remove protective cap and plug from electrical connector and receptacle.
- j. Inspect and treat electrical receptacle and connector (PARA 1-7-20).
- k. Connect electrical connector, P5, to heat exchanger.
- l. Bleed hydraulic system (PARA 7-4-65).
- m. Make sure area is clean and free of foreign material.
- n. Perform operational check of rescue hoist system (PARA 14-2-1).
- o. Install rear cowling (PARA 14-4-9).
- p. Install cuff fairing (PARA 14-4-20).

14-4-7.2 Replace Heat Exchanger **HOIST, 42315-10** .

14-4-7.2.1 Remove.

- a. Remove hoist valve module assembly (PARA 14-4-13).
- b. Remove hydraulic motor (PARA 14-4-6).
- c. Disconnect input line and return line from reducer and elbow on heat exchanger (Figure 14-4-7-2, Detail A).

NOTE

- Do not remove inner cup.

- Packing between heat exchanger and housing will offer resistance when trying to remove heat exchanger.
- d. Slide heat exchanger off adapter. Remove spacer and retaining ring.
- e. Remove packing from housing and discard.
- f. Remove elbow and packing from heat exchanger. Discard packing.
- g. Remove reducer and packing from heat exchanger. Discard packing.

14-4-7.2.2 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Install packings on elbow and reducer.
- c. Install reducer, with packing into heat exchanger port (Figure 14-4-7-2, Detail A). **TORQUE REDUCER TO 204 - 226 INCH-POUNDS.**
- d. Install elbow, with packing into heat exchanger port. **TORQUE ELBOW TO 356 - 394 INCH-POUNDS.**
- e. Coat heat exchanger/housing packing with petrolatum, Item 234, Appendix D, and install into housing.



Damage to packing will occur if heat exchanger is not installed carefully. Use care when installing heat exchanger into housing.

- f. Install retaining ring and spacer into adapter. Slide heat exchanger onto adapter.
- g. Fill heat exchanger with hydraulic fluid, Item 155, Appendix D.
- h. Connect return line to reducer. **TORQUE B-NUTS TO 204 - 226 INCH-POUNDS.**

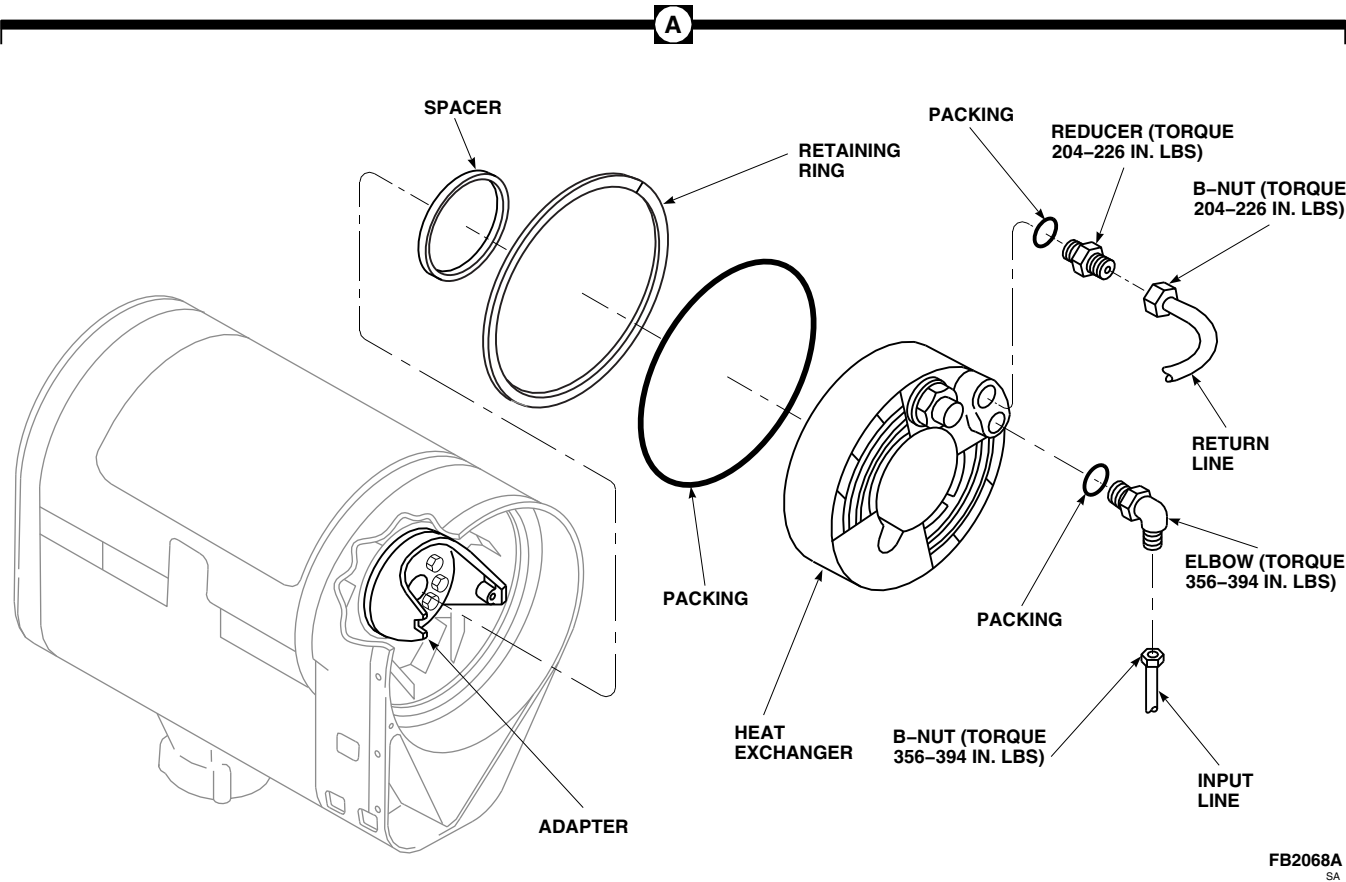
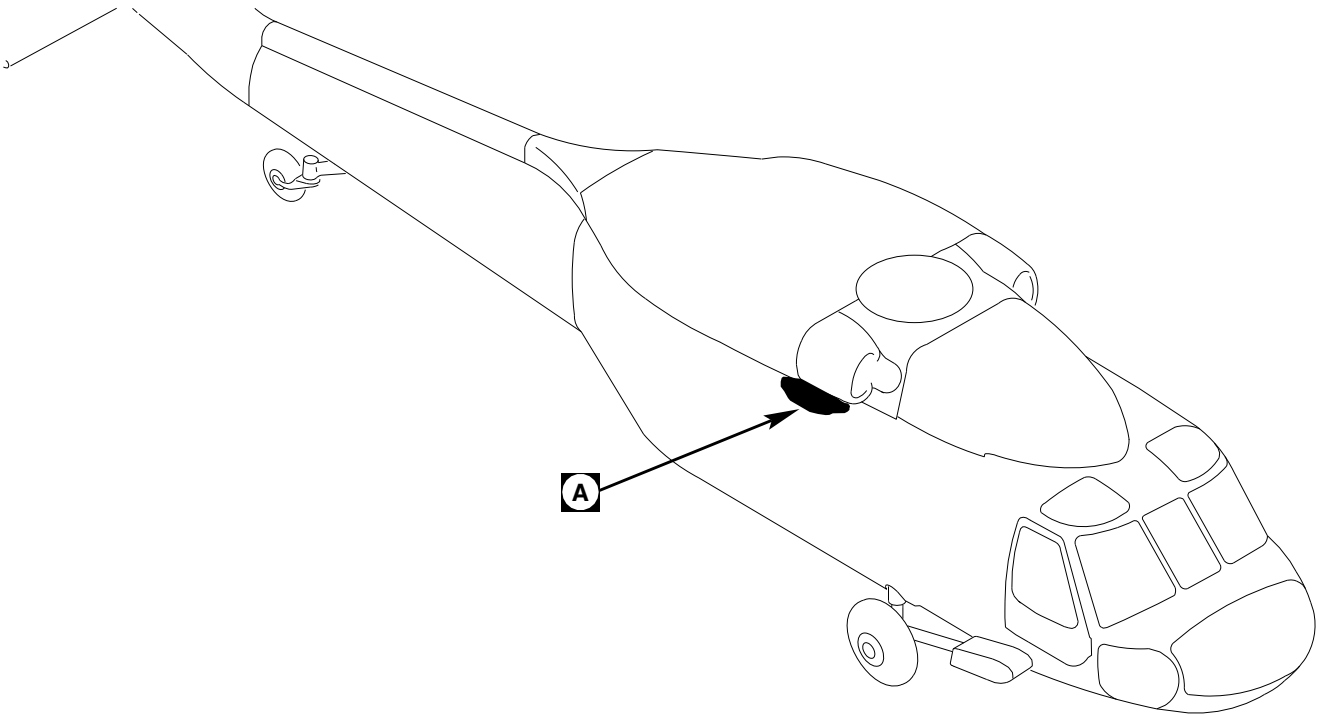


FIGURE 14-4-7-2. REPLACE HEAT EXCHANGER **HOIST, 42315-10**

- i. Connect input line to elbow. **TORQUE B-NUTS TO 356 - 394 INCH-POUNDS.**
- j. Install hydraulic motor (PARA 14-4-6).
- k. Install hoist valve module assembly (PARA 14-4-13).

14-4-8 ELAPSED TIME INDICATOR

HOIST, BL-27100-71

HOIST, 42315-10

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-8.1	Replace Elapsed Time Indicator	14-4-8-1	14-4-8.2	Replace Elapsed Time Indicator	14-4-8-2
		HOIST, BL-27100-71			HOIST, 42315-10

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-15
- PARA 1-6-16

- PARA 1-7-20
- PARA 14-2-1
- PARA 14-4-9

14-4-8.1 Replace Elapsed Time Indicator

HOIST, BL-27100-71

14-4-8.1.1 Remove.

- Remove front cowling (PARA 14-4-9).
- Disconnect electrical connector, P8, from electrical valve control (Figure 14-4-8-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws, washers, and elapsed time indicator with bracket.

14-4-8.1.2 Install.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

- Install elapsed time indicator and bracket with screws and washers.
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P8, to electrical valve control (Figure 14-4-8-1, Detail A).
- Make sure area is clean and free of foreign material.
- Perform operational check of rescue hoist (PARA 14-2-1).
- Install front cowling (PARA 14-4-9).

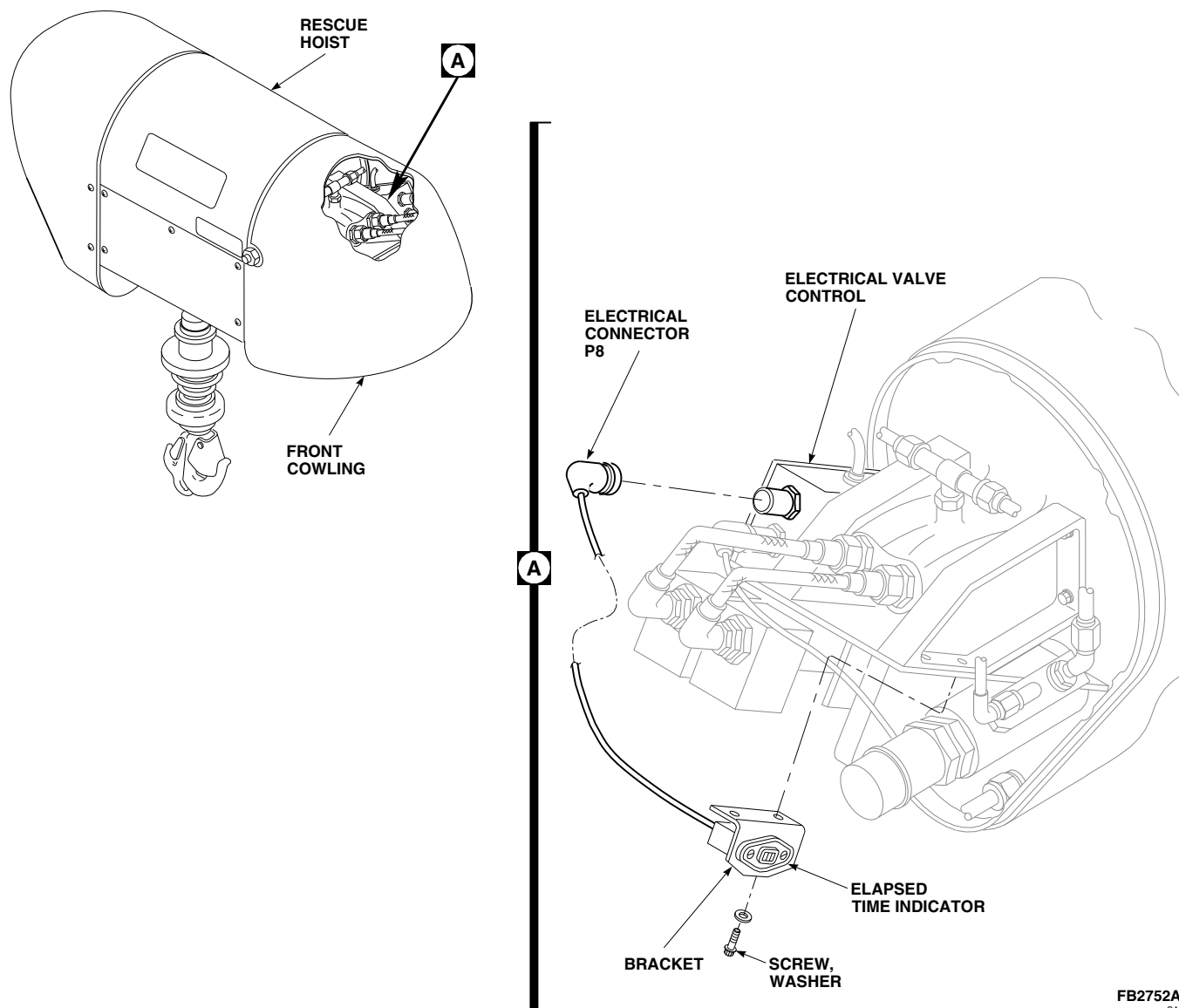
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FIGURE 14-4-8-1. REPLACE ELAPSED TIME INDICATOR **HOIST, BL-27100-71**

14-4-8.2 Replace Elapsed Time Indicator **HOIST, 42315-10**

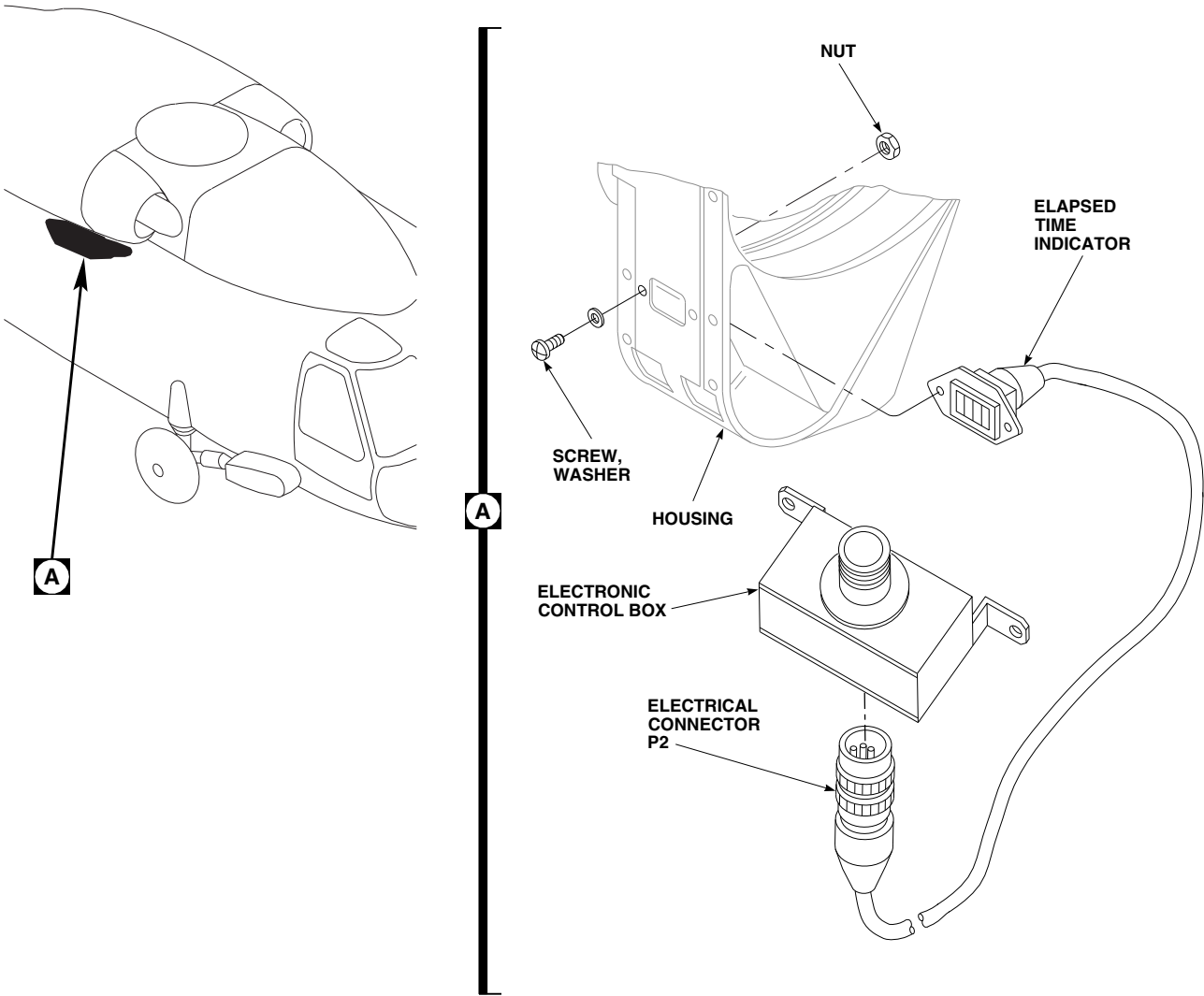
14-4-8.2.1 Remove.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Remove front cowling (PARA 14-4-9).
- Disconnect electrical connector, P2, from electronic control box (Figure 14-4-8-2, Detail A).
- Install protective caps and plugs on electrical connectors.

- Remove screws, washers, and nuts holding elapsed time indicator to housing. Remove elapsed time indicator.

14-4-8.2.2 Install.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Position elapsed time indicator in housing and install with screws, washers, and nuts (Figure 14-4-8-2, Detail A).
- Remove protective caps and plugs from electrical connectors.



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FIGURE 14-4-8-2. REPLACE ELAPSED TIME INDICATOR **HOIST, 42315-10**

- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect electrical connector, P2, to electronic control box.
- f. Make sure area is clean and free of foreign material.
- g. Perform operational check of rescue hoist (PARA 14-2-1).
- h. Install front cowling (PARA 14-4-9).

14-4-9 FRONT AND REAR COWLINGS **HOIST, BL-27100-71**
HOIST, 42315-10 .

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-9.1	Replace Front and Rear Cowlings HOIST, BL-27100-71	14-4-9-1	14-4-9.2	Replace Front and Rear Cowlings HOIST, 42315-10	14-4-9-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 14-4-20

14-4-9.1 Replace Front and Rear Cowlings **HOIST, BL-27100-71** .

14-4-9.1.1. Remove.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Remove screws and washers holding front cowling to rescue hoist and remove cowling (Figure 14-4-9-1, Detail A).
- Remove support arm fairings and cuff fairing (PARA 14-4-20).
- Remove screws and washers holding rear cowling to rescue hoist, and remove cowling.

14-4-9.1.2. Install.

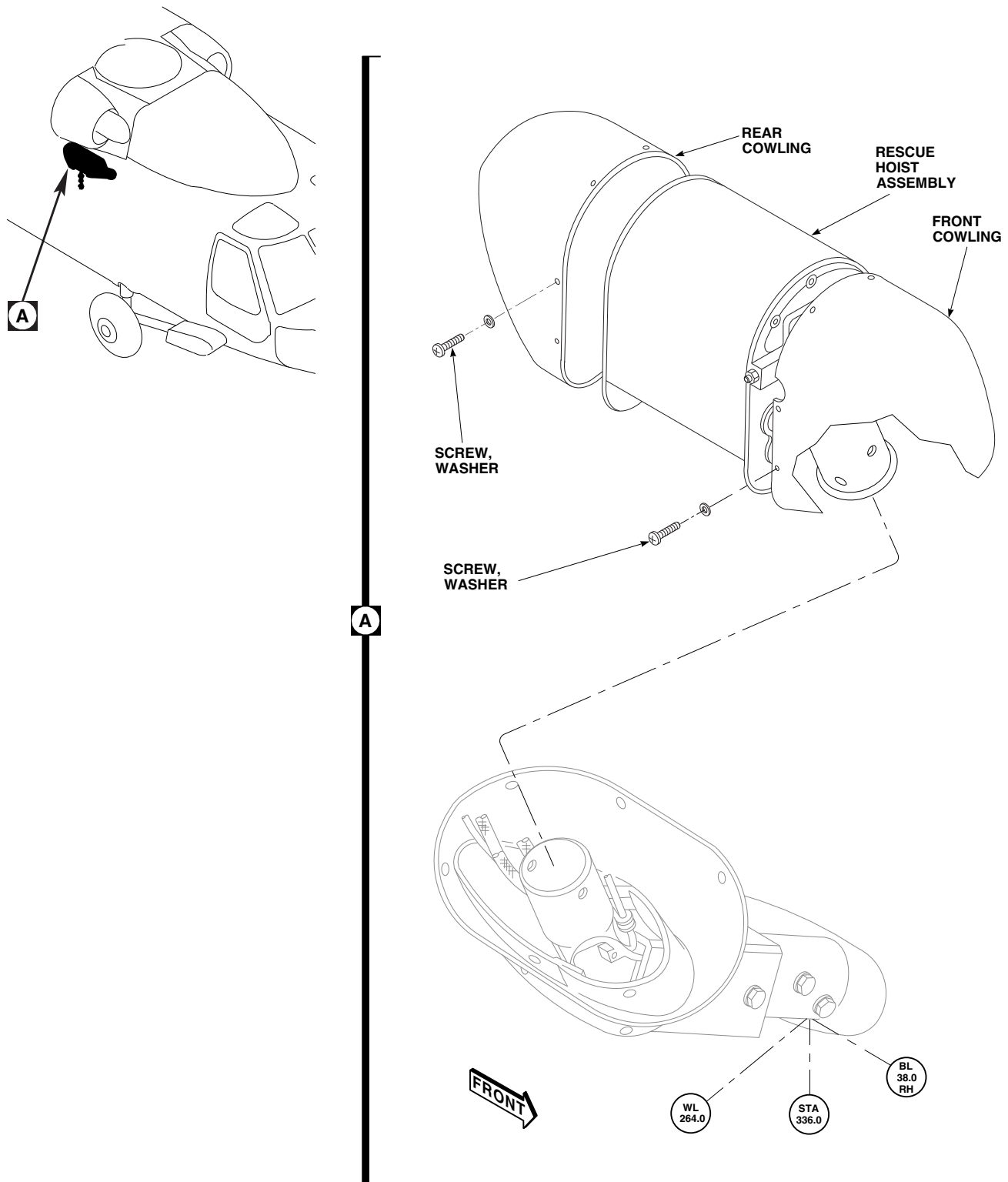
- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Make sure area is clean and free of foreign material before installing fairings and cowlings.

- Place rear cowling in position on rescue hoist and secure with screws and washers (Figure 14-4-9-1, Detail A). Lockwire, Item 197, Appendix D, screws.
- Install support arm fairings and cuff fairing (PARA 14-4-20).
- Place front cowling in position on rescue hoist and secure with screws and washers. Lockwire, Item 197, Appendix D, screws.

14-4-9.2 Replace Front and Rear Cowlings **HOIST, 42315-10** .

14-4-9.2.1. Remove.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Remove screws and washers holding front cowling to rescue hoist and remove cowling (Figure 14-4-9-2, Detail A).
- Remove support arm fairings and cuff fairing (PARA 14-4-20).



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FIGURE 14-4-9-1. REPLACE FRONT AND REAR COWLINGS HOIST, BL-27100-71

14-4-9-2

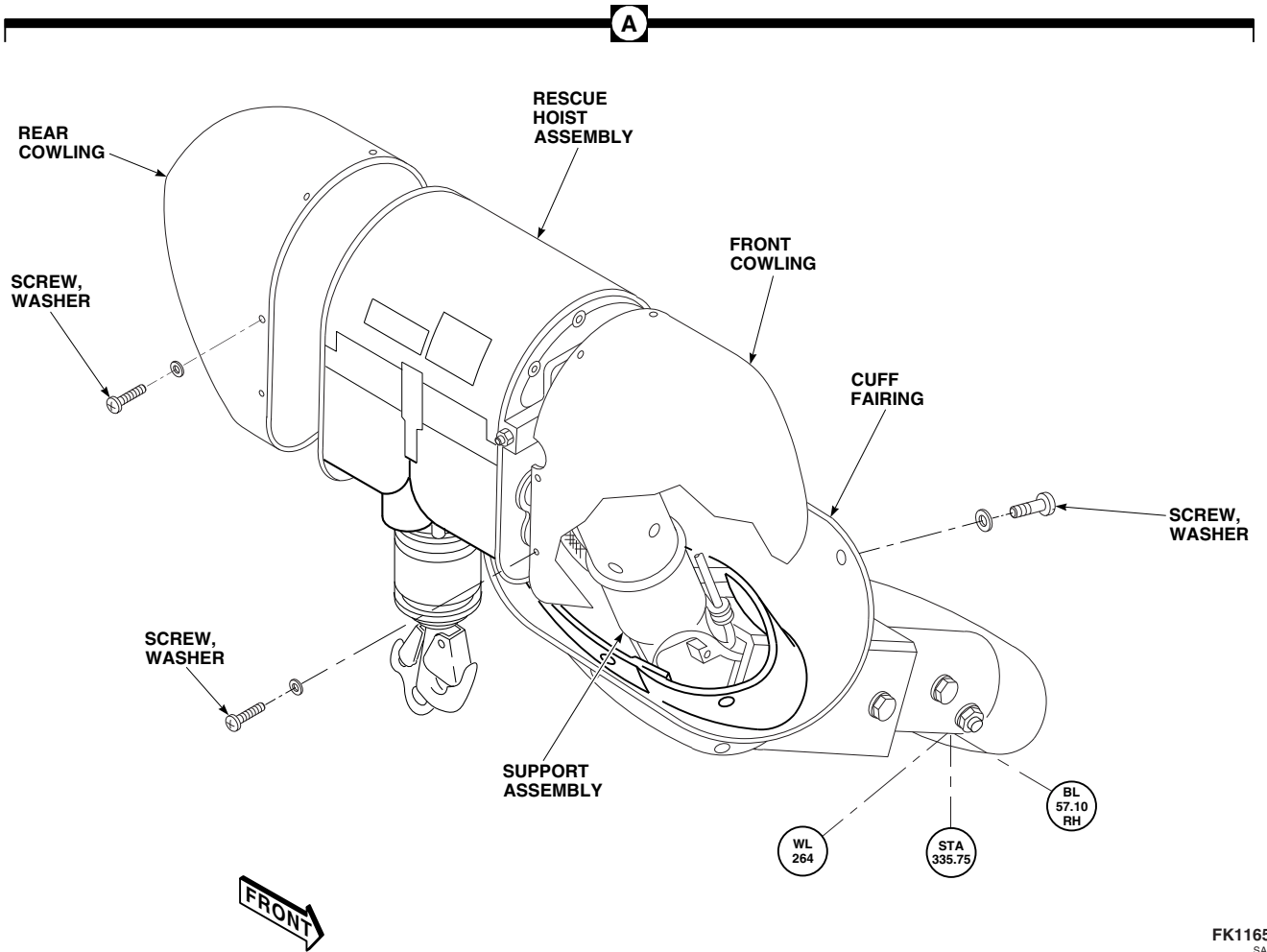
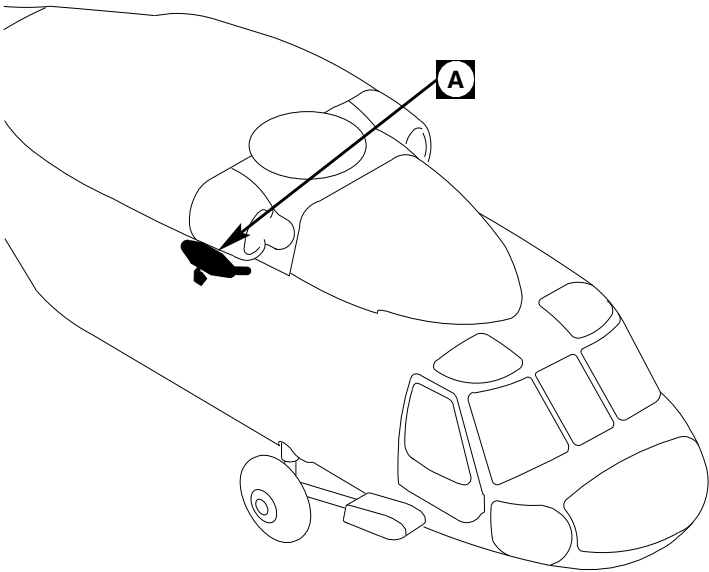


FIGURE 14-4-9-2. REPLACE FRONT AND REAR COWLINGS **HOIST, 42315-10**

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- d. Remove screws and washers holding rear cowling to rescue hoist, and remove cowling.

14-4-9.2.2. Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Make sure area is clean and free of foreign material before installing fairings and cowlings.
- c. Place rear cowling in position on rescue hoist and secure with screws and washers (Figure 14-4-9-2, Detail A). Lockwire, Item 197, Appendix D, screws.
- d. Install support arm fairings and cuff fairing (PARA 14-4-20).
- e. Place front cowling in position on rescue hoist and secure with screws and washers. Lockwire, Item 197, Appendix D, screws.

14-4-10 CONTROL VALVE **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-10.1 Replace Control Valve	14-4-10-1

INITIAL SETUP

- PARA 14-4-15

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-6-15
- PARA 1-6-16

14-4-10.1 Replace Control Valve.

14-4-10.1.2 Install.

14-4-10.1.1 Remove.

- Remove hydraulic control assembly (PARA 14-4-15).
- Remove screws, lockwashers, and washers securing control valve to mounting plate. Remove control valve (Figure 14-4-10-1, Detail B).

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Position control valve on mounting plate and secure with screws, lockwashers, and washers (Figure 14-4-10-1, Detail B).
- Install hydraulic control assembly (PARA 14-4-15).

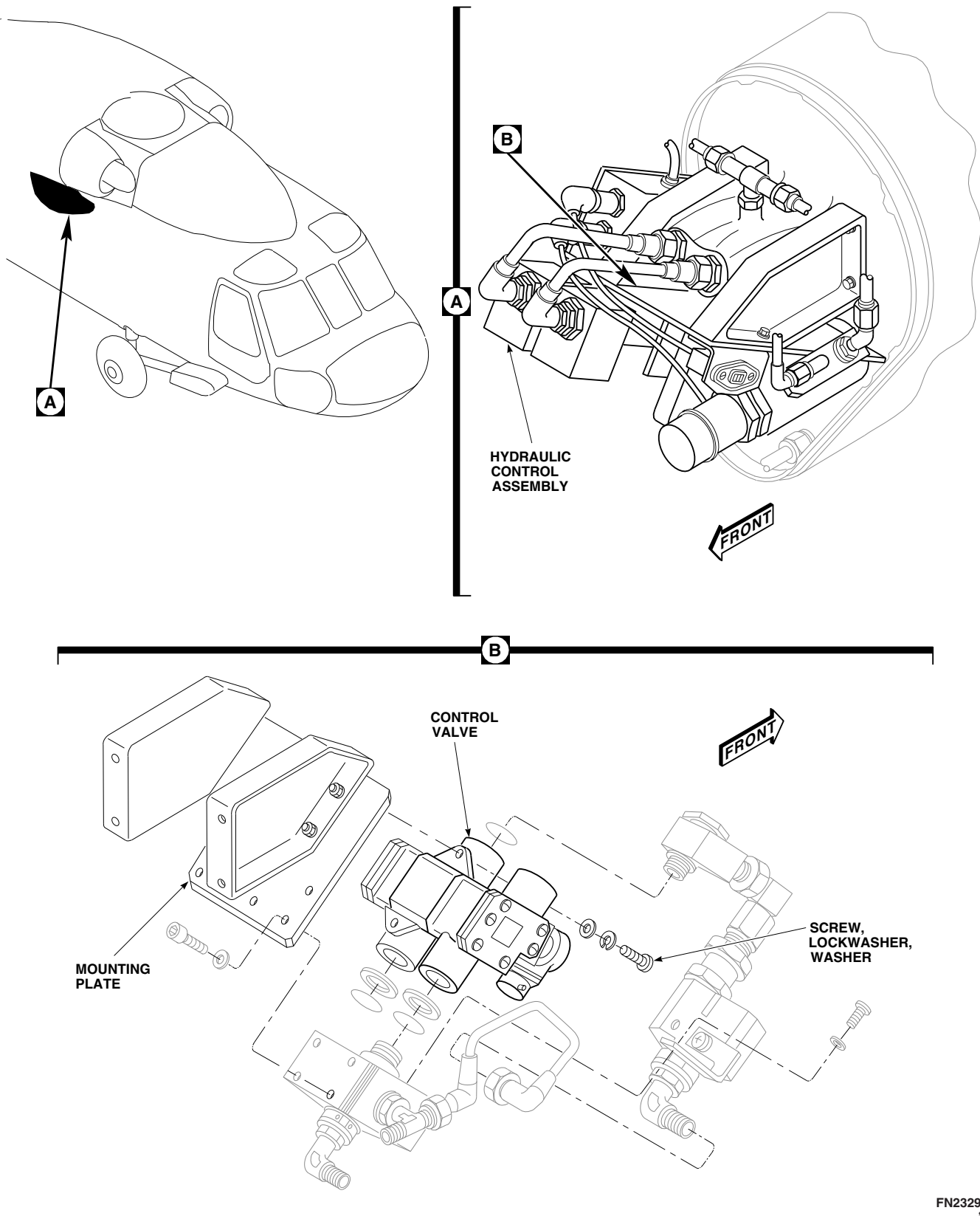


FIGURE 14-4-10-1. REPLACE CONTROL VALVE

14-4-11 ELECTRICAL VALVE CONTROL **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-11.1 Replace Electrical Valve Control	14-4-11-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-15
- PARA 1-6-16

- PARA 1-7-20
- PARA 14-2-1
- PARA 14-4-9

14-4-11.1 Replace Electrical Valve Control.

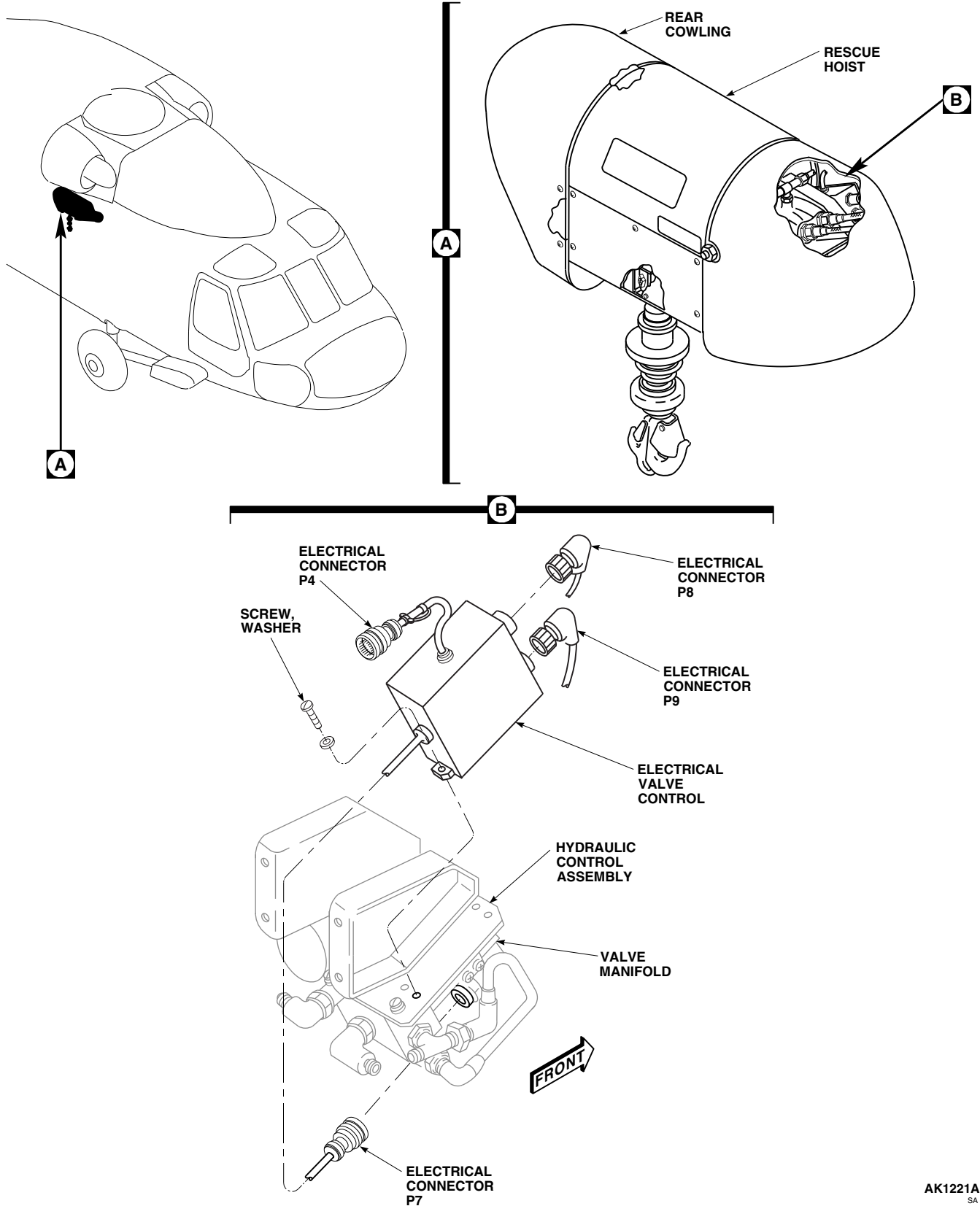
14-4-11.1.1 Remove.

- Remove front cowling (PARA 14-4-9).
- Disconnect electrical connectors as follows (Figure 14-4-11-1, Detail B):
 - P4 from J4 connector.
 - P7 from hydraulic control assembly.
 - P8 and P9 from electrical valve control.
- Install protective caps and plugs on electrical connectors.
- Remove screws and washers securing electrical valve control to valve manifold. Remove electrical valve control.

14-4-11.1.2 Install.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Position electrical valve control on valve manifold and secure with screws and washers (Figure 14-4-11-1, Detail B).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connectors as follows:
 - P4 to J4 connector.
 - P7 to hydraulic control assembly.
 - P8 and P9 to electrical valve control.

- f. Make sure area is clean and free of foreign material.
- g. Install front cowling (PARA 14-4-9).
- h. Perform operational check of rescue hoist (PARA 14-2-1).



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FIGURE 14-4-11-1. REPLACE ELECTRICAL VALVE CONTROL

14-4-12 ELECTRONIC CONTROL BOX **HOIST, 42315-10** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-12.1 Replace Electronic Control Box	14-4-12-1

INITIAL SETUP

- PARA 1-7-20

Tools

- PARA 14-2-1

- Aircraft Mechanic’s Toolkit

- PARA 14-4-9

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-15
- PARA 1-6-16

14-4-12.1 Replace Electronic Control Box.

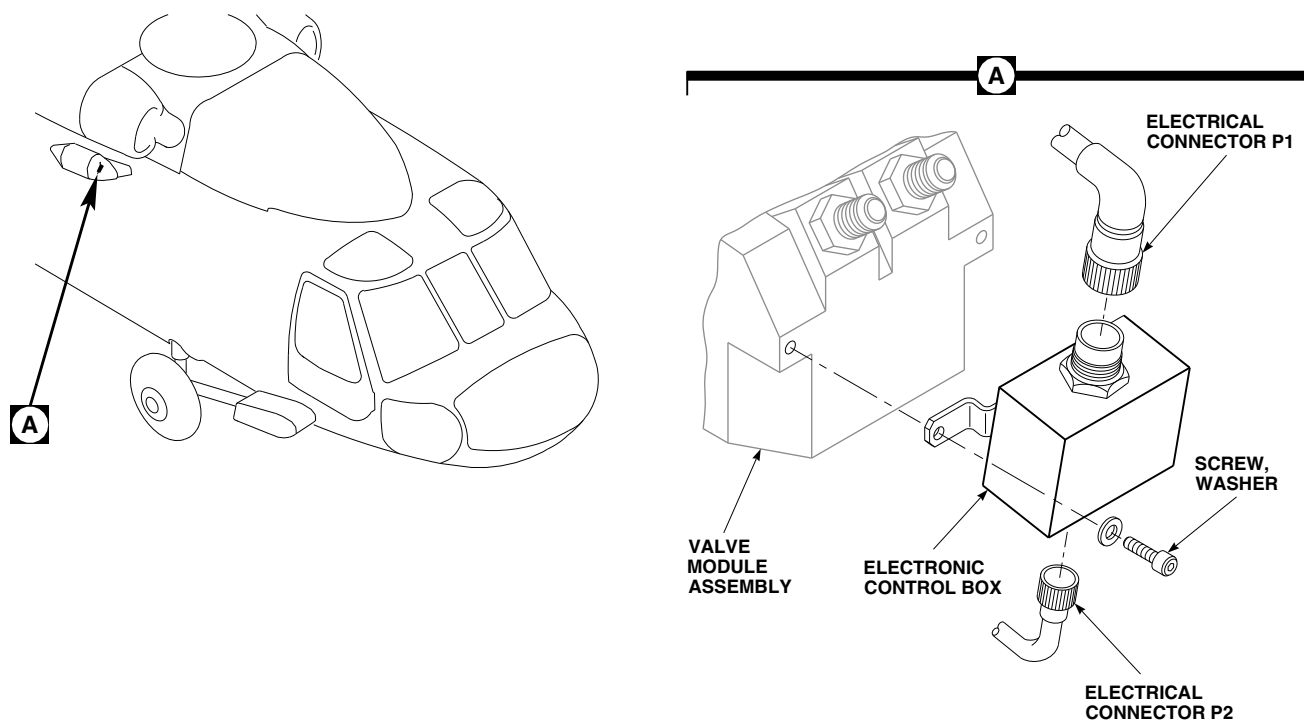
14-4-12.1.1. Remove.

- Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16).
- Remove front cowling (PARA 14-4-9).
- Disconnect electrical connectors, P1 and P2, from electronic control box (Figure 14-4-12-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws and washers holding electronic control box to valve module assembly. Remove electronic control box.

14-4-12.1.2. Install.

- Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16).

- Position electronic control box on valve module assembly and install with screws and washers (Figure 14-4-12-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connectors, P1 and P2, to electronic control box.
- Make sure area is clean and free of foreign material.
- Perform operational check of rescue hoist system (PARA 14-2-1).
- Install front cowling (PARA 14-4-9).

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SA**FIGURE 14-4-12-1. REPLACE ELECTRONIC CONTROL BOX****14-4-12-2**

14-4-13 HOIST VALVE MODULE ASSEMBLY **HOIST, 42315–10** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-13.1 Replace Hoist Valve Module Assembly	14-4-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 1-3-30
- PARA 1-7-20
- PARA 7-4-65
- PARA 14-2-1
- PARA 14-4-9
- PARA 14-4-12

14-4-13.1 **Replace Hoist Valve Module Assembly.**

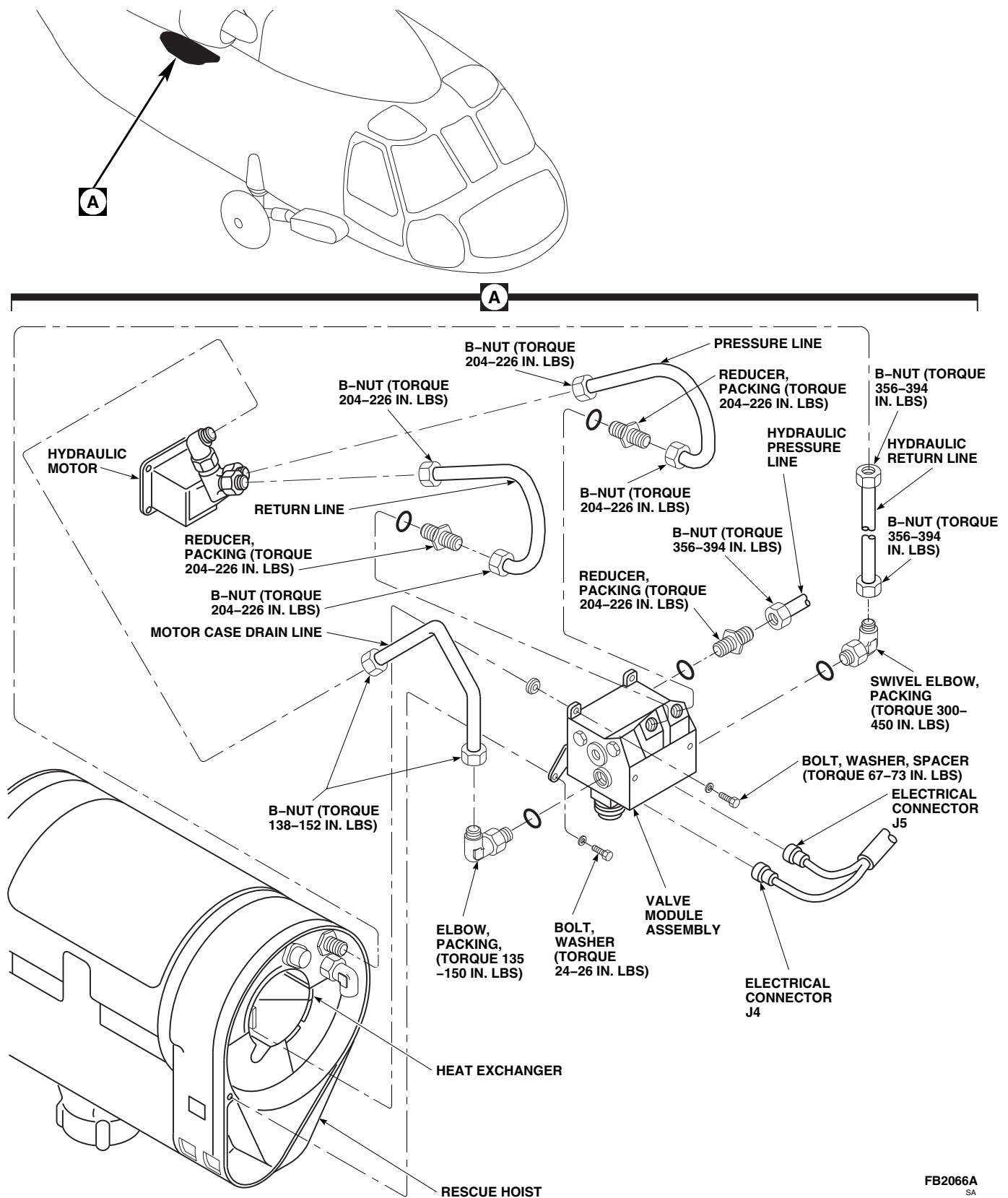
14-4-13.1.1. **Remove.**

- Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16).
- Remove front cowling (PARA 14-4-9).
- Place 1-quart container under rescue hoist to catch hydraulic fluid.
- Remove electronic control box from valve module assembly (PARA 14-4-12).
- Disconnect electrical connectors, J4 and J5, from valve module assembly (Figure 14-4-13-1, Detail A).

- Remove protective caps and plugs from electrical connectors.

NOTE

- Install protective caps and plugs as parts are removed or disconnected.
- Disconnect hydraulic lines from valve module assembly.
 - Loosen and move hydraulic lines from hydraulic motor out of the way.
 - Disconnect and remove hydraulic return line from swivel elbow.
 - Remove reducers, elbows, and packings from valve module. Discard packings.

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14-4-13-2

- k. Remove bolts, washers, and spacers holding valve module assembly to rescue hoist. Remove valve module assembly.

14-4-13.1.2. Install.

- a. Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16).

NOTE

Spacers must be installed with bevel toward valve module assembly.

- b. Install bolts, washers, and spacers into top mount holes of valve module assembly and install assembly to hoist (Figure 14-4-13-1, Detail A). **TORQUE BOLTS TO 67 - 73 INCH-POUNDS.**
- c. Install bolts and washers into lower mount holes of valve module assembly. **TORQUE BOLTS TO 24 - 26 INCH-POUNDS.**

NOTE

Remove protective caps and plugs as parts are installed or connected.

- d. Install packings on reducers, elbow, and swivel elbow.
- e. Install reducers into valve module assembly. **TORQUE REDUCERS TO 204 - 226 INCH-POUNDS.**
- f. Install elbow into outboard side of valve module assembly. **TORQUE ELBOW TO 135 - 150 INCH-POUNDS.**
- g. Install swivel elbow into inboard side of valve module assembly. **TORQUE ELBOW TO 300 - 450 INCH-POUNDS.**

- h. Connect hydraulic pressure line to inboard side of valve module assembly. **TORQUE B-NUT TO 356 - 394 INCH-POUNDS.**
- i. Connect hydraulic return line to heat exchanger port and inboard swivel elbow on valve module assembly. **TORQUE B-NUTS TO 356 - 394 INCH-POUNDS.**
- j. Connect motor case drain line to outboard elbow on valve module assembly. **TORQUE B-NUTS ON HYDRAULIC MOTOR AND VALVE TO 138 - 152 INCH-POUNDS.**
- k. Connect hydraulic lines to valve module assembly pressure and return ports. **TORQUE B-NUTS ON HYDRAULIC MOTOR AND VALVE TO 204 - 226 INCH-POUNDS.**
- l. Install electronic control box (PARA 14-4-12).
- m. Remove protective caps and plugs from electrical connectors.
- n. Inspect and treat electrical connectors (PARA 1-7-20).
- o. Install electrical connectors, J4 and J5, to valve.
- p. Make sure area is clean and free of foreign material.
- q. Service rescue hoist (PARA 1-3-30).
- r. Bleed hydraulic system (PARA 7-4-65).
- s. Perform operational check of rescue hoist system (PARA 14-2-1).
- t. Install front cowling (PARA 14-4-9).

14-4-14 HOOK HOUSING LIMIT SWITCHES **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-14.1 Replace Hook Housing Limit Switches	14-4-14-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Lockwire, Item 196, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 14-2-1

14-4-14.1 Replace Hook Housing Limit Switches.

14-4-14.1.1. Remove.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Remove screws and washers securing access panel to hoist. Remove access panel (Figure 14-4-14-1, Detail A).
- Remove screws and washers and carefully remove cable guide cover and gasket from hook housing (cable guide) (Figure 14-4-14-1, Detail B).
- Tag and disconnect wires from switches.
- Remove screws, switch actuator, insulators, and switches. Discard inoperative switch.

14-4-14.1.2. Install.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

- Position switches, insulators, and switch actuator on screws (Figure 14-4-14-1, Detail B).
- Install screws with switches, insulators, and switch actuator into hook housing (cable guide).
- Connect wires to switches and remove tags from wires.
- Install gasket and cable guide cover with screws and washers.
- Secure access panel to rescue hoist with screws and washers (Figure 14-4-14-1, Detail A). Lockwire, Item 196, Appendix D, screws.
- Make sure area is clean and free of foreign material.
- Perform operational check of rescue hoist (PARA 14-2-1).

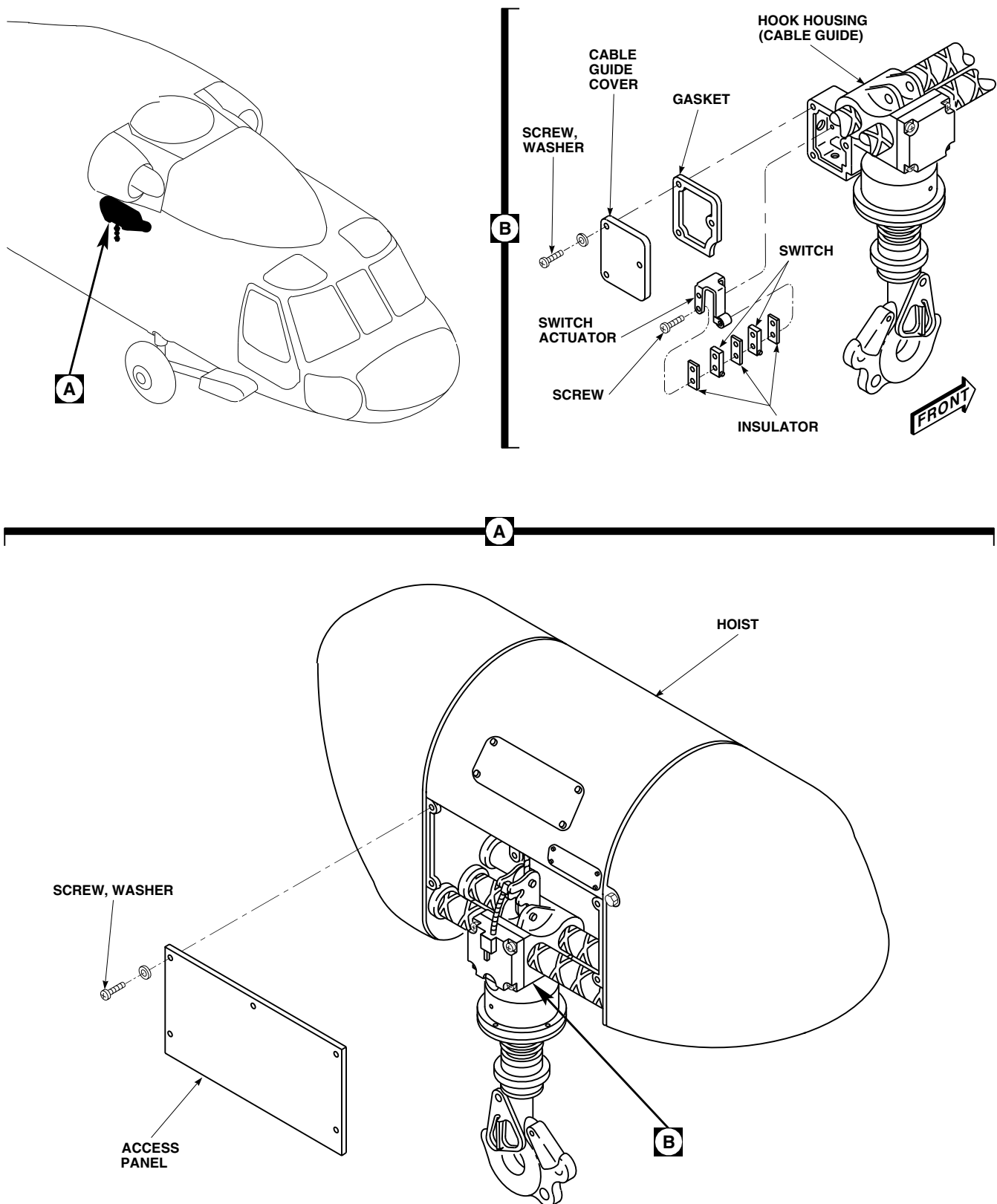


FIGURE 14-4-14-1. REPLACE HOOK HOUSING LIMIT SWITCHES

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14-4-14-2

14-4-15 HYDRAULIC CONTROL ASSEMBLY **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-15.1 Replace Hydraulic Control Assembly	14-4-15-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-13
- PARA 7-4-65
- PARA 14-2-1
- PARA 14-4-11

14-4-15.1 Replace Hydraulic Control Assembly.

14-4-15.1.1 Remove.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Disconnect hydraulic pressure and return line quick-disconnects on backup hydraulic pump module (PARA 7-4-13).
- Remove electrical valve control (PARA 14-4-11).
- Place 1-quart container and machinery towel, Item 211, Appendix D, under hydraulic control assembly to catch hydraulic fluid.
- Disconnect motor inlet and outlet hoses (with adapters) from the hydraulic control assembly (Figure 14-4-15-1, Sheet 1, Detail A). Discard packings.
- Disconnect main return tube from elbow on hydraulic control assembly.
- Disconnect case drain tube from tee on hydraulic motor (Figure 14-4-15-1, Sheet 2, Detail C) and from case drain tee (Figure 14-4-15-1, Sheet 1, Detail A). Remove tube.
- Disconnect main hydraulic input line from forward end of hoist bulkhead (Figure 14-4-15-1, Sheet 1, Detail A) and from elbow on control assembly (Figure 14-4-15-1, Sheet 1, Detail B). Remove line.

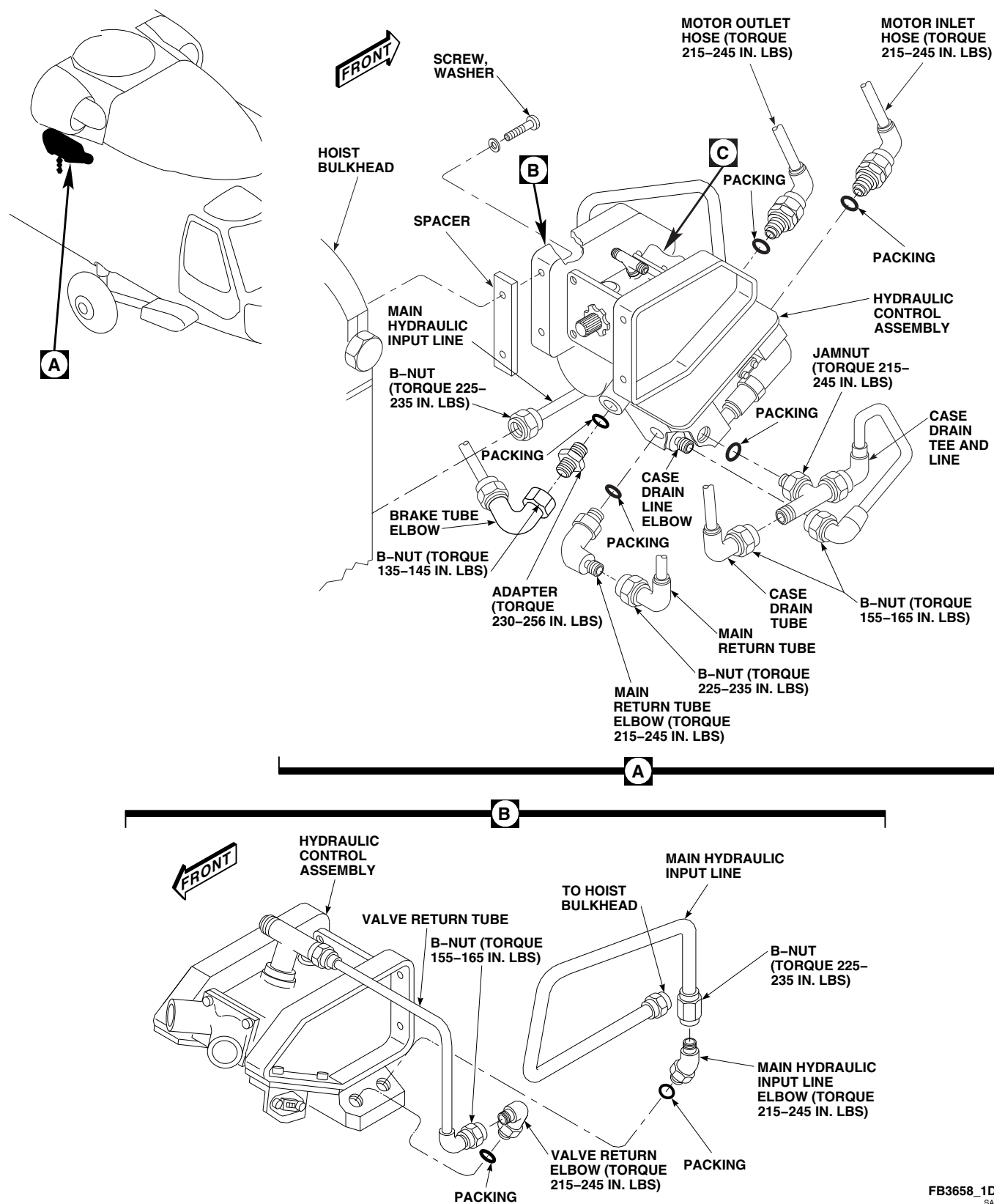
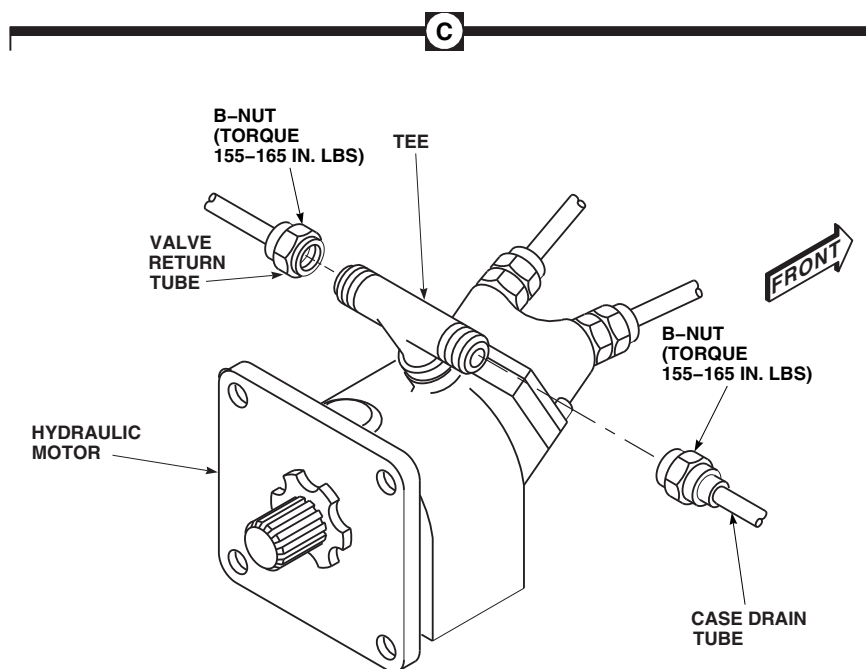


FIGURE 14-4-15-1. REPLACE HYDRAULIC CONTROL ASSEMBLY (SHEET 1 OF 2)



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FIGURE 14-4-15-1. REPLACE HYDRAULIC CONTROL ASSEMBLY (SHEET 2 OF 2)

- i. Disconnect valve return tube from elbow on control assembly (Figure 14-4-15-1, Sheet 1, Detail B) and from tee on hydraulic motor (Figure 14-4-15-1, Sheet 2, Detail C). Remove tube.
- j. Disconnect case drain line from case drain line elbow, and remove case drain tee and packing from control assembly (with case drain line attached) (Figure 14-4-15-1, Sheet 1, Detail A). Discard packing.
- k. Disconnect brake tube elbow from adapter.
- l. Remove adapter and packing from control assembly. Discard packing.
- m. Remove elbows and packings from control assembly (Figure 14-4-15-1, Sheet 1, Detail A and Detail B). Discard packings.
- n. Install protective caps and plugs.
- o. Remove screws, washers, and spacers securing control assembly support to forward end of hoist bulkhead. Remove control assembly.

14-4-15.1.2 Inspect.

Check mating surfaces on support and hoist bulkhead for corrosion, cracks and cleanliness. **NO CORROSION OR CRACKS ALLOWED.**

14-4-15.1.3 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Remove protective caps and plugs as parts are installed or connected.
- c. Coat packings and union threads with hydraulic fluid, Item 155, Appendix D, as parts are installed or connected.
- d. Install packing and main return tube elbow into control assembly. **TORQUE ELBOW TO 215 - 245 INCH-POUNDS** (Figure 14-4-15-1, Sheet 1, Detail A).
- e. Install packing and case drain tee into control assembly. Connect case drain line to case drain line elbow on control assembly. **TORQUE**

**JAMNUT TO 215 - 245 INCH-POUNDS AND
LINE B-NUT TO 155 - 165 INCH-POUNDS.**

- f. Install packing on short threaded end of adapter and install adapter into control assembly. **TORQUE ADAPTER TO 230 - 256 INCH-POUNDS.**
- g. Install brake tube elbow onto adapter. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**
- h. Install control assembly on hoist bulkhead and secure with screws, washers, and spacers.
- i. Install packing and valve return elbow into control assembly. **TORQUE ELBOW TO 215 - 245 INCH-POUNDS** (Figure 14-4-15-1, Sheet 1, Detail B).
- j. Connect valve return tube to valve return elbow on control assembly and to tee on hydraulic motor. **TORQUE B-NUTS TO 155 - 165 INCH-POUNDS** (Figure 14-4-15-1, Sheet 1, Detail B, and Sheet 2, Detail C).
- k. Install packing and main hydraulic input line elbow into control assembly. **TORQUE ELBOW TO 215 - 245 INCH-POUNDS** (Figure 14-4-15-1, Sheet 1, Detail B).
- l. Connect main hydraulic input line to forward end of hoist bulkhead and to main hydraulic input line elbow. **TORQUE B-NUTS TO 225 - 235 INCH-POUNDS** (Figure 14-4-15-1, Sheet 1, Detail A and Detail B).
- m. Connect case drain tube to case drain tee and to tee on hydraulic motor. **TORQUE B-NUTS TO 155 - 165 INCH-POUNDS** (Figure 14-4-15-1 Sheet 1, Detail A, and Sheet 2, Detail C).
- n. Install packing on motor inlet hose (with adapter) and connect inlet hose to control assembly. **TORQUE END FITTING TO 215 - 245 INCH-POUNDS** (Figure 14-4-15-1, Sheet 1, Detail A).
- o. Install packing on motor outlet hose (with adapter) and connect outlet hose to control assembly. **TORQUE END FITTING TO 215 - 245 INCH-POUNDS.**
- p. Connect main return tube to elbow on control assembly. **TORQUE B-NUT TO 225 - 235 INCH-POUNDS.**
- q. Install electrical valve control (PARA 14-4-11).
- r. Connect hydraulic pressure and return line quick-disconnects on backup hydraulic pump module (PARA 7-4-13).
- s. Bleed hydraulic system (PARA 7-4-65).
- t. Make sure area is clean and free of foreign material.
- u. Perform operational check of rescue hoist (PARA 14-2-1).

14-4-16 RESCUE HOIST FAN **HOIST, 42315-10** .

PARAGRAPH BREAKDOWN

PAGE	
14-4-16.1	Replace Rescue Hoist Fan 14-4-16-1
INITIAL SETUP	• PARA 1-7-20
Tools	• PARA 14-2-1
• Aircraft Mechanic’s Toolkit • Protective Caps and Plugs	• PARA 14-4-9
References	
• PARA 1-6-15 • PARA 1-6-16	
14-4-16.1 Replace Rescue Hoist Fan.	14-4-16.1.2. Install.
14-4-16.1.1. Remove.	a. Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16). b. Place fan collar on fan and install fan and fan collar on rescue hoist housing using screws and washers (Figure 14-4-16-1, Detail A). c. Remove protective caps and plugs from electrical connectors. d. Inspect and treat electrical connectors (PARA 1-7-20). e. Install electrical connector, J1, to interface block using screws and washers (Figure 14-4-16-1, Detail B). f. Remove protective caps and plugs on electrical connectors. g. Inspect and treat electrical connector (PARA 1-7-20). h. Connect electrical connector, P962 to J1.
a. Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16). b. Remove front cowling (PARA 14-4-9). c. Remove screws and washers holding outboard fairing to hoist (Figure 14-4-16-1, Detail A). Remove outboard fairing. d. Disconnect electrical connector, P962 from J1 (Figure 14-4-16-1, Detail B). e. Install protective caps and plugs on electrical connectors. f. Remove screws and washers holding electrical connector, J1, to interface block. Remove connector. g. Remove screws and washers holding fan and fan collar to rescue hoist housing. Remove fan and fan collar (Figure 14-4-16-1, Detail A).	

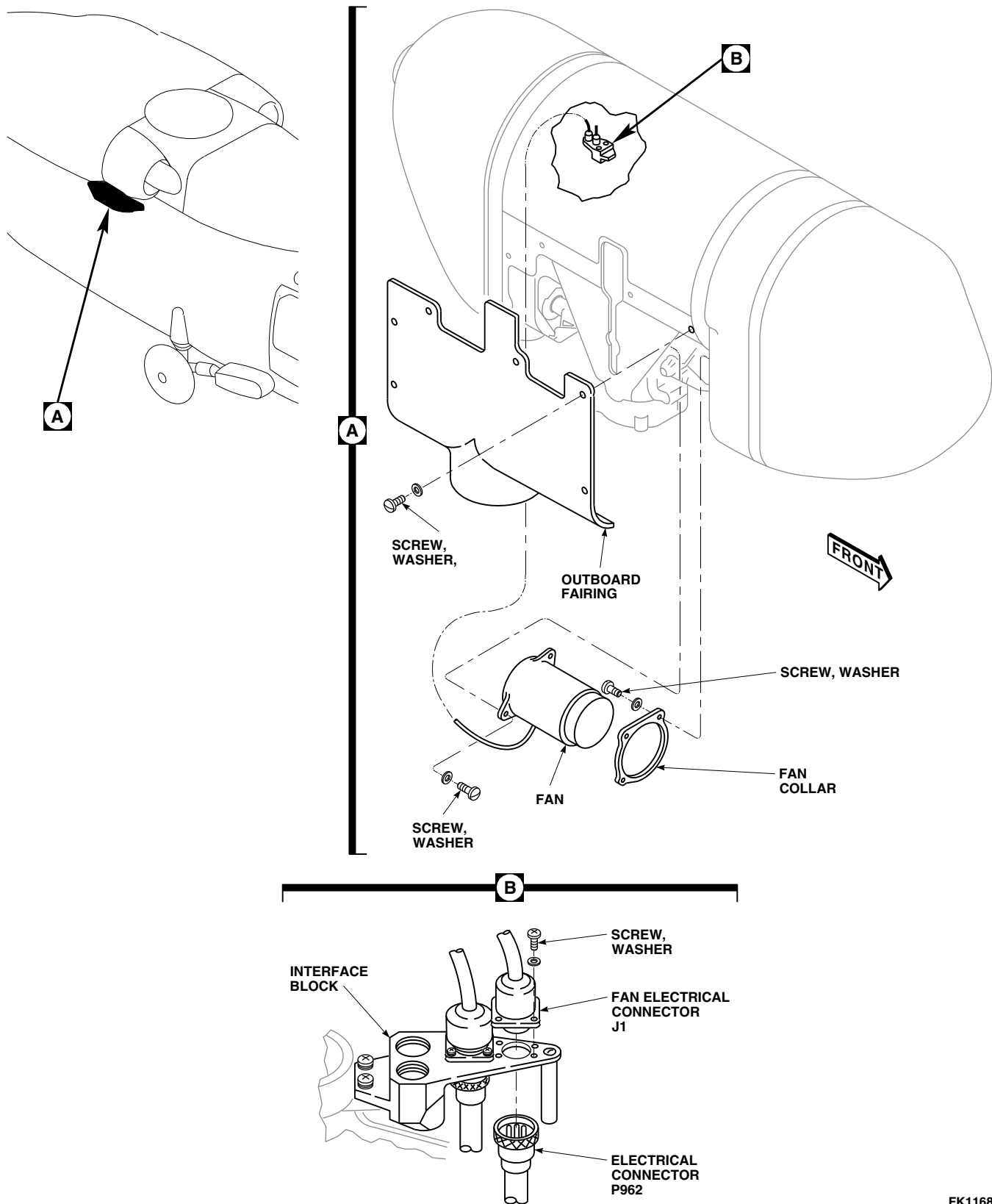


FIGURE 14-4-16-1. REPLACE RESCUE HOIST FAN

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- i. Make sure area is clean and free of foreign material.
- j. Perform operational check of rescue hoist (PARA 14-2-1).
- k. Install outboard fairing to hoist using screws and washers (Figure 14-4-16-1, Detail A).
- l. Install front cowling (PARA 14-4-9).

14-4-17 SHAFT HOUSING LIMIT SWITCHES **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-17.1 Replace Shaft Housing Limit Switches	14-4-17-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-65
- PARA 14-2-1
- PARA 14-4-9

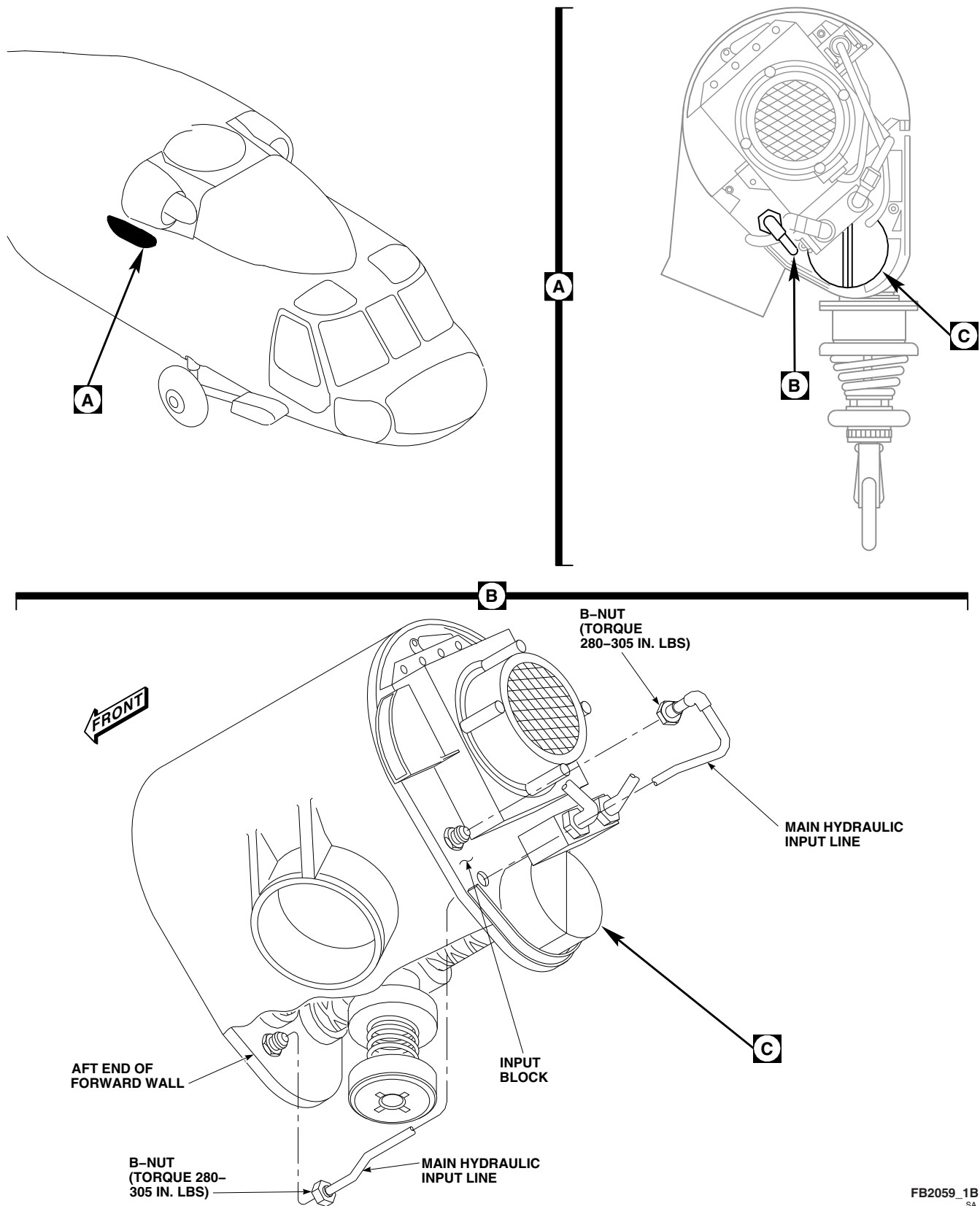
14-4-17.1 Replace Shaft Housing Limit Switches.

14-4-17.1.1 Remove.

- Remove rear cowling (PARA 14-4-9).
- Remove main hydraulic input line from input block and aft end of forward wall. Install protective caps and plugs (Figure 14-4-17-1, Sheet 1, Detail B).
- Remove screws and washers and carefully remove shaft housing adapter cover (Figure 14-4-17-1, Sheet 2, Detail C).
- Tag and disconnect wires from inoperative switch.
- Remove screws, switch actuator, insulators, and switches from shaft housing adapter bracket. Discard inoperative switch (Figure 14-4-17-1, Sheet 2, Detail D).

14-4-17.1.2 Install.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Position switches, insulators, and switch actuator on screws (Figure 14-4-17-1, Sheet 2, Detail D).
- Install screws with switches, insulators, and switch actuator into shaft housing adapter bracket.
- Connect wires to switches and remove tags from wires.
- Install shaft housing adapter cover and secure with screws and washers (Figure 14-4-17-1, Sheet 2, Detail C).
- Remove protective caps and plugs and secure main hydraulic input line to input block and aft end of the forward hoist wall (Figure 14-4-17-1, Sheet 1, Detail B). **TORQUE B-NUTS TO 280 - 305 INCH-POUNDS.**



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FIGURE 14-4-17-1. REPLACE SHAFT HOUSING LIMIT SWITCHES (SHEET 1 OF 2)

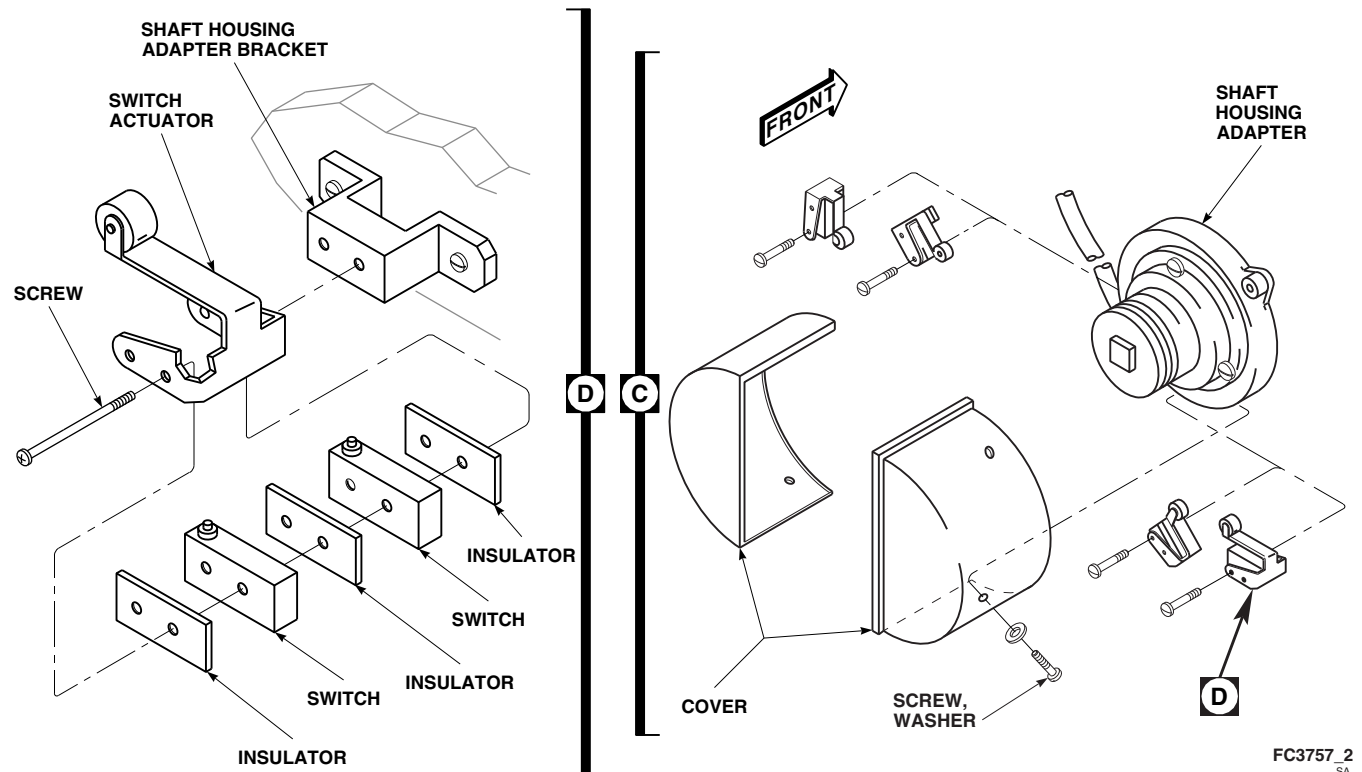


FIGURE 14-4-17-1. REPLACE SHAFT HOUSING LIMIT SWITCHES (SHEET 2 OF 2)

- g. Bleed hydraulic system (PARA 7-4-65).
- h. Install rear cowling (PARA 14-4-9).
- i. Perform operational check of rescue hoist (PARA 14-2-1).

14-4-18 SOLENOID VALVE **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-18.1 Replace Solenoid Valve	14-4-18-1

INITIAL SETUP	• PARA 1-6-16
Tools	• PARA 1-7-20
• Aircraft Mechanic’s Toolkit	• PARA 7-4-65
Materials/Parts	
• Hydraulic Fluid, Item 155, Appendix D	• PARA 14-2-1
• Protective Caps and Plugs	• PARA 14-4-9
References	
• Appendix D	
• PARA 1-6-15	

14-4-18.1 Replace Solenoid Valve.

14-4-18.1.1 Remove.

- a. Remove front cowling (PARA 14-4-9).
- b. Disconnect electrical connector, P9, from electrical valve control (Figure 14-4-18-1, Detail B).
- c. Install protective caps and plugs on electrical connectors.

NOTE

Solenoid valve and housing must be replaced as an assembly.

- d. Unscrew and remove solenoid valve and housing from valve manifold. Install plug on valve manifold to prevent fluid loss.
- e. Remove packings and rings from solenoid valve. Discard packings and rings.

14-4-18.1.2 Install.

- a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- b. Coat packings with hydraulic fluid, Item 155, Appendix D, and install with rings on solenoid valve (Figure 14-4-18-1, Detail B).
- c. Remove plug from valve manifold and install solenoid valve and housing.
- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Connect electrical connector, P9, to electrical valve control.
- g. Make sure area is clean and free of foreign material.

- h. Bleed hydraulic system (PARA 7-4-65).
- i. Install front cowling (PARA 14-4-9).
- j. Perform operational check of rescue hoist (PARA 14-2-1).

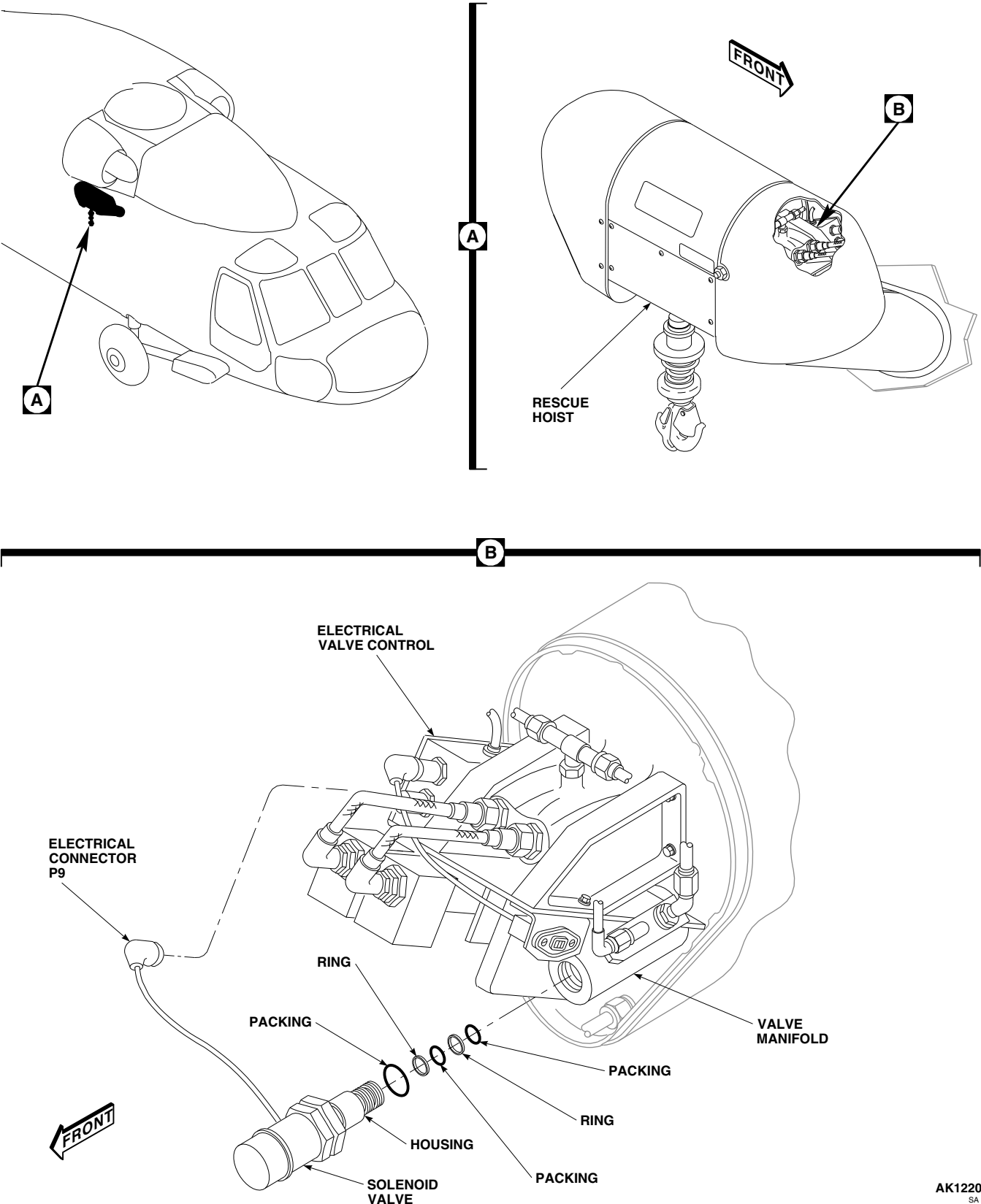


FIGURE 14-4-18-1. REPLACE SOLENOID VALVE

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14-4-19 SUPPORT ASSEMBLY **HOIST, 42315-10** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-19.1 Replace Support As- sembly	14-4-19-1

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Drain Tube, Locally-Made
- Drill Set
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Scraper
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Corrosion Resistant Coating, Item 117, Appendix D

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Paint Remover, Item 231, Appendix D
- Plastic Tubing, 3-inches long (0.133 or 0.186-inch ID) (for Drain Tube, Locally-Made)
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 296, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-1
- PARA 2-6-5
- PARA 14-4-1

14-4-19.1 Replace Support Assembly.

14-4-19.1.1 Remove.

- a. Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16).
- b. Remove rescue hoist (PARA 14-4-1).
- c. Remove screws, washers, clamps, and electrical wiring cable from support assembly (Figure 14-4-19-1, Detail A). Leave clamps attached to

cable and secure cable away from support assembly.

- d. Remove screws, washers, clamp, channel, spacer, electrical wiring, and hydraulic hose block from support assembly. Secure hoses away from support assembly.
- e. Remove bolts, washers, and nuts holding support assembly to airframe fitting, and remove support assembly.

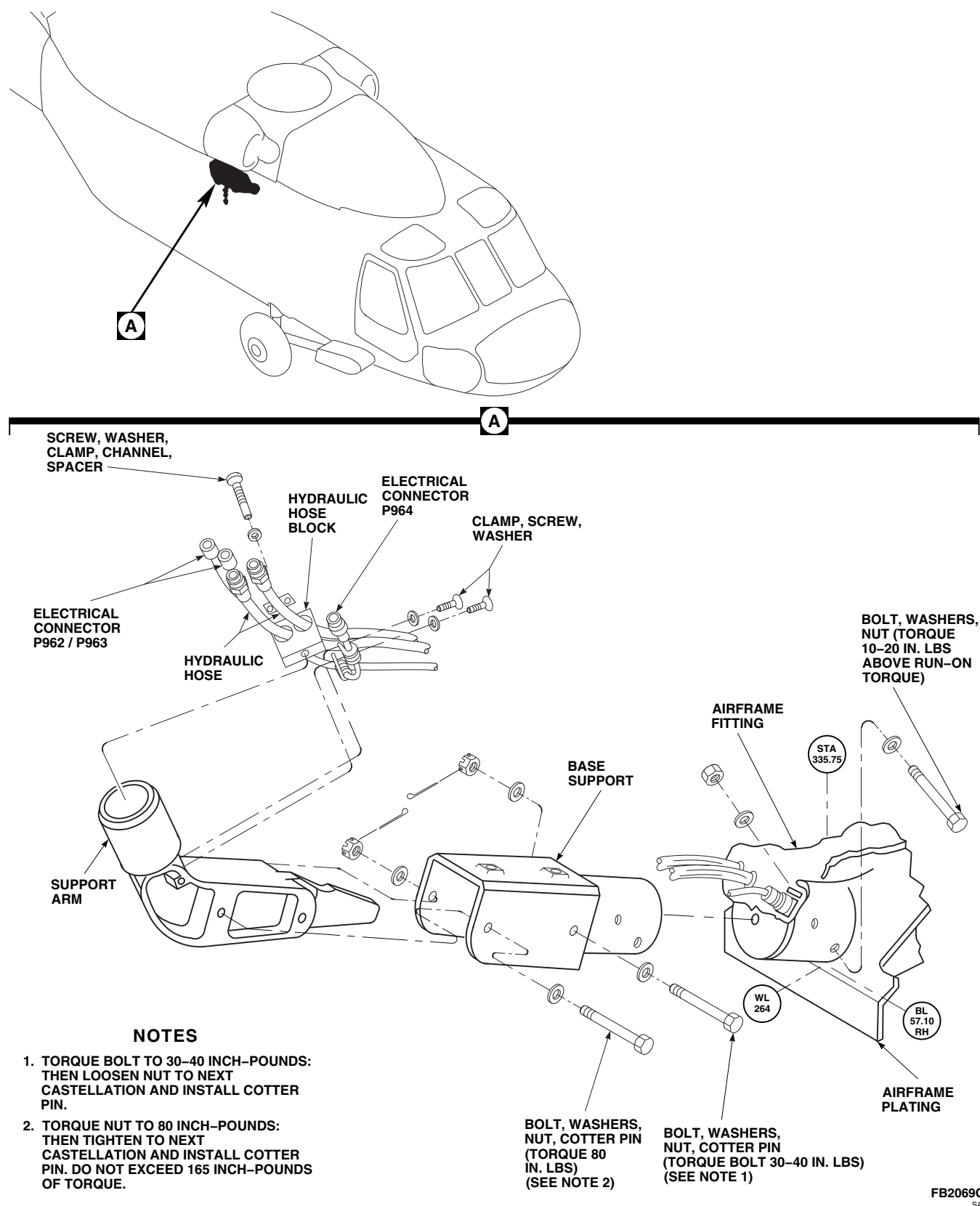


FIGURE 14-4-19-1. REPLACE SUPPORT ASSEMBLY

- f. Remove bolts, washers, and nuts, and remove support arm from base support.
- g. Remove sealing compound from base support and airframe fitting mating surface using nonmetallic scraper.

14-4-19.1.2 Inspect.

- a. Inspect base support for cracks as follows (Figure 14-4-19-2, Sheet 1):

- (1) Using paint remover, Item 231, Appendix D, remove paint, sealing compound, and primer from inside and outside surfaces of cylindrical section that mates with fitting and channel section wall, Areas B and A, respectively.

NOTE

Pay particular attention to areas around bolt holes during fluorescent penetrant inspection.

- (2) Fluorescent penetrant inspect, Type 1, Method B, C or D, Sensitivity Level 3 or 4, on cylindrical areas for cracks (ASTM E1417). **NONE ALLOWED.**

- (3) Visually inspect all other areas for cracks. **NONE ALLOWED.**

- b. Inspect base support for nicks, dents, scratches, pits, or gouges. Repair nicks, dents, scratches, pits, or gouges using abrasive cloth, Item 7, Appendix D, as follows:

- (1) Repair AREA A as follows:

- (a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.010-INCH DEEP. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750 INCH DIAMETER.**
- (b) **DAMAGE SHALL NOT BE CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLES.**

- (c) If damage or repairs are more than limits, replace base support.

- (2) Repair AREA B as follows:

- (a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.002-INCH DEEP AND 0.750-INCH DIAMETER.**

- (b) If damage or repairs are more than limits, replace base support.

- c. Inspect base support for corrosion, and repair using abrasive cloth, Item 7, Appendix D, as follows:

- (1) Repair AREA A as follows:

- (a) **BLEND OUT CORROSION UP TO 0.0050-INCH DEEP OVER 100% OF SURFACE AREA.**

- (b) If damage or repairs are more than limits, replace base support.

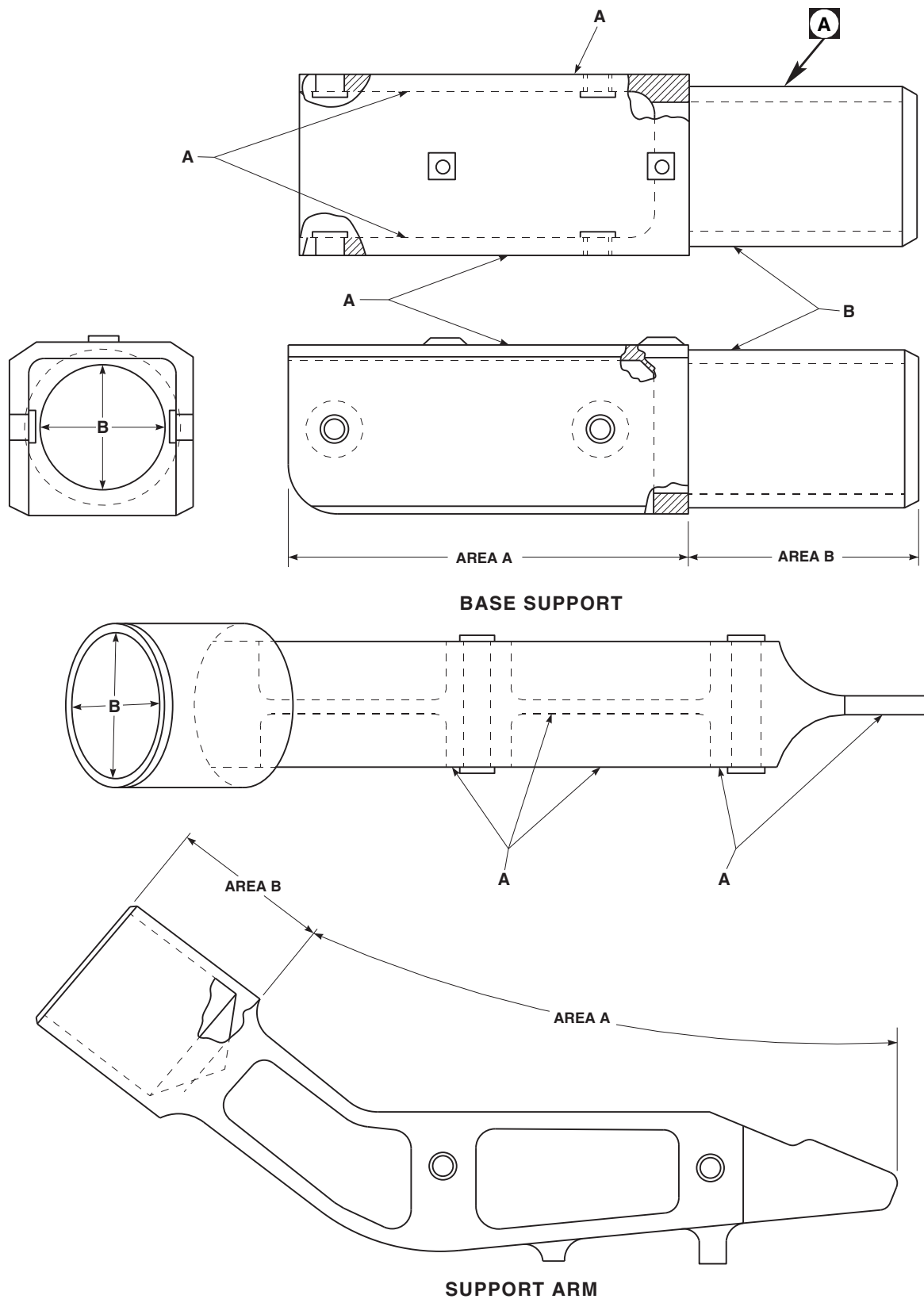
- (2) Repair AREA B as follows:

- (a) **BLEND OUT CORROSION UP TO 0.0100-INCH DEEP OVER 100% OF INNER DIAMETER, AND UP TO 0.002-INCH DEEP INTO THE WALL THICKNESS TO MINIMUM OUTSIDE DIAMETER OF 2.966-INCHES.**

- (b) If damage or repairs are more than limits, replace base support.

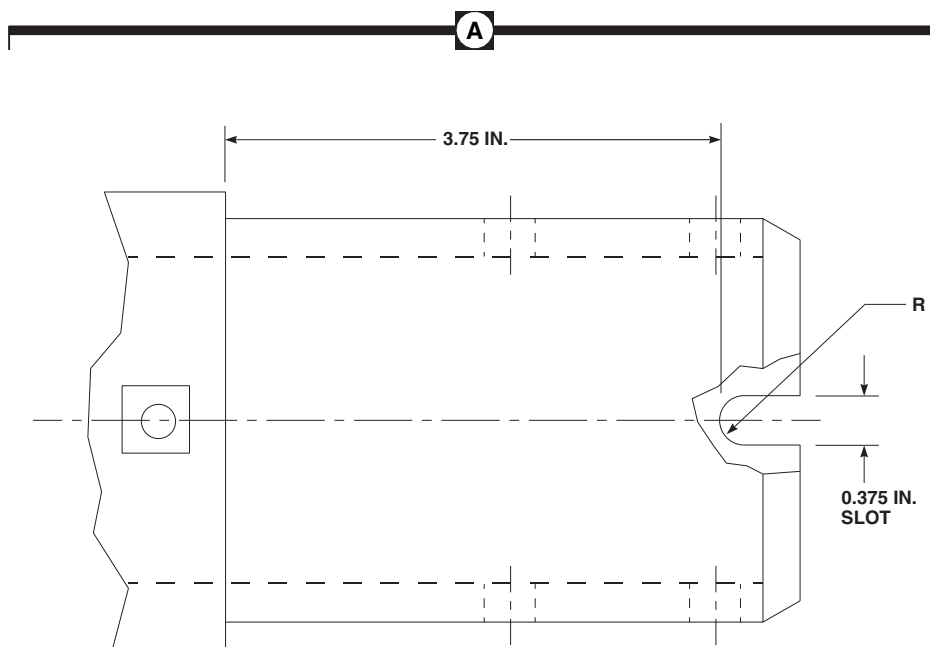
- d. Inspect base support for incorporation of 0.375-inch slot at bottom of base support. If slot is not present perform the following:

- (1) Add slot to base support as shown (Figure 14-4-19-2, Sheet 2, Detail A).
- (2) Treat repaired area with corrosion resistant coating, Item 117, Appendix D.
- (3) Using epoxy primer coating kit, Item 135, Appendix D, prime stripped and repaired area.



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FIGURE 14-4-19-2 INSPECT RESCUE HOIST (SHEET 1 OF 2)



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FIGURE 14-4-19-2 INSPECT RESCUE HOIST (SHEET 2 OF 2)

- e. Inspect support arm for cracks as follows (Figure 14-4-19-2, Sheet 1):

- (1) Using paint remover, Item 231, Appendix D, remove paint, sealing compound, and primer from inside and outside surfaces of socket section that mates with fitting and channel section wall, Areas B and A, respectively.

NOTE

Pay particular attention to areas around bolt holes during fluorescent penetrant inspection.

- (2) Fluorescent penetrant inspect, Type 1, Method B, C or D, Sensitivity Level 3 or 4, on cylindrical areas for cracks (ASTM E1417). **NONE ALLOWED.**
- (3) Visually inspect all other areas for cracks. **NONE ALLOWED.**

- f. Inspect support arm for nicks, dents, scratches, pits, or gouges. Repair nicks, dents, scratches,

pits, or gouges using abrasive cloth, Item 7, Appendix D, as follows:

- (1) Repair AREA A as follows:

- (a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.010-INCH DEEP. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.**
- (b) **DAMAGE SHALL NOT BE CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLES.**

- (c) If damage or repairs are more than limits, replace support arm.

- (2) Repair AREA B as follows:

- (a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.020-INCH DEEP OVER 100 % OF SURFACE AREA.**

BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.

- (b) If damage or repairs are more than limits, replace support arm.
- g. Inspect support arm for corrosion, and repair using abrasive cloth, Item 7, Appendix D, as follows:
 - (1) Repair AREA A as follows:
 - (a) **BLEND OUT CORROSION UP TO 0.0050-INCH DEEP OVER 100% OF SURFACE AREA.**
 - (b) If damage or repairs are more than limits, replace support arm.
 - (2) Repair AREA B as follows:
 - (a) **BLEND OUT CORROSION UP TO 0.010-INCH DEEP OVER 100% OF INNER AND OUTER DIAMETER.**
 - (b) If damage or repairs are more than limits, replace support arm.
- h. Inspect airframe fitting for cracks, corrosion, or deformation (PARA 2-6-1).
- i. Inspect airframe fitting for plugged/missing drain hole at bottom of fitting at 6 o'clock position as follows:

NOTE

If 0.188-inch diameter drain hole exists at 45° angle at bottom of fitting at 6 o'clock position. Use hole as is, with 0.133-inch diameter drain tube (Figure 14-4-19-4, Detail A).

- (1) If drain hole is plugged, clean out drain hole and install drain tube per step i. (5).
- (2) If drain hole is drilled at 90° angle in fitting, plug hole with sealing compound, Item 296, Appendix D, and drill new hole per step i. (4) (Figure 14-4-19-4, Detail A).

- (3) If drain hole is missing, drill hole per step i. (4).
- (4) If required, drill 45° angle hole as follows (Figure 14-4-19-4, Detail A and Detail B):

NOTE

Make sure not to gouge inner wall of airframe fitting when drilling hole.

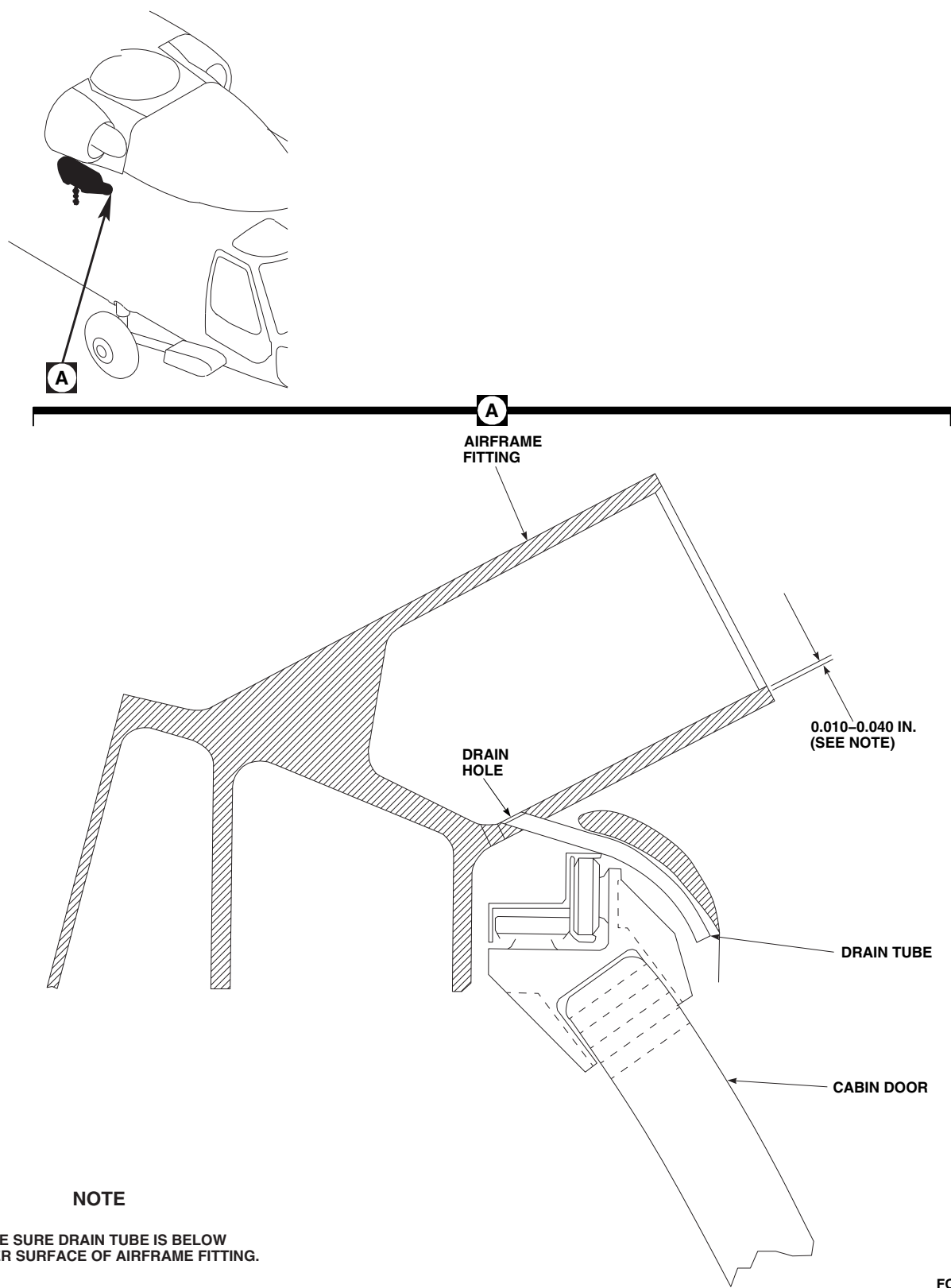
- (a) Locate hole 0.85-inch from airframe fitting base at 75° ($\pm 2^\circ$) angle.
- (b) Drill 0.250-inch diameter hole.
- (5) Install drain tube as follows:
 - (a) If required, locally make drain tube from 0.133 or 0.186-inch ID plastic tubing, 3-inches long.
 - (b) Install drain tubing into fitting with sealing compound, Item 296, Appendix D. Make sure drain tube is 0.010 - 0.040-inch below inner surface of fitting (Figure 14-4-19-3, Detail A).
 - (c) Seal drain tube to fitting with sealing compound, Item 296, Appendix D.
 - (d) Make sure drain tube is clear of sealing compound.

- j. Treat stripped and repaired areas with corrosion resistant coating, Item 117, Appendix D.
- k. Using epoxy primer coating kit, Item 135, Appendix D, prime stripped and repaired areas.

NOTE

Do not paint outside diameter of Area B, holes, bushings and inserts of support base assembly or outside diameter of Area B, holes, bushings and inserts of support arm assembly.

- l. Paint stripped and repaired areas (PARA 2-6-5).



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FIGURE 14-4-19-3. INSPECT RESCUE HOIST DRAIN TUBE

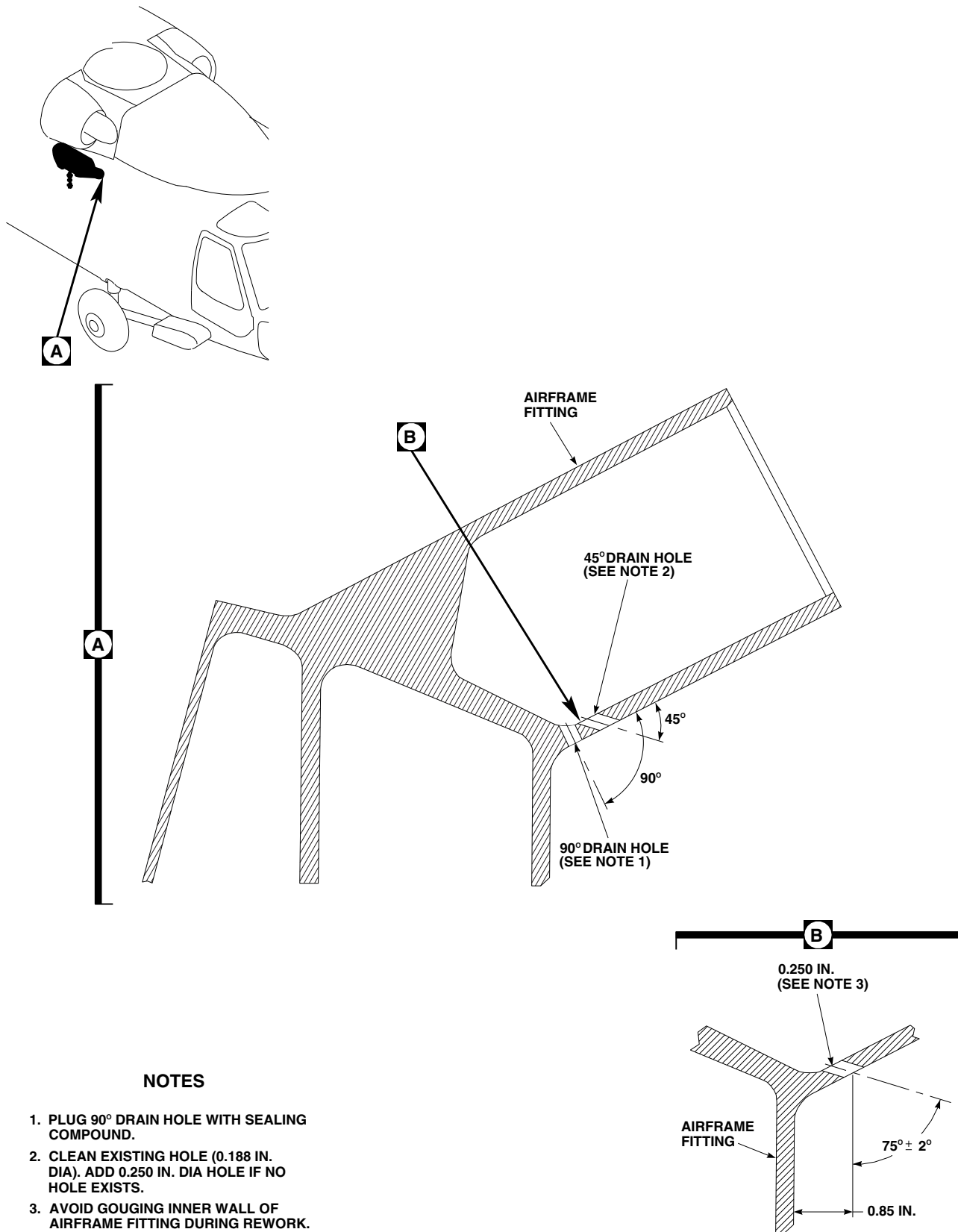


FIGURE 14-4-19-4. INSPECT RESCUE HOIST DRAIN HOLE

14-4-19.1.3 Install.

- a. Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16).
- b. Apply sealing compound, Item 296, Appendix D, onto support arm and base support mating surface.
- c. Install support arm to base support with bolts, washers, and nuts (Figure 14-4-19-1, Detail A).
- d. **TORQUE UPPER NUT TO 80 INCH-POUNDS. THEN TIGHTEN NUT TO NEXT CASTELLATION.** Install cotter pin. **DO NOT EXCEED 165 INCH-POUNDS TORQUE.**
- e. **TORQUE LOWER BOLT TO 30 - 40 INCH-POUNDS.** If required, loosen nut to next castellation. Install cotter pin.

NOTE

Avoid blocking drain hole of airframe fitting (cup area) with sealing compound.

- f. Apply sealing compound, Item 296, Appendix D, to base support and airframe fitting mating surface.

- g. Install base support in airframe fitting with bolts, washers, and nuts. **TORQUE NUTS TO 10 - 20 INCH-POUNDS ABOVE RUN-ON TORQUE.**
- h. Apply sealing compound, Item 292, Appendix D, around base support/airframe fitting mating edges and fastener locations.



Damage to electrical connectors may occur while installing hydraulic hoses. Be careful not to damage electrical connectors when installing hydraulic hoses.

- i. Install hydraulic hose block and electrical wiring to support with spacer, channel, clamp, screws, and washers.
- j. Position electrical cable wiring with clamps on support, and install with screws, washers, and bracket.
- k. Install rescue hoist (PARA 14-4-1).
- l. Make sure area is clean and free of foreign material.

14-4-20 RESCUE HOIST FAIRINGS.

PARAGRAPH BREAKDOWN

	PAGE
14-4-20.1 Replace Rescue Hoist Fairings	14-4-20-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-16
- PARA 2-6-2
- PARA 14-4-1

14-4-20.1 Replace Rescue Hoist Fairings.

14-4-20.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- If removing cuff fairing, remove rescue hoist (PARA 14-4-1).
- Remove screws and washers securing support fairing to cuff fairing (Figure 14-4-20-1, Detail A).
- Remove bolts and washers securing support fairings to support assembly. Remove fairing.
- Remove screws and washers securing cuff fairing to hoist. Remove cuff fairing.

14-4-20.1.2. Inspect/Repair.

Inspect and repair fairings (PARA 2-6-2).

14-4-20.1.3. Install.

- Turn off all electrical power (PARA 1-6-16).
- Make sure area is clean and free of foreign objects.
- Install cuff fairing and secure to hoist with screws and washers (Figure 14-4-20-1, Detail A).
- Install support fairing and secure to support with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- Secure support fairing to cuff fairing with screws and washers.
- If removed, install rescue hoist (PARA 14-4-1).
- Make sure area is clean and free of foreign objects.

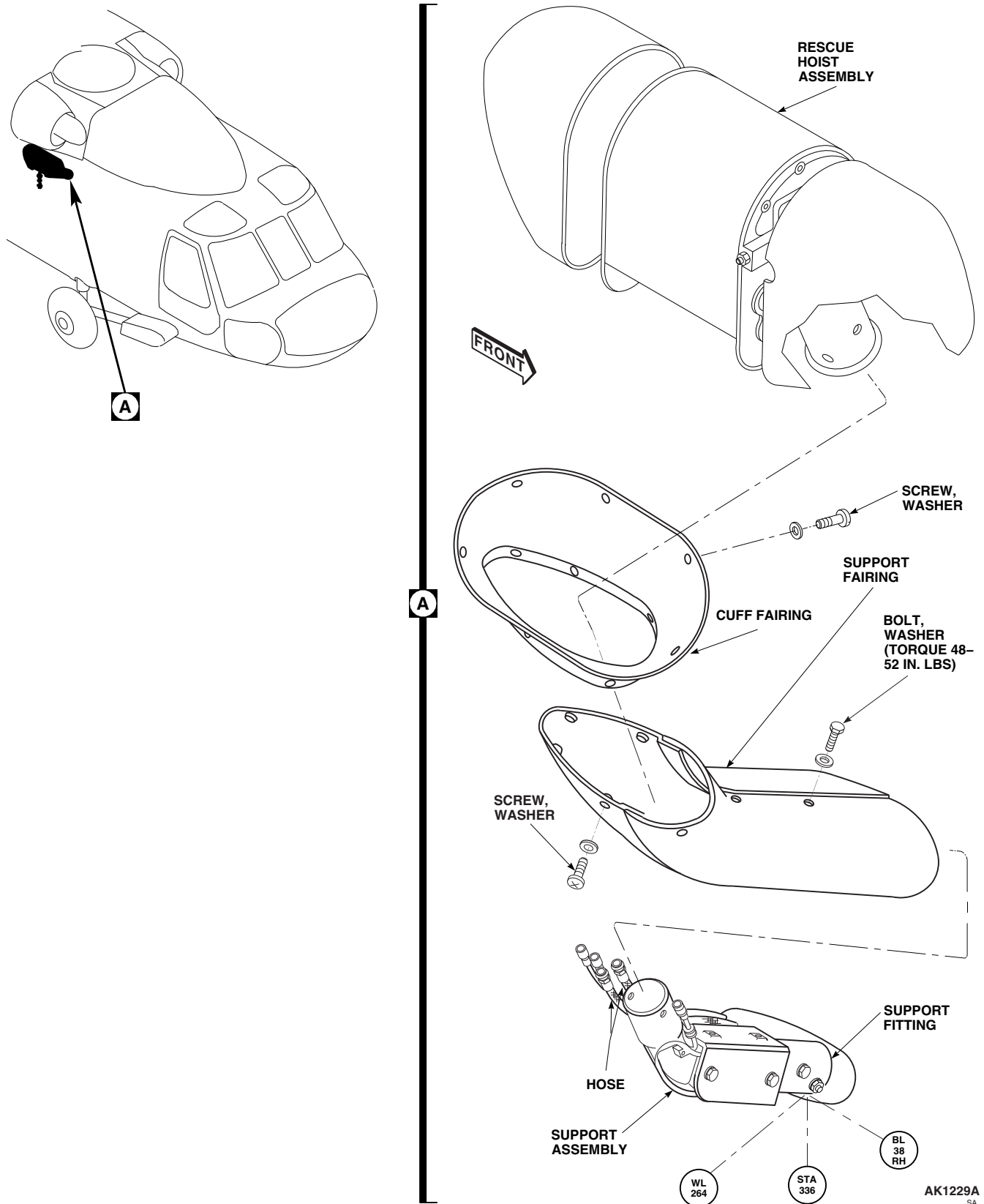


FIGURE 14-4-20-1. REPLACE RESCUE HOIST FAIRINGS

14-4-20-2

14-4-21
RESCUE HOIST/CARGO HOOK PENDANT

HOIST, BL-27100-71
HOIST, 42315-10

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-21.1	Replace Rescue Hoist/ Cargo Hook Pendant HOIST, BL-27100-71	14-4-21-1	14-4-21.2	Replace Rescue Hoist/ Cargo Hook Pendant HOIST, 42315-10	14-4-21-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 14-2-1

14-4-21.1
Replace Rescue Hoist/Cargo Hook Pendant
HOIST, BL-27100-71

14-4-21.1.1. Remove.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Remove soundproofing panels (PARA 2-4-69).
- Disconnect rescue hoist/cargo hook pendant electrical connector, P968, from J1 (Figure 14-4-21-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Unfasten Velcro hook from coil cord.
- Open clip and remove rescue hoist/cargo hook pendant from support.

14-4-21.1.2. Install.

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Position rescue hoist/cargo hook pendant in support and close clip (Figure 14-4-21-1, Detail A).
- Fasten Velcro hook to coil cord.
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect rescue hoist/cargo hook pendant electrical connector, P968, to J1.
- Perform operational check of rescue hoist system (PARA 14-2-1).
- Install soundproofing panels (PARA 2-4-69).

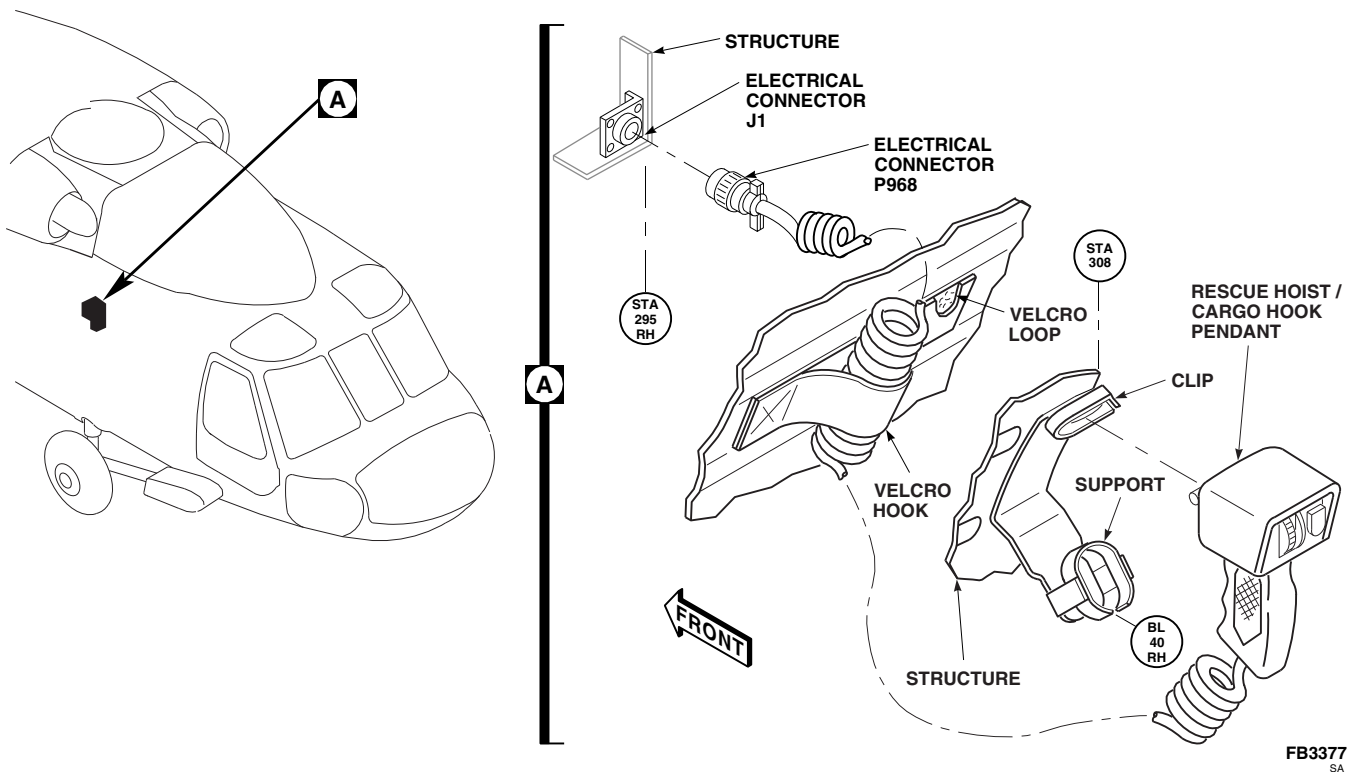


FIGURE 14-4-21-1. REPLACE RESCUE HOIST/CARGO HOOK PENDANT **HOIST, BL-27100-71**

14-4-21.2 Replace Rescue Hoist/Cargo Hook Pendant **HOIST, 42315-10** .

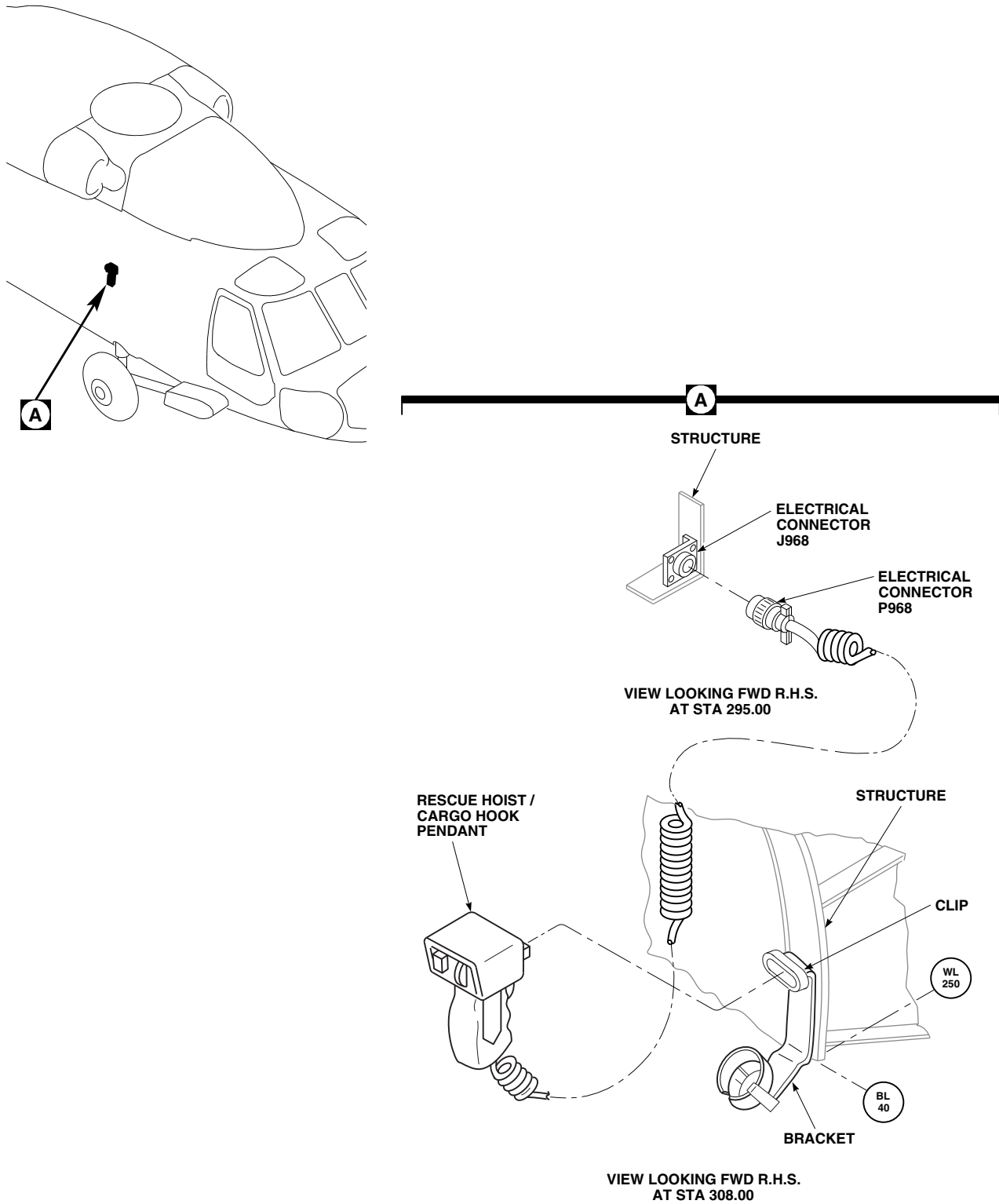
14-4-21.2.1. Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Disconnect rescue hoist/cargo hook pendant electrical connector, P968, from receptacle (Figure 14-4-21-2, Detail A). Install dummy connector in electrical connector, J968.
- c. Install protective cap on electrical connector, P968.
- d. Unlatch clip and remove rescue hoist/cargo hook pendant from bracket.

14-4-21.2.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).

- b. Position rescue hoist/cargo hook pendant in bracket and latch clip (Figure 14-4-21-2, Detail A).
- c. Remove protective cap from electrical connectors, P968.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Remove dummy connector and connect rescue hoist/cargo hook pendant electrical connector, P968, to electrical receptacle, J968.
- f. Perform operational check of rescue hoist system (PARA 14-2-1).

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SAFIGURE 14-4-21-2. REPLACE RESCUE HOIST/CARGO HOOK PENDANT **HOIST, 42315-10**

14-4-21-4

14-4-22 WIRING HARNESS **HOIST, BL-27100-71** **HOIST, 42315-10** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-22.1 Replace Wiring Harness	14-4-22-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 14-2-1
- PARA 14-4-20
- PARA 14-4-24

14-4-22.1 **Replace Wiring Harness.**

14-4-22.1.1. **Remove.**

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).
- Remove support arm fairings and cuff fairing (PARA 14-4-20).
- Tag and disconnect electrical connector, P969 from J969, on airframe fitting (Figure 14-4-22-1, Detail A).
- Tag and disconnect the following electrical connectors on hoist:
 - P962 from J962.
 - P963 from J963.
 - P964 from J964.

- Install protective caps and plugs on electrical connectors.
- Remove rescue hoist light lamp (PARA 14-4-24).
- Carefully pull hoist light wiring through grommets in support and rescue hoist light assembly.
- Remove screws and washers from clamps on wiring harness.
- Identify clamp locations and transfer clamps to replacement harness.
- Remove harness from hoist support.

14-4-22.1.2. **Install.**

- Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

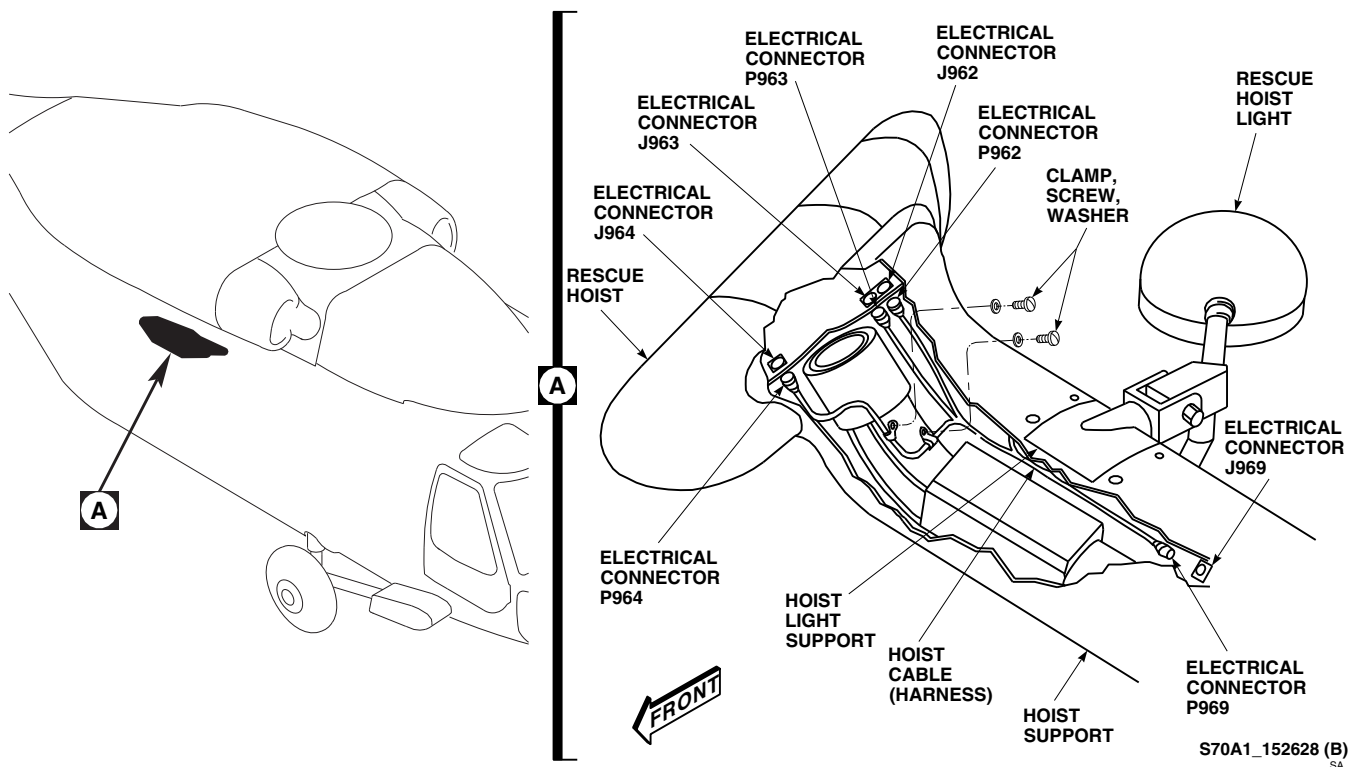


FIGURE 14-4-22-1. REPLACE WIRING HARNESS

- b. Remove connector tags from old wiring harness and transfer to replacement harness.
- NOTE**
- When installing harness to support, make sure that harness is located within clamps in such a way as to prevent damage to harness by fairings.
- c. Position harness with clamps on hoist support and install using screws and washers (Figure 14-4-22-1, Detail A).
 - d. Carefully guide hoist light wiring through grommets in support.
 - e. Install rescue hoist light lamp (PARA 14-4-24).
 - f. Remove protective caps and plugs from electrical connectors.
 - g. Inspect and treat electrical connectors (PARA 1-7-20).
 - h. Connect the following electrical connectors to hoist and remove tags. Lockwire, Item 197, Appendix D, connectors.
 - (1) P962 to J962.
 - (2) P963 to J963.
 - (3) P964 to J964.
 - i. Connect electrical connector, P969 to J969, on airframe fitting.
 - j. Install support arm fairings and cuff fairing (PARA 14-4-20).
 - k. Perform operational check of rescue hoist system (PARA 14-2-1).

14-4-23 RESCUE HOIST LIGHT ASSEMBLY **HOIST, BL-27100-71**
HOIST, 42315-10 .

PARAGRAPH BREAKDOWN

		PAGE		PAGE
14-4-23.1	Replace Rescue Hoist Light Assembly HOIST, BL-27100-71	14-4-23-1	14-4-23.2	Replace Rescue Hoist Light Assembly HOIST, 42315-10

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Spring Scale
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry Lubricant, Item 123A, Appendix D
- Grease, Item 151B, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 14-4-24

14-4-23.1 Replace Rescue Hoist Light Assembly **HOIST, BL-27100-71** .

14-4-23.1.1 Remove.

- Remove rescue hoist light lamp (PARA 14-4-24).
- Remove rescue hoist light assembly wires from conduit in light assembly (Figure 14-4-23-1, Detail A).
- Remove bolt, washers, lockwasher, nylon washers, castellated nut, and cotter pin securing rescue hoist light assembly to clevis. Remove rescue hoist light assembly from clevis (Figure 14-4-23-1, Detail A). Discard cotter pin.

14-4-23.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install rescue hoist light assembly wires in conduit of rescue hoist light assembly (Figure 14-4-23-1, Detail A).
- Install rescue hoist light assembly to clevis as follows:
 - Apply dry lubricant, Item 123A, Appendix D, to both sides of nylon washers.
 - Secure rescue hoist light assembly to clevis with bolt, washers, nylon washers, lockwasher, and castellated nut. **TORQUE NUT TO 70 - 90 INCH-POUNDS.**

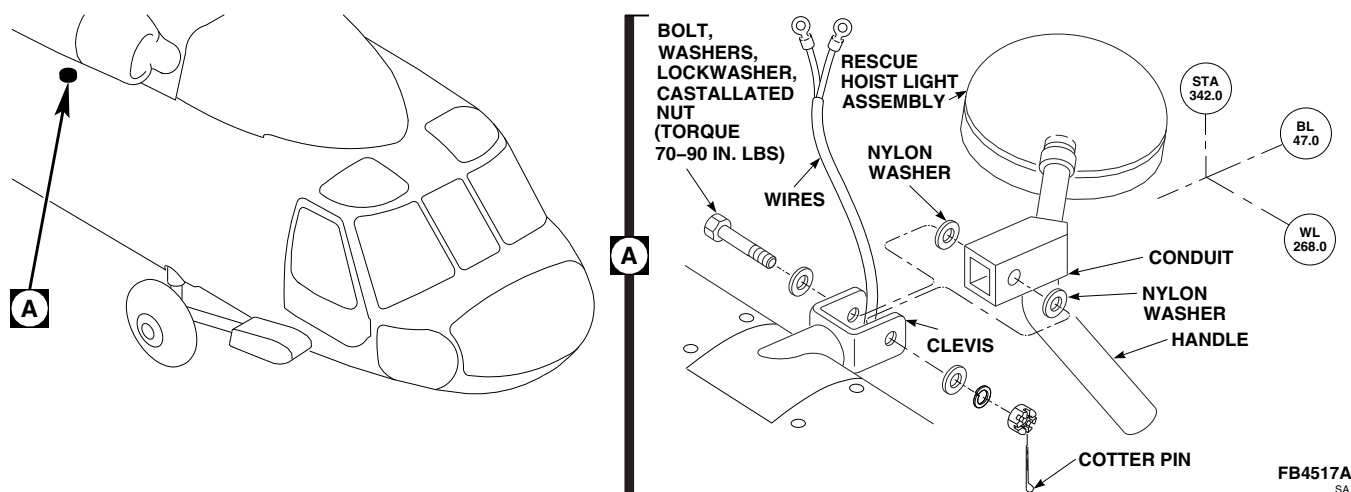


FIGURE 14-4-23-1. REPLACE RESCUE HOIST LIGHT ASSEMBLY **HOIST, BL-27100-71**

(3) Move light assembly all around to work out rough spots.

(4) **RETORQUE NUT TO 70 - 90 INCH-POUNDS.** Install cotter pin.

d. Install rescue hoist light lamp (PARA 14-4-24).

14-4-23.2 Replace Rescue Hoist Light Assembly **HOIST, 42315-10**

14-4-23.2.1 Remove.

- Remove rescue hoist light lamp (PARA 14-4-24).
- Remove wires from conduit in rescue hoist light assembly (Figure 14-4-23-2, Detail A).
- Remove bolt, washers, nylon washers, lockwasher, and nut securing rescue hoist light assembly to adapter bracket and remove rescue hoist light assembly.

14-4-23.2.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install wires in conduit of rescue hoist light assembly (Figure 14-4-23-2, Detail A).

c. Install rescue hoist light assembly to adapter bracket as follows:

- Apply grease, Item 151B, Appendix D, to both sides of nylon washers.
- Secure rescue hoist light assembly to adapter bracket with bolt, washers, nylon washers, lockwasher, and nut. **TORQUE BOLT TO 25 - 45 INCH-POUNDS.**
- Connect spring scale to handle.
- Pull handle with spring scale in all directions. **FORCE SHALL BE BETWEEN 10 - 20 POUNDS.**
- If force is within limits, tighten nut to next castellation and install cotter pin.
- If force is not between 10 - 20 pounds, adjust torque on bolt, not to exceed limits, and repeat step c. (3) and step c. (4).
- If correct force still cannot be obtained, replace bolt and repeat step c. (2) through step c. (4).

d. Install rescue hoist light lamp (PARA 14-4-24).

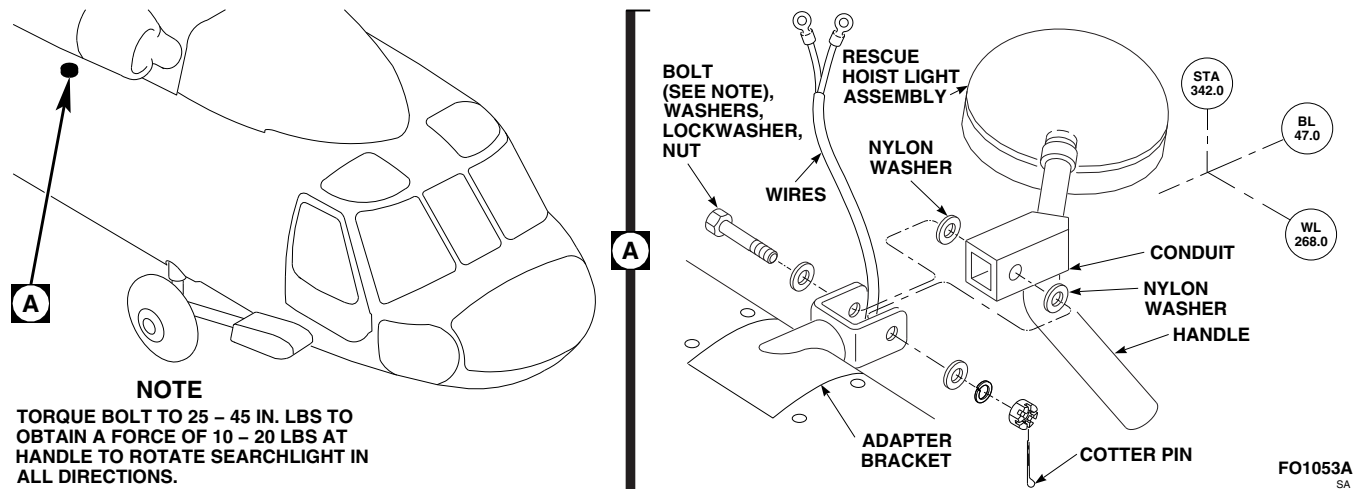


FIGURE 14-4-23-2. REPLACE RESCUE HOIST LIGHT ASSEMBLY **HOIST, 42315-10**

14-4-24
RESCUE HOIST LIGHT LAMP

HOIST, BL-27100-71

HOIST, 42315-10

PARAGRAPH BREAKDOWN

		PAGE			PAGE
14-4-24.1	Replace Rescue Hoist Light Lamp	14-4-24-1	14-4-24.2	Replace Rescue Hoist Light Lamp	14-4-24-1
	HOIST, BL-27100-71			HOIST, 42315-10	

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 14-2-1

14-4-24.1
Replace Rescue Hoist Light Lamp

HOIST, BL-27100-71

14-4-24.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Loosen wingnut screws and remove lens retainer, clear lens, lens holder, infrared lens, lens retainer, and gasket (Figure 14-4-24-1, Detail A).
- Carefully lift lamp out of housing to get to lamp terminals.
- Tag wires and remove screws and washers securing power leads to lamp terminals.

14-4-24.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Connect power leads to lamp terminals with screws and washers (Figure 14-4-24-1, Detail A). Remove tags.

- Position lamp in housing.

NOTE

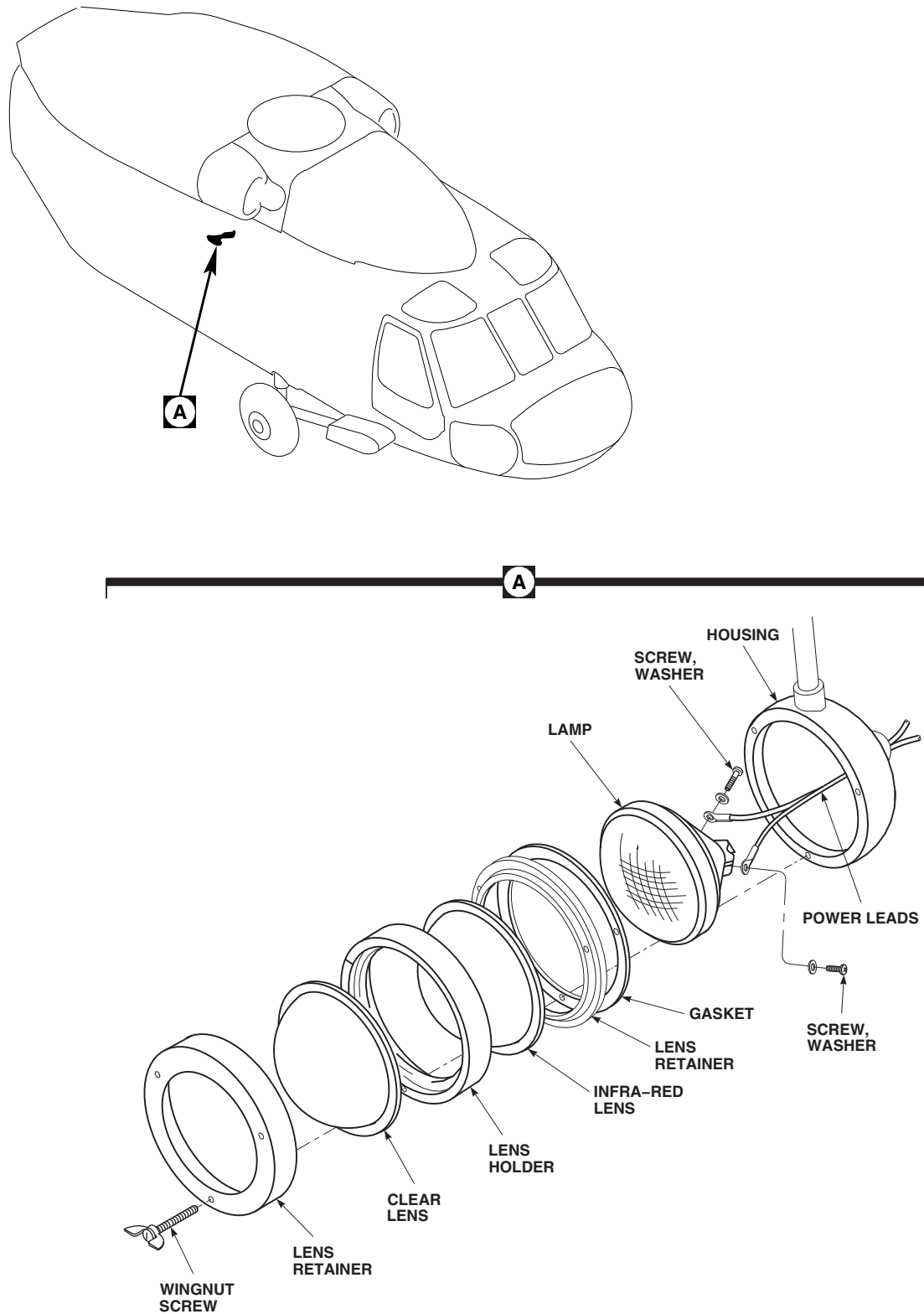
- Gasket may protrude beyond retainer approximately 0.060 inches.
- Position gasket, lens retainer, infrared lens, lens holder, clear lens, and lens retainer on housing and secure with wingnut screws.
 - Make sure area is clean and free of foreign material.
 - Perform operational check of rescue hoist system (PARA 14-2-1).

14-4-24.2
Replace Rescue Hoist Light Lamp

HOIST, 42315-10

14-4-24.2.1. Remove.

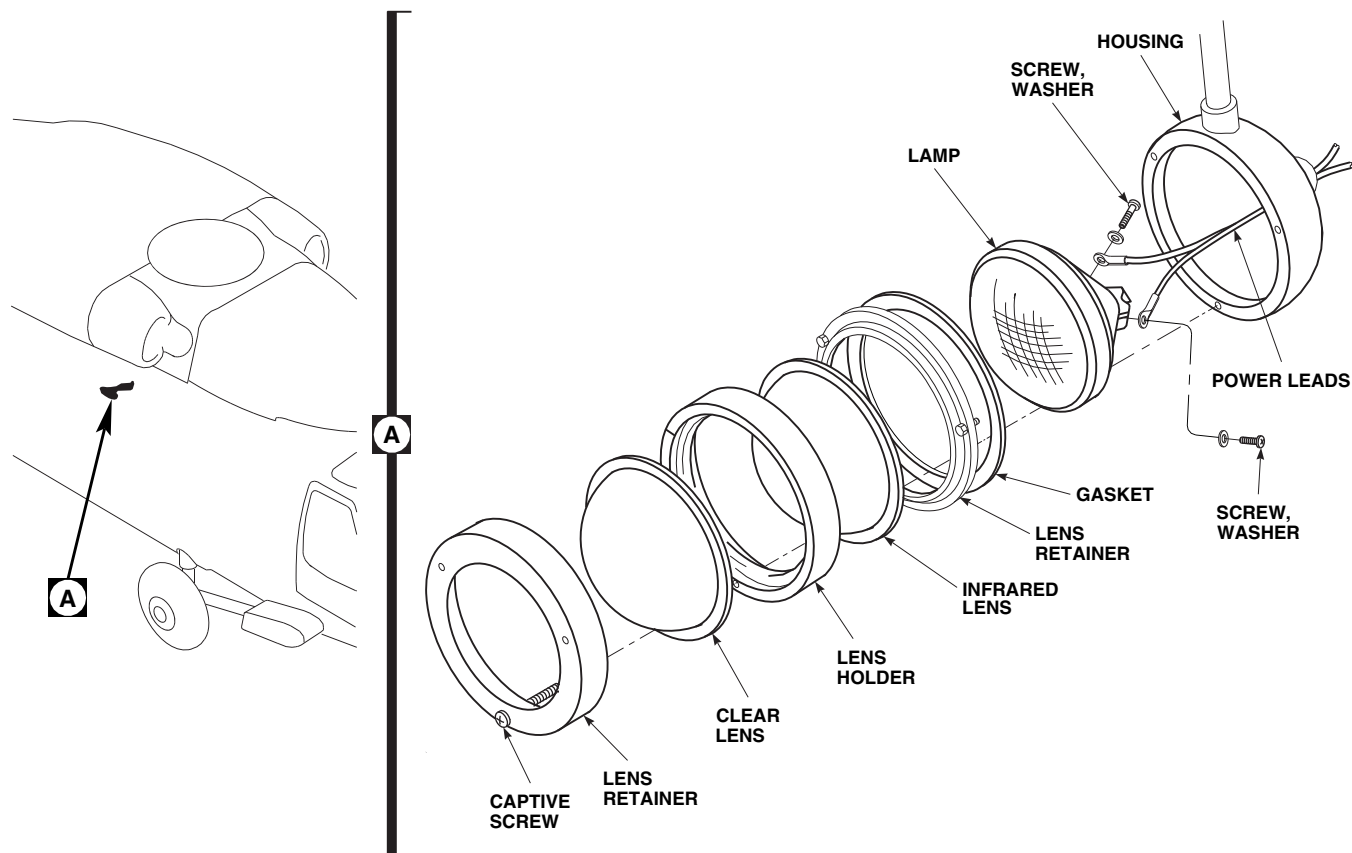
- Turn off all electrical power (PARA 1-6-16).
- Loosen captive screws and remove lens retainer, clear lens, lens holder, infrared lens,



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FIGURE 14-4-24-1. REPLACE RESCUE HOIST LIGHT LAMP **HOIST, BL-27100-71**

14-4-24-2



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FIGURE 14-4-24-2. REPLACE RESCUE HOIST LIGHT LAMP **HOIST, 42315-10**

- lens retainer, and gasket (Figure 14-4-24-2, Detail A).
- c. Carefully lift lamp out of housing to access lamp terminals.
- d. Tag wires and remove screws and washers securing power leads to lamp terminals.

14-4-24.2.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).

- b. Connect power leads to lamp terminals with screws and washers. Remove tags (Figure 14-4-24-2, Detail A).
- c. Position lamp in housing.
- d. Position gasket, lens retainer, infrared lens, lens holder, clear lens, and lens retainer on housing and secure with captive screws.
- e. Perform operational check of rescue hoist system (PARA 14-2-1).

14-4-24-3/(14-4-24-4 Blank)

14-4-25
AUX RESCUE HOIST CONTROL PANEL

HOIST, BL-27100-71

HOIST, 42315-10

PARAGRAPH BREAKDOWN

	PAGE
14-4-25.1 Replace Aux Rescue Hoist Control Panel	14-4-25-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 14-2-1

14-4-25.1 Replace Aux Rescue Hoist Control Panel.

14-4-25.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screws and washers, and remove AUX RESCUE HOIST control panel from overhead support (Figure 14-4-25-1, Detail A).
- Disconnect electrical connector, P352, from J352.
- Install protective caps and plugs on electrical connectors.
- Remove nut, lockwasher, and washer securing bonding jumper to stud on control panel.
- Remove control panel.

14-4-25.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

- Clean surfaces around mounting holes using machinery towel, Item 211, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D, for good electrical ground contact.
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector J352 to P352 (Figure 14-4-25-1, Detail A).
- Install bonding jumper to AUX RESCUE HOIST control panel stud with nut, lockwasher, and washer.
- Install AUX RESCUE HOIST control panel in overhead support with screws and washers.
- Perform operational check of rescue hoist system (PARA 14-2-1).

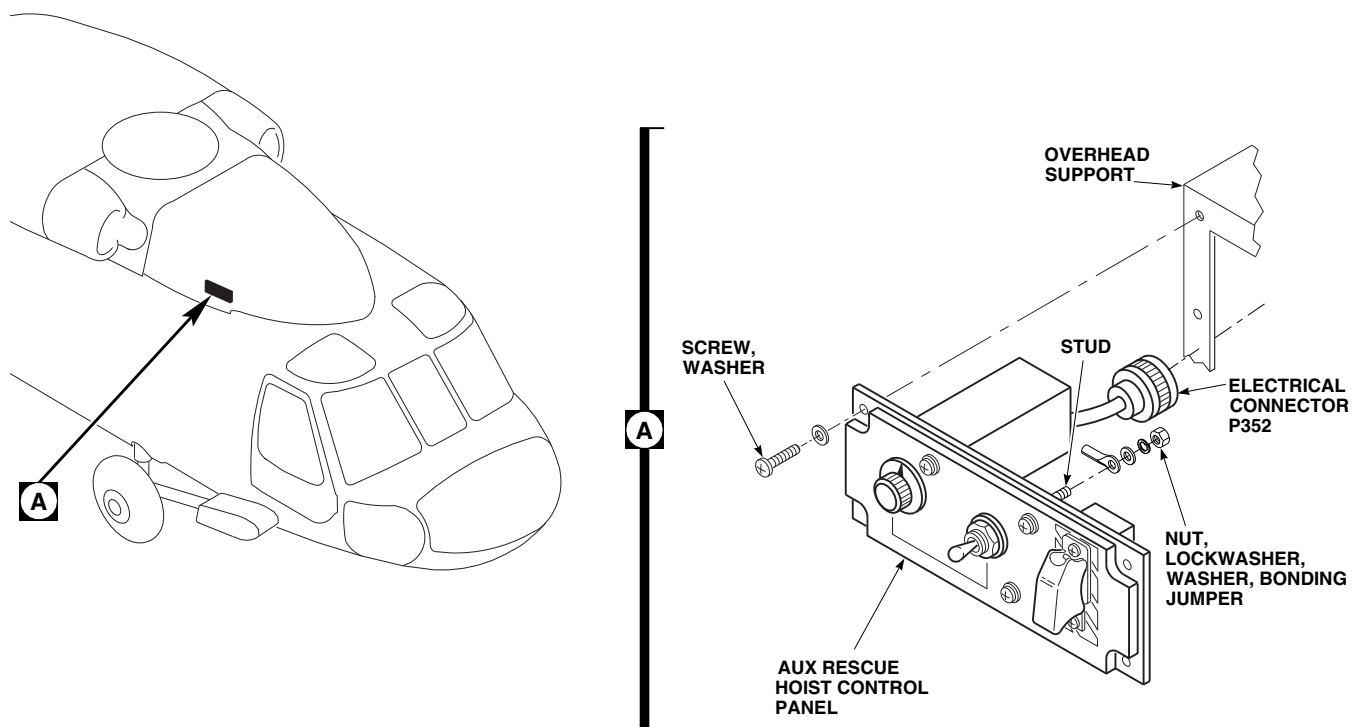
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FIGURE 14-4-25-1. REPLACE AUX RESCUE HOIST CONTROL PANEL

14-4-26 **BACKUP CONTROL PANEL** **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-26.1 Replace Backup Control Panel	14-4-26-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 195, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-6-5
- PARA 14-2-1

14-4-26.1 **Replace Backup Control Panel.**

14-4-26.1.1 **Remove.**

- Turn off all electrical power (PARA 1-6-16).
- Remove soundproofing panels (PARA 2-4-69).
- Remove bolts, lockwashers, and washers securing control panel to support pan (Figure 14-4-26-1, Detail A).
- Carefully lower control panel to gain access to electrical connectors.
- Disconnect electrical connectors, P970 from J1 and P971 from J2, from control panel.
- Install protective caps and plugs on electrical connectors.

14-4-26.1.2 **Install.**

- Turn off all electrical power (PARA 1-6-16).
- Clean mounting surface using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface using clean, dry cheesecloth, Item 90, Appendix D.
- Inspect chemical conversion coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches apply chemical conversion coating (PARA 2-6-5).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).

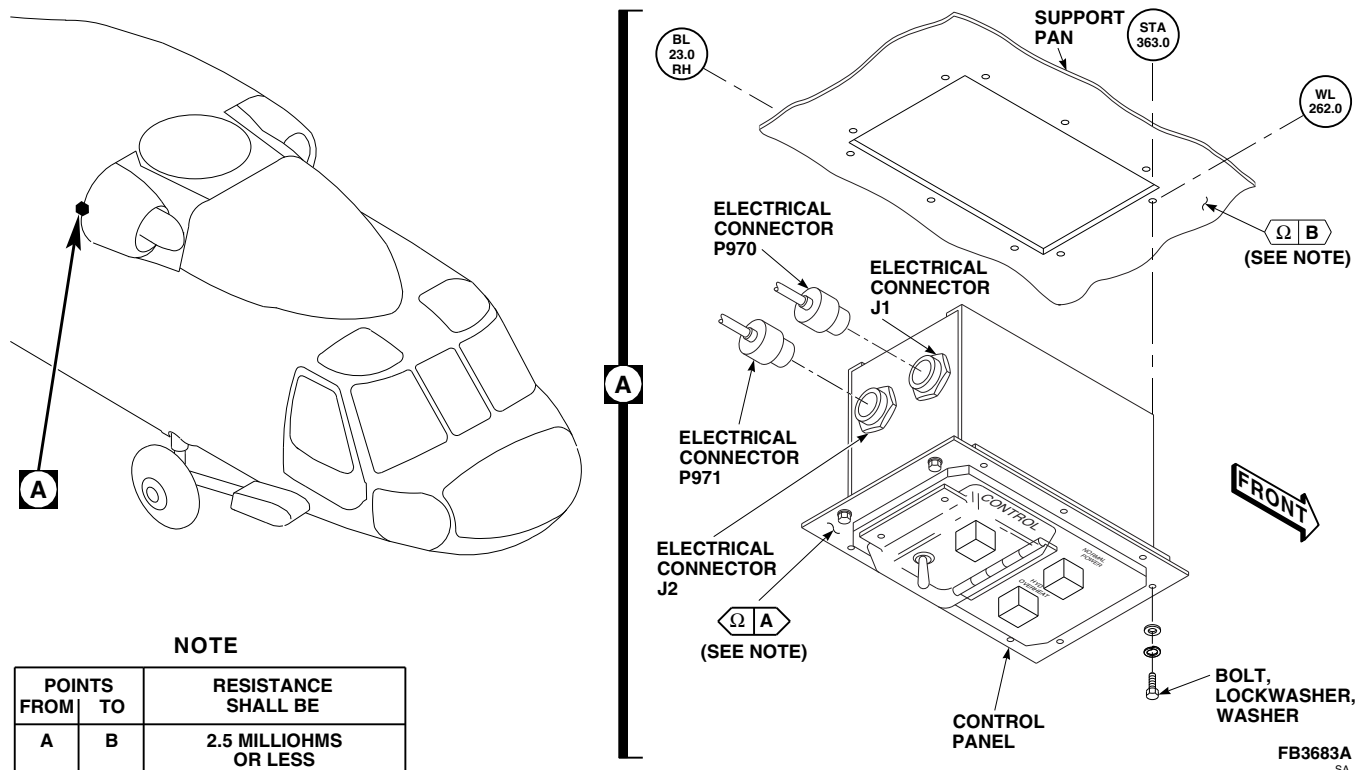


FIGURE 14-4-26-1. REPLACE BACKUP CONTROL PANEL

- | | |
|---|--|
| <p>f. Connect electrical connectors, P970 to J1 and P971 to J2, to control panel (Figure 14-4-26-1, Detail A).</p> <p>g. Position control panel on support pan and secure with bolts, lockwashers, and washers.</p> <p>h. Using lockwire, Item 195, Appendix D, lockwire switch guard cover to bolts.</p> <p>i. Perform EME operational check as follows:</p> <p>(1) Check resistance measured from control panel to support pan using digital low-</p> | <p>resistance ohmmeter. RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.</p> <p>(2) If resistance is greater than 2.5 milliohms, remove control panel and apply chemical conversion coating (PARA 2-6-5).</p> <p>j. Perform operational check of rescue hoist system (PARA 14-2-1).</p> <p>k. Make sure area is clean and free of foreign material.</p> <p>l. Install soundproofing panels (PARA 2-4-69).</p> |
|---|--|

14-4-27 RESCUE HOIST CONTROL BOX **HOIST, 42315-10** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-27.1 Replace Rescue Hoist Control Box	14-4-27-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit’s

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 195, Appendix D

- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 14-2-1

14-4-27.1 Replace Rescue Hoist Control Box.

14-4-27.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screws, drilled screws, lockwashers, and washers, and remove rescue hoist control box from overhead support.
- Disconnect electrical connectors, P965 from J1 and P966 from J2, from rescue hoist control box (Figure 14-4-27-1, Detail A).
- Install protective caps and plugs on electrical connectors.

14-4-27.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Clean surfaces around mounting holes using machinery towel, Item 211, Appendix D,

moistened with dry-cleaning solvent, Item 124, Appendix D, for good electrical ground contact.

- Position rescue hoist control box in overhead support (Figure 14-4-27-1, Detail A).
- Install forward side of rescue hoist control box onto overhead support with screws, lockwashers, and washers.
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connectors, P965 to J1 and P966 to J2.
- Install rear side of rescue hoist control box onto overhead support with drilled screws, lockwashers, and washers.

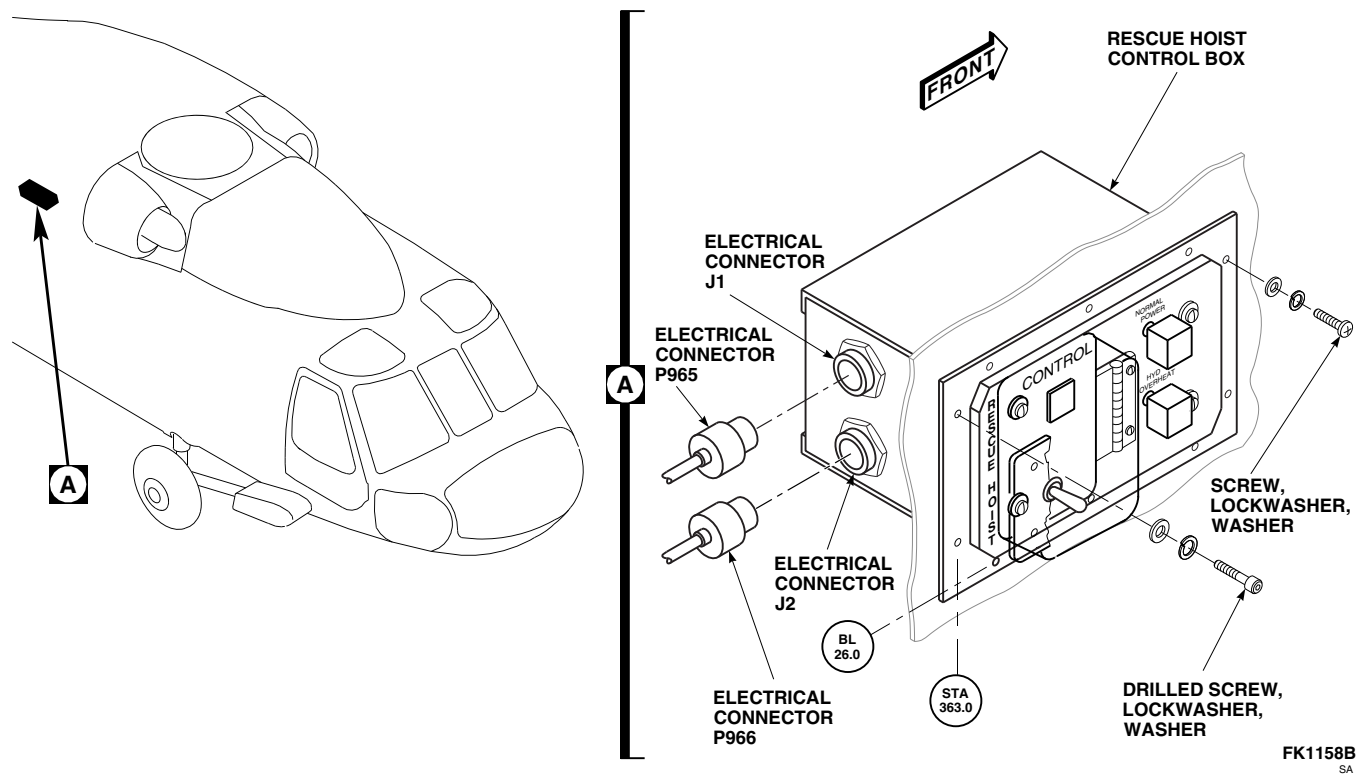


FIGURE 14-4-27-1. REPLACE RESCUE HOIST CONTROL BOX

- i. Using lockwire, Item 195, Appendix D, install safety cover over drilled screws.
- j. Perform operational check of rescue hoist system (PARA 14-2-1).

14-4-28 SWITCH GUARD COVER **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-28.1 Replace Switch Guard Cover	14-4-28-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Lockwire, Item 195, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69

14-4-28.1 Replace Switch Guard Cover.

14-4-28.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove soundproofing panels (PARA 2-4-69).
- Loosen captive screws and remove switch guard cover from control panel (Figure 14-4-28-1, Detail A).

14-4-28.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

- Install switch guard cover on control panel and secure with captive screws (Figure 14-4-28-1, Detail A).
- Lockwire, Item 195, Appendix D, switch guard cover to bolts.
- Make sure area is clean and free of foreign material.
- Install soundproofing panels (PARA 2-4-69).

14-4-29 INDICATOR AND BACKUP CONTROL POWER LAMPS **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-29.1 Replace Indicator and Backup Control Power Lamps	14-4-29-1

INITIAL SETUP

- PARA 14-2-1

Tools

- PARA 14-4-28

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 2-4-69

14-4-29.1 Replace Indicator and Backup Control Power Lamps.

NOTE

Rescue hoist control panel has three indicator lamps, POWER, NORMAL POWER and HYD OVERHEAT. All lamps are replaced the same, except as noted.

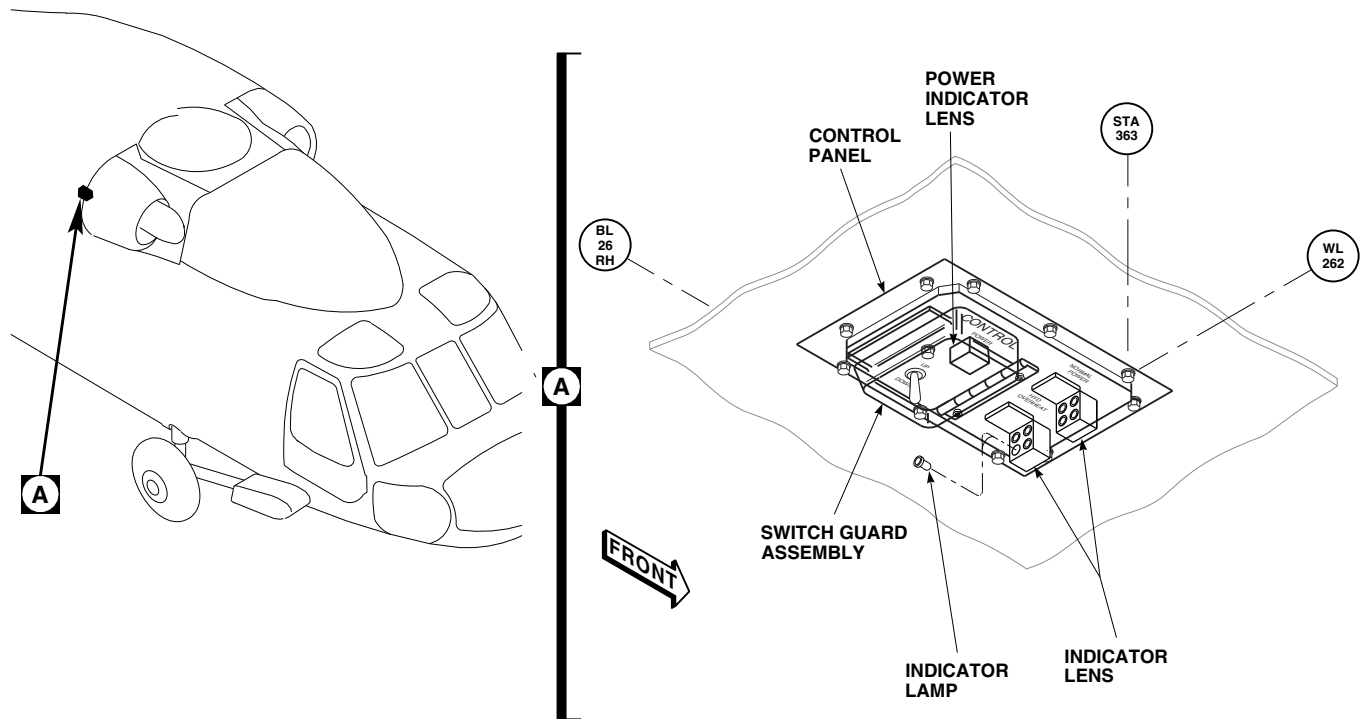
14-4-29.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove soundproofing panels (PARA 2-4-69).
- If POWER lamp is to be replaced, remove switch guard cover (PARA 14-4-28).
- Carefully pull indicator lens out from control panel and hinge lens down (Figure 14-4-29-1, Detail A).

- Remove indicator lamp from indicator lens.

14-4-29.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install indicator lamp in indicator lens (Figure 14-4-29-1, Detail A).
- Hinge lens up and push into control panel.
- If POWER lamp was replaced, install switch guard cover (PARA 14-4-28).
- Perform operational check of rescue hoist (PARA 14-2-1).
- Make sure area is clean and free of foreign material.
- Install soundproofing panels (PARA 2-4-69).



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FIGURE 14-4-29-1. REPLACE INDICATOR AND BACKUP CONTROL POWER LAMPS

14-4-30 **BACKUP CONTROL PANEL INFORMATION PLATE** **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-30.1 Replace Backup Control Panel Information Plate	14-4-30-1

INITIAL SETUP

- PARA 14-2-1

Tools

- PARA 14-4-28

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 2-4-69

14-4-30.1 **Replace Backup Control Panel Information Plate.**

- b. Position information plate on control panel and secure with captive screws (Figure 14-4-30-1, Detail A).

14-4-30.1.1. **Remove.**

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove soundproofing panels (PARA 2-4-69).
- c. Remove switch guard cover (PARA 14-4-28).
- d. Loosen captive screws and remove information plate from control panel (Figure 14-4-30-1, Detail A).

- c. Install switch guard cover (PARA 14-4-28).
- d. Perform operational check of rescue hoist (PARA 14-2-1).
- e. Make sure area is clean and free of foreign material.

14-4-30.1.2. **Install.**

- a. Turn off all electrical power (PARA 1-6-16).

- f. Install soundproofing panels (PARA 2-4-69).

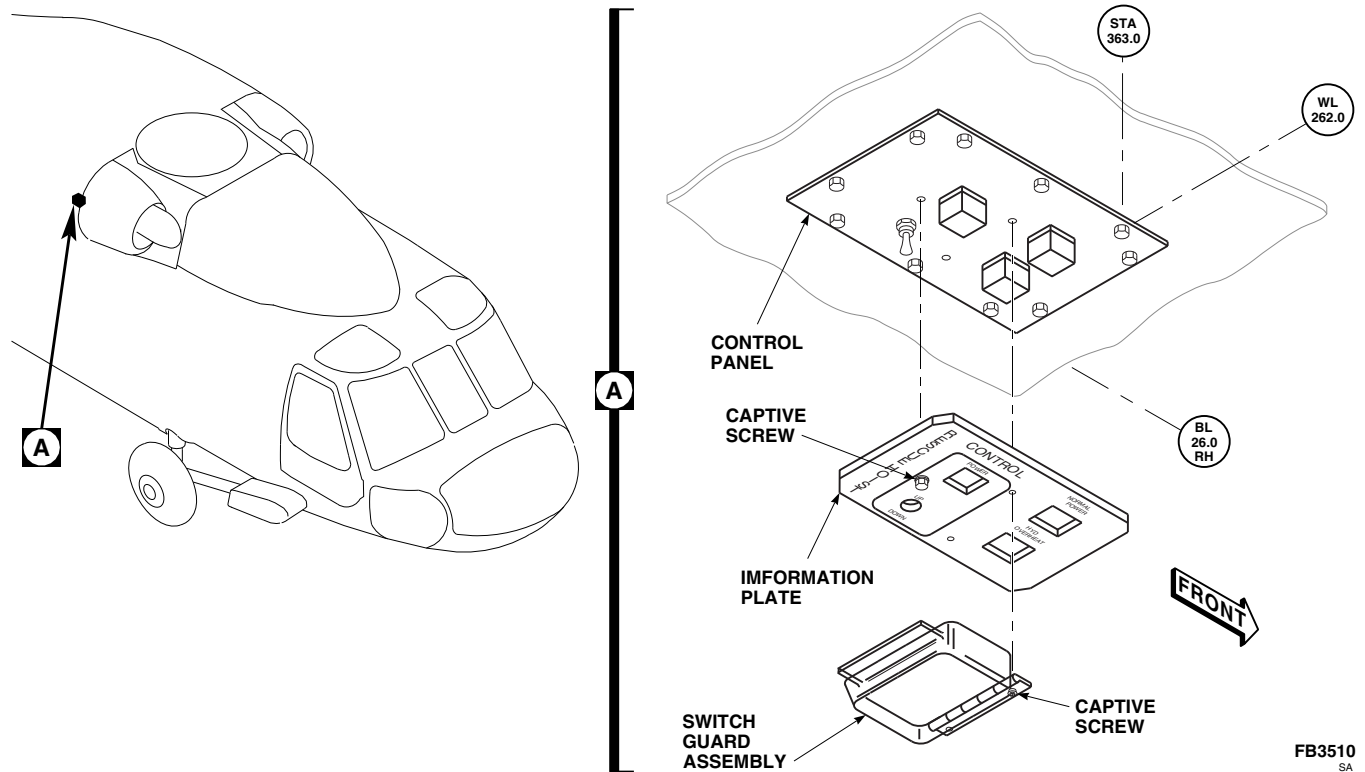


FIGURE 14-4-30-1. REPLACE BACKUP CONTROL PANEL INFORMATION PLATE

14-4-31 AUXILIARY HOIST RELAY PANEL **HOIST, BL-27100-71** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-31.1 Replace Auxiliary Hoist Relay Panel	14-4-31-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 14-2-1

14-4-31.1 **Replace Auxiliary Hoist Relay Panel.**

14-4-31.1.1 **Remove.**

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connectors, P353 from J353 and P357 from J357, from auxiliary hoist relay panel (Figure 14-4-31-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove bolts and washers, and remove auxiliary hoist relay panel from overhead support.

14-4-31.1.2 **Install.**

- Turn off all electrical power (PARA 1-6-16).
- Clean surfaces around mounting holes using machinery towel, Item 211, Appendix D,

moistened with dry-cleaning solvent, Item 124, Appendix D, for good electrical ground contact.

- Install auxiliary hoist relay panel in overhead support with bolts and washers (Figure 14-4-31-1, Detail A). **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- Remove protective caps and plugs on electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connectors, P353 to J353 and P357 to J357, to auxiliary hoist relay panel.
- Perform operational check of rescue hoist system (PARA 14-2-1).

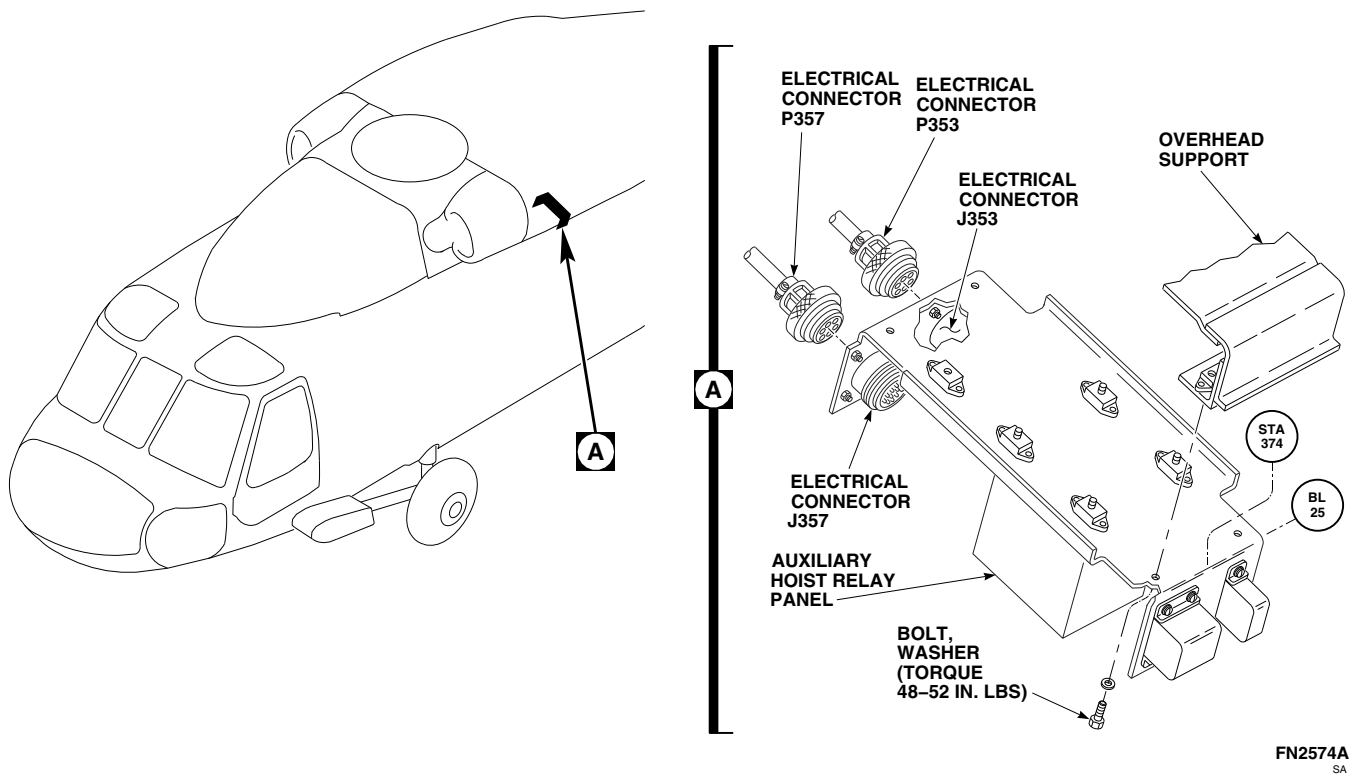


FIGURE 14-4-31-1. REPLACE AUXILIARY HOIST RELAY PANEL

14-4-32 RESCUE HOIST **HOIST BL-29900-30** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-32.1 Replace Rescue Hoist	14-4-32-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drain Tube, Locally-Made
- Drill Set
- Flashlight
- Fluorescent Penetrant Inspection Kit
- Inspection Mirror
- Nonmetallic Scraper
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Chemical Conversion Coating, Item 118, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

- Lockwire, Item 196, Appendix D
- Paint Remover, Item 231, Appendix D
- Plastic Tubing, 3-inches long (0.133 or 0.186-inch inside diameter) (for Drain Tube, Locally-Made)
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 296, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-1
- PARA 2-6-5
- PARA 14-2-1
- PARA 14-4-37
- PARA 14-4-41

14-4-32.1 Replace Rescue Hoist.

14-4-32.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

- Remove support arm fairings and connector cover (PARA 14-4-37).
- Remove explosive cartridge (PARA 14-4-41).

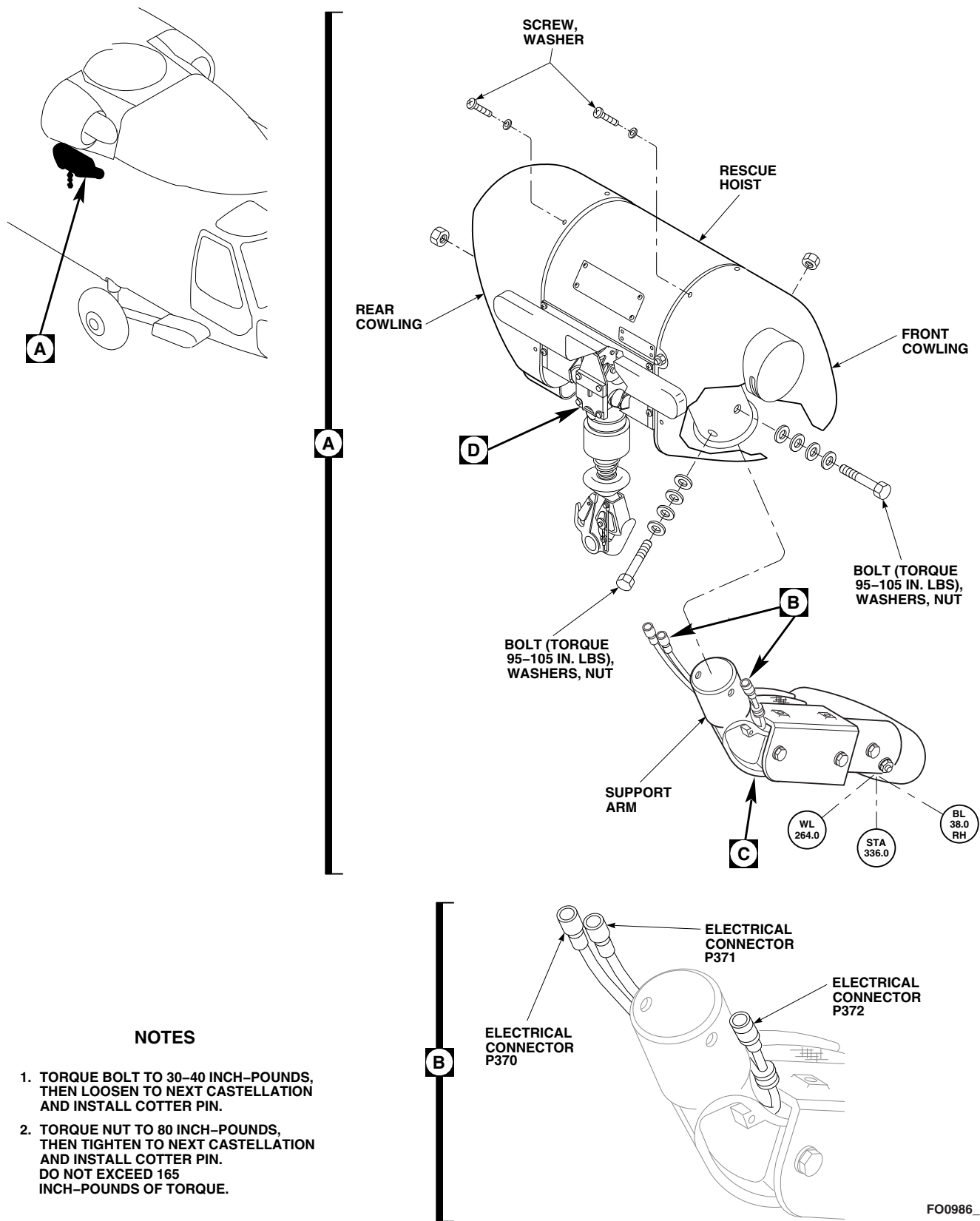
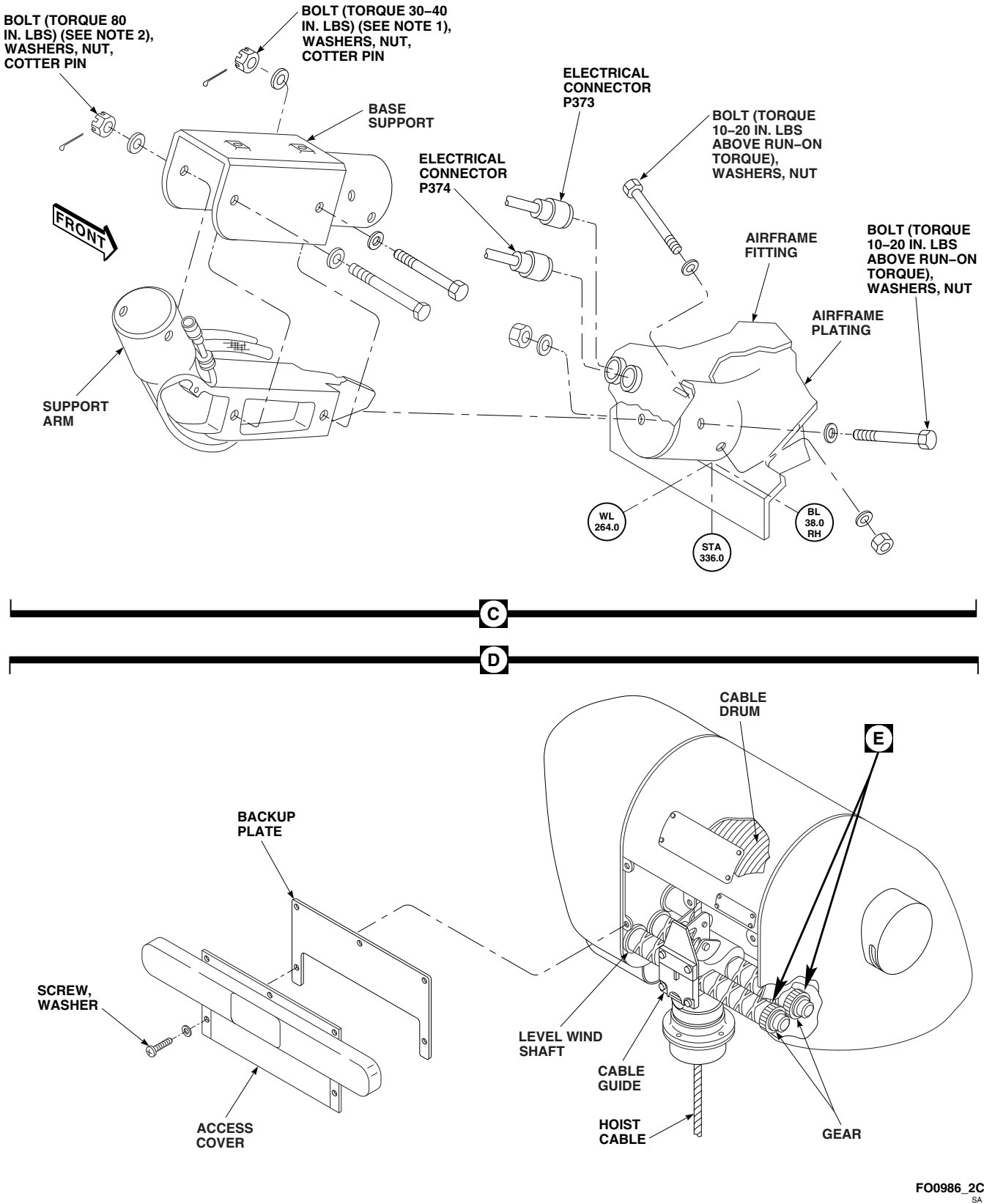
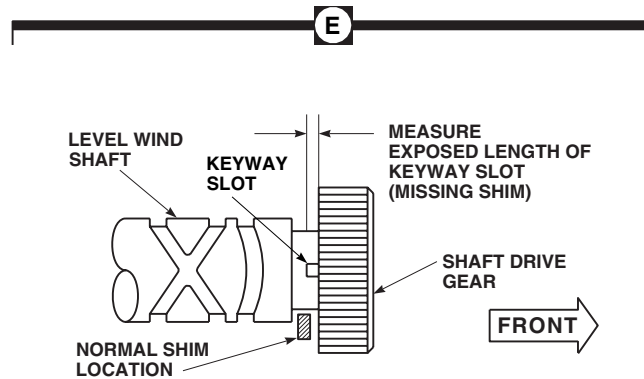


FIGURE 14-4-32-1. REPLACE RESCUE HOIST (SHEET 1 OF 3)



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- | | |
|---|---|
| d. Disconnect electrical connectors P370 from J1, P371 from J5, and P372 from J8 (Figure 14-4-32-1, Sheet 1, Detail B). | airframe fitting (Figure 14-4-32-1, Sheet 2, Detail C). |
| e. Install protective caps and plugs on electrical connectors. | g. Install protective caps and plugs on electrical connectors. |
| f. Disconnect electrical connectors, P373, from J373 and P374 from J374, located on the | h. Remove screws, washers, clamps, and electrical harness from support arm. |

WARNING

Injury to personnel and damage to equipment will result if rescue hoist is not supported during removal. Rescue hoist weighs 105 pounds. Use two assistants to support rescue hoist during removal.

- i. Remove bolts, washers, and nuts securing support arm to base support. Remove hoist and support arm from helicopter.

14-4-32.1.2 Disassemble.

- a. Remove cuff fairing from rescue hoist (PARA 14-4-37).
- b. Remove bolts, washers, and nuts securing support arm to hoist (Figure 14-4-32-1, Sheet 1, Detail A). Remove support arm.
- c. Remove bolts, washers, and nuts securing base support to airframe fitting. Remove base support (Figure 14-4-32-1, Sheet 2, Detail C).
- d. Using nonmetallic scraper, remove sealing compound from base support and airframe fitting mating surface.

14-4-32.1.3 Inspect.

- a. Turn off all electrical power (PARA 1-6-16).

NOTE

Viewed from below, the level wind shafts are two cross-cut steel shafts in parallel with each other with a cable guide between them.

- b. Visually inspect hoist to determine if one or more shims are present between the front ends of the level wind shafts and the shaft drive gears (Figure 14-4-32-1, Sheet 3, Detail E). If no shims are present, proceed as follows:
 - (1) Remove screws and washers that secure the access cover and plate to the hoist. Remove cover and backup plate (Figure 14-4-32-1, Sheet 2, Detail D).

NOTE

It may be necessary to use a flashlight and inspection mirror if the hoist is stopped with keyways facing inboard.

- (2) Measure keyway slots in the forward ends of level wind shafts (Figure 14-4-32-1, Sheet 3, Detail E).
- (3) If measurement of exposed length of keyway is $\frac{1}{8}$ -inch or less, install backup plate and access cover on rescue hoist with screws and washers (Figure 14-4-32-1, Sheet 2, Detail D). Using lockwire, Item 196, Appendix D, lockwire screws.
- (4) If measurement of exposed length of keyway is greater than $\frac{1}{8}$ -inch, replace hoist.
- c. Inspect base support for cracks as follows (Figure 14-4-32-2, Sheet 1 and Sheet 2):
 - (1) Using paint remover, Item 231, Appendix D, remove paint, sealing compound, and primer from inside and outside surfaces of cylindrical section that mates with fitting and channel section wall, Area B and Area A, respectively.

NOTE

Pay particular attention to areas around boltholes during fluorescent penetrant inspection.

- (2) Using fluorescent penetrant kit, fluorescent penetrant inspect, Type 1, Method B, Method C or Method D, Sensitivity Level 3 or Level 4, on cylindrical areas for cracks (ASTM E1417). **NONE ALLOWED.**
- (3) Visually inspect all other areas for cracks. **NONE ALLOWED.**
- d. Inspect base support for nicks, dents, scratches, pits, and gouges. **NONE ALLOWED.** If damaged, repair nicks, dents, scratches, pits, and gouges using abrasive cloth, Item 9, Appendix D, as follows:

(1) Repair AREA A as follows:

- (a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.010-INCH DEEP. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.**
- (b) **DAMAGE SHALL NOT BE CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLES.**

(c) If damage or repairs are more than limits, replace base support.

(2) Repair AREA B as follows:

- (a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.002-INCH DEEP AND 0.750-INCH DIAMETER.**

(b) If damage or repairs are more than limits, replace base support.

e. Inspect base support for corrosion, and repair using abrasive cloth, Item 9, Appendix D, as follows:

(1) Repair AREA A as follows:

- (a) **BLEND OUT CORROSION UP TO 0.0050-INCH DEEP OVER 100% OF SURFACE AREA.**

(b) If damage or repairs are more than limits, replace base support.

(2) Repair AREA B as follows:

- (a) **BLEND OUT CORROSION UP TO 0.0100-INCH DEEP OVER 100% OF INNER DIAMETER, AND UP TO 0.002-INCH DEEP INTO THE WALL THICKNESS TO MINIMUM OUTSIDE DIAMETER OF 2.966-INCHES.**

(b) If damage or repairs are more than limits, replace base support.

f. Inspect base support for incorporation of 0.375-inch slot at bottom of base support. If slot is not present, perform the following:

- (1) Add slot to base support as shown (Figure 14-4-32-2, Sheet 2, Detail A).
- (2) Treat repaired area with chemical conversion coating, Item 118, Appendix D.
- (3) Using epoxy primer coating kit, Item 135, Appendix D, prime stripped and repaired area.

g. Inspect support arm for cracks as follows:

- (1) Using paint remover, Item 231, Appendix D, remove paint, sealing compound, and primer from inside and outside surfaces of socket section that mates with fitting and channel section wall, Area B and Area A, respectively.

NOTE

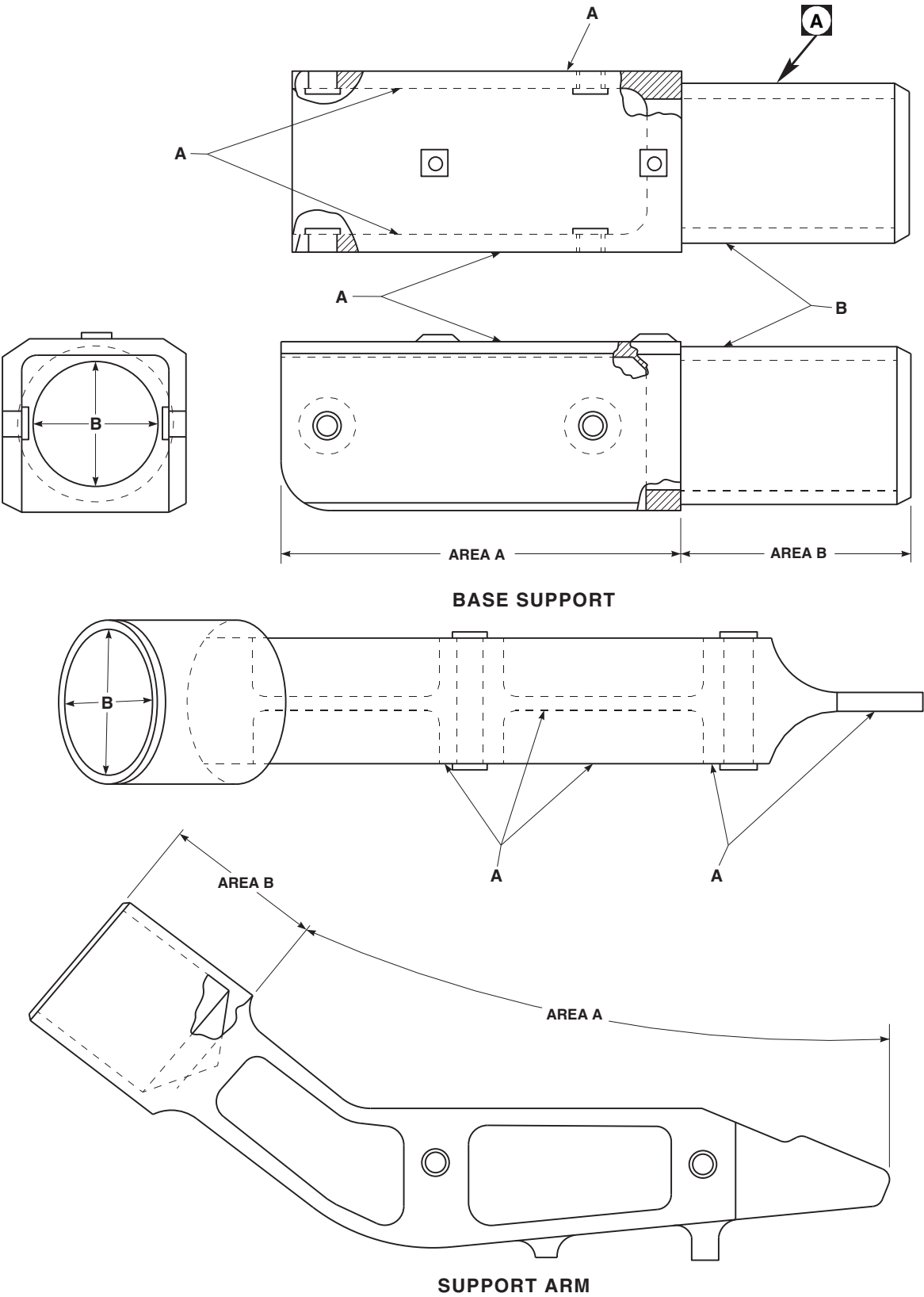
Pay particular attention to areas around boltholes during fluorescent penetrant inspection.

- (2) Using fluorescent penetrant kit, fluorescent penetrant inspect, Type 1, Method B, Method C, or Method D, Sensitivity Level 3 or Level 4, on cylindrical areas for cracks (ASTM E1417). **NONE ALLOWED.**
- (3) Visually inspect all other areas for cracks. **NONE ALLOWED.**

h. Inspect support arm for nicks, dents, scratches, pits, and gouges. Repair nicks, dents, scratches, pits, or gouges using abrasive cloth, Item 9, Appendix D, as follows:

(1) Repair AREA A as follows:

- (a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.010-INCH DEEP. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.**



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FIGURE 14-4-32-2. INSPECT RESCUE HOIST (SHEET 1 OF 2)

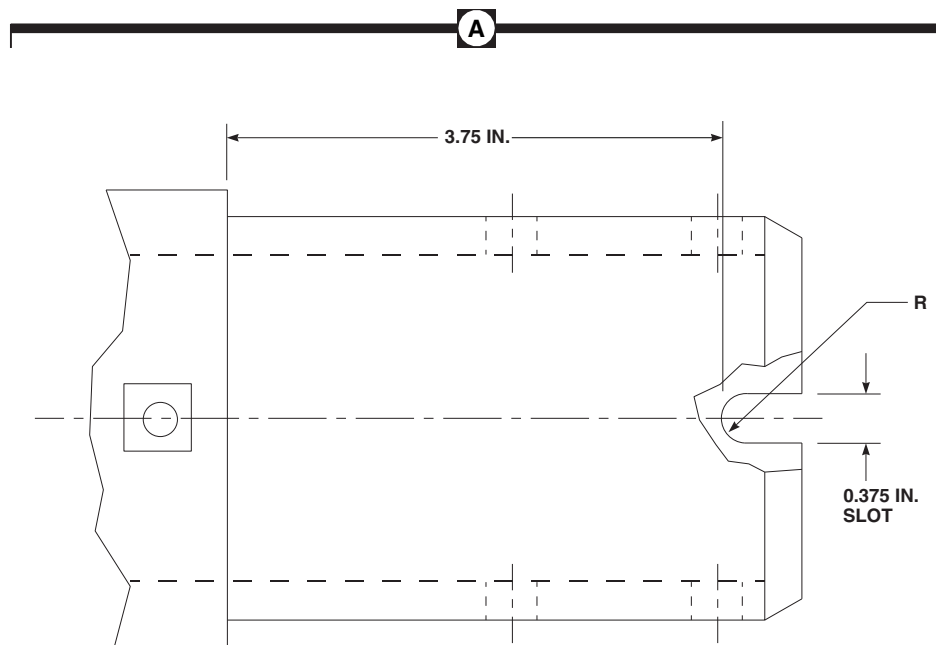
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FIGURE 14-4-32-2. INSPECT RESCUE HOIST (SHEET 2 OF 2)

(b) **DAMAGE SHALL NOT BE CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLES.**

(c) If damage or repairs are more than limits, replace support arm.

(2) Repair AREA B as follows:

(a) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.020-INCH DEEP OVER 100 % OF SURFACE AREA. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.**

(b) If damage or repairs are more than limits, replace support arm.

i. Inspect support arm for corrosion and repair using abrasive cloth, Item 9, Appendix D, as follows:

(1) Repair AREA A as follows:

(a) **BLEND OUT CORROSION UP TO 0.0050-INCH DEEP OVER 100 % OF SURFACE AREA.**

(b) If damage or repairs are more than limits, replace support arm.

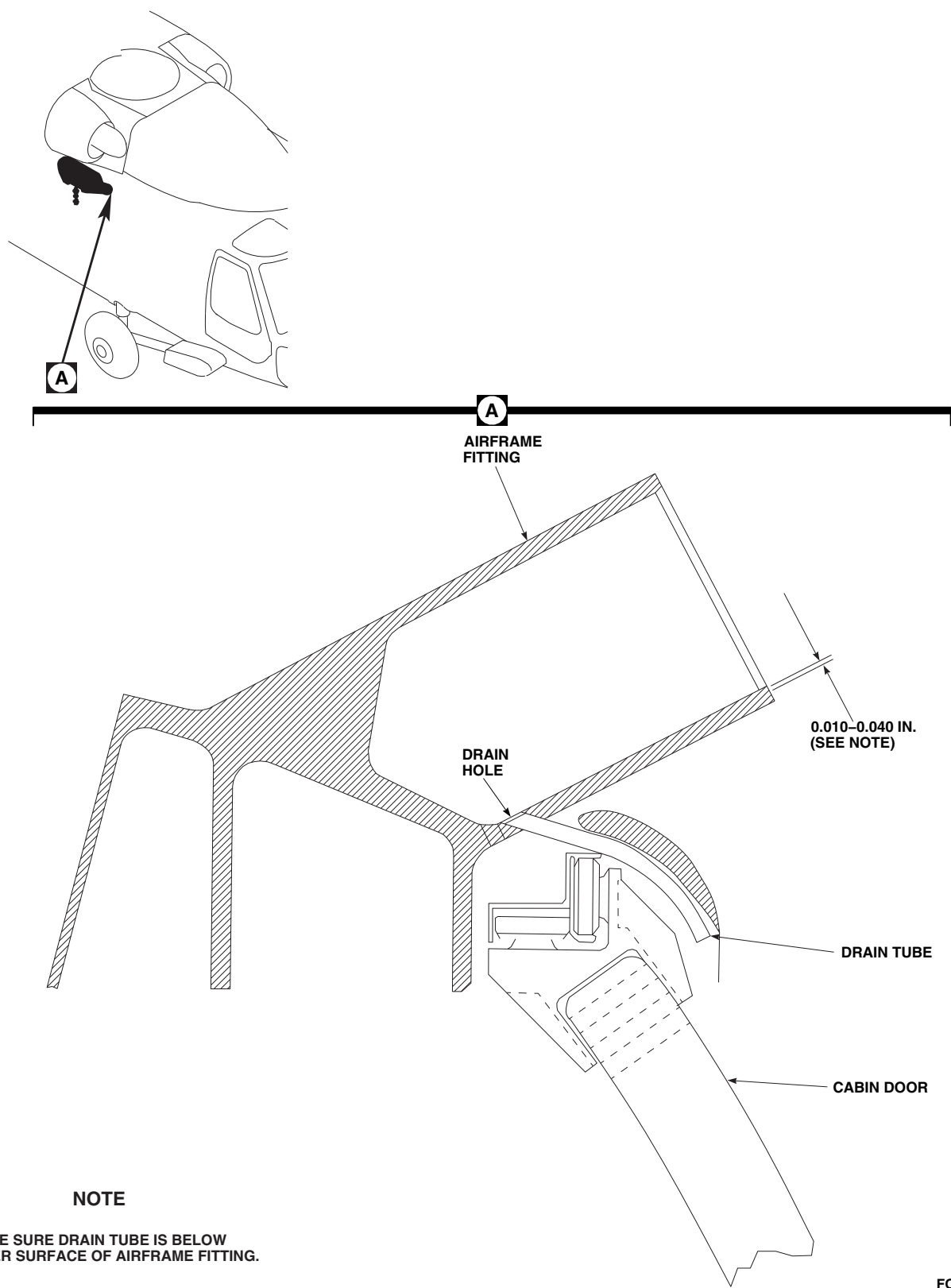
(2) Repair AREA B as follows:

(a) **BLEND OUT CORROSION UP TO 0.010-INCH DEEP OVER 100 % OF INNER AND OUTER DIAMETER.**

(b) If damage or repairs are more than limits, replace support arm.

j. Inspect airframe fitting for cracks, corrosion, and deformation (PARA 2-6-1).

k. Inspect airframe fitting for plugged/missing drain hole at bottom of fitting at 6 o'clock position as follows:



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FIGURE 14-4-32-3. INSPECT RESCUE HOIST DRAIN TUBE

NOTE

If 0.188-inch diameter drain hole exists at 45° angle at bottom of fitting at 6 o'clock position. Use hole as is, with 0.133-inch diameter drain tube (Figure 14-4-32-4, Detail A).

- (1) If drain hole is plugged, clean out drain hole and install drain tube per step 1. (5).
- (2) If drain hole is drilled at 90° angle in fitting, plug hole with sealing compound, Item 296, Appendix D, and drill new hole per step 1. (4) (Figure 14-4-32-4, Detail A).
- (3) If drain hole is missing, drill hole per step 1. (4).
- (4) If required, drill 45° angle hole as follows (Figure 14-4-32-4, Detail A and Detail B):

NOTE

Make sure not to gouge inner wall of airframe fitting when drilling hole.

- (a) Locate hole 0.85-inch from airframe fitting base at 75° ($\pm 2^\circ$) angle.
- (b) Drill 0.250-inch diameter hole.
- (5) Install drain tube as follows:
 - (a) If required, locally make drain tube from 0.133 or 0.186-inch inside diameter plastic tubing, 3-inches long.
 - (b) Install drain tubing into fitting with sealing compound, Item 296, Appendix D. Make sure drain tube is 0.010 - 0.040-inch below inner surface of fitting (Figure 14-4-32-3, Detail A).
 - (c) Seal drain tube to fitting with sealing compound, Item 296, Appendix D.
 - (d) Make sure drain tube is clear of sealing compound.

- l. Treat stripped and repaired areas with chemical conversion coating, Item 118, Appendix D.
- m. Using epoxy primer coating kit, Item 135, Appendix D, prime stripped and repaired areas.

NOTE

Do not paint outside diameter of Area B, holes, bushings and inserts of support base assembly or outside diameter of Area B, holes, bushings and inserts of support arm assembly.

- n. Paint stripped and repaired areas (PARA 2-6-5).

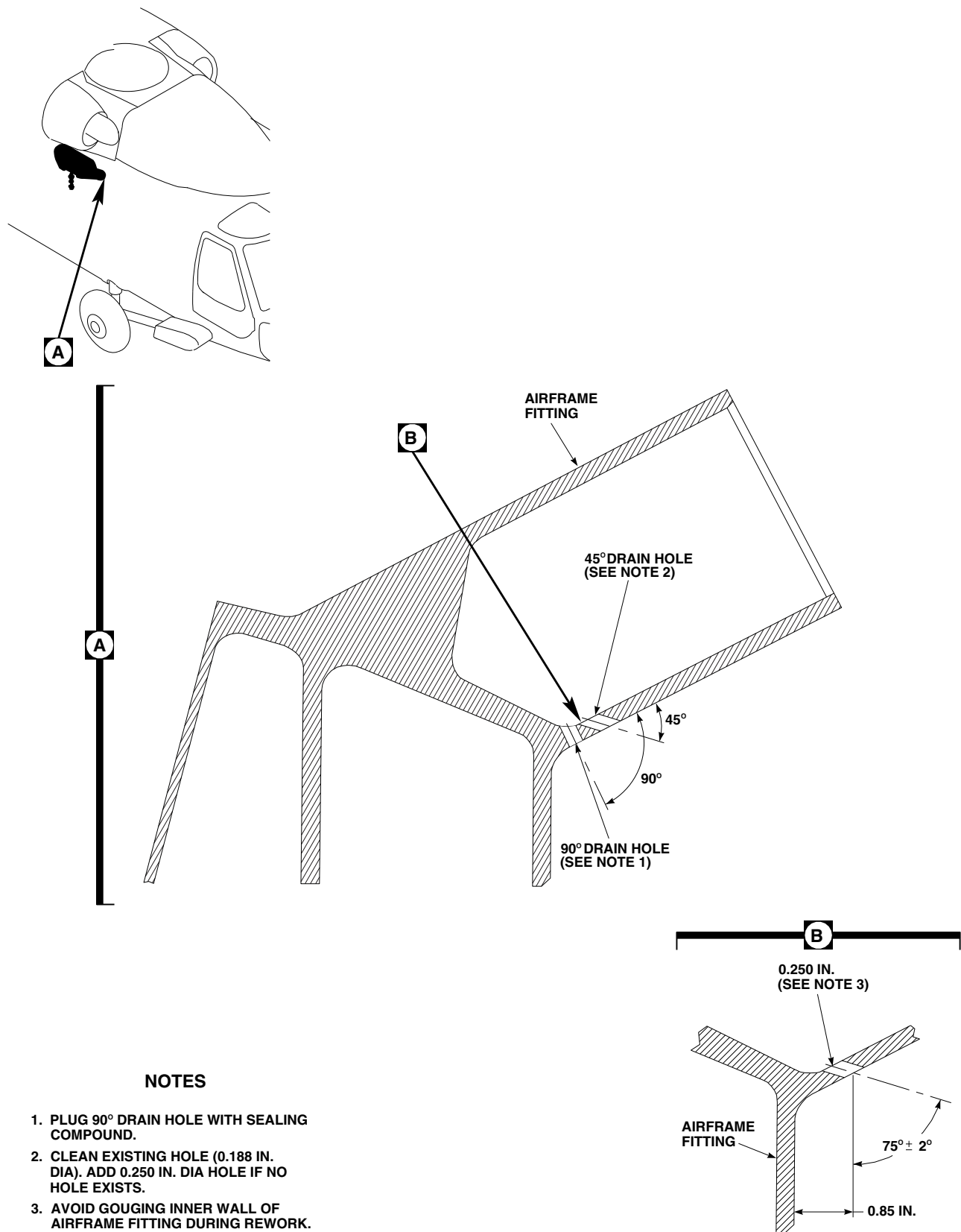
14-4-32.1.4 Assemble.

- a. Turn off all electrical power (PARA 1-6-16).

NOTE

Avoid blocking drain hole of airframe fitting (cup area) with epoxy primer.

- b. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to base support and airframe fitting mating surface. Allow primer to dry completely.
- c. Install base support in airframe fitting with bolts, washers, and nuts. **TORQUE NUTS TO 10 - 20 INCH-POUNDS ABOVE RUN-ON TORQUE** (Figure 14-4-32-1, Sheet 2, Detail C).
- d. Apply sealing compound, Item 292, Appendix D, around base support/airframe fitting mating edges and fastener locations.
- e. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer onto support arm and hoist mating surface. Allow primer to dry completely.
- f. Install support arm into hoist support with bolts, washers, and nuts. **TORQUE BOLTS TO 95 - 105 INCH-POUNDS** (Figure 14-4-32-1, Sheet 1, Detail A).
- g. Install cuff fairing on rescue hoist (PARA 14-4-37).



14-4-32.1.5 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer onto support arm and base support mating surface. Allow primer to dry completely.

WARNING

Injury to personnel and damage to equipment will result if hoist is not supported during installation. Rescue hoist weighs 105 pounds. Use two assistants to support rescue hoist during installation.

- c. Install hoist and support arm assembly to base support with bolts, washers, and nuts (Figure 14-4-32-1, Sheet 2, Detail C).
- d. **TORQUE UPPER NUT TO 80 INCH-POUNDS. THEN TIGHTEN NUT TO NEXT CASTELLATION.** Install cotter pin. **DO NOT EXCEED 165 INCH-POUNDS TORQUE.**
- e. **TORQUE LOWER BOLT TO 30 - 40 INCH-POUNDS.** If required, loosen nut to next castellation. Install cotter pin.
- f. Apply sealing compound, Item 292, Appendix D, around boltheads and nuts.
- g. Install electrical harness on support arm with screw, washer, and clamp.

- h. Install electrical harness to base support with screws, washers, and clamps.
- i. Remove protective caps and plugs from electrical connectors.
- j. Inspect and treat electrical connectors (PARA 1-7-20).
- k. Connect electrical connectors, P373, to J373 and P374 to J374, located on the airframe fitting.
- l. Remove protective caps and plugs from electrical connectors.
- m. Inspect and treat electrical connectors (PARA 1-7-20).
- n. Connect electrical connectors, P370, to J1, 371, to J5, and P372, to J8.
- o. Using lockwire, Item 196, Appendix D, lockwire connectors (Figure 14-4-32-1, Sheet 1, Detail B).
- p. Make sure area is clean and free of foreign material.
- q. Perform operational check of rescue hoist system (PARA 14-2-1).
- r. Install explosive cartridge (PARA 14-4-41).
- s. Install support fairings and connector cover (PARA 14-4-37).

14-4-33 LOAD CABLE **HOIST BL-29900-30** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-33.1 Replace Load Cable	14-4-33-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Allen Wrench, 1⁄16 x 6-in., Locally-Made
- Container, 55-Gallon
- Leather Gloves
- Light Source
- Screwdriver Attachment Socket Wrench
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Antiseize Compound, Item 75, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 14-3-2
- PARA 14-4-34
- PARA 14-4-35

14-4-33.1 Replace Load Cable.

14-4-33.1.1 Remove.

NOTE

- If cable has been cut with cable cutter, cable cutter and anvil must be replaced.
- If applicable, remove cable cutter and anvil (PARA 14-4-35).
 - Remove hook assembly (PARA 14-4-34).
 - Remove screws and washers securing access cover and backup plate to rescue hoist. Remove cover and backup plate (Figure 14-4-33-1, Sheet 1, Detail B).
 - Position rescue hoist unit so that cable windings on drum are visible.

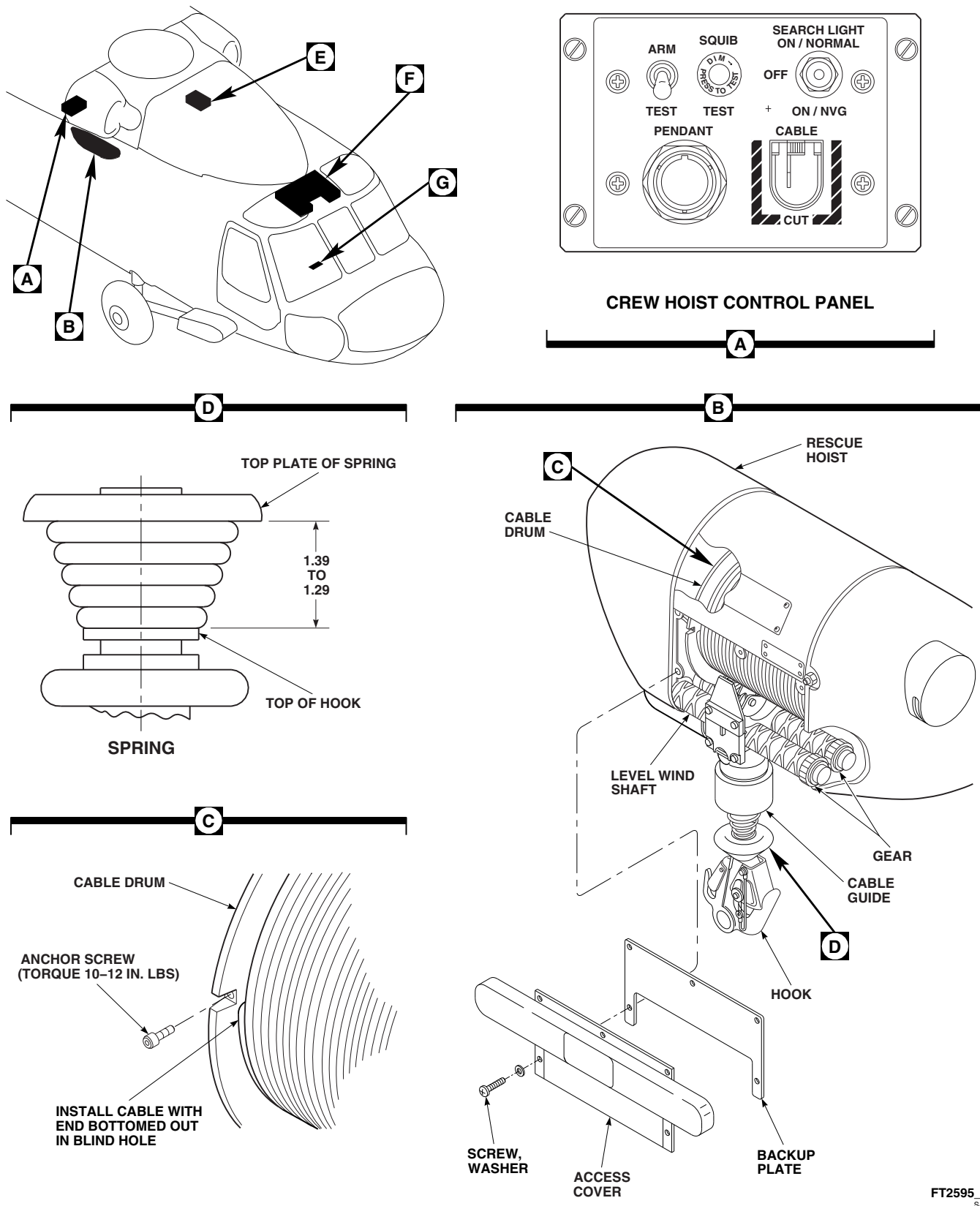
NOTE

Visually verify cable changing limit switch is aligned with access hole in front of rear cowling.

- Turn on all electrical power (PARA 1-6-16).

WARNING

- Injury to personnel and damage to equipment will occur if hands or clothing are caught in rotating machines.
- Injury to personnel will occur if gloves are not worn when reeling out cable. Leather gloves must be worn when reeling out cable.



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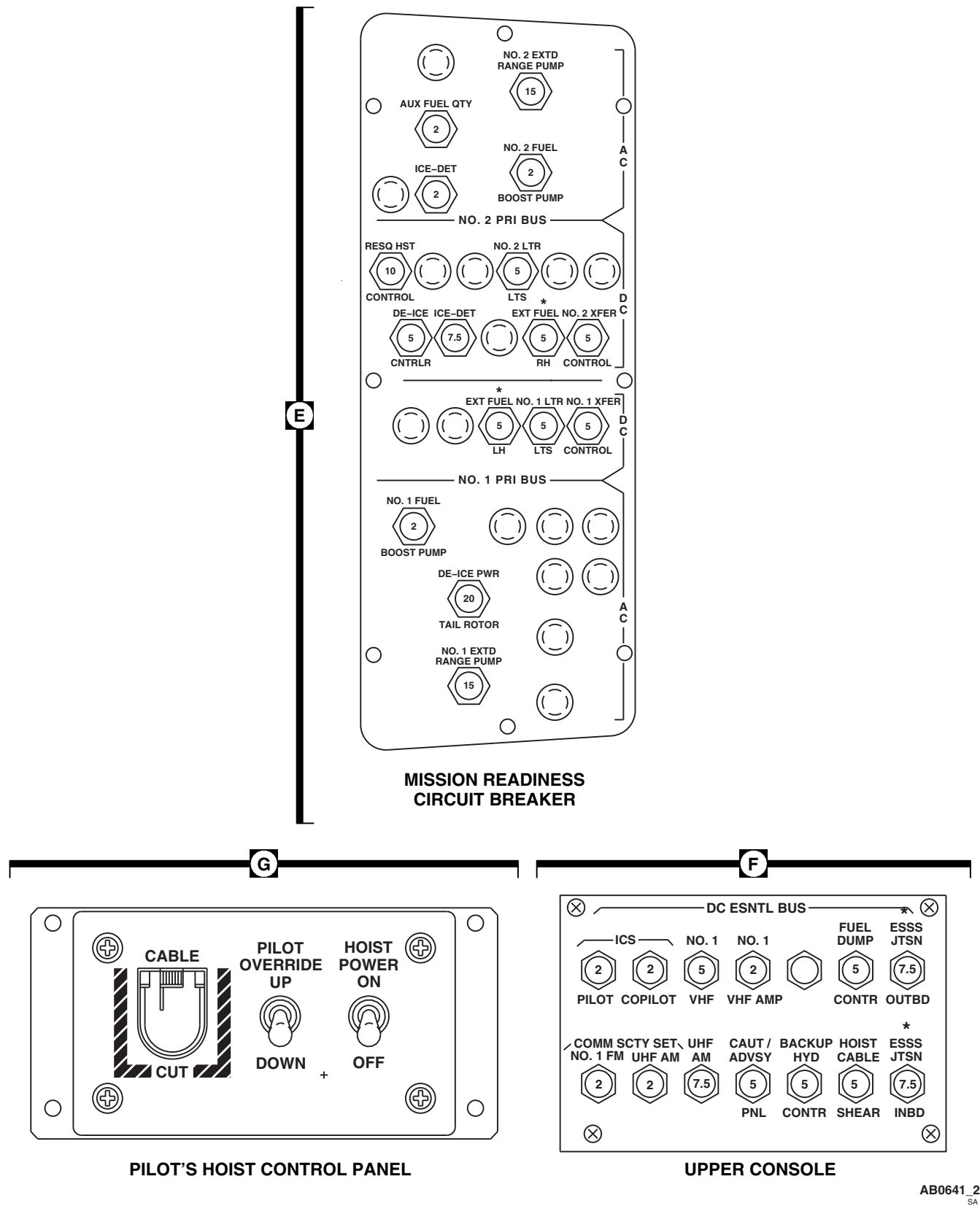


FIGURE 14-4-33-1. REPLACE LOAD CABLE (SHEET 2 OF 2)

- Injury to personnel and damage to equipment will occur if pendant control is not periodically scanned for active cable readout. Loss of readout may indicate malfunction of electronic limit switch, loss of automatic deceleration, and loss of up and down stop functions. Complete cable pay out may occur resulting in separation of cable from drum.

CAUTION

- Tension shall always be applied to cable during hoist operation.
 - Extreme care shall be taken not to allow cable to kink, twist, or ball up.
 - Care should be taken to keep cable clean for proper hoist operations.
- f. In cabin, operate rescue hoist pendant and reel out hoist cable while assistant, using leather gloves, coils cable into a clean 55-gallon container directly below rescue hoist until full-out limit switch is reached (Figure 14-4-33-1, Sheet 1, Detail A).

WARNING

Injury to personnel and damage to equipment will occur if drum is allowed to rotate beyond cable anchor set screw. Do not allow cable to begin "backwinding" on drum.

- g. Insert $\frac{1}{16}$ -inch Allen wrench (Allen wrench used for cable anchor screw) through access hole in front of rear cowling to actuate CABLE CHANGING LIMIT switch override switch. Actuate OVERRIDE switch and use the pendant control to slowly rotate the drum until no cable wraps remain on the drum and cable anchor screw is accessible.
- h. Turn off all electrical power (PARA 1-6-16).

- i. Loosen anchor screw, using $\frac{1}{16}$ -inch Allen wrench and screwdriver attachment socket wrench. Retain screw for reuse (Figure 14-4-33-1, Sheet 1, Detail C).

CAUTION

Damage to hoist will occur if electrical hoist is operated once cable has been removed. This can cause mis-synchronization of drum and level wind system.

- j. Remove cable from drum by pulling it straight down at cable guide.

14-4-33.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Thread anchor end of cable through roller housing assembly cable guide, tension roller, and pressure roller.

CAUTION

Damage to cable integrity will occur if more than two reinstallations are allowed following the initial installation.

- c. Fully insert anchor end of cable into hole in drum end flange and verify as follows:

CAUTION

Failure to conform with these requirements could compromise the integrity of load cable retention.

- (1) Verify proper insertion by viewing cable end through anchor screw hole using good light source.
- (2) Make sure that cable strands/wires only are visible through the anchor screw hole

and not the weld head on end of cable. If weld head is visible, reposition cable in hole until strands/wires only are visible.

- (3) The cable anchor screw must bear against the cable strands/wires only, not against the weld head.
- d. Secure cable to drum with anchor screw, using $\frac{1}{16}$ -inch Allen wrench and screwdriver attachment socket wrench. **TORQUE SCREW TO 10 - 12 INCH-POUNDS** (Figure 14-4-33-1, Sheet 1, Detail C).
- e. Turn on all electrical power (PARA 1-6-16).

NOTE

The first few wraps of cable to be wound on the drum are critical to proper winding of subsequent wraps.

- f. To make sure of proper initial wrapping, perform the following:
 - (1) Insert $\frac{1}{16}$ -inch Allen wrench (Allen wrench used for cable anchor screw) through access hole in front of rear cowling to actuate CABLE CHANGING LIMIT switch override switch. Actuate OVERRIDE switch.
 - (2) Apply tension to cable by pulling downward with gloved hands while operating the pendant in the raise direction at very slow speeds. The first wrap of cable must lie in the first groove of the drum. Subsequent wraps must lie in their respective grooves (2nd wrap in 2nd groove, etc.).
 - (3) Observe for proper wrapping during this operation.
- g. If proper wrapping is not being performed, perform the following:
 - (1) Turn off all electrical power (PARA 1-6-16).
 - (2) Loosen cable anchor screw using $\frac{1}{16}$ -inch Allen wrench and screwdriver attachment

socket wrench. Remove from drum flange. Retain screw for reuse.

- (3) Secure cable to drum with anchor screw, using $\frac{1}{16}$ -inch Allen wrench and screwdriver attachment socket wrench. **TORQUE SCREW TO 10 - 12 INCH-POUNDS.**
- (4) Turn on all electrical power (PARA 1-6-16).
- (5) Apply tension to cable by pulling downward with gloved hands while operating the pendant in the raise direction at very slow speeds. The first wrap of cable must lie in the first groove of the drum. Subsequent wraps must lie in their respective grooves (2nd wrap in 2nd groove, etc.).
- h. While maintaining tension on cable (gloved hands) wind cable on drum until 5 to 6 full wraps have been completed.

WARNING

Injury to personnel and damage to equipment will occur if tension is not maintained using gloved hands during this final operation to prevent loosening of cable wraps on drum. If cable is allowed to wrap loosely on drum, cable fouling problems may result during subsequent operation of hoist.

- i. Operate rescue hoist pendant and wind remaining cable on drum and observe for proper wrapping (Figure 14-4-33-1, Sheet 1, Detail A).
- j. Verify and test cable installation as follows:
 - (1) Make sure installation step a. through step i. are complied with when installing cable.
 - (2) Check that hoist limit settings and cable securing are still correct.
 - (3) Run hoist through one cycle with no load on cable at slow speed. Make sure cable

runs freely, level wind is adjusted properly, and tensioning system is operating properly.

- (4) Check cable for twisting.
- (5) Cycle hoist several times with light load on cable, applying load with gloved hands to cable. This allows cable to make adjustments to the actual operating conditions of the hoist (PARA 14-3-2).
- (6) Perform in-flight load test with a minimum load of 300 pounds for 2 to 3 full-length cycles (290 ft of usable cable on hoist). Check the cable speed and time required to travel the full cable stroke (PARA 14-3-2).

NOTE

If cable cutter assembly was replaced, complete the installation of cable cutter assembly cotter pin (PARA 14-4-35).

- k. Turn off all electrical power (PARA 1-6-16).
- l. Lubricate threads of screws with antiseize compound, Item 75, Appendix D.
- m. Install access cover and backup plate on rescue hoist and secure with screws and washers (Figure 14-4-33-1, Sheet 1, Detail B). Lock-wire, Item 196, Appendix D, screws.
- n. Install hook assembly (PARA 14-4-34).
- o. If applicable, install cable cutter and anvil (PARA 14-4-35).
- p. Make sure area is clean and free of foreign material.

14-4-34 HOOK ASSEMBLY **HOIST BL-29900-30** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-34.1 Replace Hook Assembly	14-4-34-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Punch

References

- PARA 1-6-16
- PARA 1-7-21
- PARA 14-4-41

14-4-34.1 Replace Hook Assembly.

14-4-34.1.1 Remove.

- Remove explosive cartridge (PARA 14-4-41).
- Turn on all electrical power (PARA 1-6-16).
- Operate rescue hoist unit in extend direction until cable is approximately 4 to 5 feet from full-in position.
- Turn off all electrical power (PARA 1-6-16).
- Separate bumper assembly from bushing guide nut and slide bumper up cable toward drum (Figure 14-4-34-1, Sheet 1, Detail B).
- Remove cotter pins and setscrews from bushing guide nut.

- Unscrew hook housing from bushing guide nut and remove hook and hook housing. Remove bumper from hook housing.
- Using punch, remove spring pin securing nut to hook and remove nut.
- Remove thrust washer, ball bearing, hook housing, and washer from hook.

CAUTION

Damage to hook assembly will occur if hook was used in a salt water environment or exposed to corrosive substances. The hook assembly must be washed with fresh water and be free of any foreign objects.

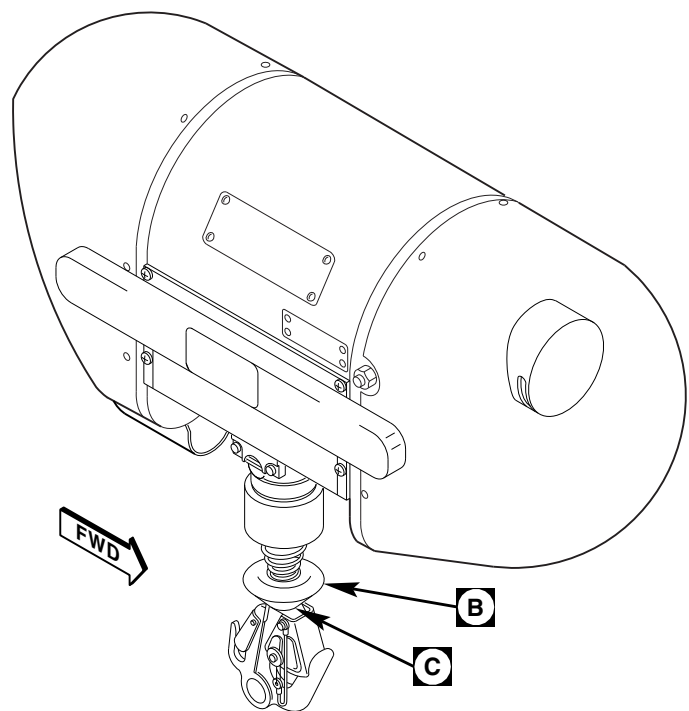
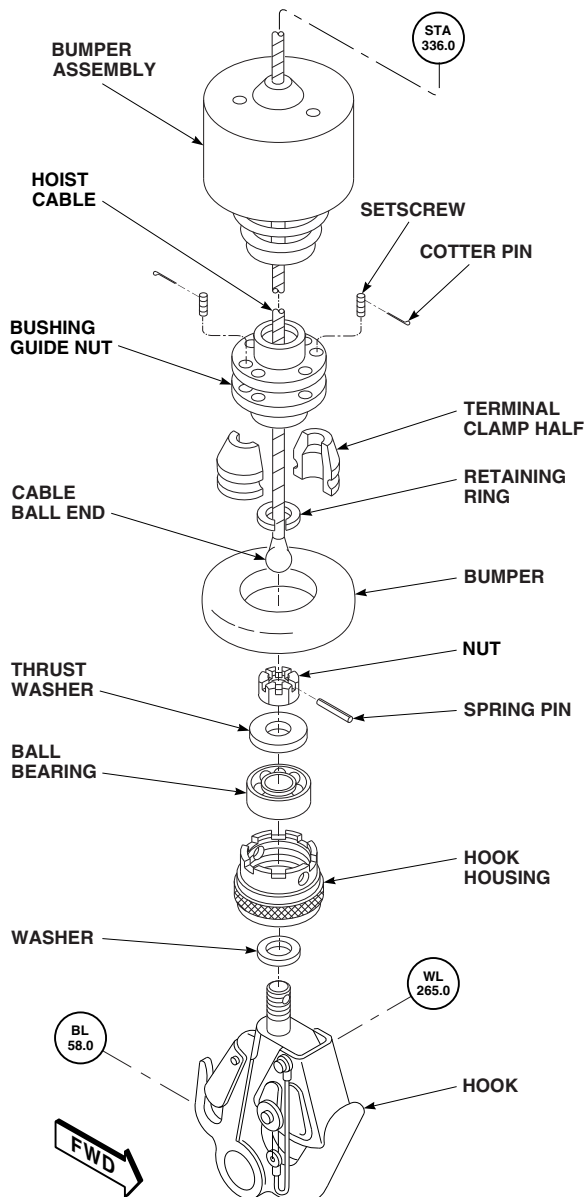
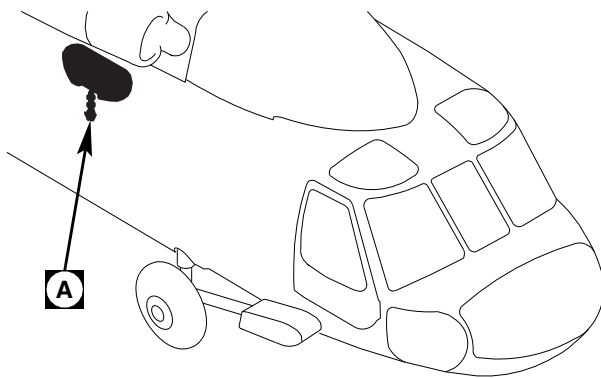
WARNING

Injury to personnel and damage to equipment will occur if hook is not supported when loosening bushing guide nut. Hook with attached parts will fall free when threads of bushing guide nut disengage from housing.

NOTE

Terminal clamp halves are a matched set. Keep halves together after removal.

- Slide bushing guide nut up cable to expose terminal clamp halves. Remove retaining ring and terminal clamp halves. Slide bushing guide nut off cable.



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FIGURE 14-4-34-1. REPLACE HOOK ASSEMBLY (SHEET 1 OF 2)

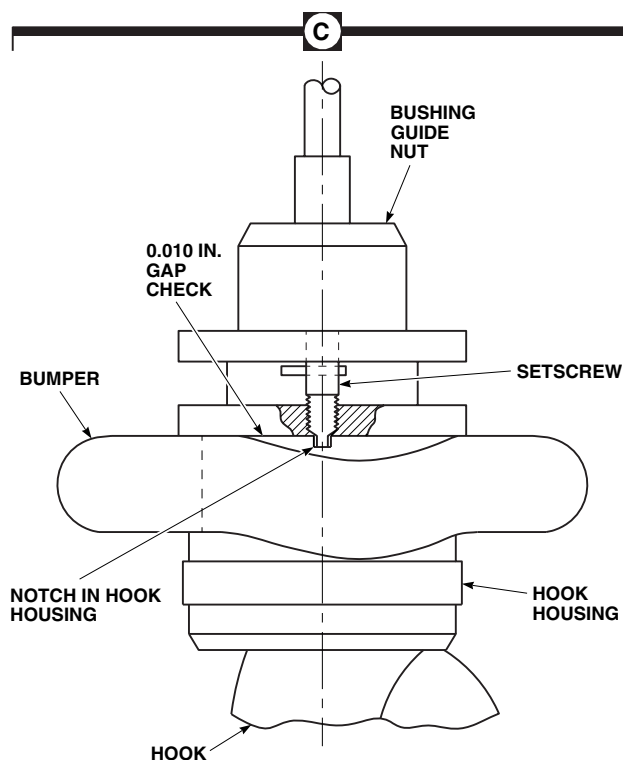


FIGURE 14-4-34-1. REPLACE HOOK ASSEMBLY (SHEET 2 OF 2)

14-4-34.1.2 Inspect.

- Check all components for wear or damage.
NONE ALLOWED.
- Inspect ball bearing for looseness, roughness, or stiffness caused by wear, deterioration of lubricants, or corrosion.
- Inspect spring pin for corrosion, cracks, and correct part number (PARA 1-7-21).
- Replace any worn or damaged components.

secure with retaining ring. Slide bushing guide nut down over terminal clamp halves.

- Place washer, hook housing, ball bearing, and thrust washer on hook and secure with nut.
- Tighten nut on hook until nut bottoms. Back nut off one-half turn and align it with hole in hook.

NOTE

Unpainted spring pin may be used until all stock is gone, but it must be replaced after every exposure to salt water.

14-4-34.1.3 Install.

- Turn off all electrical power (PARA 1-6-16).
- Push bumper over cable ball end on to hoist cable (Figure 14-4-34-1, Sheet 1, Detail B).

NOTE

Terminal clamp halves are matched set and must be replaced as set.

- Slide bushing guide nut over cable ball end. Position terminal clamp halves over cable and

- Check hook assembly for free rotation. If binding is felt, loosen nut until binding stops, and align nut with hole in hook again. Install spring pin in hole in hook.
- Install bumper on hook housing.
- Screw hook tightly onto bushing guide nut.

CAUTION

Make sure hook housing is screwed in all the way and after contact is made, back off to nearest castellation only. Setscrew must bottom into castellation of hook housing to lock.

- i. Align setscrew holes with notches in hook housing. Install setscrew in bushing guide nut and tighten until it contacts and locks into hook housing. Back off setscrew only enough to install cotter pins through slot in setscrews.

WARNING

If gap between faces of hook housing and bushing guide nut is greater than

0.010-inch or hook housing can be backed off bushing guide nut, hook may separate from cable during operation.

- j. Pull back bumper and using feeler gage, check gap between faces of hook housing and bushing guide nut. **GAP TO BE NO GREATER THAN 0.010-INCH.** If gap is greater than 0.010-inch or hook housing can be backed off bushing guide nut, check setscrews for proper installation (Figure 14-4-34-1, Sheet 2, Detail C).
- k. Push spring on bushing guide nut (Figure 14-4-34-1, Sheet 1, Detail B).
- l. Turn on all electrical power (PARA 1-6-16).
- m. Reel in hoist cable.
- n. Install explosive cartridge (PARA 14-4-41). **I**

14-4-35 CABLE CUTTER AND ANVIL **HOIST BL-29900-30** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-35.1 Replace Cable Cutter and Anvil	14-4-35-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 14-4-41

14-4-35.1 Replace Cable Cutter and Anvil.

14-4-35.1.1 Remove.

- Remove explosive cartridge (PARA 14-4-41).
- Remove cable cutter anvil from cartridge housing (Figure 14-4-35-1, Detail A).
- Remove cable cutter by applying low air pressure (50 - 80 psi) to cartridge housing.
- Remove severed cotter pin from cartridge housing.
- Remove packing from cartridge housing. Discard packing.

14-4-35.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

WARNING

Injury to personnel and damage to equipment will occur if electrical power

is not disconnected from aircraft when handling squib (cartridge) activated device.

NOTE

Prior to installation of cable cutter, make sure cable guide passageways and cable cutter are clean and free of excess discharge residue.

- Install packing into cartridge housing.
- Install cable cutter into cartridge housing ensuring alignment of cotter pin set holes (Figure 14-4-35-1, Detail A).

NOTE

The load cable must be reinstalled and the cable guide moved away from the hoist housing before complete installation of the cotter pin into the cable cutter set hole.

- Install cotter pin in set hole.
- Install cable cutter anvil and lockwire, Item 196, Appendix D.

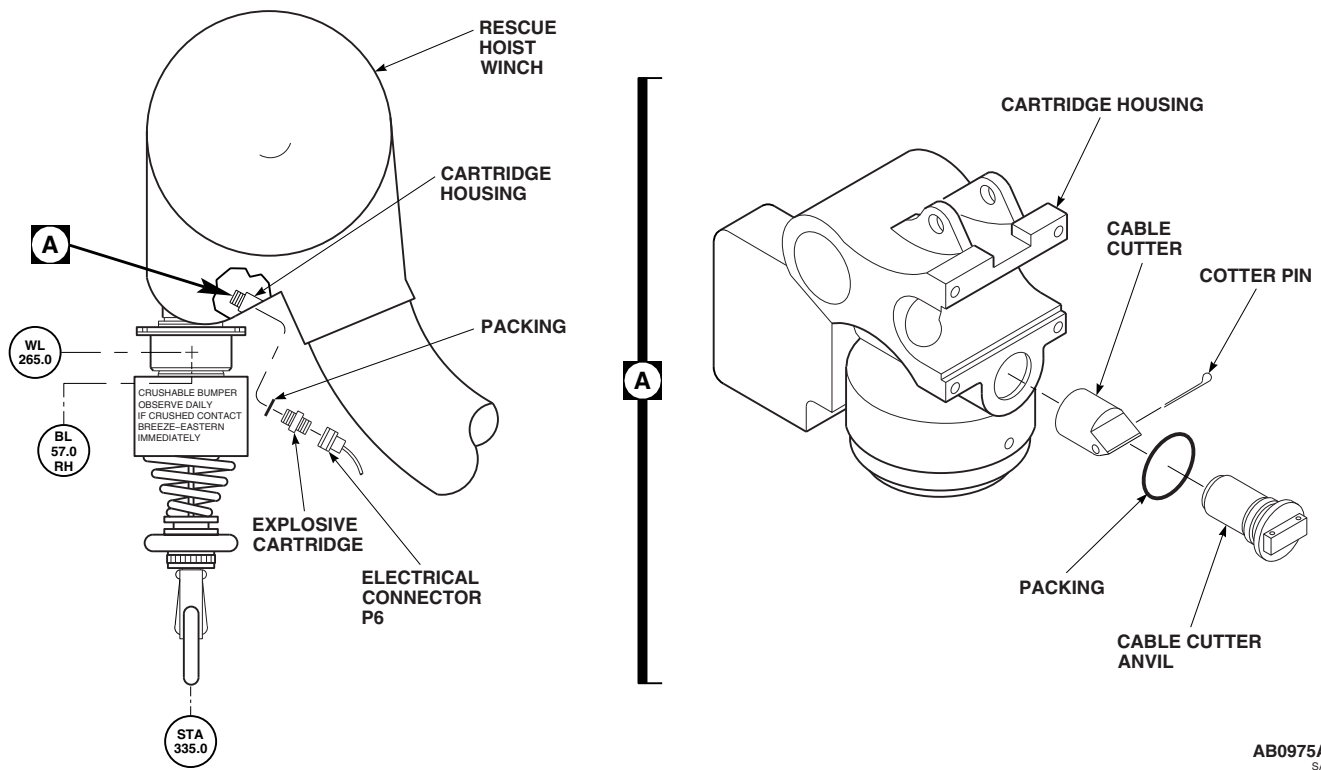


FIGURE 14-4-35-1. REPLACE CABLE CUTTER AND ANVIL

- f. Install explosive cartridge (PARA 14-4-41).

14-4-36 RESCUE HOIST PENDANT **HOIST BL-29900-30** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-36.1 Replace Rescue Hoist Pendant	14-4-36-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 14-2-1

14-4-36.1 Replace Rescue Hoist Pendant.

14-4-36.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector from crew hoist control panel (Figure 14-4-36-1, Detail A).
- Install protective cap and plug on electrical connectors.
- Open clip and remove rescue hoist pendant from support.

14-4-36.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Position rescue hoist pendant in support and close clip (Figure 14-4-36-1, Detail A).
- Remove protective cap and plug from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector to crew hoist control panel.
- Perform operational check of rescue hoist system (PARA 14-2-1).

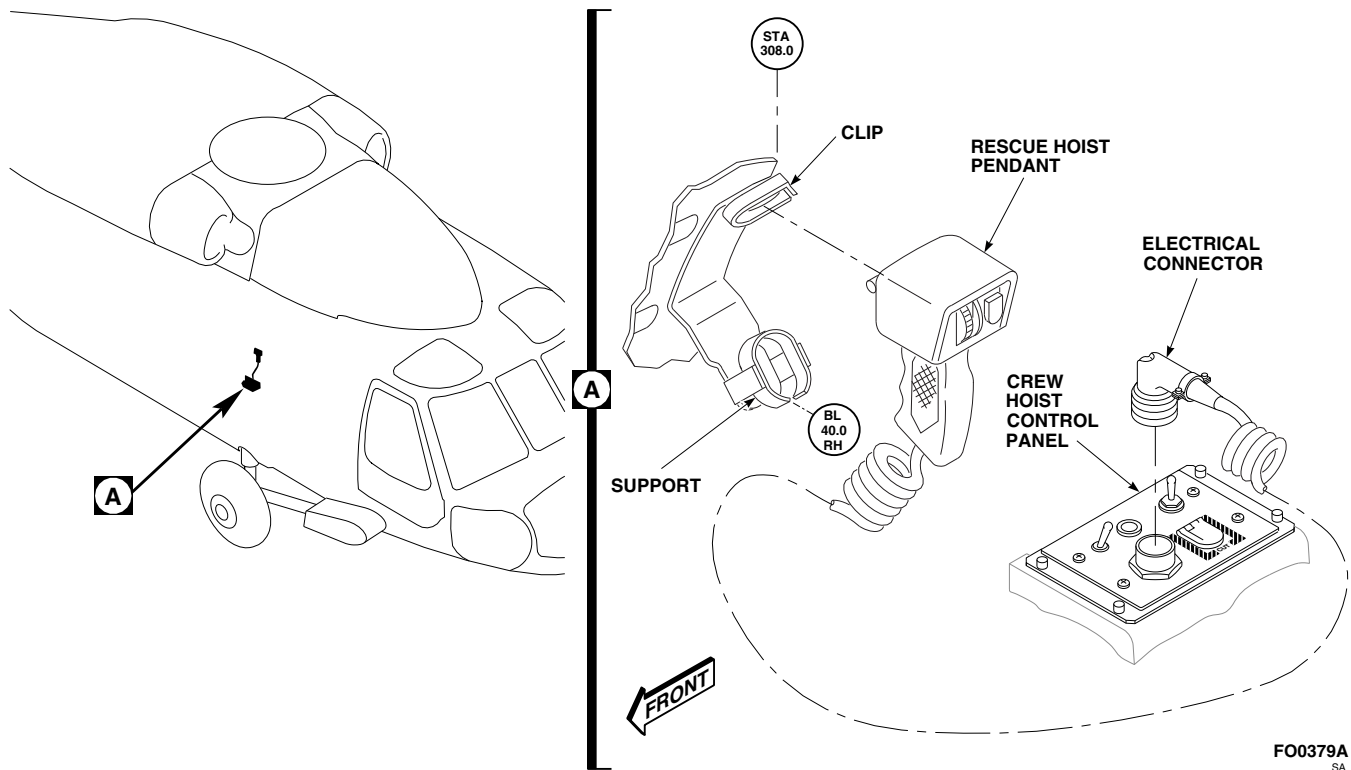


FIGURE 14-4-36-1. REPLACE RESCUE HOIST PENDANT

14-4-37 SUPPORT FAIRINGS AND COWLINGS **HOIST BL-29900-30** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-37.1 Replace Support Fairings and Cowlings	14-4-37-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 75, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 14-4-32

14-4-37.1 Replace Support Fairings and Cowlings.

14-4-37.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- If removing cuff fairing, remove rescue hoist (PARA 14-4-32).
- Remove screws and washers securing access cover and backup plate to rescue hoist. Remove cover and plate (Figure 14-4-37-1, Detail A).
- Remove screws, bolts, and washers securing connector cover, support arm fairings and cuff fairing to rescue hoist. Remove cover and fairings.
- Remove screws and washers holding front cowling to rescue hoist and remove cowling.
- Remove screws and washers holding rear cowling to rescue hoist and remove cowling.

14-4-37.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Make sure area is clean and free of foreign material before installing fairings and cowlings.
- Lubricate threads of screws with antiseize compound, Item 75, Appendix D.
- Place rear cowling in position on rescue hoist and secure with screws and washers. Lockwire, Item 196, Appendix D, screws (Figure 14-4-37-1, Detail A).
- Lubricate threads of screws with antiseize compound, Item 75, Appendix D.
- Place front cowling in position on rescue hoist and secure with screws and washers. Lockwire, Item 196 Appendix D, screws.
- Lubricate threads of screws with antiseize compound, Item 75, Appendix D.

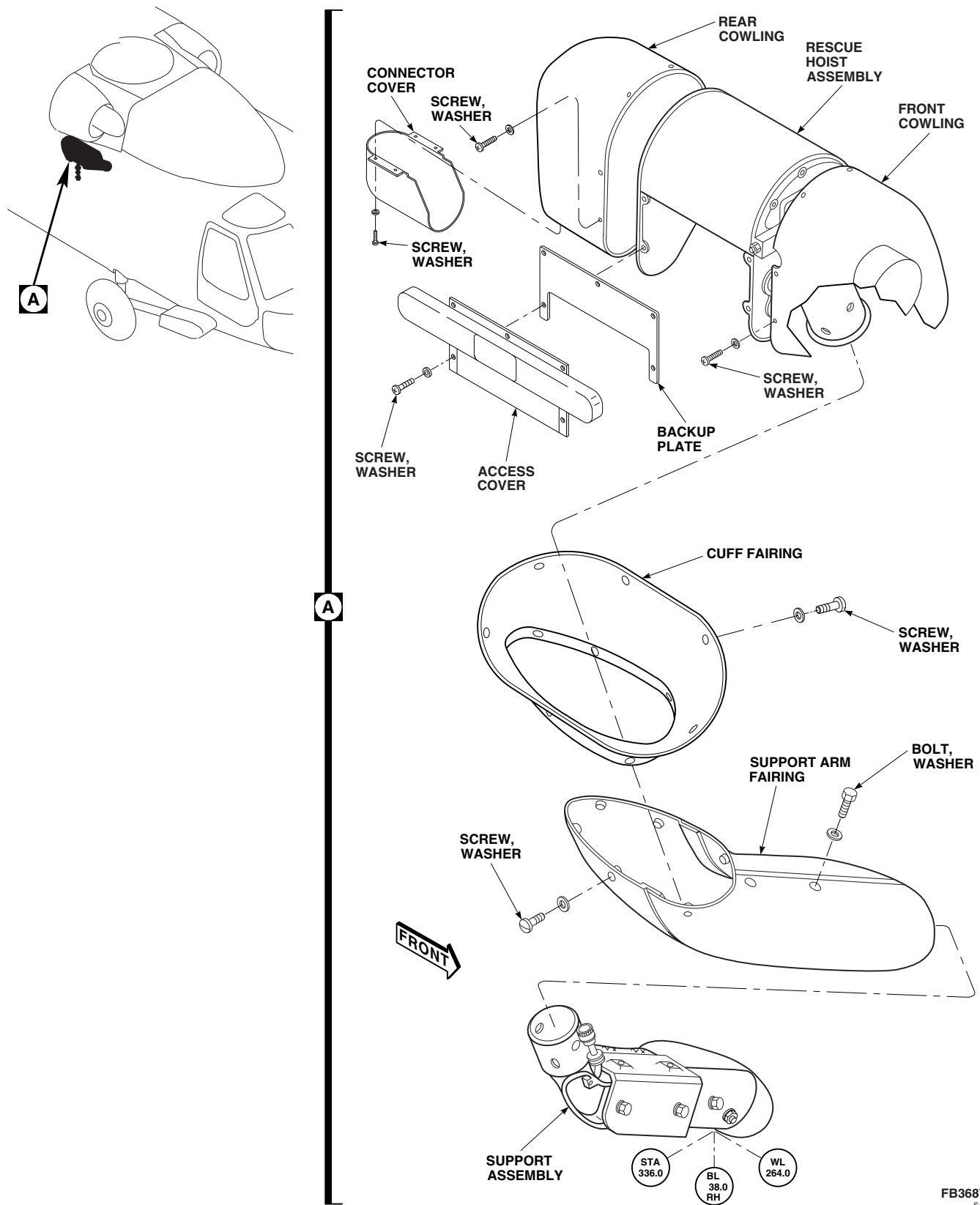
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FIGURE 14-4-37-1. REPLACE SUPPORT FAIRINGS AND COWLINGS

14-4-37-2

- h. Secure cuff fairing to rescue hoist with screws and washers.
- i. Lubricate threads of screws with antiseize compound, Item 75, Appendix D.
- j. Install connector cover on rescue hoist and secure with screws and washers. Lockwire, Item 196 Appendix D, screws.
- k. Install support arm fairings to cuff fairing with screws and washers.
- l. Lubricate threads of screws with antiseize compound, Item 75, Appendix D.
- m. Position backup plate and access cover on rescue hoist and secure with screws and washers. Lockwire, Item 196, Appendix D, screws.
- n. If removed, install rescue hoist (PARA 14-4-32).

14-4-38 PILOT’S HOIST CONTROL PANEL **HOIST BL–29900–30** .

PARAGRAPH BREAKDOWN

		PAGE
14-4-38.1	Replace Pilot’s Hoist Control Panel	14-4-38-1
14-4-38.2	Replace Pilot’s Hoist Control Panel Information Plate	14-4-38-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 14-2-1

14-4-38.1 Replace Pilot’s Hoist Control Panel.

14-4-38.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Loosen turnlock fasteners securing pilot’s hoist control panel to center console (Figure 14-4-38-1, Detail A).
- Carefully lift control panel to gain access to electrical connector.
- Disconnect electrical connector, P1005R, from control panel.
- Install protective cap and plug on electrical connectors.
- Remove control panel from center console.

14-4-38.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

- Remove protective cap and plug from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P1005R, to pilot’s hoist control panel (Figure 14-4-38-1, Detail A).
- Position control panel in center console and secure with turnlock fasteners.
- Make sure area is clean and free of foreign material.
- Perform operational check of rescue hoist system (PARA 14-2-1).

14-4-38.2 Replace Pilot’s Hoist Control Panel Information Plate.

14-4-38.2.1. Remove.

- Turn off all electrical power (PARA 1-6-16).

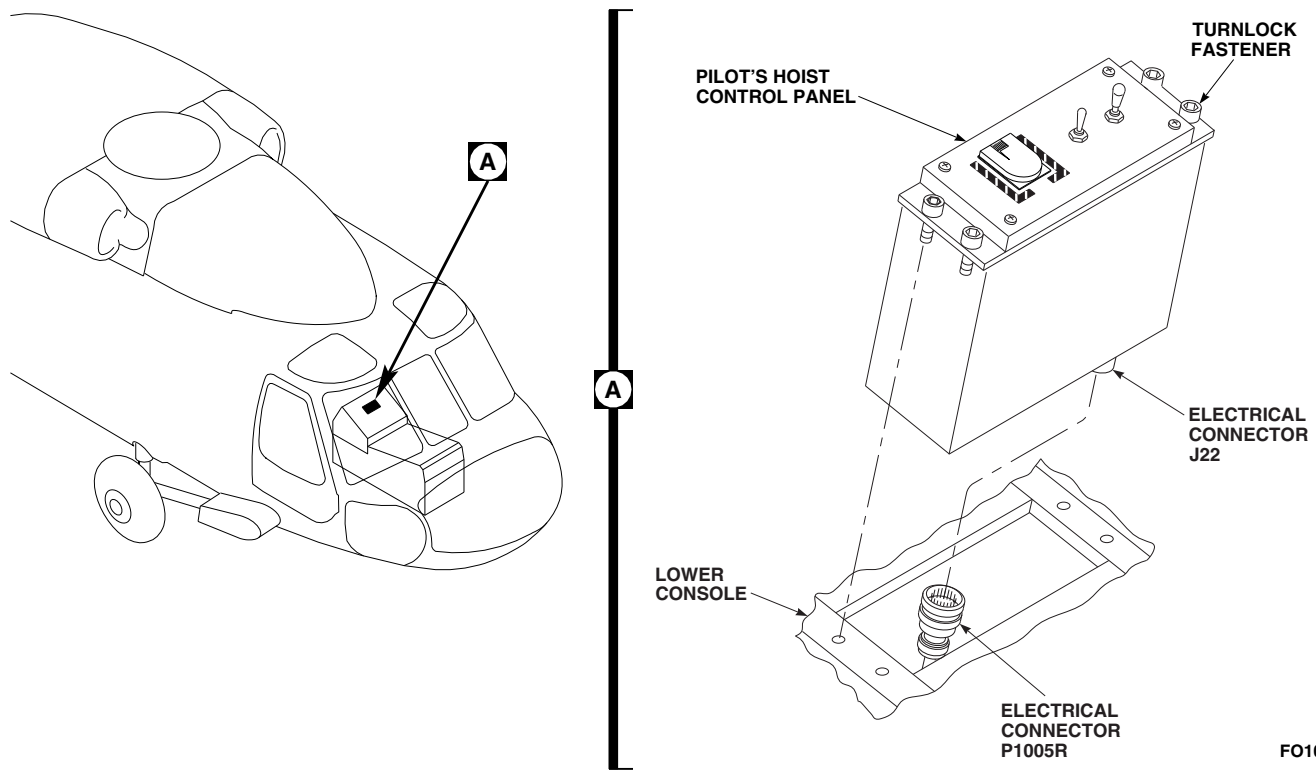


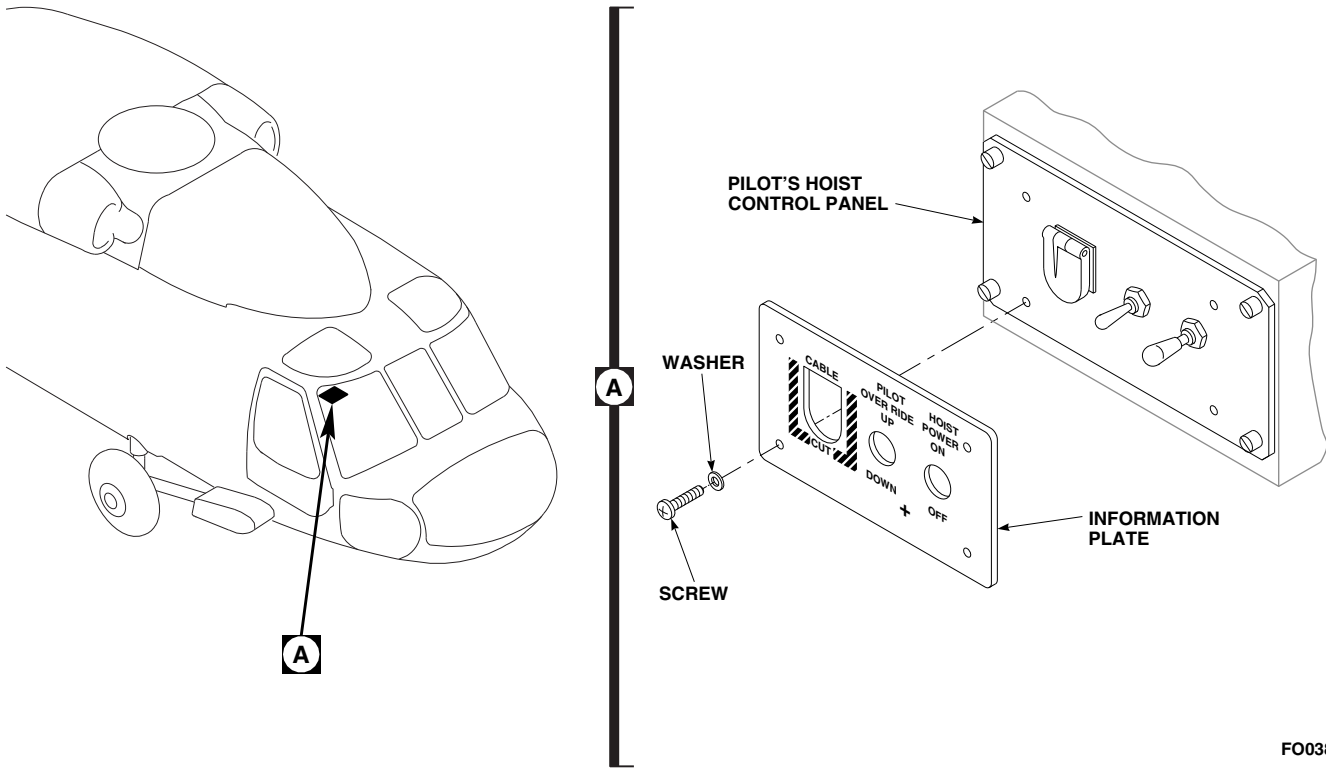
FIGURE 14-4-38-1. REPLACE PILOT'S HOIST CONTROL PANEL

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- b. Remove screws and washers securing information plate to pilot's hoist control panel. Remove information plate (Figure 14-4-38-2, Detail A).

14-4-38.2.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Position information plate on pilot's hoist control panel and secure with screws and washers (Figure 14-4-38-2, Detail A).
- c. Make sure area is clean and free of foreign material.
- d. Perform operational check of utility systems rescue hoist (PARA 14-2-1).



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FIGURE 14-4-38-2. REPLACE PILOT'S HOIST CONTROL PANEL INFORMATION PLATE

14-4-39 CREW HOIST CONTROL PANEL **HOIST BL-29900-30** .

PARAGRAPH BREAKDOWN

		PAGE
14-4-39.1	Replace Crew Hoist Control Panel	14-4-39-1
14-4-39.2	Replace Crew Hoist Control Panel Information Plate	14-4-39-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 14-2-1

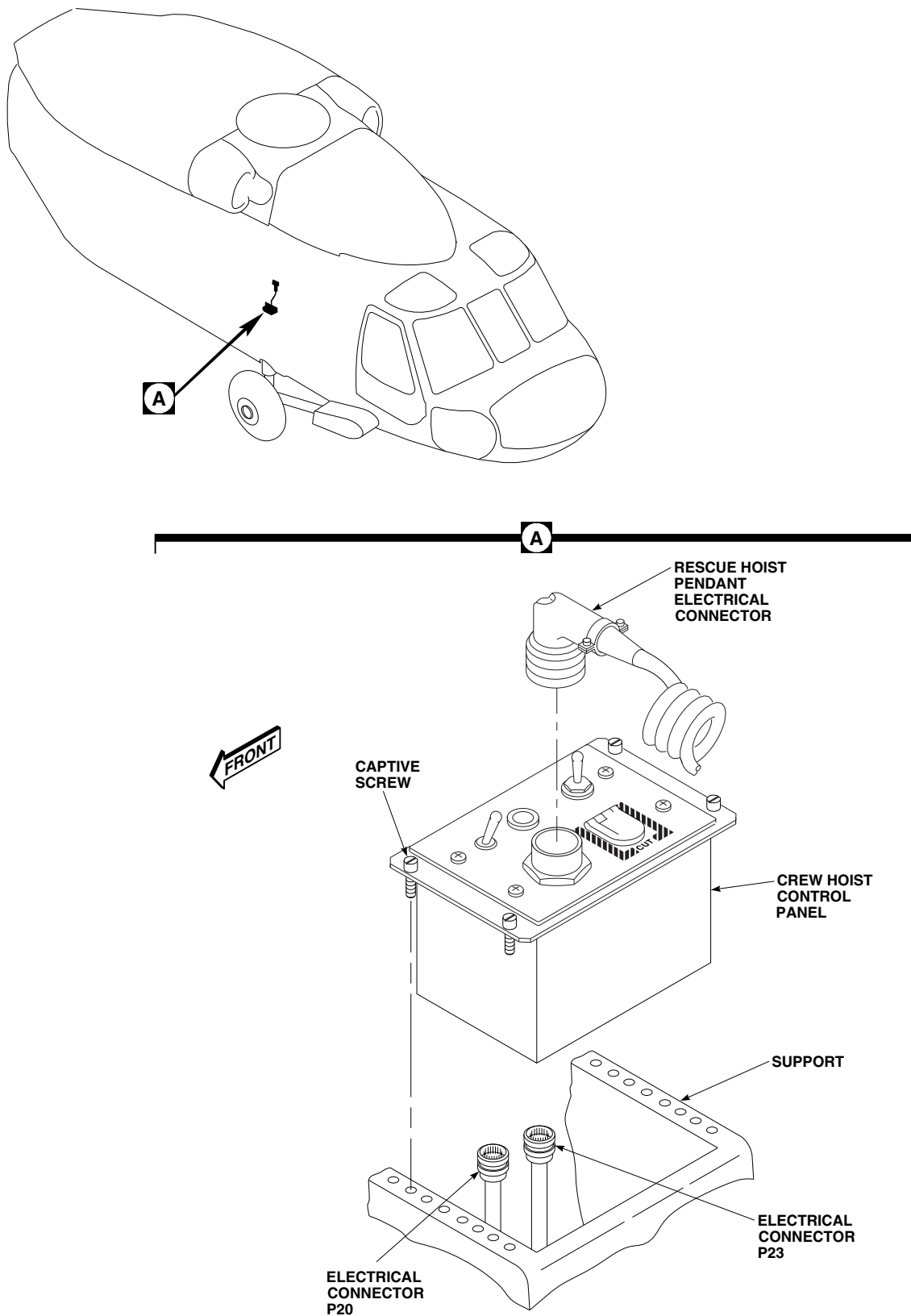
14-4-39.1 Replace Crew Hoist Control Panel.

14-4-39.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove soundproofing panel (PARA 2-4-69).
- Disconnect rescue hoist pendant electrical connector from crew hoist control panel (Figure 14-4-39-1, Detail A).
- Install protective cap and plug on electrical connectors.
- Disconnect electrical connectors, P20 and P23, from control panel.
- Install protective caps and plugs on electrical connectors.
- Loosen captive screws and remove control panel from support.

14-4-39.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Position crew hoist control panel in support and secure with captive screws (Figure 14-4-39-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connectors, P20 and P23, to control panel.
- Install soundproofing panel (PARA 2-4-69).
- Remove protective cap and plug from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).



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FIGURE 14-4-39-1. REPLACE CREW HOIST CONTROL PANEL

14-4-39-2

Change 1

- i. Connect rescue hoist pendant electrical connector to control panel.
- j. Perform operational check of rescue hoist system (PARA 14-2-1).

14-4-39.2 Replace Crew Hoist Control Panel Information Plate.

14-4-39.2.1. Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Disconnect rescue hoist pendant electrical connector from crew hoist control panel (Figure 14-4-39-2, Detail A).
- c. Install protective cap and plug on electrical connectors.
- d. Remove screws and washers securing information plate to control panel. Remove information plate from control panel.

14-4-39.2.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Position information plate on crew hoist control panel and secure with screws and washers (Figure 14-4-39-2, Detail A).
- c. Remove protective cap and plug from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect rescue hoist pendant electrical connector to control panel.
- f. Make sure area is clean and free of foreign material.
- g. Perform operational check of utility systems rescue hoist (PARA 14-2-1).

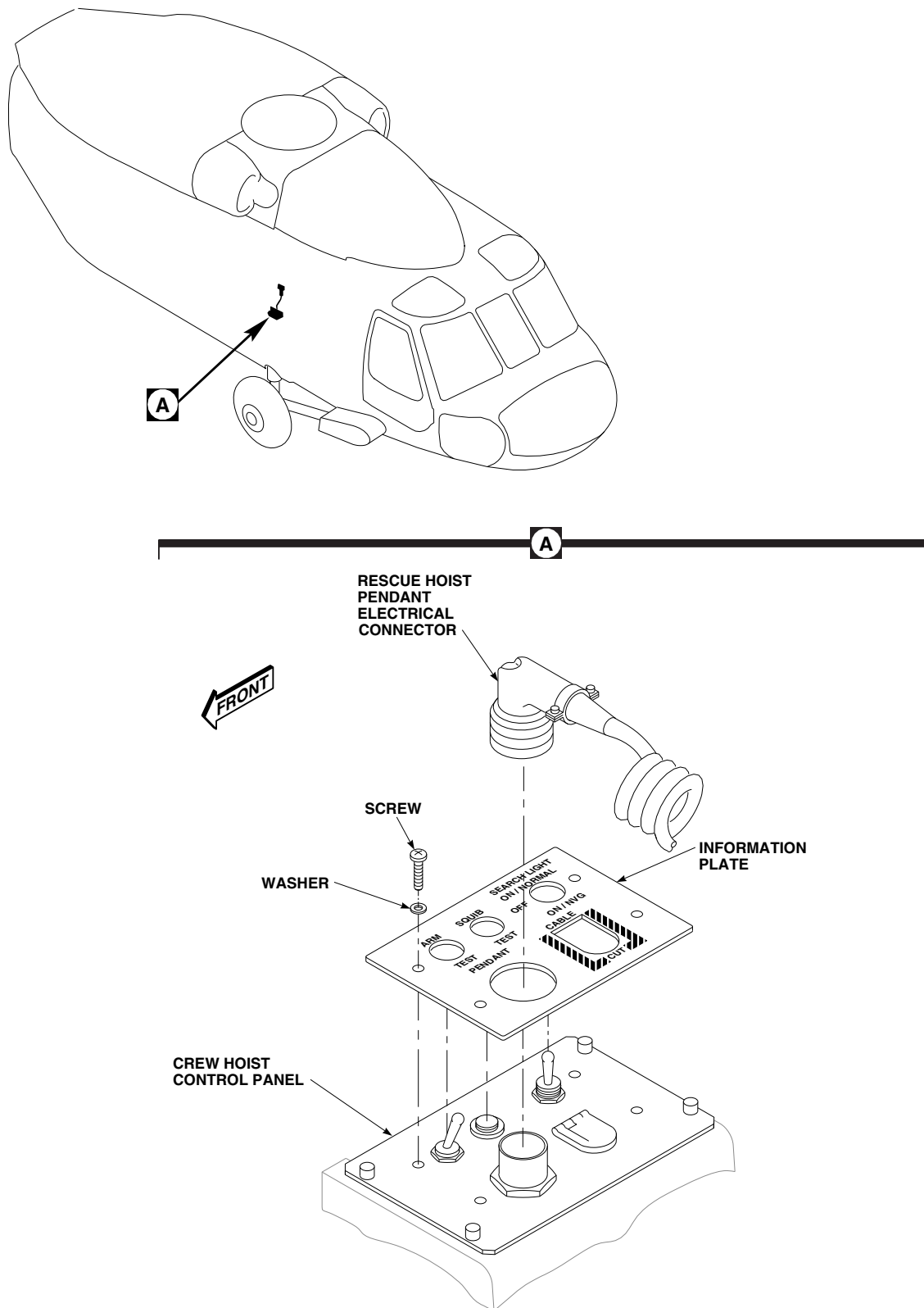
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FIGURE 14-4-39-2. REPLACE CREW HOIST CONTROL PANEL INFORMATION PLATE

14-4-40 LIMIT SWITCH **HOIST BL-29900-30** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-40.1 Set/Check Limit Switch	14-4-40-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-6-16
- PARA 14-4-33

14-4-40.1 Set/Check Limit Switch.

NOTE

Electrical power must be applied to the rescue hoist unit when performing the set and check procedures.

14-4-40.1.1 Set.

- Turn on all electrical power (PARA 1-6-16).
- Before doing any performance testing, the limit switches must be set properly for operation.
- If the cable is installed, operate the rescue hoist unit in the lower mode and unwind cable from the drum.
- Mark the cable at the following points:
 - 4.5 ft from the weld end of the cable.
 - 6 ft from the weld end of the cable.
 - 16 ft from the weld end of the cable.
 - 10 ft from the ball end of the cable.
- Install load cable (PARA 14-4-33).
- Operate rescue hoist unit in the raise mode. Wind cable onto the drum until 10 ft from the

ball end of the cable is reached. Remove the cam locking plate on the cyclocentric assembly (Figure 14-4-40-1, Section A-A) and adjust cam D so that switch S4 actuates within 12-inches of this point.

- Apply a minimum load to the rescue hoist unit. Operate the rescue hoist unit in the lower mode and pay out cable until the 16 ft mark on cable is reached at the output point. Set CAM C so that switch S3 actuates within 12-inches of this point.
- Operate the rescue hoist unit in the lower mode and payout cable until the 6 ft mark on the cable is at the hoist cable output point. Set cam B so that switch S2 actuates within 6-inches of this point.
- Actuate the cable changing switch on the electrical control panel.
- Using the pendant, operate the rescue hoist unit carefully in the lower mode using the control toggle switch, until the 4.5 foot mark on the cable is at the hoist cable output point. Set cam A to actuate switch S1 within 6-inches of this point.
- With the cable changing switch actuated, raise using the pendant control slowly until 6 wraps of cable are on the drum.

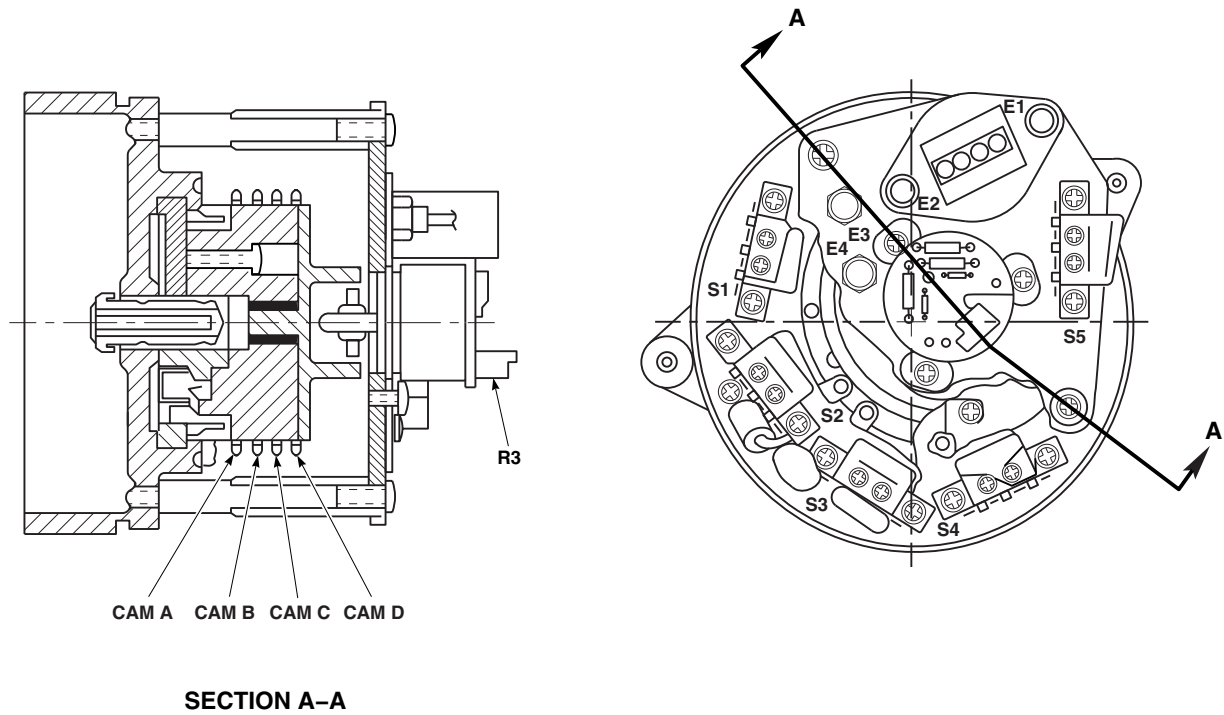


TABLE-FUNCTION (FOR REFERENCE)

FUNCTION	CAM	SWITCH	SWITCH ACTUATION
EMERGENCY FULL OUT	A	S1	3 ± 1/2 WRAPS LEFT ON DRUM
FULL OUT	B	S2	4 ± 1/2 WRAPS LEFT ON DRUM
DECELERATION (LOWERING)	C	S3	10 FEET ± 2 FEET FULLY REELED OUT
DECELERATION (HOISTING)	D	S4	10 FEET ± 2 FEET FULLY REELED IN

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FIGURE 14-4-40-1. SET/CHECK LIMIT SWITCH

14-4-40-2

Change 4

14-4-40.1.2 Check.

- a. Turn on all electrical power (PARA 1-6-16).
- b. Check limit switch operation as follows (from the full-in position):
- c. Operate the rescue hoist unit over its full cable travel in both the raise and lower modes. Verify that the limit switches operate as follows:
 - (1) Intermediate lower limit switch S3 shall actuate when the extending cable is 10 ± 2 ft from the fully reeled-out position.
 - (2) Full-out limit switch S2 shall actuate when the cable has been extended to the point where 3.5 to 4.5 wraps of cable remain on the drum.
- d. Actuate the cable changing switch and using the pendant slowly rotate the cable in lower

mode until the mark that was previously applied to the cable appears. Continue approximately $\frac{1}{2}$ -wrap further. Release the cable changing switch. 28 vdc power to the system shall be disconnected as indicated by no payout displays.



Damage to cable will occur if cable is allowed to begin backwinding on drum. Using pendant control, it is imperative that slow speed be used at this point to avoid backwinding cable on drum.

- e. Intermediate hoist limit switch S4 shall actuate when the retracting cable is 10 ± 2 ft from the fully reeled-in position.

14-4-41 EXPLOSIVE CARTRIDGE **HOIST BL-29900-30** .

PARAGRAPH BREAKDOWN

	PAGE
14-4-41.1 Replace Explosive Cartridge	14-4-41-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit
- Shorting Jumper
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Marking Pen, Item 211C, Appendix D

- Pressure Cartridge Installation Kit, KT-444
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-4-34
- PARA 9-4-106
- PARA 14-2-1

14-4-41.1 Replace Explosive Cartridge.

14-4-41.1.1 Inspect.

- Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).
- Visually inspect hoist fairing and electrical connector, P6, for damage. **NONE ALLOWED.** If damage to fairing is found, remove fairing from flight operations until fairing can be repaired or replaced. If damage is found on electrical components, replace damaged parts per applicable PARA in technical manual.

- Inspect backshell of electrical connector, P6, for proper orientation and clearance as follows:
 - Backshell of electrical connector, P6, must be oriented downward, in the 6 o’clock position (± 30 minutes tolerance) (Figure 14-4-41-1, View A-A).
 - Operate hoist per the appropriate PARA in technical manual and verify minimum clearance of 0.125-inch between hoist fairing and backshell of electrical connector, P6.

- d. If backshell of electrical connector, P6, is oriented correctly and meets minimum 0.125-inch clearance, inspection is complete.
- e. If backshell of electrical connector, P6, is oriented incorrectly or required clearance of 0.125-inch minimum is not met, perform the following:
 - (1) Remove explosive cartridge (PARA 14-4-41.1.2).
 - (2) Install cartridge in cartridge housing to determine initial key orientation (Figure 14-4-41-2, Detail B). **NOTE POSITION OF KEY.**
 - (3) If key is located in desired position "A" (10:30 o'clock position, ± 30 minutes tolerance), then no shims are required. Remove cartridge, then proceed to step e. (5).
- (4) If key is not located in desired position "A" (10:30 o'clock position, ± 30 minutes tolerance), shim(s) are required. Perform the following:
 - (a) Refer to Table 14-4-41-1 to determine the type and quantity of shim(s) required. Obtain shim(s) from pressure cartridge installation kit, KT-444.
 - (b) Remove cartridge from cartridge housing (Figure 14-4-41-2, Detail A).
 - (c) Install shim(s) on cartridge as determined in step e. (4) (a).
- (5) Install packing.

Table 14-4-41-1. Shim Requirements

Key Position (See Note)	Shim (0.005 in.) Qty.	Shim (0.010 in.) Qty.	Shim (0.031 in.) Qty.
A (Desired Position)	-	-	-
B	1	-	-
C	-	1	-
D	1	1	-
E	-	2	-
F	1	2	-
G	-	-	1
H	1	-	1
I	-	1	1
J	1	1	1

Note: Key positions
are $\pm 18^\circ$.

14-4-41.1.2 Remove.

- a. Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16).
- b. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

WARNING

- Injury to personnel will result if explosive cartridge accidentally fires. Explosive cartridges may be fired by radio or radar frequencies and static electricity. Short circuit explosive cartridge connector pins to prevent accidental firing.
 - Injury to personnel and damage to equipment will result if cartridges are inadvertently activated. All cartridges and cartridge actuated devices shall be handled as live ammunition. Explosive cartridges which have been fired may retain an explosive residue capable of presenting a hazardous condition.
- c. Disconnect electrical connector, P6, from explosive cartridge (Figure 14-4-41-2, Detail A).
 - d. Install protective cap and plug on electrical connectors.
 - e. Install suitable shorting jumper on cartridge receptacle.
 - f. Remove explosive cartridge and packing from cartridge housing. Discard packing.

WARNING

Use dry-cleaning solvent in open area or in well-ventilated compartment, and keep clear of open flames. Dry-cleaning solvent is flammable. Repeated contact

with the skin can cause dermatitis. Wearing of nonporous gloves and apron is recommended.

- g. Clean inside of cartridge housing with dry-cleaning solvent, Item 124, Appendix D.

14-4-41.1.3 Install.

- a. Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16).
- b. On replacement explosive cartridge, perform the following:
 - (1) Remove and discard packing.
 - (2) Remove and discard collar.

CAUTION

Damage to explosive cartridge will result if part is scribed, scratched, or vibroetched. Damage to corrosion resistant surface will result. Do not scribe, scratch, or vibroetch service life date on explosive cartridge.

- c. Using marking pen, Item 211C, Appendix D, mark each side (four places) with service life date (month and year).
- d. Install cartridge in cartridge housing. **TORQUE CARTRIDGE TO 80 - 90 INCH-POUNDS** (Figure 14-4-41-2, Detail A).
- e. Using lockwire, Item 197, Appendix D, lock-wire cartridge.

WARNING

Injury to personnel and damage to equipment will result if cartridge is fired by stray voltage. When installing cartridge, do not connect electrical connector until stray voltage check has been performed.

- f. Perform stray voltage check (PARA 14-2-1).

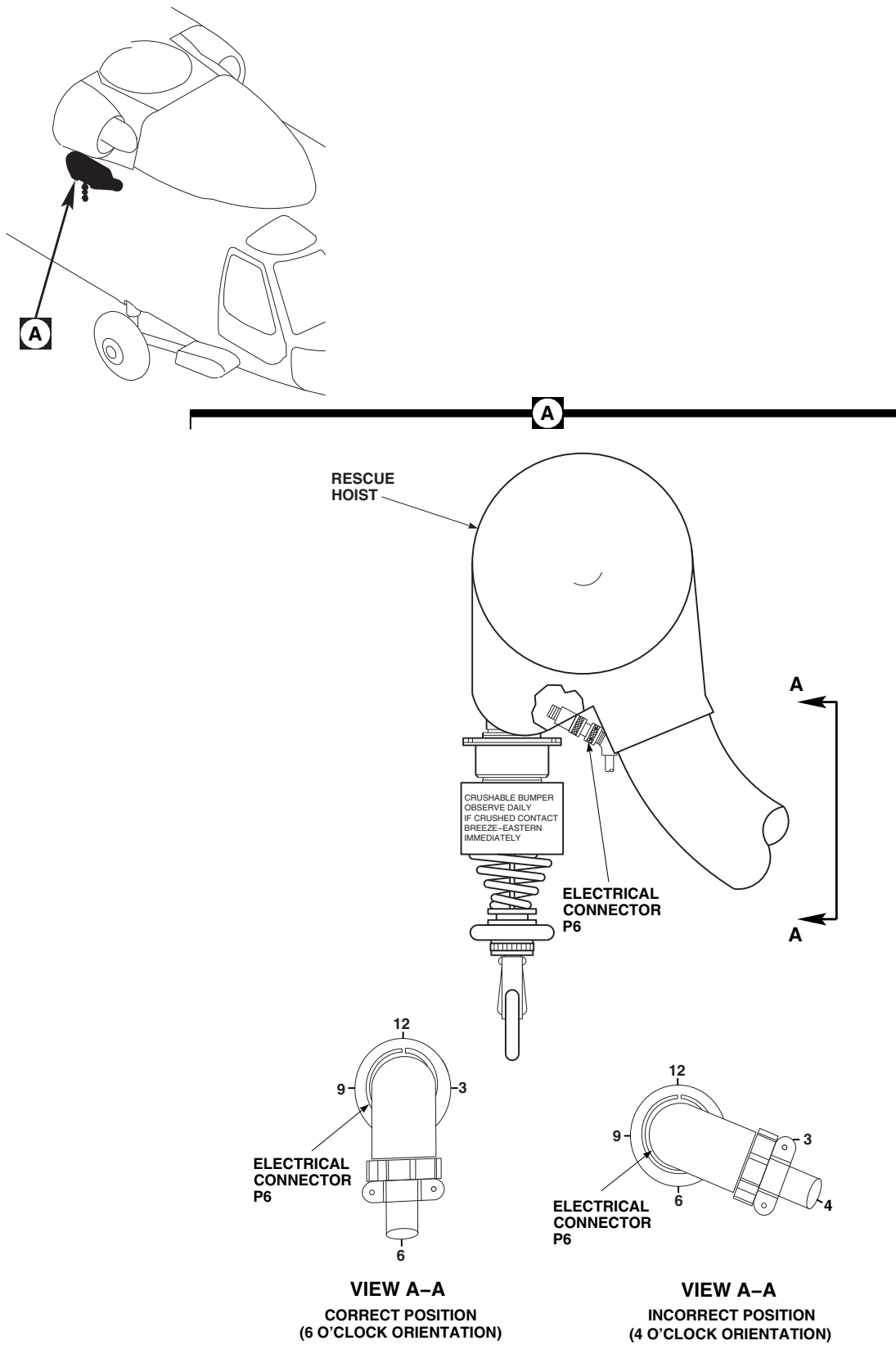


FIGURE 14-4-41-1. INSPECT EXPLOSIVE CARTRIDGE ELECTRICAL CONNECTOR, P6

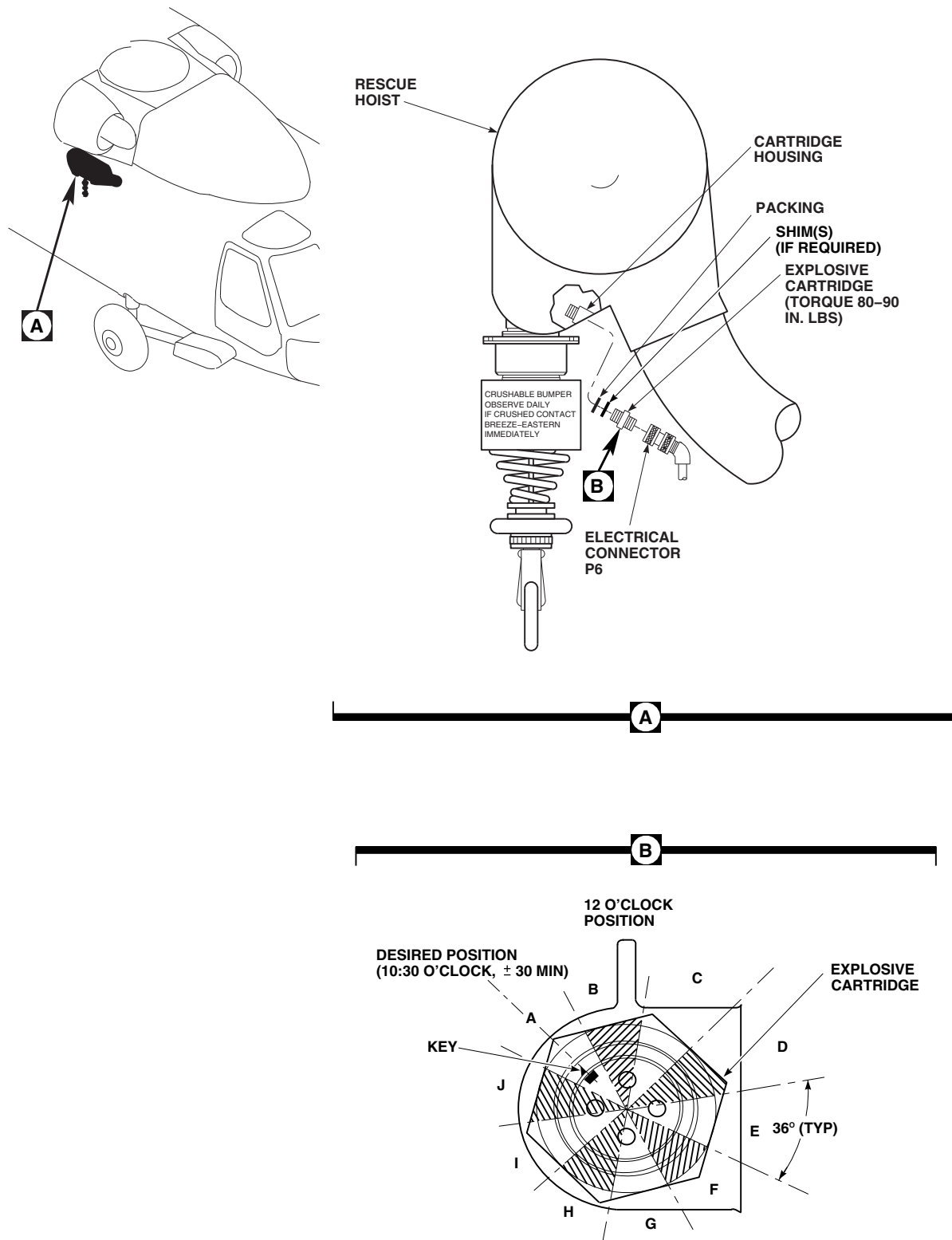
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- g. Remove shorting jumper from cartridge receptacle.
- h. Remove protective cap and plug from electrical connectors.
- i. Inspect and treat electrical connectors (PARA 1-7-20).
- j. Connect electrical connector, P6, to explosive cartridge. Make sure backshell connector is orientated to 6 o'clock position ($\pm$ 30 minutes tolerance) (Figure 14-4-41-1, View A-A).
- k. Using lockwire, Item 197, Appendix D, lockwire electrical connector to cartridge.
- l. Connect battery connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).



To prevent damage to electrical connector backshell and hoist fairing during operational check, make sure that a minimum clearance of 0.125-inch exists between connector backshell and fairing. If clearance is not present, repeat the necessary steps in PARA 14-4-41.1.1.

- m. Perform operational check of rescue hoist system (PARA 14-2-1).



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FIGURE 14-4-41-2. REPLACE EXPLOSIVE CARTRIDGE

SECTION 14-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

PARAGRAPH	TITLE	
14-5-1	Base Support Bushings (BENCH)	I
14-5-2	Support Arm Bushings (BENCH)	I

14-5-1 **BASE SUPPORT BUSHINGS (BENCH).**

PARAGRAPH BREAKDOWN

	PAGE
14-5-1.1 Replace Base Support Bushings (BENCH)	14-5-1-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Aircraft Repairer’s Toolkit
- Arbor Press
- Reamer Set, Hand

Materials/Parts

- Bolt, AN5-7
- Bolt, NAS428-5-44
- Bushings, NAS538-6P024
- Bushings, NAS77-5-024P

- Cloth, Cleaning, Item 102, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Dry Ice, Item 125, Appendix D
- Nut, AN316-5
- Primer, Item 258, Appendix D
- Washers, NAS1149D0563K
- Washers, NAS1149D0663K

References

- Appendix D
- PARA 14-4-1

14-5-1.1 **Replace Base Support Bushings (BENCH).**

14-5-1.1.1 **Remove.**

- Remove and disassemble rescue hoist (PARA 14-4-1).

NOTE

Bushing replacement is to be done in pairs.

- Remove bushings as follows:

- Install washer, NAS1149D0563K, on bolt, NAS428-5-44, flush against head (Figure 14-5-1-1).
- Install bushing, NAS77-5-024P, on bolt with flange flush against washer.
- Slide bolt through bushing bore of base support.
- Install a second bushing, NAS77-5-024P, on bolt with flange outward.
- Install a second washer, NAS1149D0563K, on bolt and secure with nut.

- (6) Tighten nut to draw out bushings from base support.
- (7) Repeat step b. (1) through step b. (6) for remaining bushings.

- c. Clean base support and bushing bores with dry-cleaning solvent, Item 124, Appendix D.
- d. Dry parts with cleaning cloth, Item 102, Appendix D.

14-5-1.1.2 Inspect/Repair.

- a. Inspect base support for damage (PARA 14-4-1).
- b. Inspect base support bushing bores for corrosion, and repair as follows:

NOTE

When line-reaming, the front and rear bushing measurement of 5.125 inches center to center must be maintained.

- (1) To remove corrosion and permit standard bushing replacement, line-ream bushing bore to 0.4993 - 0.4996 inch diameter, using hand reamer set.
- (2) To permit corrosion removal that exceeds standard bushing bore limits, line-ream bushing bore to 0.6300 or 0.7000 inch diameter, using hand reamer set. This increase in bore diameter will require the use of an oversized bushing or straight sleeve/flanged bushing combination.
- (3) If corrosion is still present after increasing bushing bore to 0.7000 inch diameter, base support must be replaced.

14-5-1.1.3 Install.

NOTE

- Replacement bushings should be chilled in dry ice, Item 125, Appendix D, for at least 30 minutes prior to installation.

- Full circumferential seating of bushing/sleeve in bore must be achieved with interference fit.

- a. For replacement of standard and oversized bushings, install as follows (Figure 14-5-1-1):

- (1) If a straight sleeve/flanged bushing combination is required, install sleeve bushing into bushing bore at this time using arbor press. Line-ream sleeve bushing, to 0.4993 - 0.4996-inch diameter, using hand reamer set.

- (2) Install washer, NAS1149D0663K, on bolt, AN5-7, flush against head.

- (3) Apply primer, Item 258, Appendix D, to bushing outside diameter and install bushing, NAS538-6P024, on bolt with flange flush against washer.

- (4) Slide bolt through bushing bore from inside of base support.

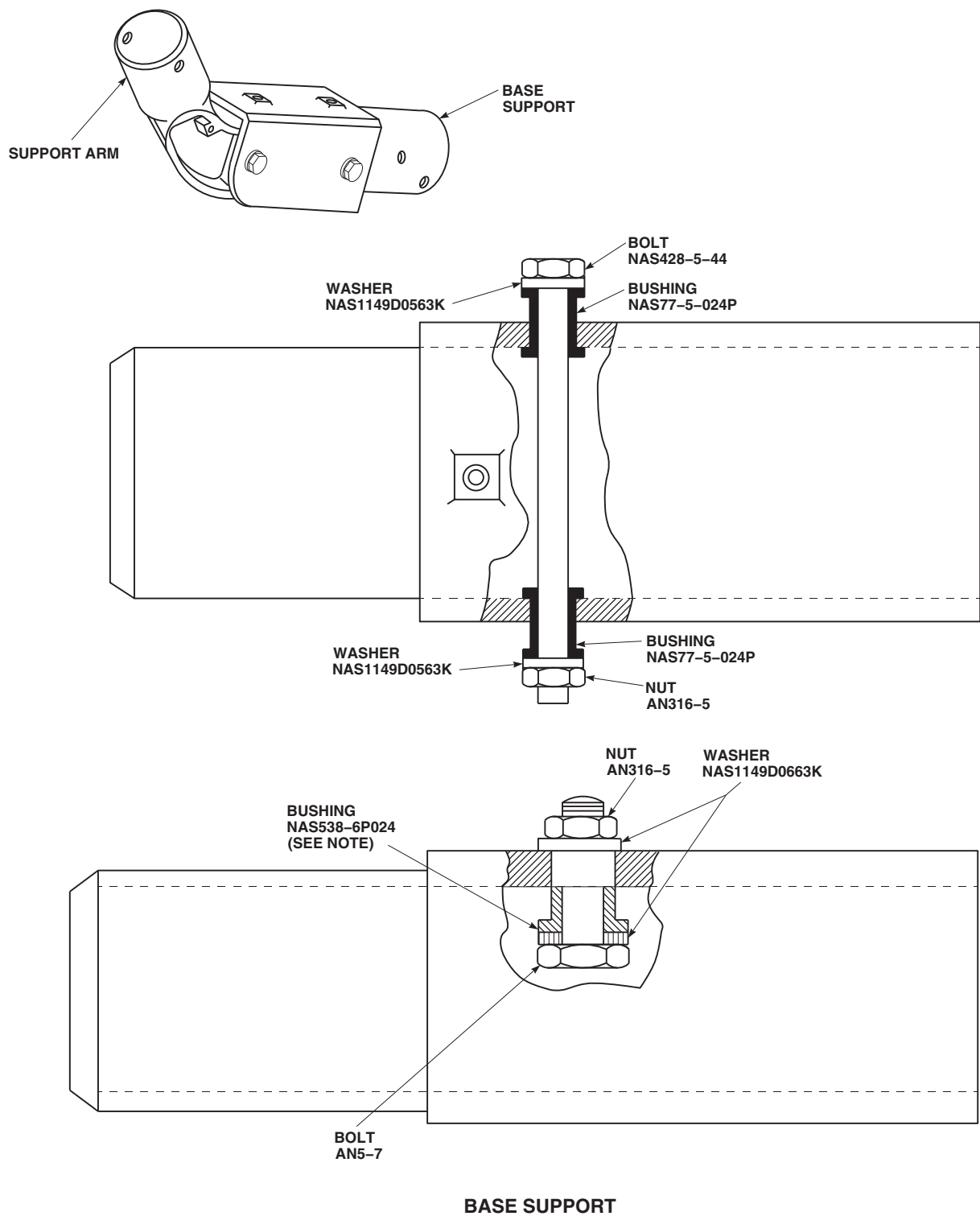
- (5) Install a second washer, NAS1149D0663K, on bolt and secure with nut.

- (6) Tighten nut to draw bushing into base support.

- (7) Repeat step a. (1) through step a. (6) for remaining bushings.

- (8) Line-ream bushings, to 0.3765 - 0.3780-inch diameter, using hand reamer set. Distance between front and rear bushings should be 5.125 inches center of hole to center of hole.

- b. Assemble and install rescue hoist (PARA 14-4-1).



NOTE

OVERSIZED BUSHING OR SLEEVE
BUSHING MAY BE REQUIRED.

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FIGURE 14-5-1-1. REPLACE BASE SUPPORT BUSHINGS (BENCH)

14-5-2 SUPPORT ARM BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
14-5-2.1 Replace Support Arm Bushings (BENCH)	14-5-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Aircraft Repairer’s Toolkit
- Drift Punch, Soft Material
- Reamer Set, Hand

Materials/Parts

- Bolt, NAS428-5-34
- Bushings, NAS538-6P032

- Cloth, Cleaning, Item 102, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Dry Ice, Item 125, Appendix D
- Nut, AN316-5
- Primer, Item 258, Appendix D
- Washers, NAS1149F0632P

References

- Appendix D
- PARA 14-4-1

14-5-2.1 Replace Support Arm Bushings (BENCH).

14-5-2.1.1 Remove.

- Remove and disassemble rescue hoist (PARA 14-4-1).

NOTE

Bushing replacement is to be done in pairs.

- Tap out bushings from opposite sides using suitable soft material drift punch (Figure 14-5-2-1).
- Clean support arm and bushing bores with dry-cleaning solvent, Item 124, Appendix D.

- Dry parts with cleaning cloth, Item 102, Appendix D.

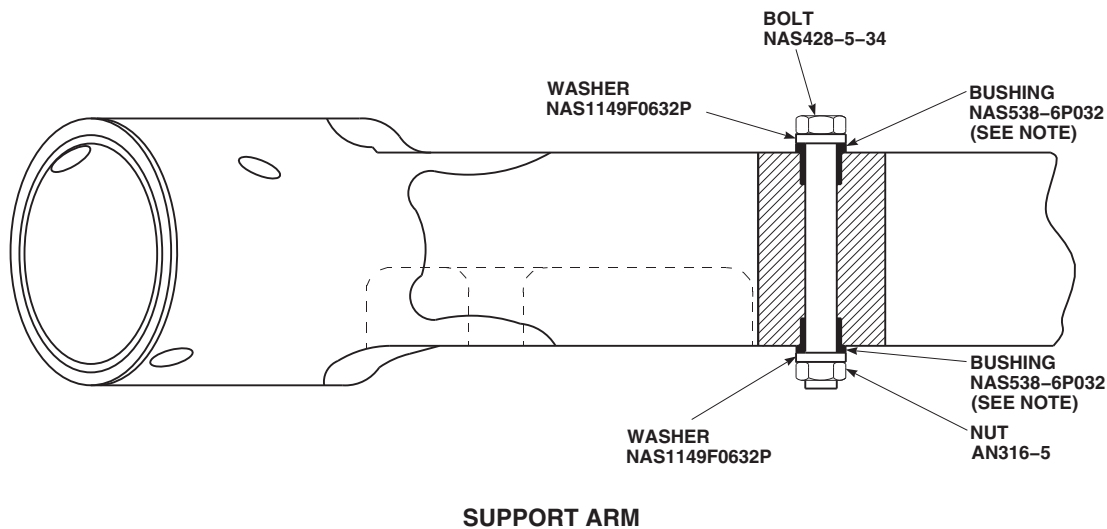
14-5-2.1.2 Inspect/Repair.

- Inspect support arm for damage (PARA 14-4-1).
- Inspect support arm bushing bores for corrosion, and repair as follows:

NOTE

When line-reaming, the forward and aft bushing measurement of 5.125 inches center to center must be maintained.

- To remove corrosion and permit standard bushing replacement, line-ream bushing

**NOTE**

OVERSIZED BUSHING OR SLEEVE
BUSHING MAY BE REQUIRED.

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FIGURE 14-5-2-1. REPLACE SUPPORT ARM BUSHINGS (BENCH)

bore to 0.4993 - 0.4996 inch diameter, using hand reamer set.

- (2) To permit corrosion removal that exceeds standard bushing bore limits, line-ream bushing bore to 0.6000 inch diameter, using hand reamer set. This increase in bore diameter will require the use of an oversized bushing.
- (3) If corrosion is still present after increasing bushing bore to 0.6000 inch diameter, support arm must be replaced.

14-5-2.1.3 Install.

NOTE

- Replacement bushings should be chilled in dry ice, Item 125, Appendix D, for at least 30 minutes prior to installation.
- Full circumferential seating of bushing/sleeve in bore must be achieved with interference fit.

- a. For replacement of standard and oversized bushings, install as follows (Figure 14-5-2-1):

- (1) Install washer on bolt flush against head.
- (2) Apply primer, Item 258, Appendix D, to bushing outside diameter and install bushing on bolt with flange flush against washer.
- (3) Slide bolt through bushing bore in support arm.
- (4) Apply primer, Item 258, Appendix D, to outside diameter of opposite bushing and install bushing on bolt with flange outward.
- (5) Install a second washer flush against bushing flange and secure with nut.
- (6) Tighten nut to draw bushings into support arm.
- (7) Repeat step a. (1) through step a. (6) for remaining bushings.

14-5-2-2 Change 4

- (8) Line-ream bushings to 0.3765 - 0.3780 inch diameter, using hand reamer set. Distance between forward and aft bushings should be 5.125 inches center of hole to center of hole.
- b. Assemble and install rescue hoist (PARA 14-4-1).

